CAMBRIDGE INTERNATIONAL · AS & A LEVEL PHYSICS

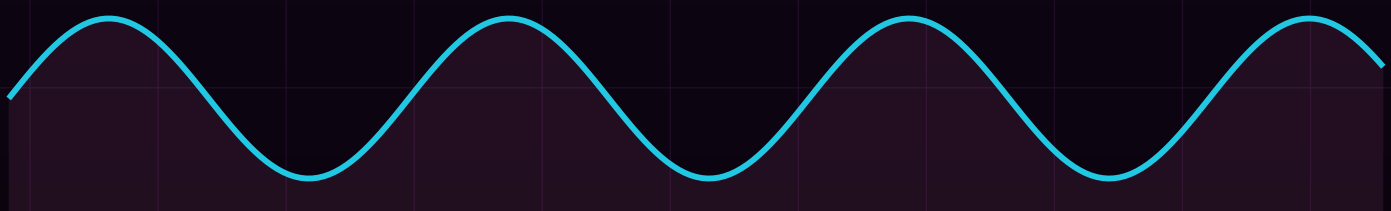
9702

PAPER 1

22 / 23

MULTIPLE CHOICE · COMPLETE COLLECTION 2022–2023

WITH ANSWER KEYS



PHYSICS (9702) · 14 PAPERS

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March 2022 · Paper 12

9702_m22_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2022**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What could **not** be a measurement of a physical quantity?

A 10 K **B** 11 $J N^{-1} m^{-1}$ **C** 17 $Pa m^3 N^{-1}$ **D** 25 T m

2 A computer memory stick is labelled as having a storage capacity of 128 GB.

The letter B stands for byte, which is a unit.

What is the equivalent storage capacity?

A 1.28×10^8 B

B 1.28×10^{11} B

C 1.28×10^{14} B

D 1.28×10^{17} B

3 A man of mass 75.2 kg uses a set of weighing scales to measure his mass three times. He obtains the following readings.

	mass / kg
reading 1	80.2
reading 2	80.1
reading 3	80.2

Which statement describes the precision and accuracy of the weighing scales?

A not precise to $\pm$ 0.1 kg and accurate to $\pm$ 0.1 kg

B not precise to $\pm$ 0.1 kg and not accurate to $\pm$ 0.1 kg

C precise to $\pm$ 0.1 kg and accurate to $\pm$ 0.1 kg

D precise to $\pm$ 0.1 kg and not accurate to $\pm$ 0.1 kg

4 Which statement about scalar and vector quantities is correct?

A A scalar quantity has direction but not magnitude.

B A scalar quantity has magnitude but not direction.

C A vector quantity has direction but not magnitude.

D A vector quantity has magnitude but not direction.

5 How can the acceleration of an object be determined?

A from the area under a displacement–time graph

B from the area under a velocity–time graph

C from the gradient of a displacement–time graph

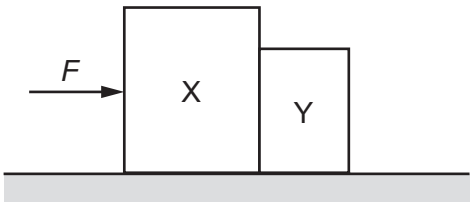
D from the gradient of a velocity–time graph

6 A sprinter takes a time of 11.0 s to run a 100 m race. She first accelerates uniformly from rest, reaching a speed of 10 m s^{-1}. She then runs at a constant speed of 10 m s^{-1} until the finish line.

What is the uniform acceleration of the sprinter for the first part of the race?

A 0.5 m s^{-2} **B** 0.91 m s^{-2} **C** 1.7 m s^{-2} **D** 5.0 m s^{-2}

7 A single horizontal force *F* is applied to a block X which is in contact with a separate block Y, as shown.



The blocks remain in contact as they accelerate along a horizontal frictionless surface. Air resistance is negligible. X has a greater mass than Y.

Which statement is correct?

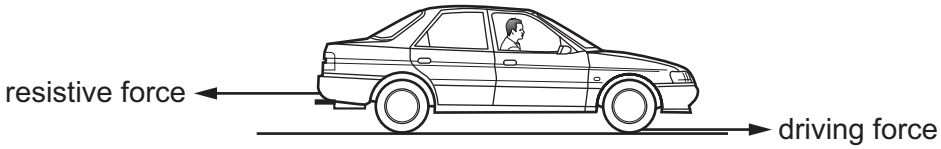
A The acceleration of X is equal to force *F* divided by the mass of X.

B The force that X exerts on Y is equal to *F*.

C The force that X exerts on Y is less than *F*.

D The force that X exerts on Y is less than the force that Y exerts on X.

8 A car of mass 750 kg has a horizontal driving force of 2.0 kN acting on it. It has a forward horizontal acceleration of 2.0 m s^{-2}.

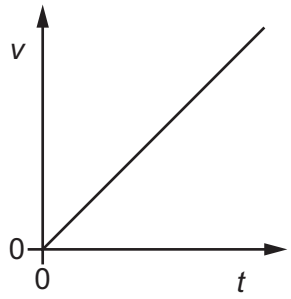


What is the resistive force acting horizontally?

A 0.50 kN **B** 1.5 kN **C** 2.0 kN **D** 3.5 kN

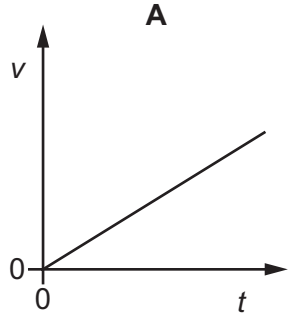
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9 An object falls freely from rest in a vacuum. The graph shows the variation with time t of the velocity v of the object.

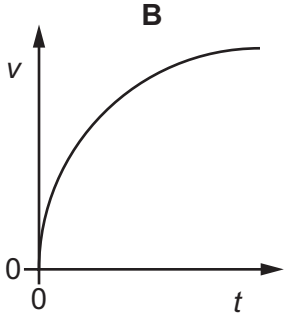


Which graph, **using the same scales**, represents the object falling in air?

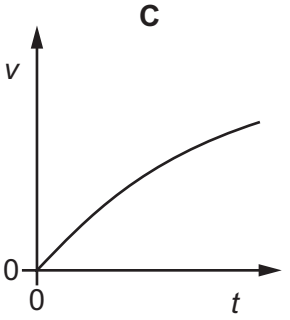
A



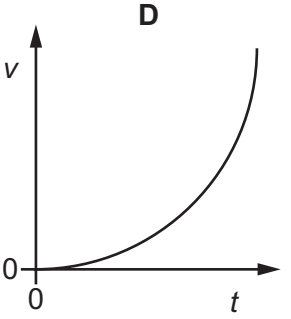
B



C



D

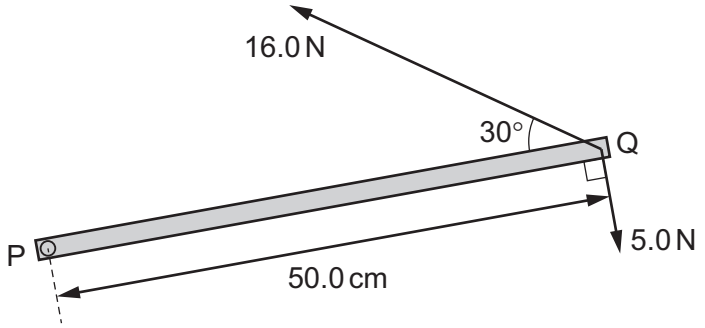


10 A rock of mass $2m$, travelling in deep space at velocity v, explodes into two parts of equal mass, one of which is then stationary.

What is the kinetic energy of the moving part after the explosion?

A $\frac{1}{2}mv^2$ **B** mv^2 **C** $\frac{3}{2}mv^2$ **D** $2mv^2$

11 A horizontal metal bar PQ of length 50.0 cm is hinged at end P. The diagram shows the metal bar viewed from above.



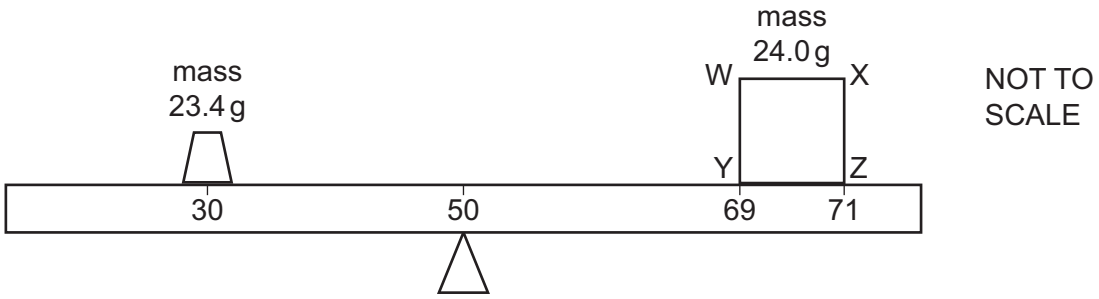
Two forces of 16.0 N and 5.0 N are in the horizontal plane and act on end Q, as shown.

What is the resultant moment about P due to the two forces?

A 1.5 N m **B** 4.4 N m **C** 6.5 N m **D** 9.4 N m

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12 A cube WXZY has sides of length 2.0 cm and mass 24.0 g. The cube rests on a metre rule of negligible mass. The geometrical centre of the cube is vertically above the 70.0 cm mark on the scale of the rule.



The cube has a non-uniform density so that its centre of gravity is **not** at its geometrical centre. The centre of gravity of the cube is in the plane of the diagram.

The rule rests on a pivot at the 50.0 cm mark. A mass of 23.4 g is placed vertically above the 30.0 cm mark. The rule is horizontal and in equilibrium.

What can be determined about the position of the centre of gravity of the cube?

A It must be somewhere along a horizontal line that is 0.5 cm from line WX.

B It must be somewhere along a horizontal line that is 0.5 cm from line YZ.

C It must be somewhere along a vertical line that is 0.5 cm from line WY.

D It must be somewhere along a vertical line that is 0.5 cm from line XZ.

13 A rigid sphere is held at rest on the sea bed. When the sphere is released, it rises to the surface of the sea. The seawater has a uniform density.

Which statement about the sphere, from its release until it reaches the surface, is correct?

A The sphere always moves with constant acceleration.

B The sphere always moves with constant velocity.

C The upthrust on the sphere always decreases.

D The upthrust on the sphere is always constant.

14 What is a unit for density?

A $N\,m^{-3}$ **B** $g\,mm^{-1}$ **C** $kg\,cm^{-2}$ **D** $\mu g\,mm^{-3}$

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15 The total energy input E_{in} in a process is partly transferred to useful energy output U and partly transferred to energy that is wasted W.

What is the efficiency of the process?

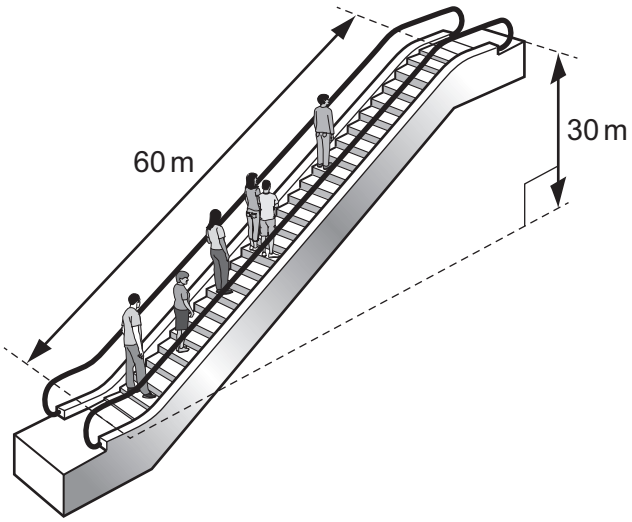
A $\frac{U}{E_{in}} \times 100\%$

B $\frac{W}{E_{in}} \times 100\%$

C $\frac{U}{W} \times 100\%$

D $\frac{U+W}{E_{in}} \times 100\%$

16 An escalator is 60 m long and lifts passengers through a vertical height of 30 m, as shown.



To drive the escalator against the forces of friction when there are no passengers requires a power of 2.0 kW.

The escalator is used by passengers of average mass 60 kg and the power to overcome friction remains constant.

How much power is required to drive the escalator when it is carrying 20 passengers and is travelling at 0.75 m s^{-1}?

A 4.4 kW **B** 6.4 kW **C** 8.8 kW **D** 10.8 kW

17 A rock of mass 40 kg is released from rest from a height of 20 m above the surface of a planet.

The rock has a kinetic energy of 32 kJ when it hits the surface of the planet. The planet does not have an atmosphere.

What is the weight of the rock on the surface of the planet?

A 1.6 N **B** 390 N **C** 1.6 kN **D** 64 kN

18 A metal wire is stretched. The wire obeys Hooke's law.

Which quantity has a value that does **not** change?

A extension

B strain

C stress

D Young modulus

19 An object is stretched until it reaches the elastic limit.

Which statement must describe the stress on the object when it is at the elastic limit?

A It is the maximum stress for which the object obeys Hooke's law.

B It is the maximum stress that can be applied to the object before it has elastic deformation.

C It is the maximum stress that can be applied to the object before it has plastic deformation.

D It is the maximum stress the object can withstand before it breaks.

20 Which statement about progressive waves is correct?

A They are always transverse waves.

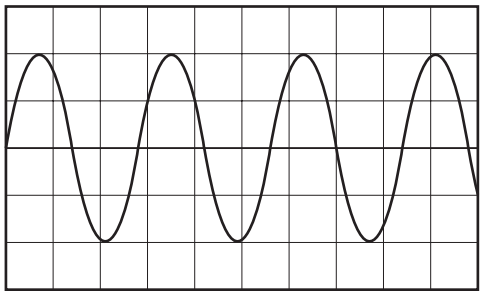
B They can exist in solids but not liquids.

C They decrease in frequency as their speed increases.

D They transfer energy away from their source.

21 A cathode-ray oscilloscope (CRO) is used to determine the frequency of a sound wave.

The diagram shows the waveform on the screen.



The time-base setting is $5.0\,\text{ms}\,\text{div}^{-1}$.

What is the best estimate of the frequency of the sound wave?

A 50 Hz **B** 71 Hz **C** 100 Hz **D** 143 Hz

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22 The warning signal on an ambulance has a frequency of 600 Hz. The speed of sound is 330 $m\,s^{-1}$. The ambulance is travelling with a constant velocity of 25 $m\,s^{-1}$ towards an observer. The ambulance passes, and then moves away from the observer with no change in velocity.



Which overall change in observed frequency takes place between the times at which the ambulance is a long way behind the observer and when it is a long way in front of the observer?

A 49 Hz **B** 84 Hz **C** 91 Hz **D** 98 Hz

23 Brief pulses of red, blue and green light are emitted from the Sun at the same time.

The pulses travel the same distance to reach Mars. Assume that the pulses travel in a vacuum for the full duration of their journey.

In which order would these pulses of light arrive at Mars?

A all arrive at the same time

B blue first, then green, then red

C red first, then blue, then green

D red first, then green, then blue

24 Two coherent progressive waves from different sources meet at a point.

Which condition **must** be satisfied for there to be zero resultant amplitude at the point where the waves meet?

A The two waves must be emitted from their sources with the same intensity.

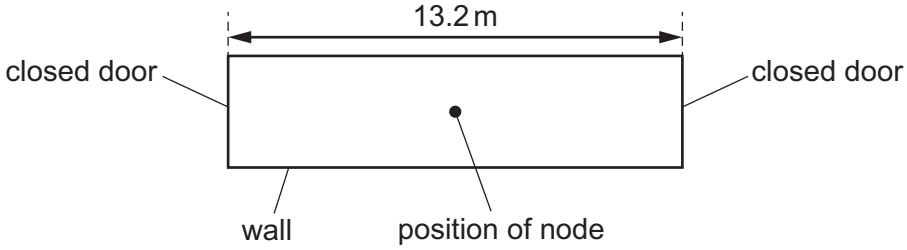
B The two waves must be in phase with each other at the point.

C The two waves must be travelling in opposite directions.

D The two waves must have the same amplitude at the point.

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25 A corridor is 13.2 m long and has closed doors that reflect sound at both ends. The speed of sound in the air in the corridor is 330 m s^{-1}.



What is the lowest frequency of sound that could create a stationary wave in the corridor with a node halfway along it?

A 0.040 Hz **B** 13 Hz **C** 25 Hz **D** 50 Hz

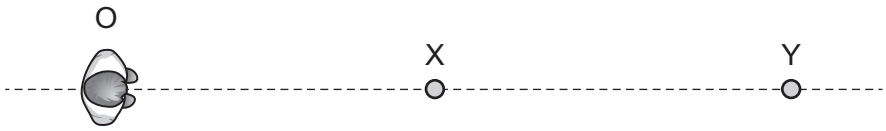
26 Water waves of wavelength λ are formed in a ripple tank. The waves are diffracted as they pass through a narrow gap of width d (d is greater than λ).

Which gap width and which wavelength will cause the largest decrease in the amount of diffraction?

	gap width	wavelength
A	$\frac{1}{2}d$	$\frac{1}{2}\lambda$
B	$\frac{1}{2}d$	2λ
C	$2d$	$\frac{1}{2}\lambda$
D	$2d$	2λ

27 Two loudspeakers X and Y emit sound waves that are in phase and of wavelength 0.75 m.

An observer O is able to stand anywhere on a straight line that passes through X and Y, as shown. The observer stands at a point where the sound waves from X and Y meet in phase.



What could be the distances OY and XY?

	distance OY / m	distance XY / m
A	1.25	3.50
B	2.00	2.75
C	2.75	2.00
D	3.25	1.50

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28 Light of a single wavelength is incident normally on a diffraction grating.

The resulting diffraction pattern is displayed on a screen.

Which change makes the first orders of intensity maxima further apart from each other on the screen?

A placing the screen closer to the diffraction grating

B using a diffraction grating with less separation between adjacent slits

C using a diffraction grating with more slits but keeping the same separation between adjacent slits

D using light with a shorter wavelength

29 For a current-carrying wire, the current can be calculated using the equation shown.

$$I = Anvq$$

What is the meaning of n?

A the number of charge carriers in the wire

B the number of charge carriers multiplied by the volume of the wire

C the number of charge carriers per unit length of the wire

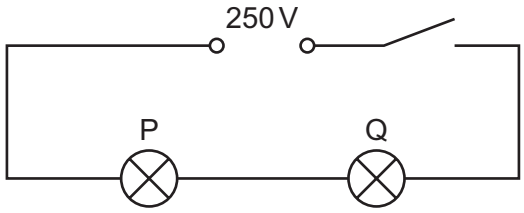
D the number of charge carriers per unit volume of the wire

30 The number of free electrons passing a point in a wire in 24 hours is 6.0×10^{23}.

What is the average current in the wire?

A 6.3 pA **B** 1.1 A **C** 67 A **D** 4.0 kA

31 In the circuit shown, lamp P is rated 250 V, 50 W and lamp Q is rated 250 V, 200 W. The two lamps are connected in series to a 250 V power supply.



Assume that the resistance of each lamp remains constant.

Which statement most accurately describes what happens when the switch is closed?

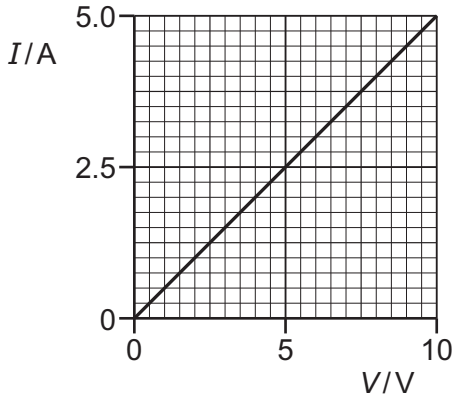
A Lamp P emits four times as much power as lamp Q.

B Lamp P emits twice as much power as lamp Q.

C Lamp Q emits four times as much power as lamp P.

D Lamp Q emits twice as much power as lamp P.

32 A piece of wire has a length of 0.80 m and a diameter of 5.0×10^{-4} m. The I–V characteristic of the wire is shown.



What is the resistivity of the metal from which the wire is made?

A $1.2 \times 10^{-7}\,\Omega\,\text{m}$

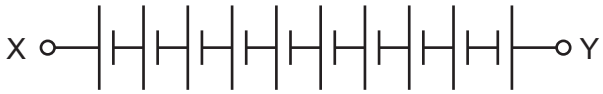
B $1.6 \times 10^{-7}\,\Omega\,\text{m}$

C $4.9 \times 10^{-7}\,\Omega\,\text{m}$

D $2.0 \times 10^{-6}\,\Omega\,\text{m}$

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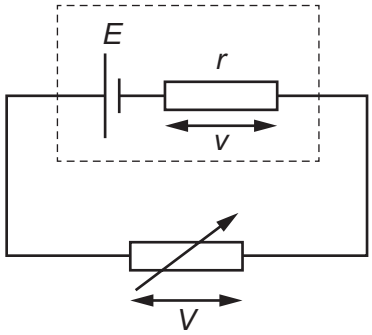
33 Ten cells, each of electromotive force (e.m.f.) 1.5 V, are connected together, as shown.



What is the combined e.m.f. between terminals X and Y?

A 8 V **B** 9 V **C** 12 V **D** 15 V

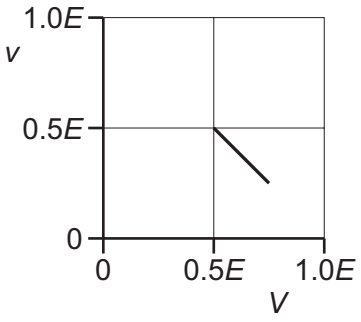
34 A cell of electromotive force (e.m.f.) *E* and internal resistance *r* is connected to a variable resistor, as shown.



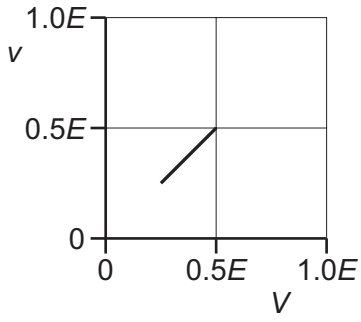
The resistance of the variable resistor is gradually increased from *r* to 3*r*.

Which graph shows the variation of the potential difference (p.d.) *v* across the internal resistance with the p.d. *V* across the variable resistor?

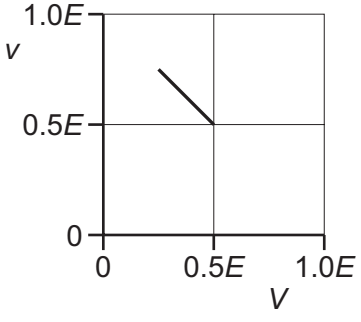
A



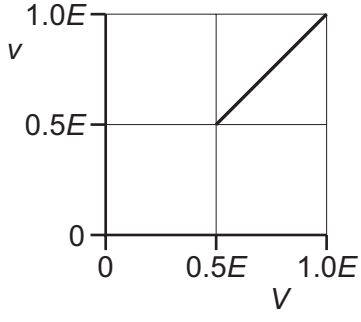
B



C



D

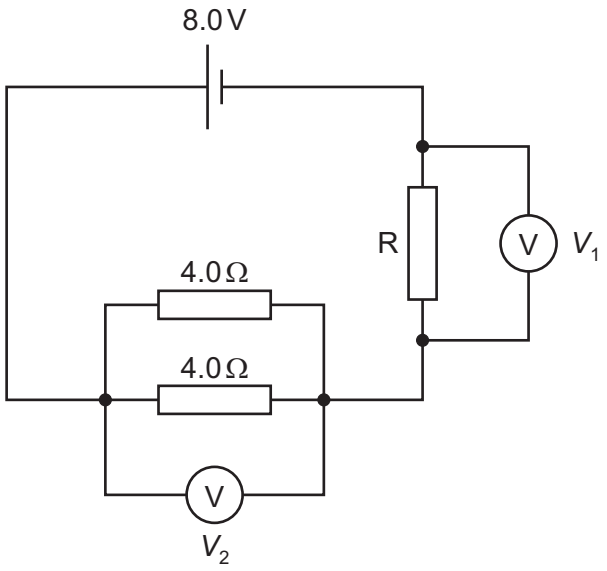


35 Each of Kirchhoff's two laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	the algebraic sum of currents into a junction is zero	charge
B	the algebraic sum of currents into a junction is zero	energy
C	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	charge
D	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	energy

36 A cell has an electromotive force (e.m.f.) of 8.0 V and negligible internal resistance. The cell forms part of a circuit, as shown.

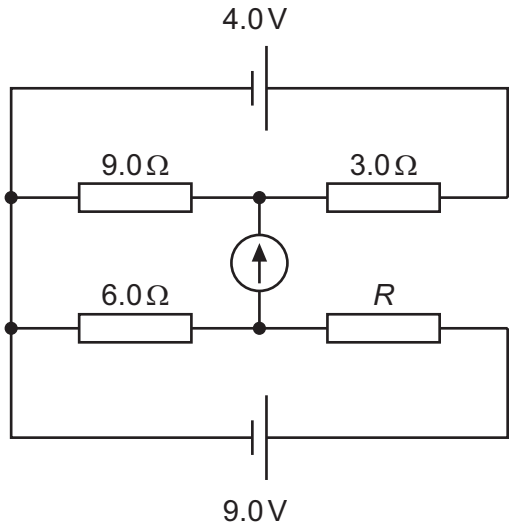


The reading V_1 is 4.0 V and the reading V_2 is also 4.0 V.

What is the resistance of resistor R?

A 0.50 Ω **B** 2.0 Ω **C** 4.0 Ω **D** 8.0 Ω

37 In the circuit shown, the cells have negligible internal resistance and the reading on the galvanometer is zero.



What is the value of resistor *R*?

A 2.0 Ω **B** 6.0 Ω **C** 12 Ω **D** 18 Ω

38 When α-particles are directed at gold leaf:

1 almost all α-particles pass through without deflection

2 a few α-particles are deviated through large angles.

What are the reasons for these effects?

	1	2
A	most α-particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α-particles bounce back from the gold surface
B	most α-particles miss all gold atoms	a few α-particles bounce off gold atoms
C	the gold nucleus is very small so most α-particles miss all nuclei	occasionally the path of an α-particle is close to a nucleus
D	the positive charge in an atom is not concentrated enough to deflect an α-particle	occasionally an α-particle experiences many small deflections in the same direction

39 A nucleus X is radioactive and decays into a nucleus Y.

X and Y are isotopes of the same element.

Which combination of particles could have been emitted during the decay process?

A 1 α-particle and 1 β^- particle

B 1 α-particle and 2 β^- particles

C 2 α-particles and 1 β^- particle

D 2 α-particles and 2 β^- particles

40 A positively charged meson consists of a quark and an antiquark.

What could be the quark and antiquark?

A charm and antiup

B down and antitop

C strange and antibottom

D up and antistrange

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Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2022**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	B	**1**
2	B	**1**
3	D	**1**
4	B	**1**
5	D	**1**
6	D	**1**
7	C	**1**
8	A	**1**
9	C	**1**
10	D	**1**
11	A	**1**
12	C	**1**
13	D	**1**
14	D	**1**
15	A	**1**
16	B	**1**
17	C	**1**
18	D	**1**
19	C	**1**
20	D	**1**
21	B	**1**
22	C	**1**
23	A	**1**
24	D	**1**
25	C	**1**
26	C	**1**
27	D	**1**
28	B	**1**

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Question	Answer	Marks
29	D	**1**
30	B	**1**
31	A	**1**
32	C	**1**
33	C	**1**
34	A	**1**
35	A	**1**
36	B	**1**
37	C	**1**
38	C	**1**
39	B	**1**
40	D	**1**

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June 2022 · Paper 11

9702_s22_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2022**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which term represents a physical quantity?

A metre

B percentage uncertainty

C quark flavour

D spring constant

2 Which two units are identical when expressed in terms of SI base units?

A $\mathrm{J\,C^{-1}}$ and $\mathrm{kg\,m^2\,A^{-1}\,s^{-2}}$

B $\mathrm{J\,s}$ and $\mathrm{kg\,m^2\,s^{-1}}$

C $\mathrm{N\,m}$ and $\mathrm{kg\,m^3\,s^{-2}}$

D $\mathrm{N\,s}$ and $\mathrm{kg\,m\,s^{-3}}$

3 A value for the acceleration of free fall on Earth is given as $(10 \pm 2)\,\mathrm{m\,s^{-2}}$.

Which statement is correct?

A The value is accurate but not precise.

B The value is both precise and accurate.

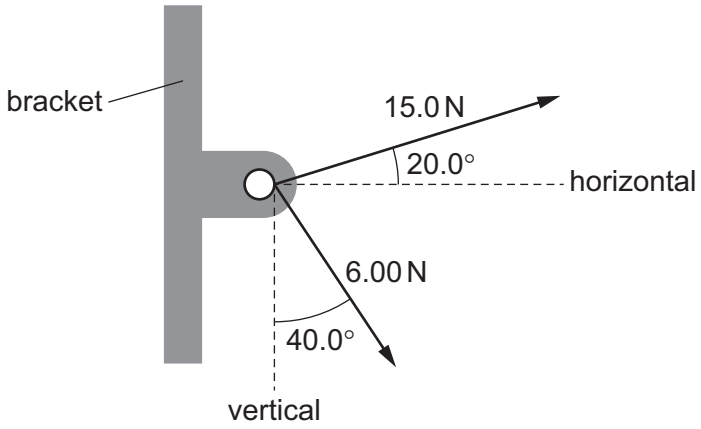
C The value is neither precise nor accurate.

D The value is precise but not accurate.

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[Turn over

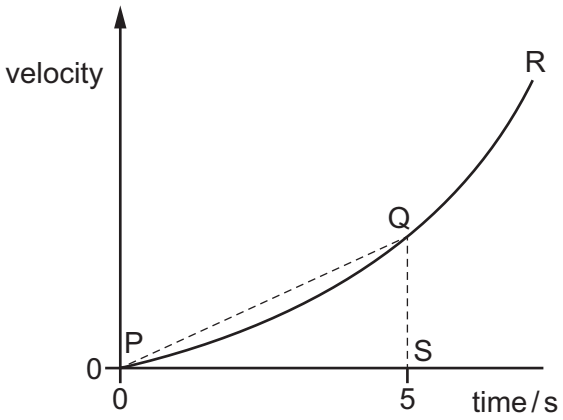
4 Two cables are attached to a bracket and exert forces as shown.



What are the magnitudes of the horizontal and vertical components of the resultant of the two forces?

	horizontal component / N	vertical component / N
A	9.73	0.534
B	9.73	10.2
C	18.0	0.534
D	18.0	10.2

5 The curved line PQR is the velocity–time graph for a car starting from rest.



What is the average acceleration of the car over the first 5 s?

A the area below the curve PQ

B the area of the triangle PQS

C the gradient of the straight line PQ

D the gradient of the tangent at Q

6 A ball is thrown horizontally with a speed of $10.0\,m\,s^{-1}$ above horizontal ground. The ball hits the ground after a time of 3.0 s.

Air resistance is negligible.

What is the speed of the ball just before it hits the ground?

A $10\,m\,s^{-1}$ **B** $29\,m\,s^{-1}$ **C** $31\,m\,s^{-1}$ **D** $39\,m\,s^{-1}$

7 An object is moving along the ground in a straight line at a constant speed.

Which statement about the resultant force acting on the object is correct?

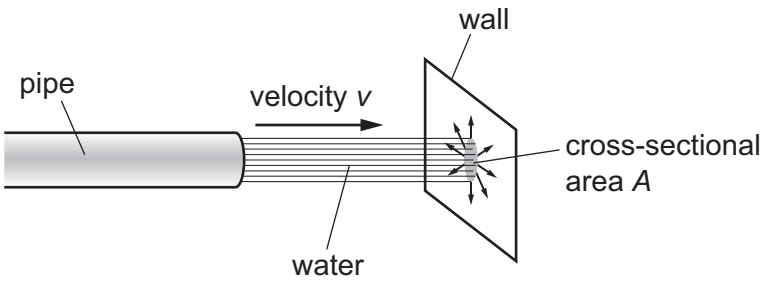
A The resultant force acting on the object is equal to its weight.

B The resultant force acting on the object is equal to the product of its mass and its velocity.

C The resultant force acting on the object is equal to the resistive force.

D The resultant force acting on the object is equal to zero.

8 Water flows out of a pipe and hits a wall.



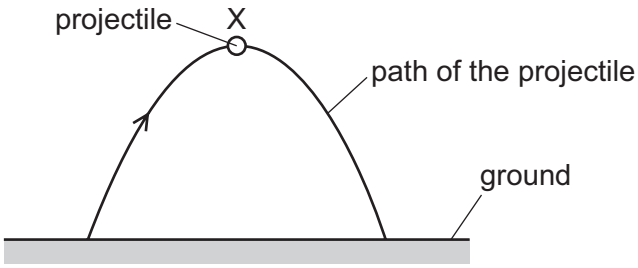
When the jet of water hits the wall, it has horizontal velocity v and cross-sectional area A.

The density of the water is ρ. The water does not rebound from the wall.

What is the force exerted on the wall by the water?

A $\frac{\rho v}{A}$ **B** $\frac{\rho v^2}{A}$ **C** $\rho A v$ **D** $\rho A v^2$

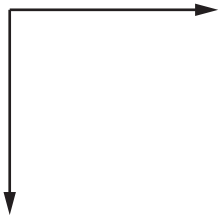
9 A projectile is launched at an angle above horizontal ground and travels through the air.



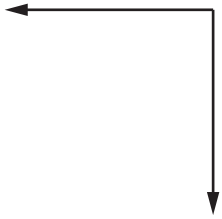
The projectile reaches its maximum height at position X. Assume that no upthrust acts on the projectile.

Which diagram shows the **directions** of the force or forces acting on the projectile at position X?

A



B



C

D



10 What is a statement of the principle of conservation of momentum?

A A force is equal to the rate of change of momentum of the object upon which it acts.

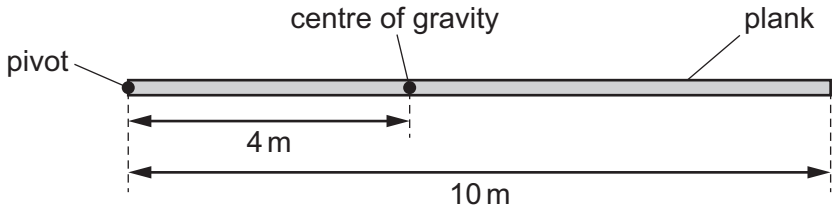
B In a perfectly elastic collision, the relative momentum of the objects before impact is equal to their relative momentum after impact.

C The momentum of an object is the product of the mass of the object and its velocity.

D The total momentum of a system of interacting objects remains constant, providing no resultant external force acts on the system.

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11 A horizontal wooden plank is pivoted at one end, as shown.



The plank has a mass of 100 kg and a length of 10 m. The centre of gravity of the plank is a distance of 4 m from the pivot.

What is the moment of the weight of the plank about the pivot?

A 4×10^2 N m **B** 5×10^2 N m **C** 4×10^3 N m **D** 5×10^3 N m

12 When **must** an object be in equilibrium?

A when no resultant force acts on the object

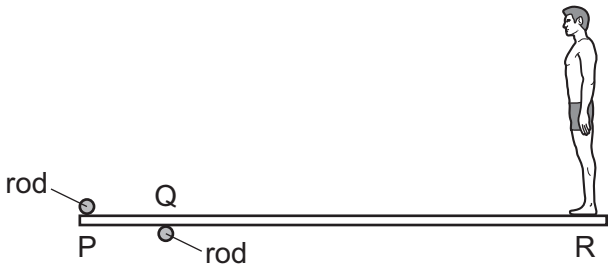
B when no resultant force and no resultant torque act on the object

C when no resultant torque acts on the object

D when the upward force on the object is equal and opposite to its weight

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13 A uniform diving board is held by two fixed rods at points P and Q. A person stands at end R of the diving board, as shown.

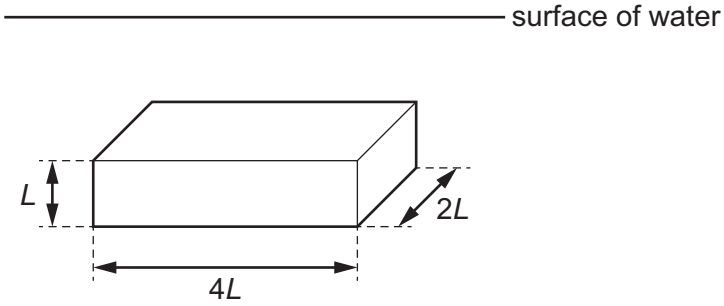


The forces exerted by the rods on the board are vertical. The board remains in equilibrium as the person slowly moves towards point Q from end R.

Which row describes the changes to the magnitudes of the forces exerted by the rods on the board?

	force at P	force at Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

14 A solid block has sides of length L, $2L$ and $4L$. The block is submerged in water of uniform density so that the faces with the largest area are horizontal, as shown.



The upthrust acting on the block is U.

The block is now rotated to a new position so that the faces with the smallest area are horizontal. The block remains fully submerged in the water.

What is the upthrust acting on the block in its new position?

A $\frac{U}{4}$ **B** U **C** $2U$ **D** $4U$

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15 In a large container in an oil refinery, three oils of different densities are mixed. No chemical activity occurs.

The mixture consists of:

1200 kg of oil of density 1100 kg m^{-3}

1500 kg of oil of density 860 kg m^{-3}

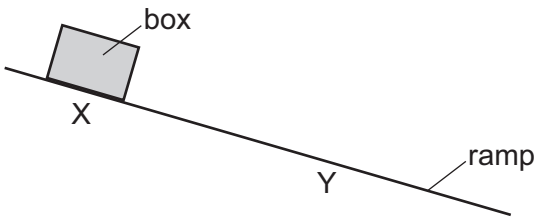
4000 kg of oil of density 910 kg m^{-3}.

What is the density of the mixture?

A 927 kg m^{-3} **B** 933 kg m^{-3} **C** 957 kg m^{-3} **D** 1045 kg m^{-3}

16 A box slides down a rough ramp.

The change in the gravitational potential energy of the box is 16 J as it moves between positions X and Y. The box has 24 J of kinetic energy at X and 35 J of kinetic energy at Y.



How much work is done against the frictional force?

A 5 J **B** 19 J **C** 27 J **D** 43 J

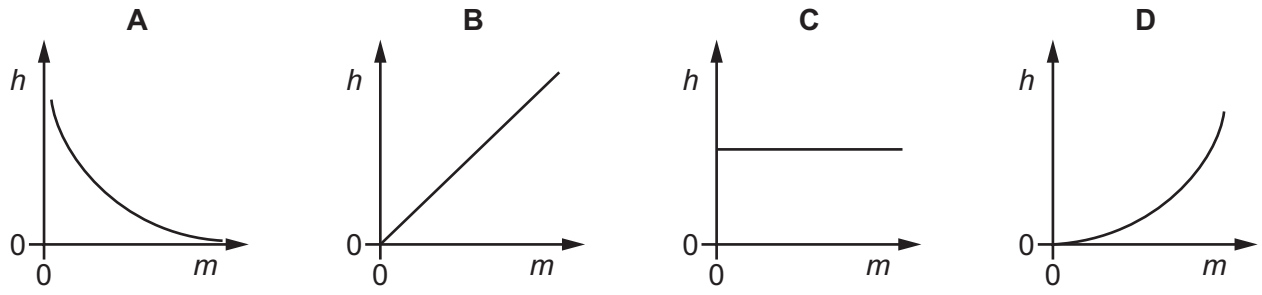
17 The total energy supplied to an electric motor is E. Energy Q is wasted and the remaining energy does useful work.

What is the efficiency of the motor?

A $\frac{Q}{E}$ **B** $\left(\frac{Q}{E}\right) - 1$ **C** $1 - \left(\frac{Q}{E}\right)$ **D** $\frac{(1-Q)}{E}$

18 Objects with different masses are placed on the horizontal surface of a table. The objects are then raised to different heights above the table. The gain in gravitational potential energy of each object is the same.

Which graph best shows the variation of the height *h* of the objects above the table with their mass *m*?



19 Two wires, P and Q, are made from the same metal and hang vertically from a steel girder.

Wire Q has half the length and twice the diameter of wire P.

Identical masses are attached to the bottom of each wire. Both wires obey Hooke's law as they are stretched by the weight of the masses.

What is the ratio $\frac{\text{extension of wire P}}{\text{extension of wire Q}}$?

A $\frac{8}{1}$ **B** $\frac{4}{1}$ **C** $\frac{1}{1}$ **D** $\frac{1}{2}$

20 Which statement about elastic and plastic deformation **must** be correct?

A Elastic deformation and plastic deformation are proportional to the applied force.

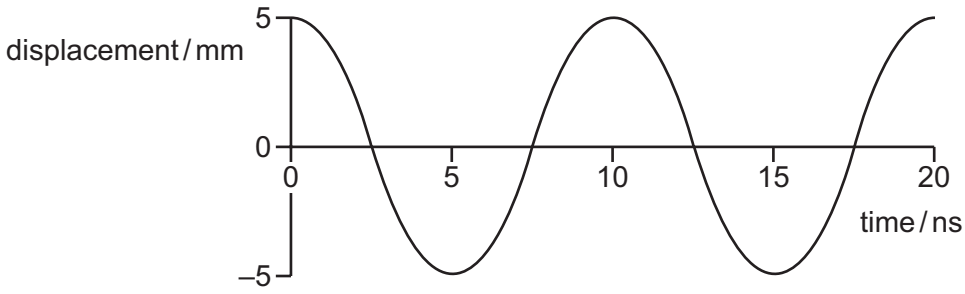
B Elastic deformation and plastic deformation cause no change in volume.

C Elastic deformation causes heating of the material but plastic deformation does not.

D Elastic deformation is reversible but plastic deformation is not.

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21 The graph shows the variation with time of the displacement of a particle as a progressive wave passes.



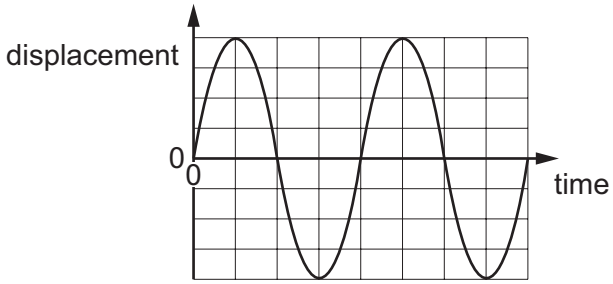
What are the frequency and the amplitude of the wave?

	frequency / MHz	amplitude / mm
A	100	5
B	200	5
C	100	10
D	200	10

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[Turn over

22 The graph shows the variation of the displacement of an air particle with time as a sound wave passes through air.

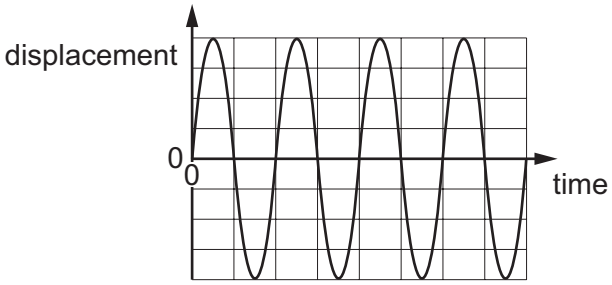


The intensity of the sound is halved while the frequency remains constant.

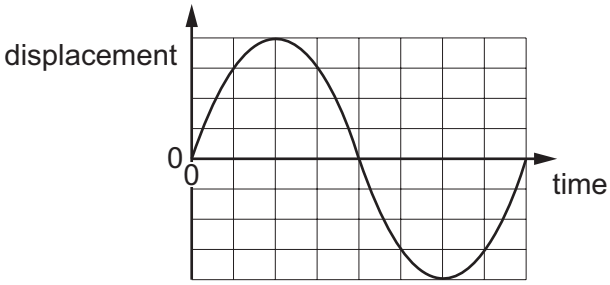
The four graphs below are drawn to the same scale as the graph above.

Which graph shows the displacement of the air particle?

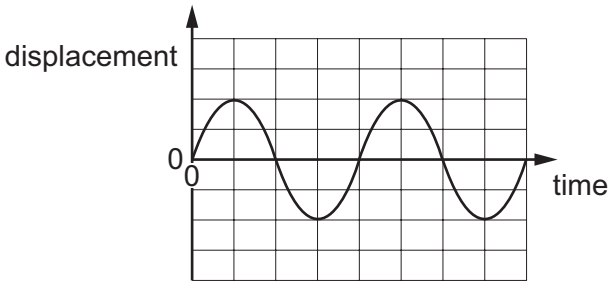
A



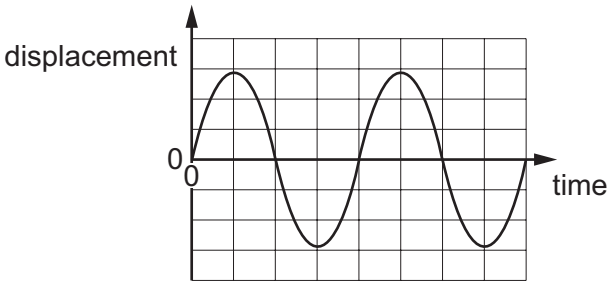
B



C



D



23 Which statement is correct?

A Gases cannot transmit longitudinal waves.

B Longitudinal sound waves cannot form stationary waves.

C Solids can transmit both transverse and longitudinal waves.

D Transverse waves cannot pass through a vacuum.

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24 A car is travelling at a constant velocity directly towards a man standing in the middle of the road.

The driver sounds the car's horn as a warning. The horn emits a sound wave of constant frequency.

The frequency of the sound heard by the man is different from the frequency of the sound emitted by the horn.

Which statement is correct?

A The frequency of the sound emitted by the horn is greater than the frequency of the sound heard by the man.

B The frequency of the sound heard by the man depends on the distance between the car and the man.

C The sound waves continually accelerate as they move from the horn to the man.

D The wavelength of the sound heard by the man is less than the wavelength of the sound emitted by the horn.

25 Which statement about electromagnetic waves is correct?

A A wave of wavelength 5.0×10^{-6} m is invisible to the human eye.

B They can all travel at different speeds in free space.

C They cannot be polarised.

D They consist of vibrating atoms.

26 A stationary wave is set up on a string that is stretched between two fixed points that are 48 cm apart.

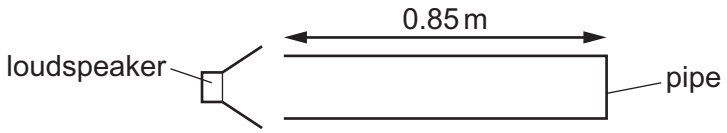
At one instant, the appearance of the string is as shown.



What is the wavelength of the stationary wave?

A 16 cm **B** 32 cm **C** 48 cm **D** 72 cm

27 A pipe, closed at one end, has a loudspeaker at the open end. For some frequencies of sound from the loudspeaker, a stationary sound wave is formed in the air within the pipe with an antinode at the open end of the pipe.



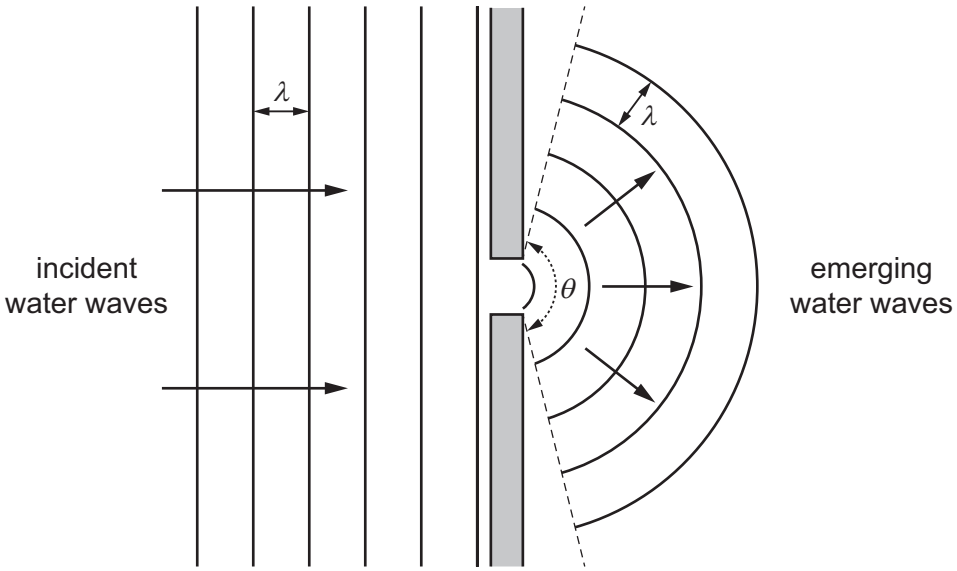
The length of the pipe is 0.85 m.

The speed of sound in air is 340 m s^{-1}.

Which frequency of sound from the loudspeaker would **not** produce a stationary wave?

A 100 Hz **B** 200 Hz **C** 300 Hz **D** 500 Hz

28 Water waves of wavelength λ are incident normally on an obstacle with a narrow gap. The width of the gap is equal to λ. The waves from the gap emerge over an angle θ, as shown.



The gap is slowly widened.

Which changes, if any, occur to θ and to the wavelength of the emerging waves?

	θ	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

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29 Light of a single frequency passes through two narrow slits and produces an interference pattern on a screen some distance away. The interference fringes are very close together.

Which change would **increase** the distance between the fringes?

A Increase the brightness of the light source.

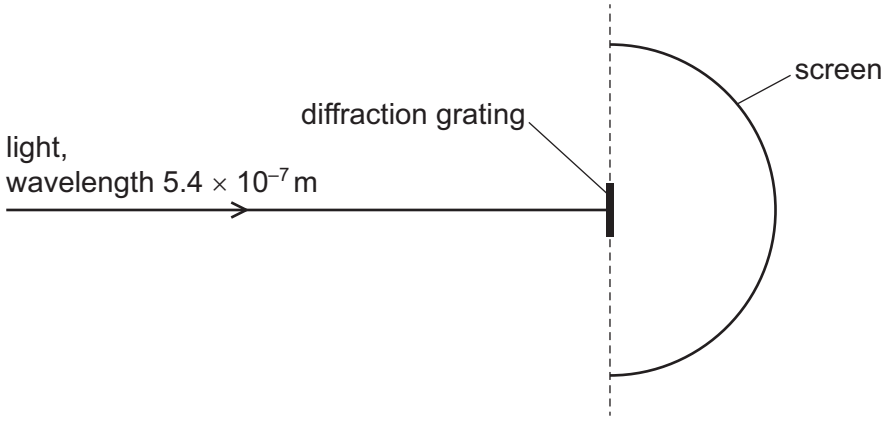
B Increase the distance between the slits and the screen.

C Increase the distance between the two slits.

D Increase the frequency of the light used.

30 Light of wavelength 5.4×10^{-7} m is incident normally on a diffraction grating.

The separation between adjacent lines in the grating is 2.0×10^{-6} m. The light that emerges from the grating falls on a semicircular screen, as shown in the view from above.



VIEW FROM ABOVE

The grating is at the centre of the semicircle, and the lines of the grating are vertical.

How many bright dots are formed on the screen?

A 3 **B** 4 **C** 6 **D** 7

31 A straight copper wire of diameter 0.42×10^{-3} m has a number density of free electrons of 8.5×10^{28} m^{-3}. In a given time interval, a charge of 0.15 C moves through the wire.

What is the average displacement of the free electrons along the wire in this time interval?

A 3.3×10^{-8} m **B** 2.0×10^{-5} m **C** 8.0×10^{-5} m **D** 2.5×10^{-4} m

32 What is the definition of the potential difference (p.d.) across a component?

A the electrical power supplied to the component

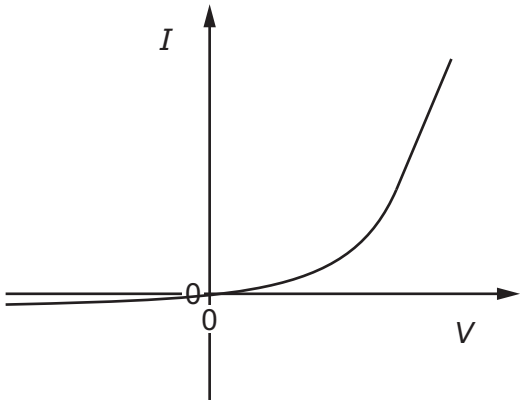
B the energy transferred to the component per unit charge

C the product of the current in the component and its resistance

D the voltage across the component

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33 The graph shows the *I*–*V* characteristic for a semiconductor diode.



Which statement can be deduced from the graph?

A Above a certain positive potential difference the diode obeys Ohm's law.

B Current is directly proportional to potential difference when the current in the diode is in one direction.

C The diode has zero resistance when the current in the diode is in one direction.

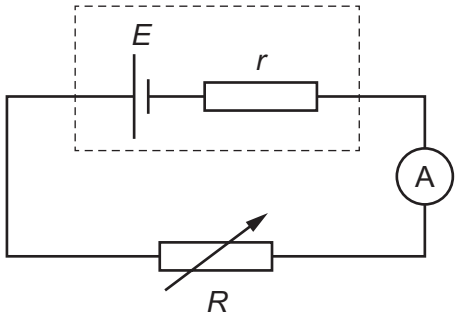
D The resistance of the diode depends upon the potential difference across it.

34 A wire has a resistance of 30 Ω. A second wire is made from the same material, has the same mass and is three times as long as the first wire.

What is the resistance of the second wire?

A 10 Ω **B** 30 Ω **C** 90 Ω **D** 270 Ω

35 A cell has internal resistance *r* and electromotive force (e.m.f.) *E*. The cell is connected in series with an ammeter and a variable resistor of resistance *R*.



When *R* is 10 Ω the ammeter reads 0.3 A.

When *R* is 5 Ω the ammeter reads 0.4 A.

What is the value of *E*?

A 0.5 V **B** 2 V **C** 3 V **D** 6 V

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36 The sum of the currents entering a junction in an electrical circuit is always equal to the sum of the currents leaving the junction.

Why is this?

A It is a consequence of the conservation of charge.

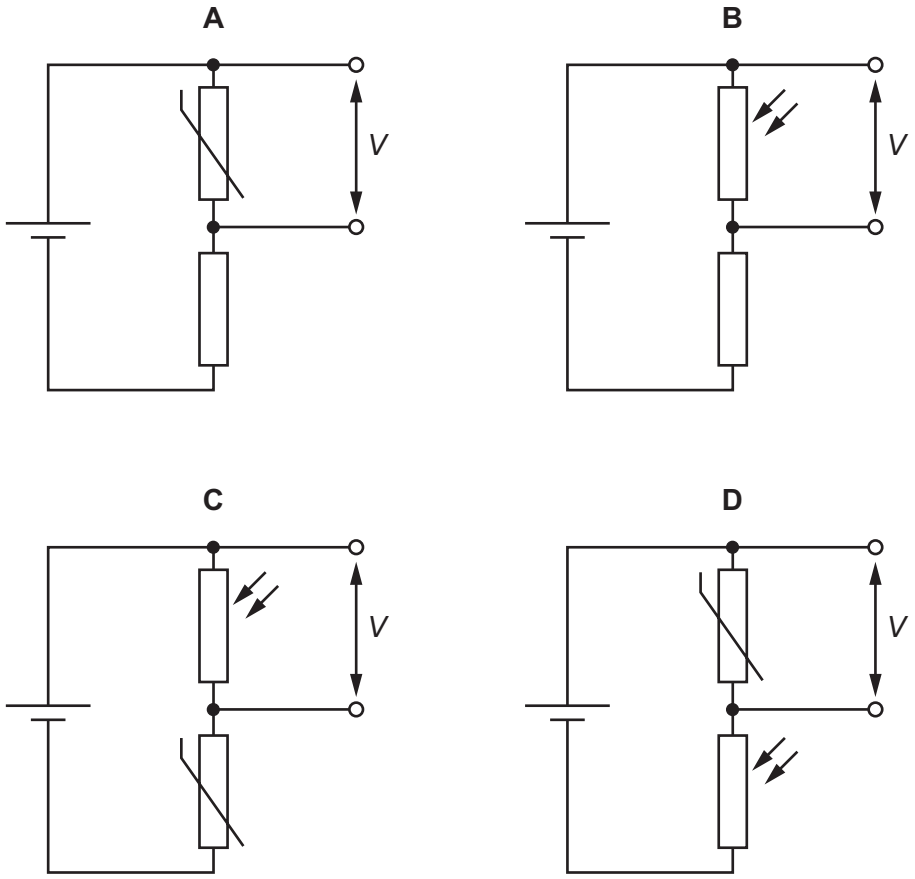
B It is a consequence of the conservation of electromotive force.

C It is a consequence of the conservation of energy.

D It is a consequence of the conservation of potential difference.

37 In the circuits shown, the temperature remains constant.

In which circuit does the potential difference (p.d.) *V* increase with increasing light intensity?



38 Carbon-14 decays into nitrogen-14 by emitting a β^- particle.

Which statement explains why the β^- particles are emitted with a range of different kinetic energies?

A The carbon-14 nuclei have slightly different masses.

B The emitted β^- particles have a range of different masses.

C The energy released in the decay process is different for each carbon-14 nucleus that decays.

D The energy released in the decay process is shared between the nitrogen-14 nucleus, a β^- particle and an antineutrino.

39 A nucleus of a radioactive element emits an α-particle, then a β^- particle and then another β^- particle.

Which statement describes the final element that is produced?

A It is a different element of higher proton number than the original element.

B It is a different element of lower nucleon number than the original element.

C It is an isotope of the original element.

D It is the same element as the original element but with a different proton number.

40 How many hadrons, baryons and mesons are there in a nucleus of $^{9}_{4}\text{Be}$?

	hadrons	baryons	mesons
A	9	4	5
B	9	5	4
C	9	9	0
D	13	9	0

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Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2022**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

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Question	Answer	Marks
1	D	**1**
2	B	**1**
3	A	**1**
4	C	**1**
5	C	**1**
6	C	**1**
7	D	**1**
8	D	**1**
9	B	**1**
10	D	**1**
11	C	**1**
12	B	**1**
13	A	**1**
14	B	**1**
15	A	**1**
16	A	**1**
17	C	**1**
18	A	**1**
19	A	**1**
20	D	**1**
21	A	**1**
22	D	**1**
23	C	**1**
24	D	**1**
25	A	**1**
26	B	**1**
27	B	**1**
28	A	**1**

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Question	Answer	Marks
29	B	**1**
30	D	**1**
31	C	**1**
32	B	**1**
33	D	**1**
34	D	**1**
35	D	**1**
36	A	**1**
37	D	**1**
38	D	**1**
39	C	**1**
40	C	**1**

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June 2022 · Paper 12

9702_s22_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2022**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which estimate is reasonable?

A 1×10^{-3} kg for the mass of a grain of sand

B 1×10^{-2} m^3 for the volume of a tennis ball

C 1×10^{0} J for the work done lifting an apple from waist height to head height

D 1×10^{4} W for the power of a light bulb in a house

2 What is the symbol for the SI base unit of temperature?

A C **B** K **C** °C **D** °K

3 Which statement about systematic errors is **not** correct?

A A systematic error can be caused by using an incorrectly calibrated instrument.

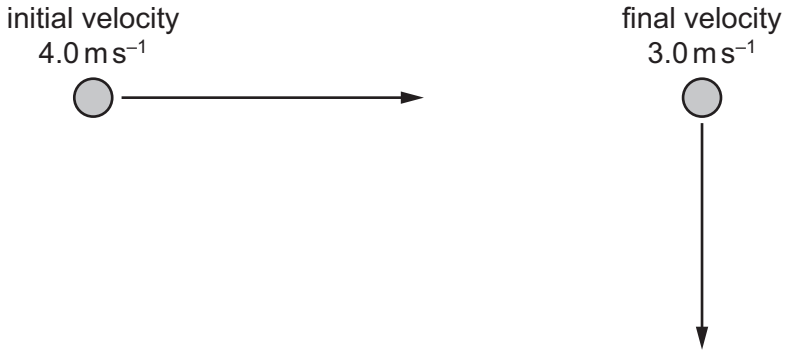
B One particular type of systematic error can affect all the measurements by the same amount.

C The effect of a systematic error can be reduced by repeating and averaging the measurements.

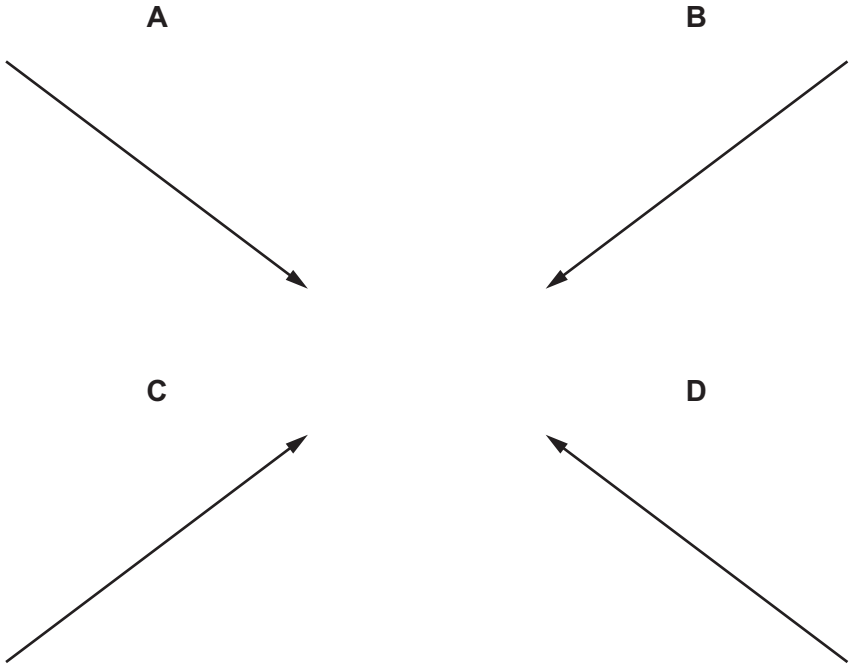
D Zero error is a type of systematic error.

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4 An object is moving with an initial velocity of $4.0\,m\,s^{-1}$ to the right. The velocity of the object changes so that its final velocity is $3.0\,m\,s^{-1}$ downwards, as shown.

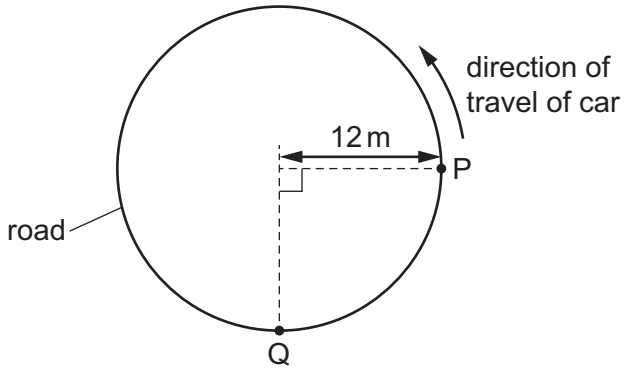


Which arrow represents the change in velocity of the object?



5 A car travels anticlockwise along a horizontal circular road of radius 12 m, as shown.

The car takes a time of 4.0 s to move from position P to position Q.



What is the magnitude of the average velocity of the car for the journey from P to Q?

A $4.2\,m\,s^{-1}$ **B** $4.7\,m\,s^{-1}$ **C** $6.0\,m\,s^{-1}$ **D** $14\,m\,s^{-1}$

6 The water surface in a deep well is 78.0 m below the top of the well. A person at the top of the well drops a heavy stone down the well.

Air resistance is negligible. The speed of sound in the air is $330\,m\,s^{-1}$.

What is the time interval between the person dropping the stone and hearing it hitting the water?

A 3.75 s **B** 3.99 s **C** 4.19 s **D** 4.22 s

7 Which statement is **not** a requirement of a pair of forces that obey Newton's third law of motion?

A The forces act in opposite directions.

B The forces act on different objects.

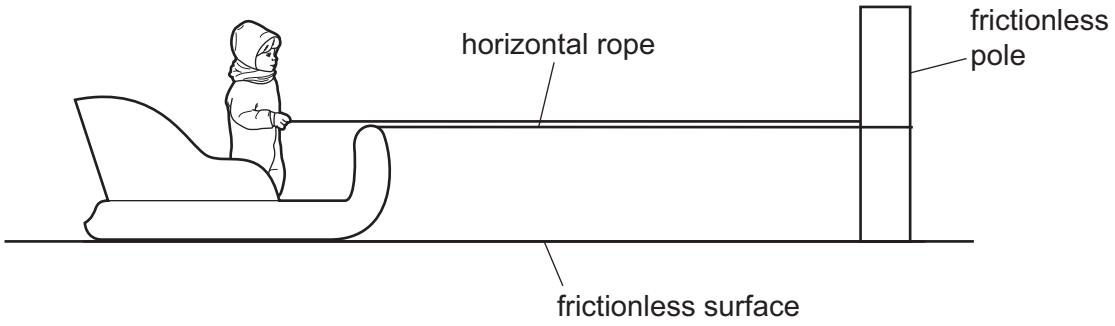
C The forces act on objects in contact.

D The forces are of equal magnitude.

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8 A child of mass 20 kg stands on the rough surface of a sledge of mass 40 kg. The sledge can slide on a horizontal frictionless surface.

One end of a rope is attached to the sledge. The rope passes around a fixed frictionless pole, and the other end of the rope is held by the child, as shown.



The rope is horizontal. The child pulls on the rope with a horizontal force of 12 N. This causes the child and the sledge to move with equal acceleration towards the pole.

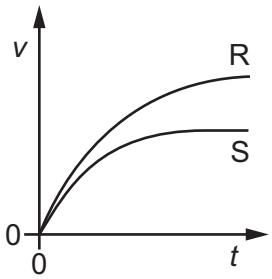
What is the frictional force between the child and the sledge?

A 4.0 N **B** 6.0 N **C** 8.0 N **D** 12 N

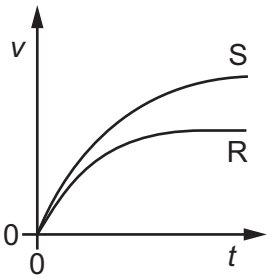
9 A stone S and a foam rubber ball R are identical spheres of equal volume. They are released from rest at time $t = 0$ and fall vertically through the air. Both reach terminal velocity.

Which graph best shows the variation with time t of the speed v of the stone and of the rubber ball?

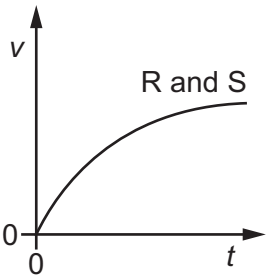
A



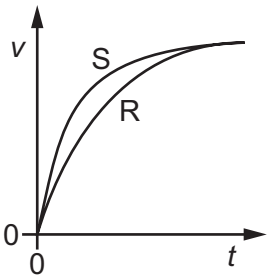
B



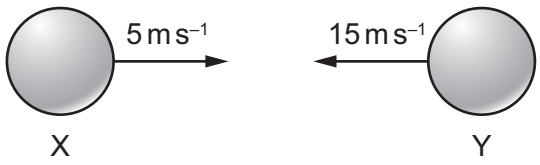
C



D



10 Two balls X and Y are moving towards each other with speeds of $5\,m\,s^{-1}$ and $15\,m\,s^{-1}$ respectively.



They make a perfectly elastic head-on collision and ball Y moves to the right with a speed of $7\,m\,s^{-1}$.

What is the speed and direction of ball X after the collision?

A $3\,m\,s^{-1}$ to the left

B $13\,m\,s^{-1}$ to the left

C $3\,m\,s^{-1}$ to the right

D $13\,m\,s^{-1}$ to the right

11 Two forces form a couple.

Which statement describes the two forces?

A They are in the same direction.

B They are perpendicular to each other.

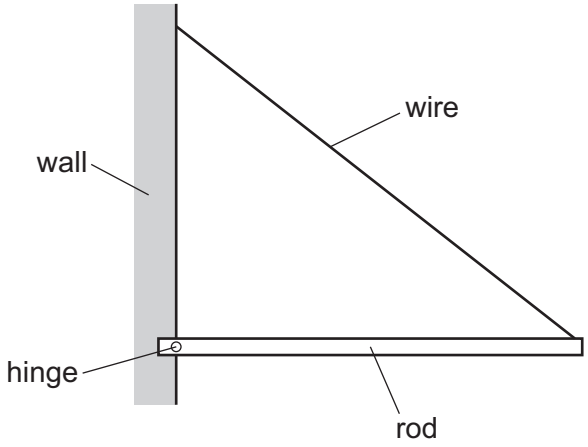
C They have the same magnitude.

D They pass through the same point.

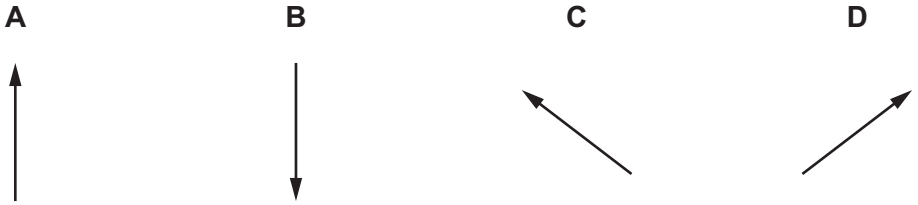
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[Turn over

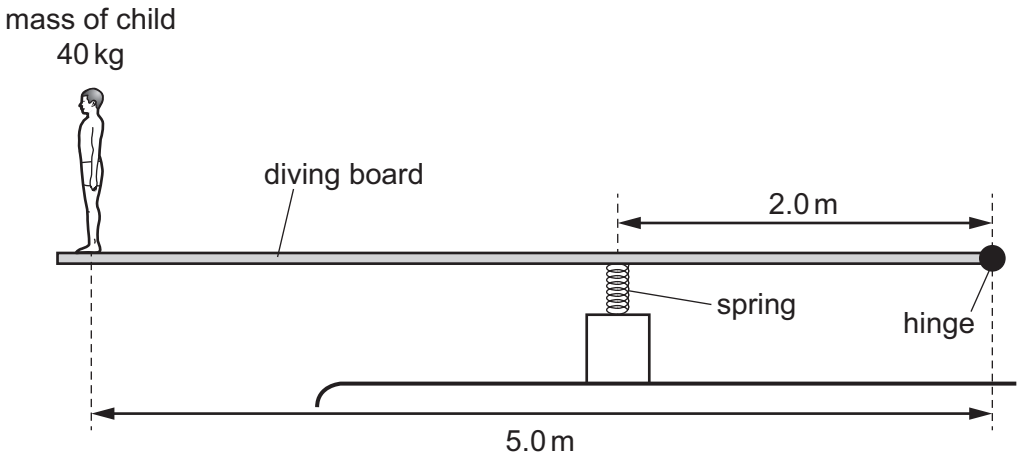
12 A uniform rod is attached by a hinge at one end to a wall. The other end of the rod is supported by a wire so that the rod is horizontal and in equilibrium.



Which arrow shows the direction of the force on the rod from the hinge?



13 A diving board of length 5.0 m is hinged at one end and supported 2.0 m from this end by a spring of spring constant 10 kN m^{-1}. A child of mass 40 kg stands at the far end of the board.



What is the extra compression of the spring caused by the child standing on the end of the board?

A 1.0 cm **B** 1.6 cm **C** 9.8 cm **D** 16 cm

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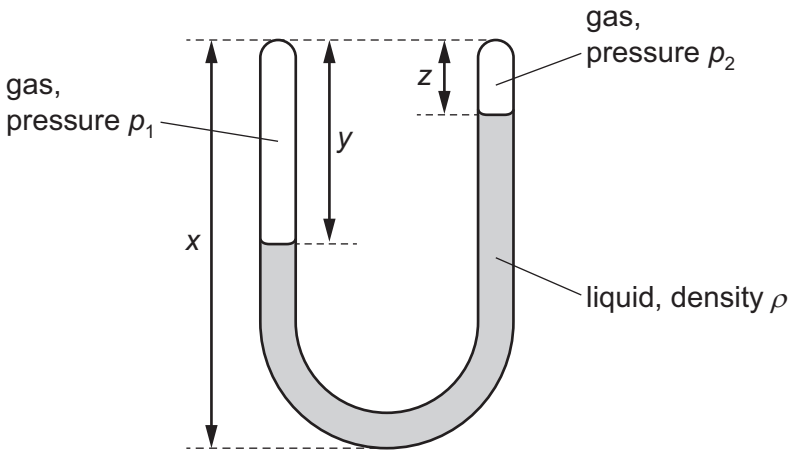
14 A granite rock at the surface of the Earth has density ρ. The rock is transported to the surface of another planet.

The acceleration of free fall on the surface of the other planet is twice that on the surface of the Earth.

What is the density of the rock on the other planet?

A 0.5ρ **B** ρ **C** 2ρ **D** 4ρ

15 A closed U-shaped tube contains a stationary liquid of density ρ. One side of the tube contains a gas at pressure p_1 and the other side contains a gas at pressure p_2, as shown.



The acceleration of free fall is g.

Which equation is correct?

A $p_1 = \rho g y$

B $p_2 = \rho g(x - z)$

C $p_1 - p_2 = \rho g(y - z)$

D $p_1 + p_2 = \rho g x$

16 Which product of two quantities is equal to power?

A force × distance

B force × velocity

C work done × time

D work done × velocity

17 Researchers have developed a new type of filament lamp with an efficiency of 40%. Old-type filament lamps have an efficiency of 5.0%. The two types of lamp produce the same useful output power.

What is the ratio $\frac{\text{input power to new type of lamp}}{\text{input power to old type of lamp}}$?

A 0.13 **B** 0.63 **C** 1.6 **D** 8.0

18 A student attempts to derive the formula for kinetic energy E_K. She begins by considering an object of mass m which is initially at rest. A constant force F applied to the object causes it to accelerate to final velocity v in displacement s. The kinetic energy gained by the object is equal to the work done on the object by the force F.

Which equation would the student **not** need in order to derive the formula for E_K?

A $F = ma$ **B** $W = Fs$ **C** $E = \frac{1}{2}Fs$ **D** $v^2 = u^2 + 2as$

19 A metal wire obeys Hooke's law and has a Young modulus of 2.0×10^{11} Pa. The wire has an original length of 1.6 m and a diameter of 0.48×10^{-3} m.

What is the spring constant of the wire?

A $7.2 \times 10^3\,\text{N m}^{-1}$

B $2.3 \times 10^4\,\text{N m}^{-1}$

C $2.9 \times 10^4\,\text{N m}^{-1}$

D $9.0 \times 10^4\,\text{N m}^{-1}$

20 A wire is being stretched by a tensile force.

Which statement about the elastic limit **must** be correct?

A The deformation is plastic after the elastic limit has been reached.

B The deformation is plastic until the elastic limit is reached.

C The extension is proportional to the tensile force after the elastic limit has been reached.

D The extension is proportional to the tensile force until the elastic limit is reached.

21 Which statement is correct for **all** types of progressive wave?

A The distance from a peak to the next trough is equal to a wavelength.

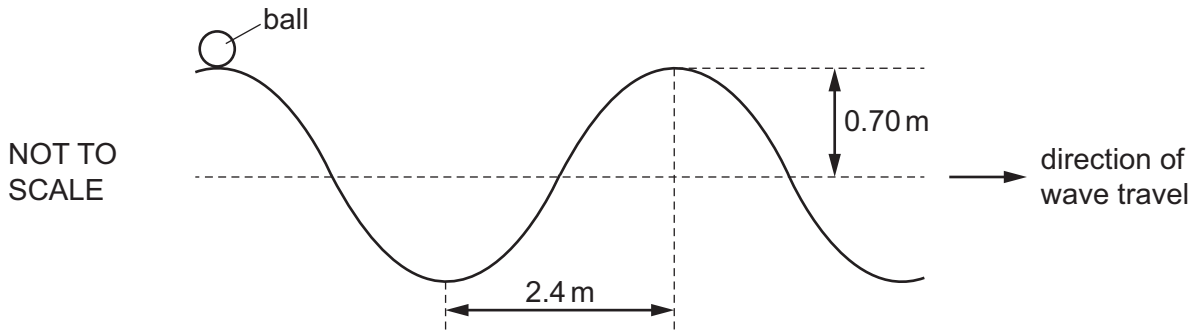
B They can be demonstrated in ripple tanks.

C They consist of vibrating atoms.

D They transfer energy from one position to another.

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22 A transverse water wave is moving along the surface of some water. This causes a ball to move vertically without moving horizontally as it floats upon the surface. At one instant, the ball is at the position shown.



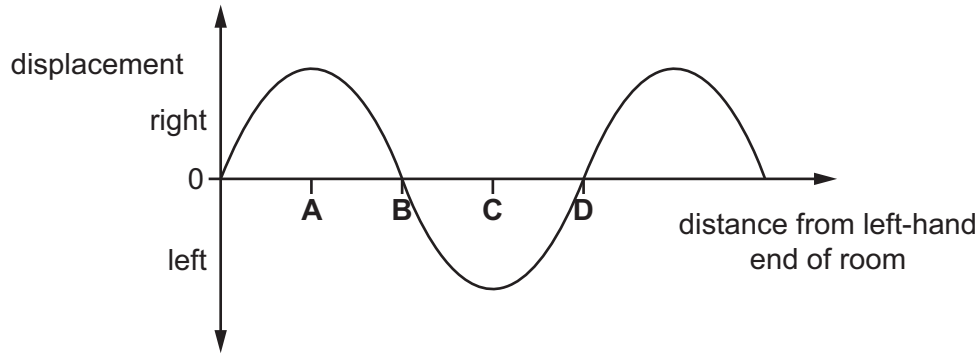
The wave has a frequency of 0.20 Hz and an amplitude of 0.70 m. The distance between a trough and an adjacent peak is 2.4 m.

What is the distance travelled by the ball in a time of 20 s?

A 5.6 m **B** 9.6 m **C** 11.2 m **D** 19.2 m

23 A sound wave travels from left to right across a room. The variation with distance across the room of the displacement of the air molecules at one instant is shown.

At which distance will the air pressure be lowest?



24 A source emits a sound wave of a single frequency. The Doppler effect causes a different frequency of sound to be heard by a stationary observer.

What is a requirement for the Doppler effect to occur?

A a source that is moving as it produces the sound wave

B a source that produces a polarised sound wave

C a source that produces a sound wave of changing amplitude

D a source that produces a sound wave of changing frequency

25 An electromagnetic wave in free space has a frequency of 2.5×10^{14} Hz.

Which region of the electromagnetic spectrum includes this frequency?

A infrared

B microwave

C ultraviolet

D X-ray

26 Two polarising filters are placed next to each other so that their planes are parallel.

The first polarising filter has its transmission axis at an angle of 50° to the vertical.

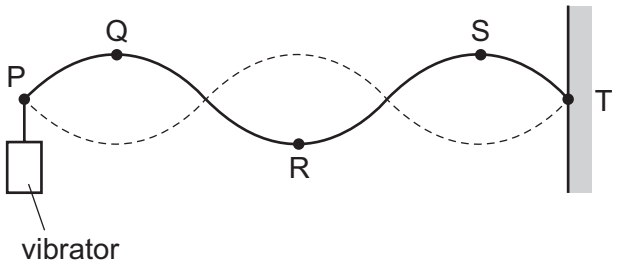
The second polarising filter has its transmission axis at an angle of 20° to the vertical. The angle between the transmission axes of the two polarising filters is 30°.

A beam of vertically polarised light of intensity $8.0\,\text{W m}^{-2}$ is incident normally on the first polarising filter.

What is the intensity of the light that is transmitted from the second polarising filter?

A zero **B** $2.5\,\text{W m}^{-2}$ **C** $2.9\,\text{W m}^{-2}$ **D** $6.0\,\text{W m}^{-2}$

27 A stationary wave on a stretched string is set up between two points P and T.



Which statement about the stationary wave is correct?

A Point R is at a node.

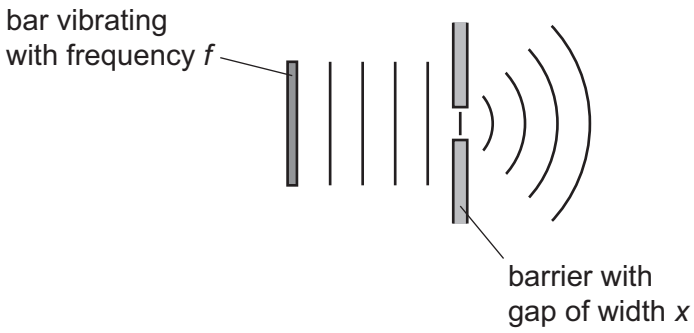
B Points Q and S vibrate in phase.

C The distance between P and T is three wavelengths.

D The wave transfers energy from P to T.

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28 A bar vibrates with frequency f to produce water waves in a ripple tank.



The waves pass through a gap of width x in a barrier so that diffraction occurs.

Which combination of vibration frequency and gap width will produce the smallest angle of diffraction?

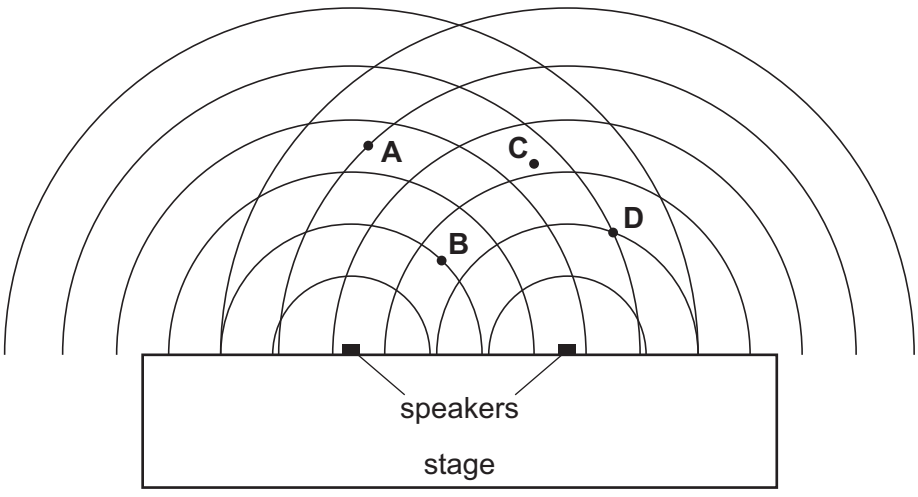
	vibration frequency	gap width
A	$\frac{f}{2}$	$\frac{x}{2}$
B	$\frac{f}{2}$	$2x$
C	$2f$	$\frac{x}{2}$
D	$2f$	$2x$

29 An outdoor concert has two large speakers beside the stage for broadcasting music.

In order to test the speakers, they are made to emit sound of the same wavelength and the same amplitude.

The curved lines in the diagram represent wavefronts.

Where is the loudest sound heard?



30 The equation

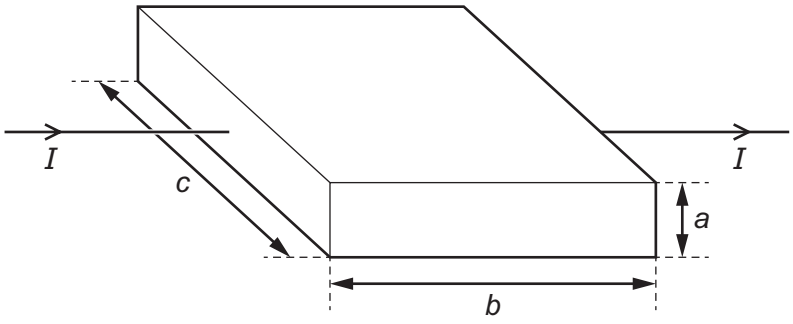
$$\lambda = \frac{d \sin\theta}{n}$$

is used to calculate the wavelength λ of light in an experiment that uses a diffraction grating. The light from the diffraction grating is displayed on a screen.

What do the symbols *n* and *d* represent?

	n	*d*
A	number of slits in the grating	distance between adjacent slits in the grating
B	number of slits in the grating	distance from grating to screen
C	order of intensity maximum	distance between adjacent slits in the grating
D	order of intensity maximum	distance from grating to screen

31 The diagram shows a metal block.



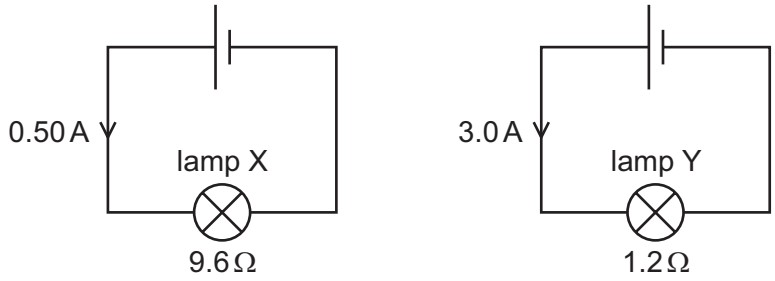
The block has sides of length *a*, *b* and *c* as shown, and its volume is *V*. Each charge carrier has a charge –*q* and the number density of the charge carriers in the metal is *n*. It takes each charge carrier an average time of *t* to pass through the block.

What is an expression for the current *I*?

A $I = nqabc$ **B** $I = \frac{nqV}{t}$ **C** $I = \frac{nqbc}{t}$ **D** $I = nqaV$

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32 The circuit diagrams show two lamps X and Y each connected to a cell. The current in lamp X is 0.50 A and its resistance is 9.6 Ω. The current in lamp Y is 3.0 A and its resistance is 1.2 Ω.



What is the ratio $\frac{\text{power in lamp X}}{\text{power in lamp Y}}$?

A 0.22 **B** 0.75 **C** 1.3 **D** 4.5

33 The intensity of light incident on a light-dependent resistor (LDR) is increased. The temperature of a thermistor is increased. In each case, the current in the component is maintained at a constant value.

What happens to the potential difference across each component?

	LDR	thermistor
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

34 An iron wire has length 8.0 m and diameter 0.50 mm. The wire has resistance R.

A second iron wire has length 2.0 m and diameter 1.0 mm.

What is the resistance of the second wire?

A $\frac{R}{16}$ **B** $\frac{R}{8}$ **C** $\frac{R}{2}$ **D** R

35 A cell with constant electromotive force (e.m.f.) is connected across a fixed resistor. Over time, the internal resistance of the cell increases.

Which change occurs as the internal resistance of the cell increases?

A a decrease in the charge of each charge carrier

B a decrease in the potential difference measured across the cell

C an increase in the energy dissipated per unit time in the fixed resistor

D an increase in the number of charge carriers leaving the cell per unit time

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36 Kirchhoff's first and second laws are consequences of the conservation of different quantities.

What are those quantities?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	energy
B	energy	current
C	current	charge
D	energy	charge

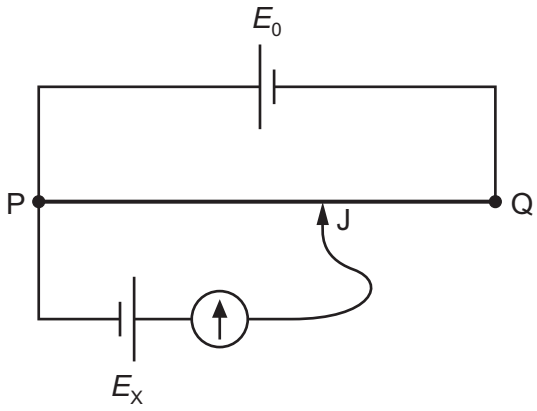
37 A potentiometer circuit is used to determine the electromotive force (e.m.f.) E_X of a cell. The circuit includes a second cell of known e.m.f. E_0 and negligible internal resistance, and a uniform resistance wire PQ of known length.

E_X is less than E_0.

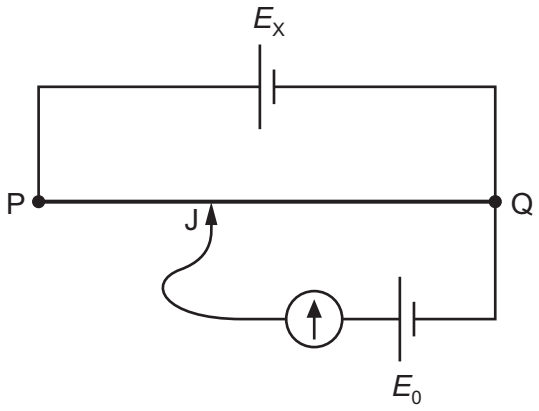
The movable connection J can be positioned anywhere along the length of the resistance wire.

Which circuit is suitable for determining E_X?

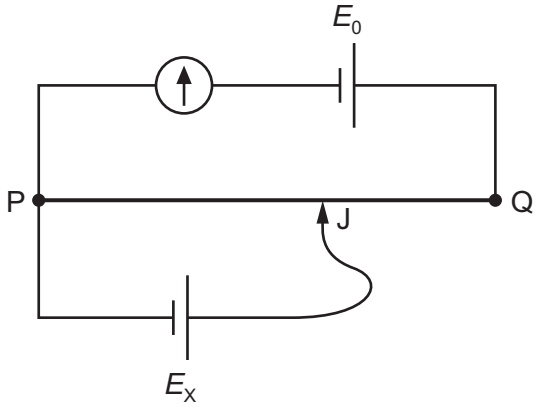
A



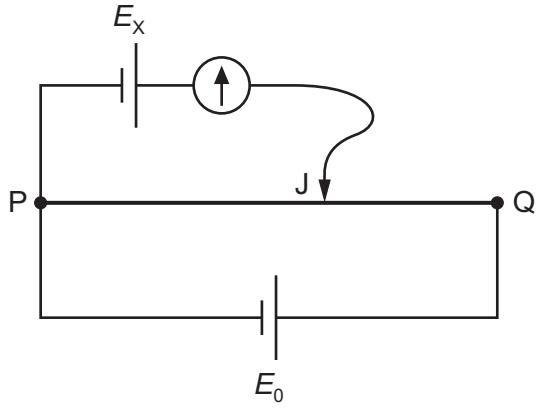
B



C



D



38 In an experiment on α-particle scattering, α-particles are directed at a thin gold foil. Most of the α-particles pass straight through the foil or are deflected by a small angle. A small number of α-particles are deflected by a large angle.

Which statement **cannot** be deduced from this experiment?

A Atoms are mostly empty space.

B Most of the mass of an atom is concentrated in the nucleus.

C The nucleus of an atom contains protons.

D The nucleus of an atom is small compared to the size of an atom.

39 Four nuclei are represented below.

$$^{28}_{14}\mathrm{E} \qquad ^{25}_{15}\mathrm{G} \qquad ^{25}_{12}\mathrm{M} \qquad ^{24}_{13}\mathrm{Q}$$

Which statement about these nuclei is correct?

A An uncharged atom of element Q has 24 orbital electrons.

B Nuclei G and M are isotopes of the same element.

C When E absorbs a neutron and then emits an α-particle, E transforms into M.

D When M emits a β^- particle, M transforms into Q.

40 A neutron is composed of one up (u) quark and two down (d) quarks. When a neutron decays to a proton, a beta-particle is emitted.

What is the change in the quark structure of the neutron due to the emission of the beta-particle?

(The symbol for a neutrino is ν_e and for an antineutrino is $\overline{\nu_e}$.)

A $\mathrm{d} \rightarrow \mathrm{u} + \beta^- + \nu_e$

B $\mathrm{d} \rightarrow \mathrm{u} + \beta^- + \overline{\nu_e}$

C $\mathrm{u} \rightarrow \mathrm{d} + \beta^+ + \nu_e$

D $\mathrm{u} \rightarrow \mathrm{d} + \beta^+ + \overline{\nu_e}$

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Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2022**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

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Question	Answer	Marks
1	C	**1**
2	B	**1**
3	C	**1**
4	B	**1**
5	A	**1**
6	D	**1**
7	C	**1**
8	A	**1**
9	B	**1**
10	B	**1**
11	C	**1**
12	D	**1**
13	C	**1**
14	B	**1**
15	C	**1**
16	B	**1**
17	A	**1**
18	C	**1**
19	B	**1**
20	A	**1**
21	D	**1**
22	C	**1**
23	D	**1**
24	A	**1**
25	A	**1**
26	B	**1**
27	B	**1**
28	D	**1**

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Question	Answer	Marks
29	D	**1**
30	C	**1**
31	B	**1**
32	A	**1**
33	D	**1**
34	A	**1**
35	B	**1**
36	A	**1**
37	D	**1**
38	C	**1**
39	C	**1**
40	B	**1**

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June 2022 · Paper 13

9702_s22_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2022**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

IB22 06_9702_13/3RP
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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which pair of quantities are physical quantities?

A charge and ampere

B efficiency and kilogram

C pascal and strain

D period and potential difference

2 Which list of unit prefixes decreases in magnitude from left to right?

A centi, deci, milli

B deci, milli, centi

C pico, kilo, milli

D kilo, milli, pico

3 The drag coefficient C_d is a number with no units. It is used to compare the drag on different cars at different speeds. C_d is given by the equation

$$C_d = \frac{2F}{v^n \rho A}$$

where F is the drag force on the car, ρ is the density of the air, A is the cross-sectional area of the car and v is the speed of the car.

What is the value of n?

A 1 **B** 2 **C** 3 **D** 4

4 A micrometer screw gauge is used to measure the diameter of a small uniform steel sphere. The measurement of the diameter is 5.00 mm $\pm$ 0.01 mm.

What is the percentage uncertainty in the calculated volume of the sphere, using these values?

A 0.2% **B** 0.4% **C** 0.6% **D** 1.2%

5 Forces of magnitudes 2 N, 4 N and 7 N combine to produce a resultant force.

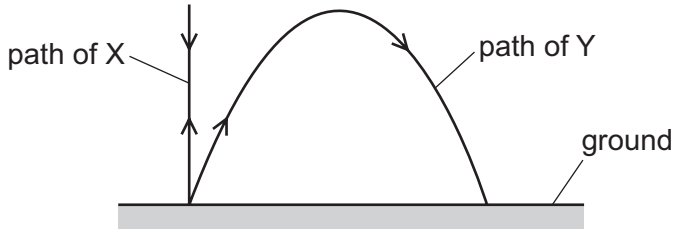
The magnitudes of the three forces are fixed, but the forces may act in any direction in the same plane.

What is **not** a possible magnitude of the resultant force?

A 0 N **B** 5 N **C** 8 N **D** 13 N

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6 Two projectiles, X and Y, are fired into the air from the same place on level ground and reach the same maximum height, as shown.



Projectile X is fired vertically upwards and projectile Y is fired at an angle to the horizontal.

Air resistance is negligible.

Which statement is correct?

A X and Y are at rest at their maximum heights.

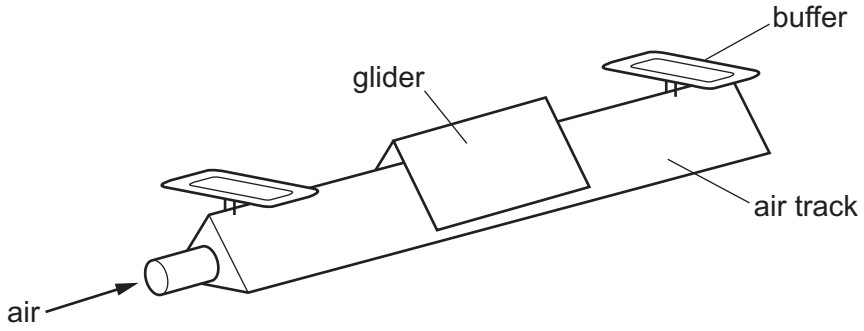
B X and Y are fired with the same speed.

C X and Y take the same time to return to the ground.

D X and Y travel the same distance.

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7 A small glider moves along a horizontal air track as shown.

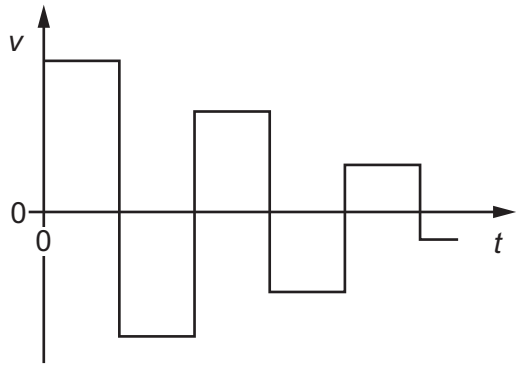


At each end of the air track, the glider has a perfectly elastic collision with a fixed buffer.

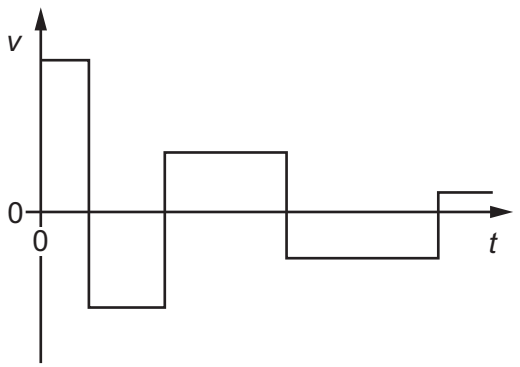
The glider moves at a constant speed between collisions.

Which graph represents the variation with time t of the velocity v of the glider as it moves between the two buffers?

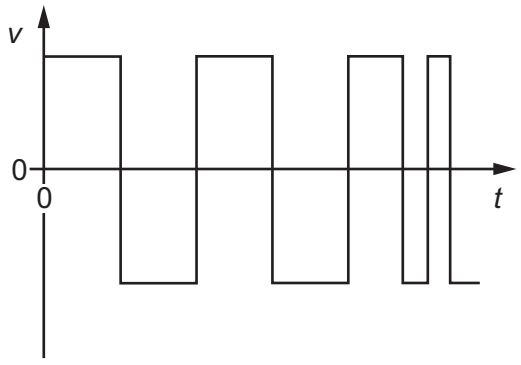
A



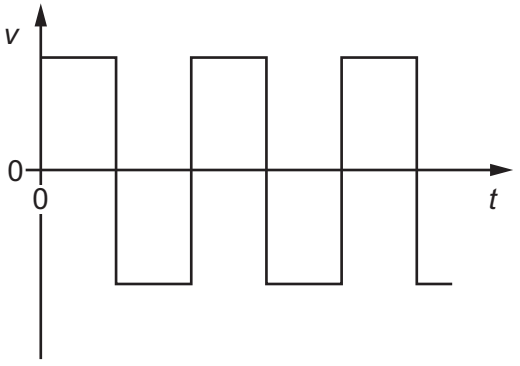
B



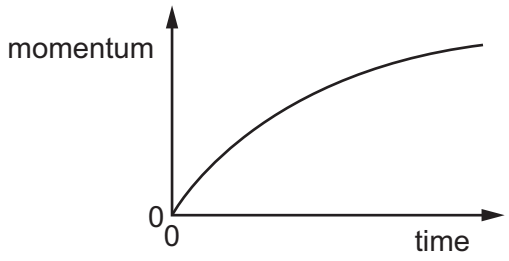
C



D



8 A car accelerates from rest. The graph shows the variation of the momentum of the car with time.



What is the meaning of the gradient of the graph at a particular time?

A the kinetic energy of the car

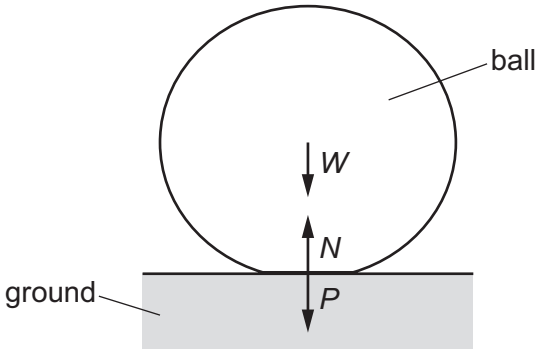
B the rate of change of kinetic energy of the car

C the resultant force on the car

D the velocity of the car

9 A ball is dropped onto horizontal ground and bounces vertically upwards. When the ball is in contact with the ground, the following forces act:

- the weight *W* of the ball
- the contact force *P* exerted on the ground by the ball
- the contact force *N* exerted on the ball by the ground.



When the ball is in contact with the ground, the ball is momentarily stationary.

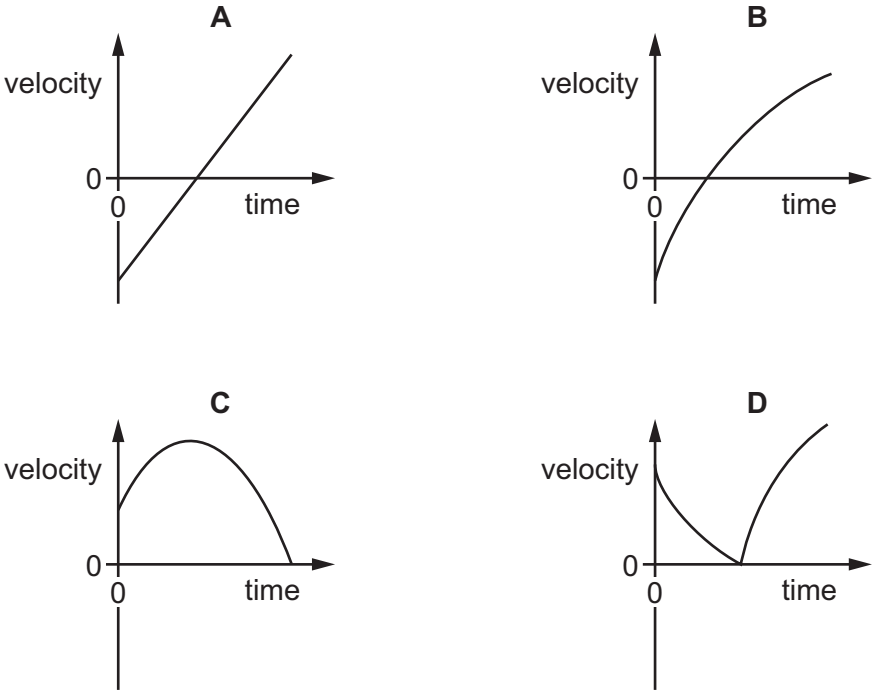
At this instant, which relationship is correct?

A $N = P + W$ **B** $N > P + W$ **C** $N = W$ **D** $N > W$

10 A person stands on the edge of a high cliff that is next to the sea. The person throws a stone vertically upwards. Air resistance acts on the stone.

The stone eventually hits the sea.

Which velocity–time graph best shows the motion of the stone from when it is released until it hits the sea?



11 Skaters of masses 80 kg and 40 kg move directly towards each other and collide.

Before the collision, the heavier skater is moving to the right at a speed of $2.0\,m\,s^{-1}$ and the lighter skater is moving to the left at a speed of $1.0\,m\,s^{-1}$.

After the collision, the heavier skater moves to the right at a speed of $0.80\,m\,s^{-1}$.

What is the relative speed of separation of the two skaters?

A $0.6\,m\,s^{-1}$ **B** $1.4\,m\,s^{-1}$ **C** $2.2\,m\,s^{-1}$ **D** $2.6\,m\,s^{-1}$

12 Which statement describes the two forces in a couple?

A They act in the same direction.

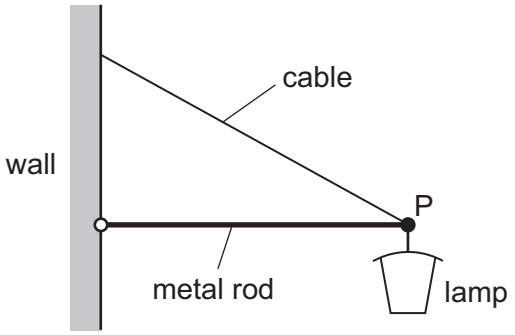
B They act through the same point.

C They produce zero resultant force.

D They produce zero resultant moment.

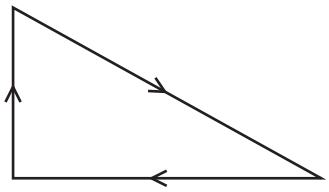
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13 A street lamp is fixed to a wall by a metal rod and a cable.

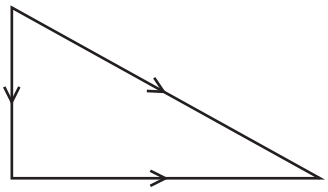


Which vector triangle could represent the forces acting on the end of the rod at point P?

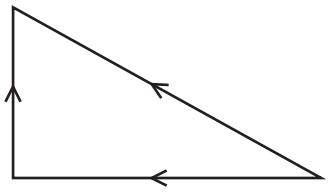
A



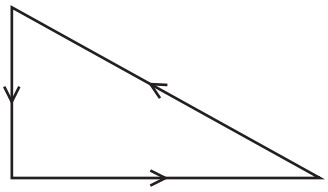
B



C

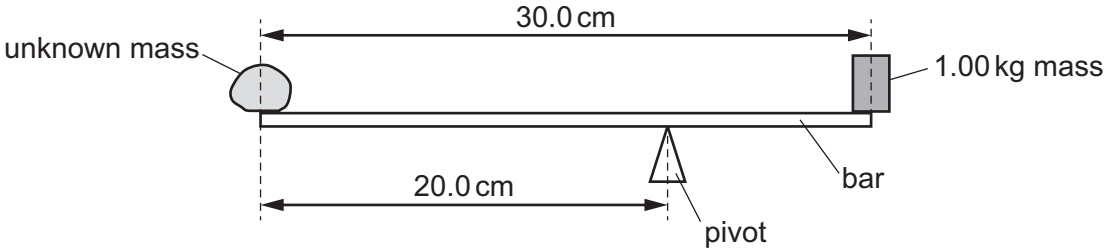


D



14 An unknown mass and a 1.00 kg mass are fixed at opposite ends of a bar. The bar has negligible mass and a length of 30.0 cm.

The bar balances when supported by a pivot placed 20.0 cm from the unknown mass, as shown.



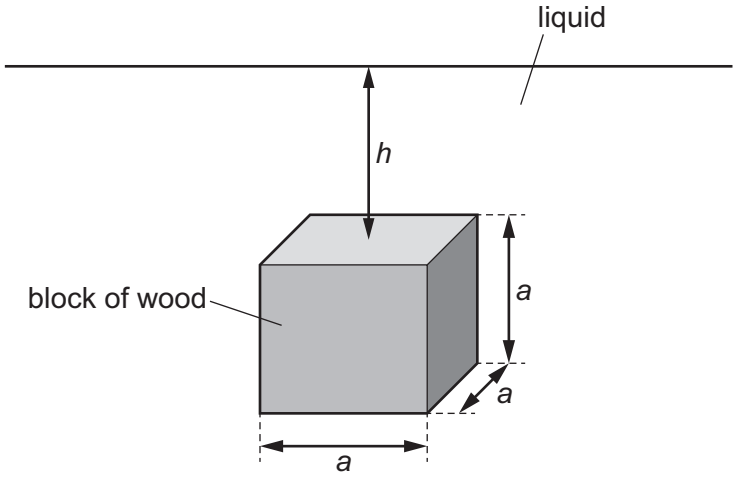
What is the unknown mass?

A 333 g **B** 500 g **C** 667 g **D** 1000 g

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15 A block of wood of density ρ_w has sides of length a.

The block is immersed in a liquid of density ρ_L. The top surface of the block is at a depth h below the surface of the liquid.



The acceleration of free fall is g.

What is the upthrust acting on the block from the liquid?

A $\rho_L a^3 g$ **B** $\rho_w a^3 g$ **C** $\rho_L hg$ **D** $\rho_L ag$

16 A technical article about diesel engines expresses the energy available from diesel fuel both as $41.8\,\text{MJ}\,\text{kg}^{-1}$ and as $34.9\,\text{GJ}\,\text{m}^{-3}$.

What is the density of diesel fuel?

A $8.35 \times 10^2\,\text{kg}\,\text{m}^{-3}$

B $1.20 \times 10^3\,\text{kg}\,\text{m}^{-3}$

C $8.35 \times 10^5\,\text{kg}\,\text{m}^{-3}$

D $1.20 \times 10^6\,\text{kg}\,\text{m}^{-3}$

17 What is meant by the efficiency of a system?

A the difference between the useful energy output from the system and the total energy input

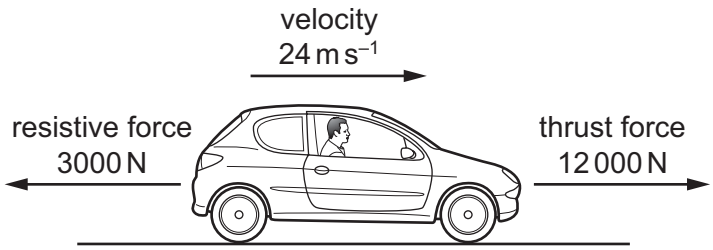
B the difference between the useful energy output from the system and the wasted energy output

C the ratio of the useful energy output from the system to the total energy input

D the ratio of the useful energy output from the system to the wasted energy output

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18 A car of weight 15 000 N is travelling along a horizontal road.

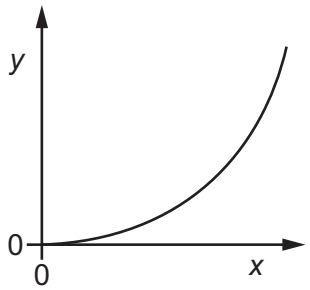


At one instant, the thrust force acting on the car from the engine is 12 000 N and the resistive force acting on the car is 3000 N. The velocity of the car at this instant is 24 m s^{-1}.

What is the power output from the engine?

A 72 kW **B** 220 kW **C** 290 kW **D** 360 kW

19 The diagram shows the variation of a quantity *y* with a quantity *x* for objects in a uniform gravitational field.



What could *x* and *y* represent?

	x	*y*
A	mass for different objects moving at the same speed	kinetic energy
B	speed for an object of constant mass	kinetic energy
C	vertical distance fallen for an object of constant mass	change of gravitational potential energy
D	mass for different objects falling the same vertical distance	change of gravitational potential energy

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20 A steel wire has a length of 300 cm and a cross-sectional area of 0.50 mm^2. The Young modulus of steel is 2.0×10^{11} Pa.

One end of the wire is attached to a fixed point. A load of 10 N is hung from the other end. The wire obeys Hooke's law.

What is the extension of the wire?

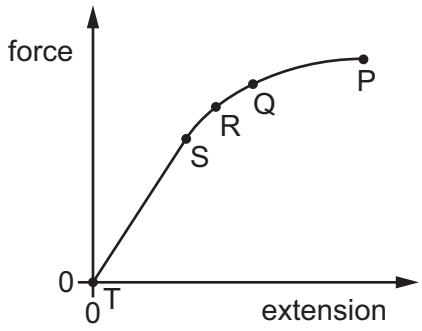
A 3.0×10^{-7} m

B 3.0×10^{-5} m

C 3.0×10^{-4} m

D 3.0×10^{-2} m

21 The extension of a copper wire is measured for different forces applied to the wire. A graph is plotted to show the variation of the force on the wire against extension. The maximum force is applied at point P.



Which statement **must** be correct?

A Point R is the limit of proportionality.

B The elastic potential energy of the wire at point S is given by the area under the graph between points T and S.

C There is no plastic deformation between points Q and P.

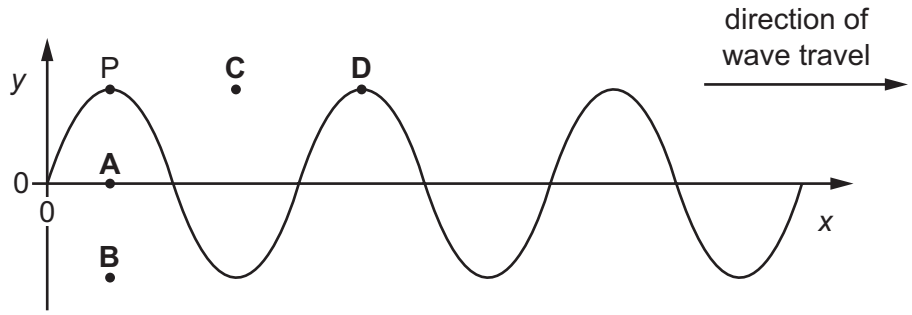
D The wire obeys Hooke's law up to a point between R and Q.

22 The variation with distance x of the displacement y of a transverse wave on a rope is shown at time $t = 0$.

The wave has a frequency of 0.5 Hz.

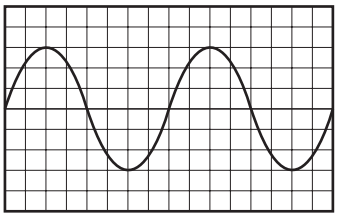
A point P on the rope is marked. The diagram shows the original position of P and four new positions.

What is the position of P at time $t = 1$ s?



23 A cathode-ray oscilloscope (CRO) is connected to a microphone which detects sound of constant frequency.

The trace on the screen of the CRO is shown.



Which property of the sound wave is measured using **only** information from the CRO?

A amplitude

B period

C speed

D wavelength

24 A transverse wave and a longitudinal wave both travel in the same direction down a long stretched spring.

Which statement is **not** correct for these two forms of wave?

A The displacement measurements for the particles of the two waves are made at right angles to each other.

B The energy transferred by the two waves is in the same direction.

C The velocities of the two waves are in the same direction.

D The wavelength measurements for the two waves are made at right angles to each other.

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25 A man standing next to a stationary train hears sound of frequency 400 Hz emitted from the train's horn. The train then moves directly away from the man and sounds its horn when it has a speed of 50 m s^{-1}. The speed of sound in the air is 340 m s^{-1}.

What is the difference in frequency of the sound heard by the man on the two occasions?

A 51 Hz **B** 69 Hz **C** 350 Hz **D** 470 Hz

26 Which list of regions of the electromagnetic spectrum is in order of increasing wavelength from left to right?

A gamma-ray → ultraviolet → infrared

B infrared → microwave → ultraviolet

C microwave → X-ray → infrared

D X-ray → ultraviolet → gamma-ray

27 The principle of superposition states that a certain quantity is added when two or more waves meet at a point.

What is this quantity?

A amplitude

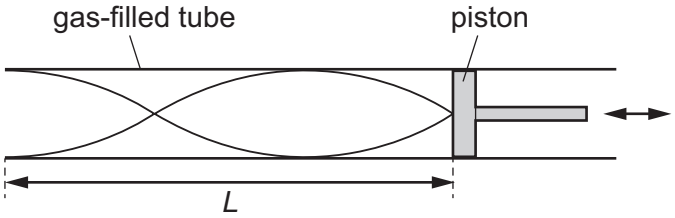
B displacement

C intensity

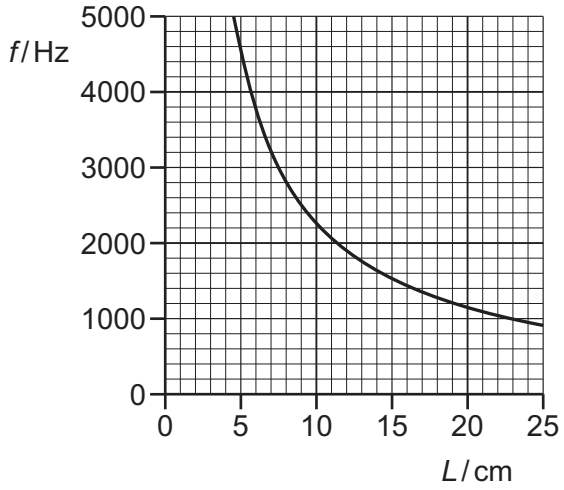
D wavelength

28 A stationary sound wave is formed in a gas-filled tube of length L, which is closed at one end by a piston. The length of the tube can be altered by moving the piston.

The length of the tube and the frequency of the sound are varied so that the stationary wave always has two antinodes and two nodes, as shown.



The graph shows the variation of the frequency f of the stationary sound wave with the length L of the tube.



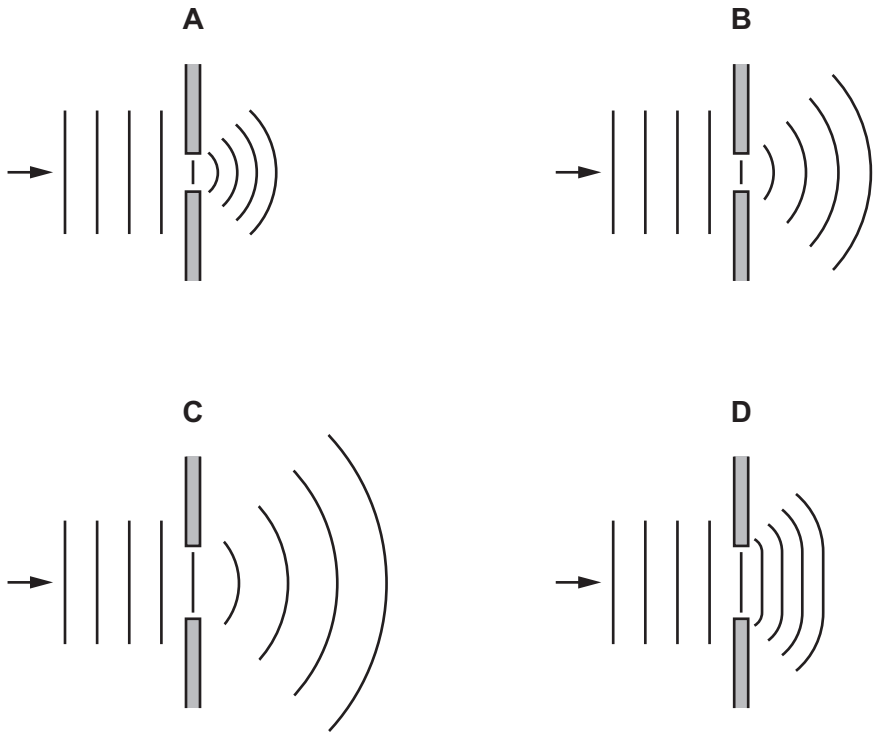
What is the speed of sound in the gas in the tube?

A $150\,m\,s^{-1}$ **B** $230\,m\,s^{-1}$ **C** $300\,m\,s^{-1}$ **D** $340\,m\,s^{-1}$

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29 The diagrams show the diffraction of water waves in a ripple tank as they pass through a gap between two barriers.

Which diagram is correct?



30 A beam of light from a laser is incident normally on a double slit. Interference fringes are seen on a screen placed parallel to the double slit.

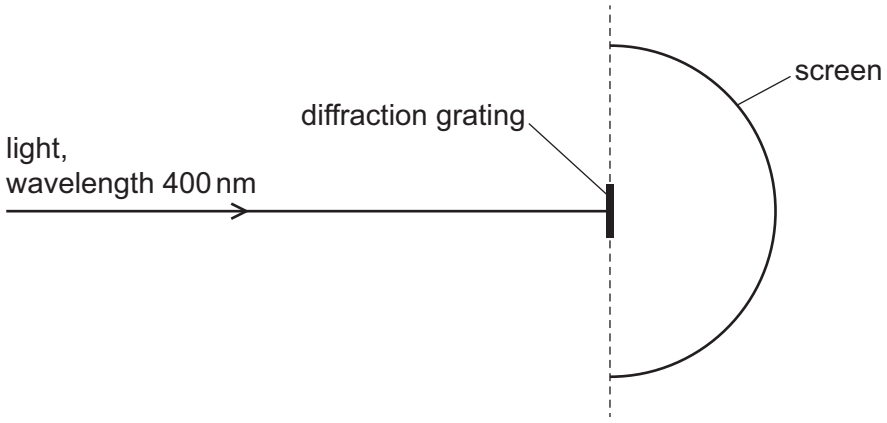
The separation of the two slits is a. The distance between the slits and the screen is D. The distance between the centres of two adjacent bright fringes is x.

D and a are both halved.

What is the distance between the centres of two adjacent bright fringes after these changes?

A $\frac{x}{2}$ **B** x **C** $2x$ **D** $4x$

31 A beam of light of wavelength 400 nm is incident normally on a diffraction grating that has 300 lines per millimetre. The light passes through the grating and produces a series of maxima which are observed on a semicircular screen, as shown.



VIEW FROM ABOVE

What is the total number of maxima observed on the screen?

A 8 **B** 9 **C** 16 **D** 17

32 Two wires, X and Y, are made from the same metal.

The diameter of wire Y is twice that of wire X.

Wire X, wire Y and a battery are connected in series.

What is the ratio $\frac{\text{average drift speed of free electrons in wire X}}{\text{average drift speed of free electrons in wire Y}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{2}{1}$ **D** $\frac{4}{1}$

33 A resistor has resistance R. When the potential difference (p.d.) across the resistor is V, the current in the resistor is I. The power dissipated in the resistor is P. Work W is done when charge Q flows through the resistor.

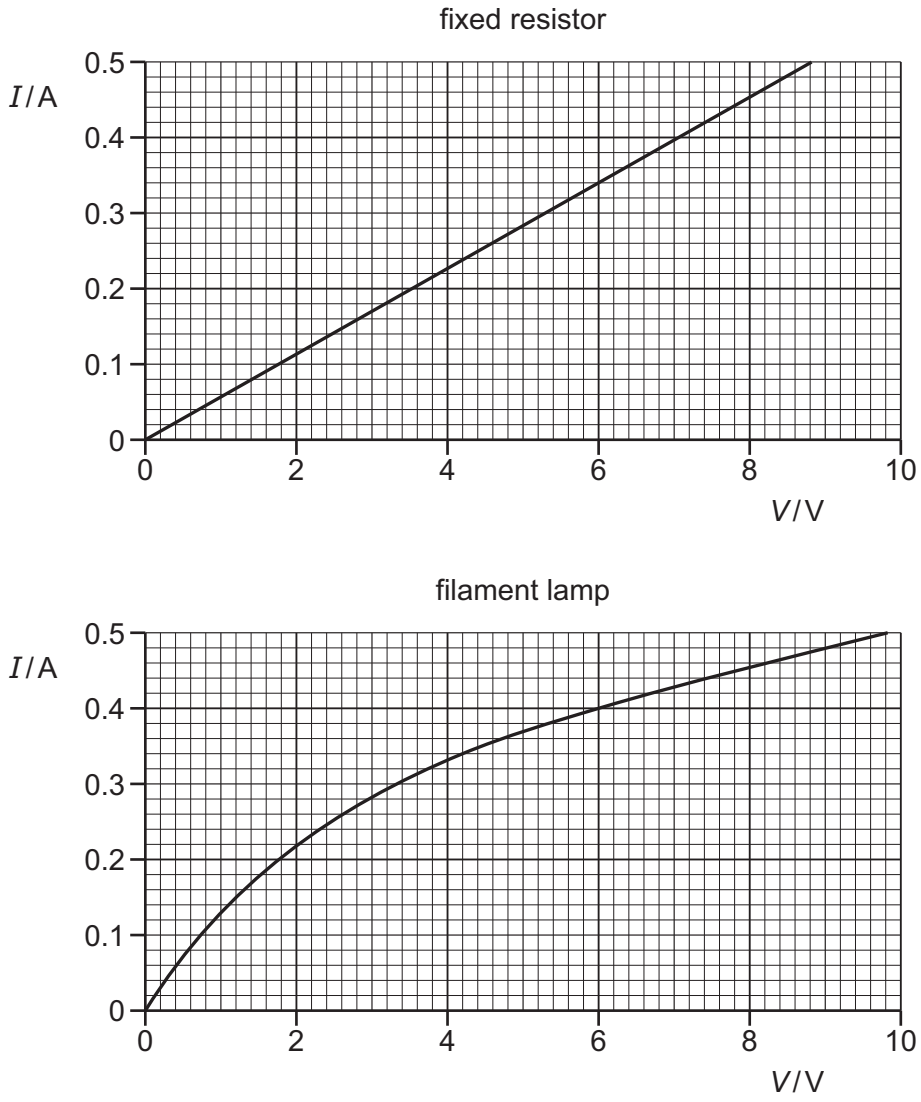
What is **not** a valid relationship between these variables?

A $I = \frac{P}{V}$ **B** $Q = \frac{W}{V}$ **C** $R = \frac{P}{I^2}$ **D** $R = \frac{V}{P}$

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34 A fixed resistor and a filament lamp are connected in series to a power supply.

The I–V characteristics for the two components are shown.



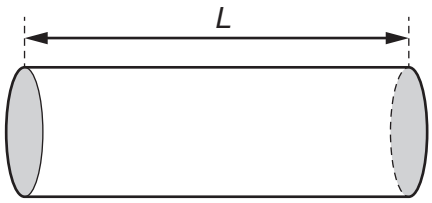
The current in the fixed resistor is 0.34 A.

What is the resistance of the filament lamp?

A 0.081 Ω **B** 12 Ω **C** 15 Ω **D** 18 Ω

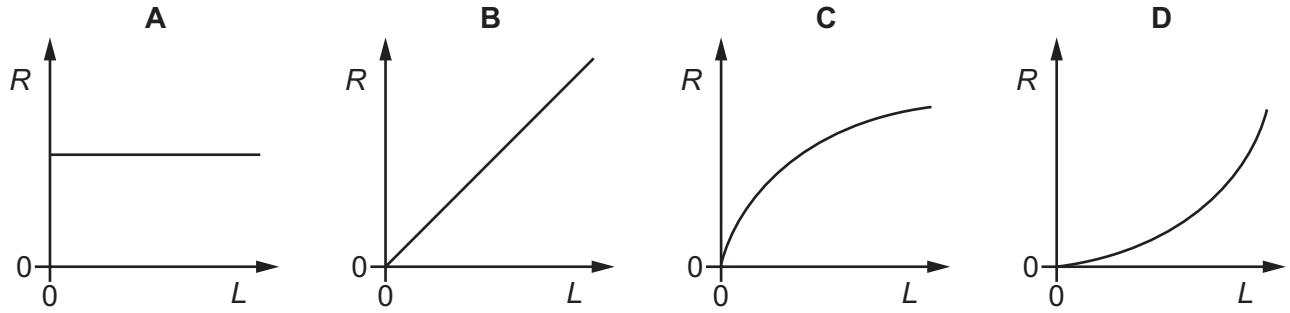
35 A piece of conducting putty (modelling clay) of constant resistivity is formed into a cylindrical shape.

The resistance *R* between its flat ends (shaded) is measured.



The same volume of putty is re-formed into cylinders of different lengths *L*, and the resistance *R* between the flat ends is measured for each value of *L*.

Which graph best shows the variation of *R* with *L*?



36 The diagram shows the symbol for a component that may be used in an electrical circuit.



Which component is represented by this circuit symbol?

A buzzer

B electric bell

C loudspeaker

D microphone

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37 Which row correctly describes Kirchhoff's laws?

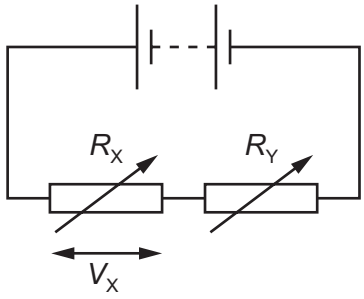
	Kirchhoff's first law	physics principle applied for first law	Kirchhoff's second law	physics principle applied for second law
A	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy
B	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge
C	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge
D	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy

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38 A potential divider circuit is formed by connecting a battery of negligible internal resistance in series with two variable resistors, as shown.



The variable resistors have resistances R_X and R_Y.

V_X is the potential difference (p.d.) across the variable resistor with resistance R_X.

R_X and R_Y are both changed at the same time.

Which combination of changes **must** cause V_X to increase?

	R_X	R_Y
A	larger	larger
B	larger	smaller
C	smaller	larger
D	smaller	smaller

39 An actinium nucleus has a nucleon number of 227 and a proton number of 89. It decays to form a radium nucleus, emitting a β^- particle and an α-particle in the process.

What are the nucleon number and the proton number of this radium nucleus?

	nucleon number	proton number
A	223	87
B	223	88
C	224	87
D	225	86

40 Which statement is **not** correct?

A A meson consists of three quarks.

B A proton is a baryon.

C A quark is a fundamental particle.

D There are six flavours (types) of quark.

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Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2022**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

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Question	Answer	Marks
1	D	**1**
2	D	**1**
3	B	**1**
4	C	**1**
5	A	**1**
6	C	**1**
7	D	**1**
8	C	**1**
9	D	**1**
10	B	**1**
11	A	**1**
12	C	**1**
13	D	**1**
14	B	**1**
15	A	**1**
16	A	**1**
17	C	**1**
18	C	**1**
19	B	**1**
20	C	**1**
21	B	**1**
22	B	**1**
23	B	**1**
24	D	**1**
25	A	**1**
26	A	**1**
27	B	**1**
28	C	**1**

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Question	Answer	Marks
29	B	**1**
30	B	**1**
31	D	**1**
32	D	**1**
33	D	**1**
34	B	**1**
35	D	**1**
36	A	**1**
37	A	**1**
38	B	**1**
39	B	**1**
40	A	**1**

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November 2022 · Paper 11

9702_w22_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2022**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What is needed to accurately represent all physical quantities?

A a base unit and a number

B a unit and a number expressed in standard form (scientific notation)

C a unit and a numerical magnitude

D an SI unit and a numerical magnitude

2 A voltmeter connected across a resistor in a circuit reads 3.6 V.

What could be the current in the resistor and the resistance of the resistor?

	current	resistance
A	150 mA	0.24 kΩ
B	15 mA	2.4 kΩ
C	1.5 mA	0.24 MΩ
D	15 μA	240 kΩ

3 In an experiment to determine the acceleration of free fall *g*, the time *t* taken for a ball to fall through distance *s* is measured. The percentage uncertainty in the measurement of *s* is 2%. The percentage uncertainty in the measurement of *t* is 3%.

The value of *g* is determined using the equation shown.

$$g = \frac{2s}{t^2}$$

What is the percentage uncertainty in the calculated value of *g*?

A 1% **B** 5% **C** 8% **D** 11%

4 Which quantity is a vector?

A momentum

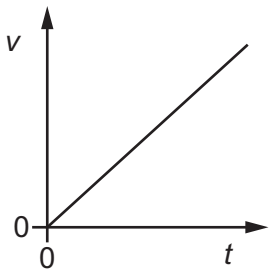
B speed

C temperature

D Young modulus

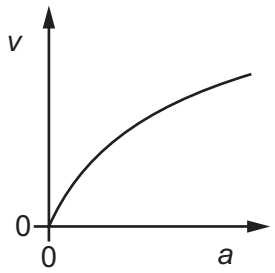
5 A particle accelerates from rest.

The graph shows the variation of the velocity v of the particle with time t.

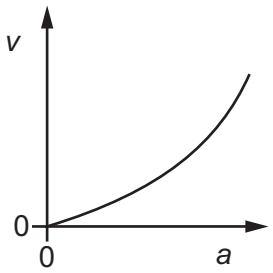


Which graph shows the variation of the velocity v with the acceleration a of the particle?

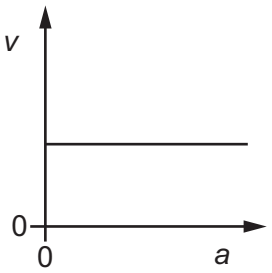
A



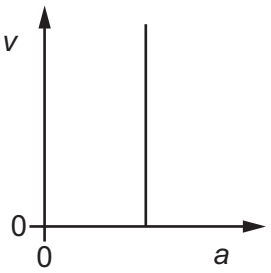
B



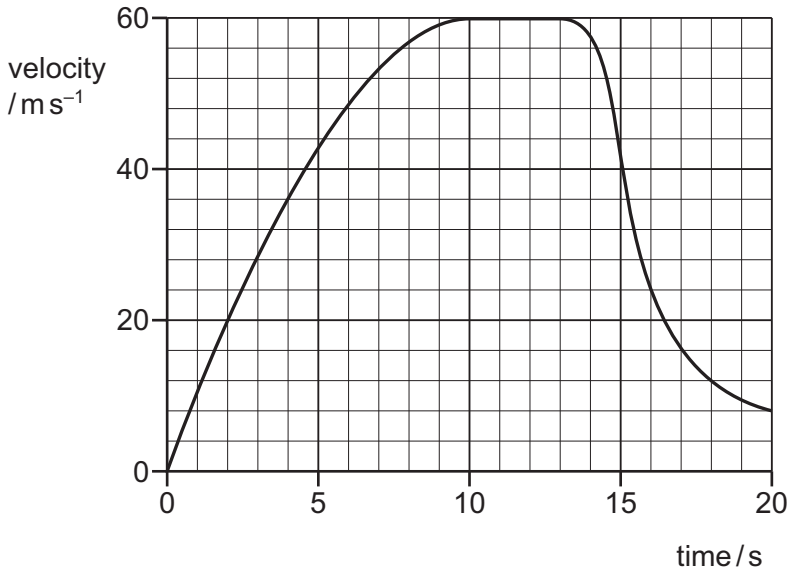
C



D



6 The graph shows the vertical velocity of a parachutist during the first 20 s of her jump.



Approximately how far does she fall **before** opening the parachute?

A 390 m **B** 570 m **C** 710 m **D** 770 m

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7 What is the definition of linear momentum?

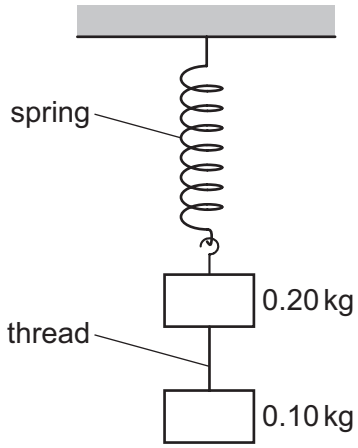
A force per unit time

B product of force and time

C product of velocity and mass

D velocity per unit mass

8 A mass of 0.20 kg is suspended from the lower end of a light spring. A second mass of 0.10 kg is suspended from the first mass by a thread. The arrangement is allowed to come into static equilibrium and then the thread is cut.



Immediately after the thread is cut, what is the upward acceleration of the 0.20 kg mass?

A $4.9\,m\,s^{-2}$ **B** $6.5\,m\,s^{-2}$ **C** $9.8\,m\,s^{-2}$ **D** $15\,m\,s^{-2}$

9 A snowflake and a raindrop are in still air. They both fall from rest at the same time and from the same height, far above the ground.

The snowflake and raindrop contain the same mass of water. Assume that there is no evaporation or melting. Also assume that, for a given speed, the drag force acting on the snowflake is greater than the drag force acting on the raindrop.

Which statement about the snowflake and raindrop is correct?

A The raindrop takes more time than the snowflake to reach terminal velocity.

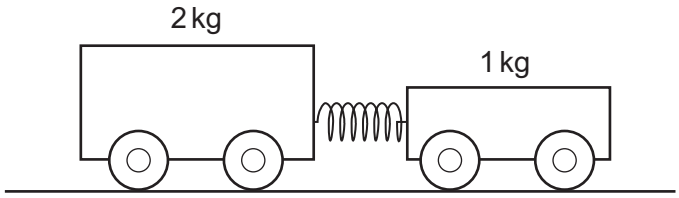
B The raindrop takes more time than the snowflake to reach the ground.

C They reach the same terminal velocity.

D They take the same amount of time to reach the ground.

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10 Two trolleys are held together on a horizontal surface with a compressed spring between them.



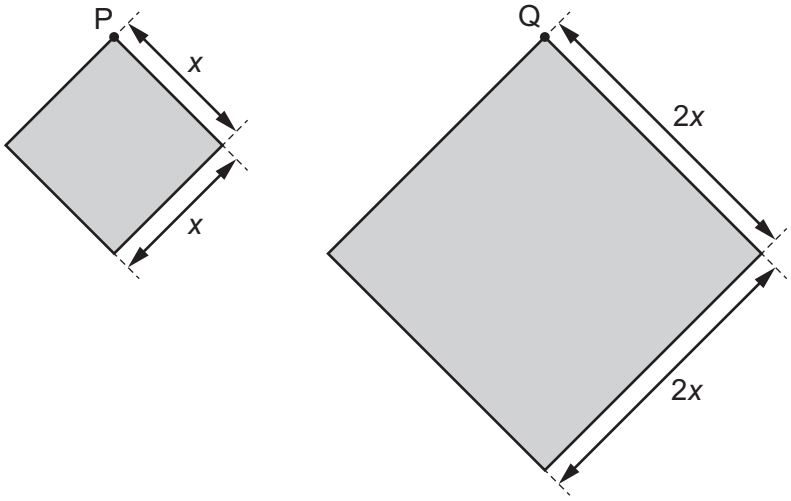
When they are released, the trolleys lose contact with the spring. The trolley of mass 2 kg moves to the left at a final speed of $2\,m\,s^{-1}$.

How much elastic potential energy was stored in the spring?

A 4 J **B** 6 J **C** 8 J **D** 12 J

11 A square board, of side length x, hangs freely from a nail P, as shown.

The board has uniform thickness and is made from material of uniform density.



A second square board, of side length $2x$, is made of the same material and has the same thickness as the original board. This second board is then hung from a nail Q. Nails P and Q are at the same height.

What is the vertical distance between the positions of the centres of gravity of the two boards?

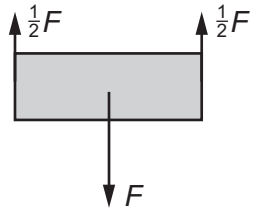
A 0 **B** $\frac{x}{\sqrt{2}}$ **C** x **D** $x\sqrt{2}$

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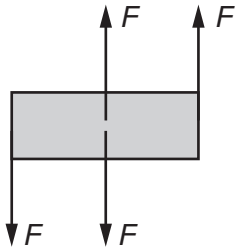
12 Forces are applied to a rigid object. The forces all act in the same plane.

In which diagram is the object in equilibrium?

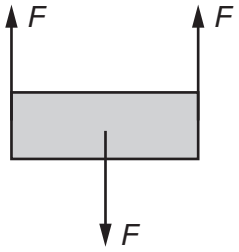
A



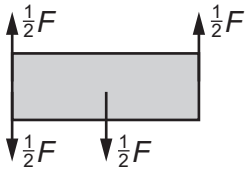
B



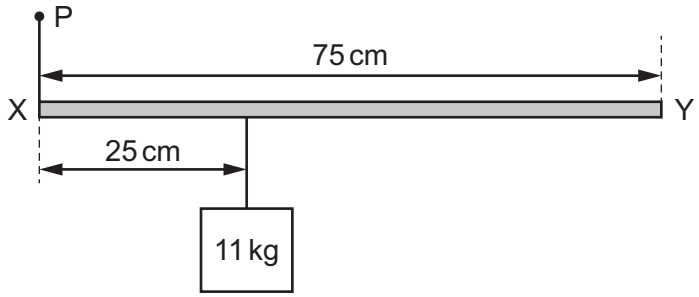
C



D



13 A rigid rod XY has negligible mass and length 75 cm. The rod is suspended from a fixed point P by a string attached to end X. An object of mass 11 kg is suspended by a string that is attached to the rod at a distance of 25 cm from end X, as shown.



Which vertically upward force acting on end Y of the rod would hold the rod horizontally in equilibrium?

A 3.7 N **B** 33 N **C** 36 N **D** 320 N

14 The density of water is 1.0 g cm^{-3} and the density of glycerine is 1.3 g cm^{-3}.

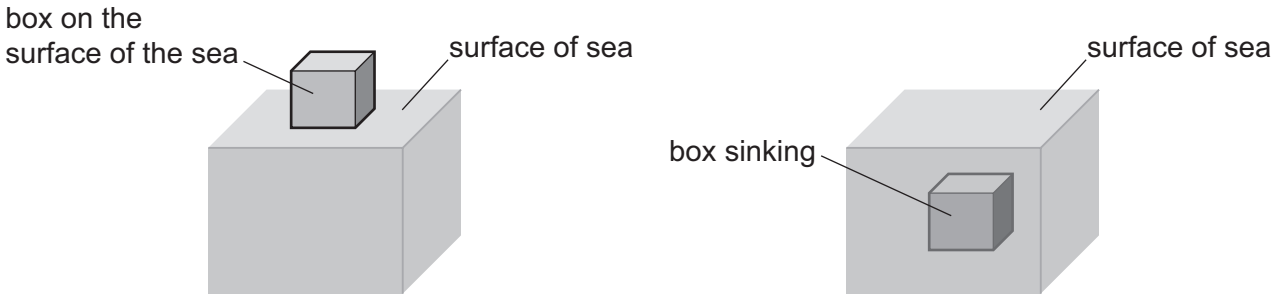
Water is added to a measuring cylinder containing 40 cm^3 of glycerine so that the density of the mixture is 1.1 g cm^{-3}. Assume that the mixing process does not change the total volume of the liquid.

What is the volume of water added?

A 40 cm^3 **B** 44 cm^3 **C** 52 cm^3 **D** 80 cm^3

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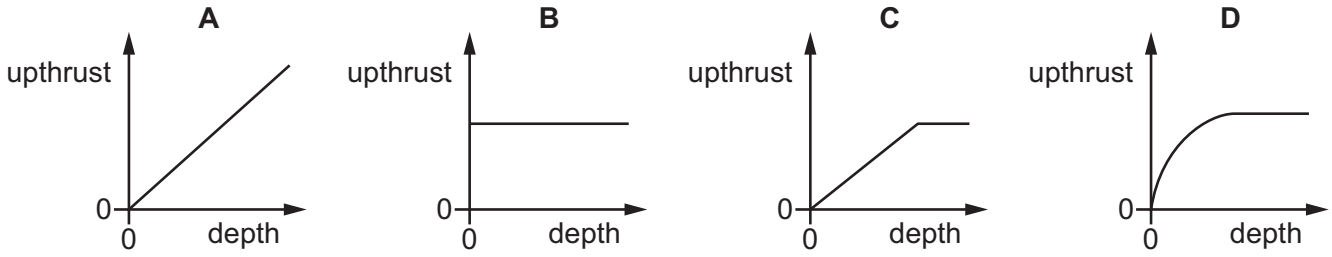
15 A box, in the shape of a cube, falls from a ship into the sea. The box lands with its lower face level with the surface of the sea.



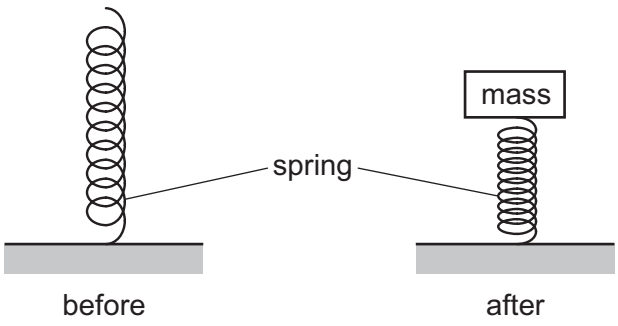
The box begins to sink, becomes totally submerged and then sinks deeper into the sea.

As the box sinks, its lower face is always parallel to the surface of the sea.

Which graph best represents the variation of the upthrust acting on the box with the depth of its lower face below the surface of the sea?



16 A spring is compressed by a mass, as shown.



Which statement describes the changes to the energy of the spring when it is compressed by the mass?

A The spring gains both gravitational potential energy and elastic potential energy.

B The spring gains gravitational potential energy and loses elastic potential energy.

C The spring loses both gravitational potential energy and elastic potential energy.

D The spring loses gravitational potential energy and gains elastic potential energy.

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17 A man of mass 75 kg runs up a staircase consisting of 30 steps. Each step is 20 cm high.

The man takes a time of 7.0 s to run from the bottom of the staircase to the top.

What is the average rate of increase of gravitational potential energy of the man?

A 64 W **B** 450 W **C** 630 W **D** 4400 W

18 An alpha-particle has 2.2×10^{-13} J of kinetic energy.

What is the speed of the alpha-particle?

A $4.1 \times 10^{6}\,m\,s^{-1}$

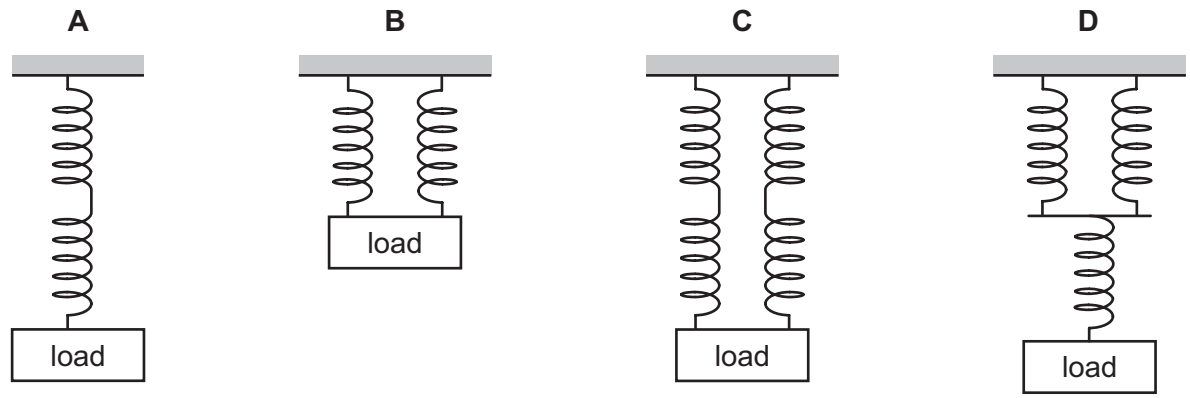
B $5.8 \times 10^{6}\,m\,s^{-1}$

C $8.1 \times 10^{6}\,m\,s^{-1}$

D $1.2 \times 10^{7}\,m\,s^{-1}$

19 Identical springs are joined in four arrangements.

Which arrangement has the same spring constant as a single spring?



20 An unstretched spring has a length of 2.0 cm.

The spring is then stretched within its limit of proportionality by a tensile force of 1.5 N so that the elastic potential energy stored in the spring is 0.045 J.

What is the stretched length of the spring?

A 3.0 cm **B** 5.0 cm **C** 6.0 cm **D** 8.0 cm

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21 A signal generator, amplifier and loudspeaker are used to produce different sound waves in the air of a room. The relationships between the properties of these waves are investigated.

Which relationship is **not** correct?

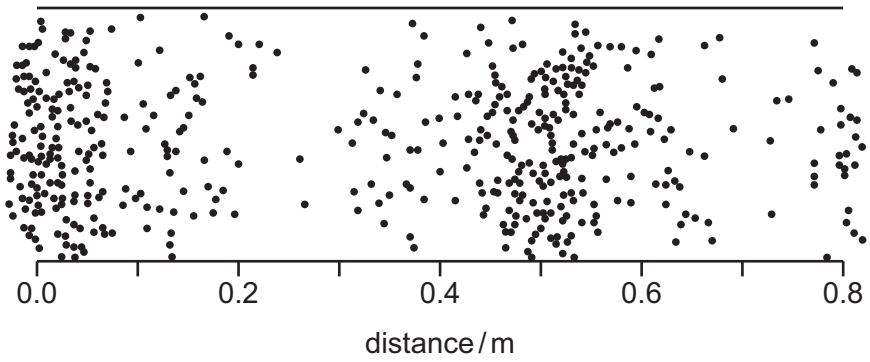
A Amplitude is proportional to wavelength.

B Frequency is inversely proportional to wavelength.

C Intensity is proportional to amplitude squared at a given frequency.

D Period is equal to the reciprocal of frequency.

22 When a guitar string is plucked, it causes a longitudinal sound wave in the air, as shown.



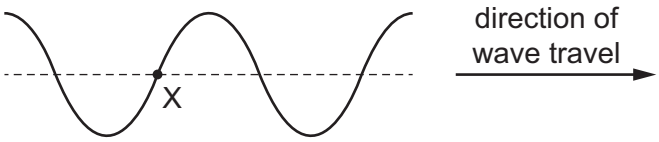
The speed of sound in the air is $340\,m\,s^{-1}$.

What is the approximate frequency of the sound wave shown?

A 430 Hz **B** 680 Hz **C** 1100 Hz **D** 1400 Hz

23 A transverse wave travels along a rope. The diagram shows the rope at time $t = 0$.

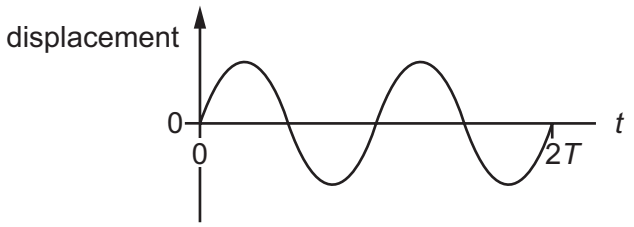
The wave is travelling from left to right. The period of the wave is T.



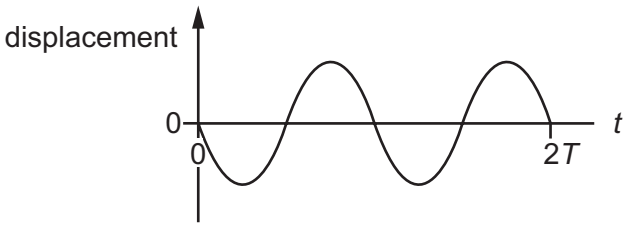
One particle of the rope is labelled X.

Which graph shows the variation with time of the displacement of particle X between $t = 0$ and $t = 2T$?

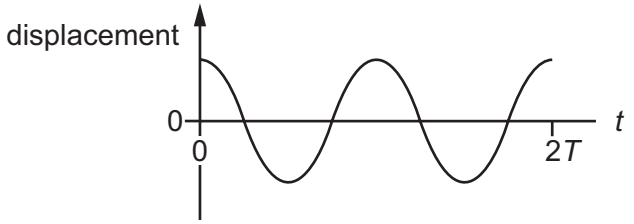
A



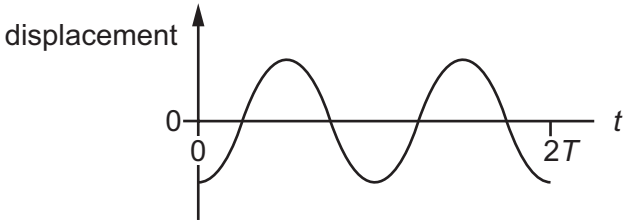
B



C



D



24 A jet aircraft travels at a speed of $0.80v$, where v is the speed of sound. The aircraft directly approaches a stationary observer. The frequency of sound emitted by the aircraft is 100 Hz.

Which frequency does the observer hear?

A 56 Hz **B** 180 Hz **C** 400 Hz **D** 500 Hz

25 A telescope detects and analyses some electromagnetic radiation of wavelength 2 cm.

Which type of telescope is it?

A microwave telescope

B optical telescope

C radio telescope

D X-ray telescope

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26 Two waves of the same type overlap.

When does the principle of superposition apply?

A always

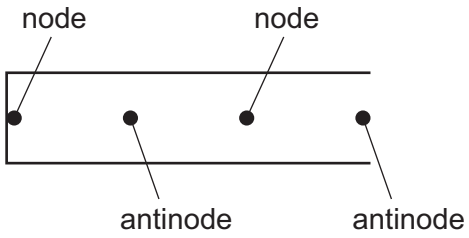
B only when the waves have the same amplitude

C only when the waves have the same frequency

D only when the waves travel in opposite directions

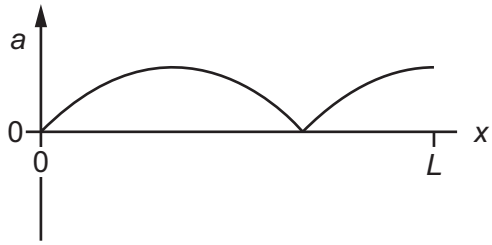
27 A stationary sound wave is formed in a tube of length *L* that is closed at one end.

The diagram shows the positions of the nodes and antinodes of the stationary wave.

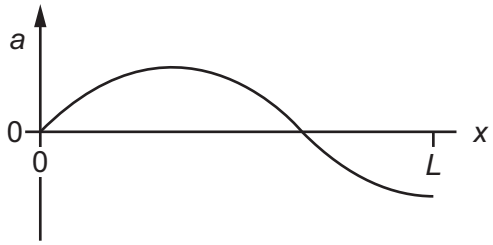


Which graph shows the variation of the amplitude *a* of the wave with distance *x* measured from the closed end of the tube?

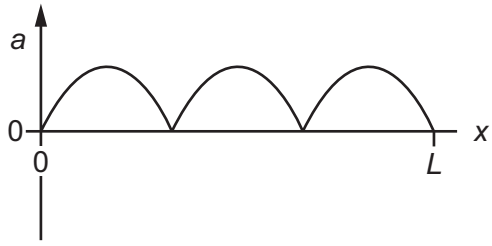
A



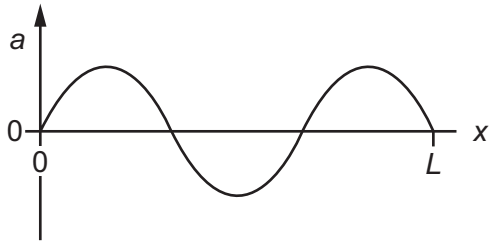
B



C

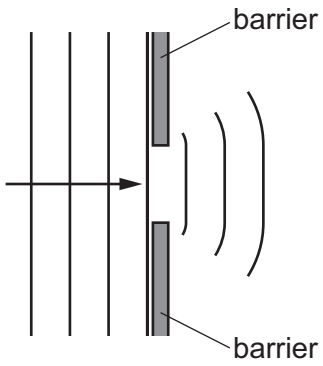


D



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28 A wave on the surface of water passes through a gap between two barriers and is diffracted, as shown.



What happens when the frequency of the wave is halved?

A Less diffraction is observed.

B More diffraction is observed.

C There is no diffraction.

D The same amount of diffraction is observed.

29 Observable interference fringes are produced using light from a double slit. The intensity of the light emerging from each slit is initially the same.

The intensity of the light emerging from one of the slits is now reduced.

How does this affect the interference pattern?

A The bright fringes and the dark fringes all become brighter.

B The bright fringes and the dark fringes all become darker.

C The bright fringes become brighter and the dark fringes become darker.

D The bright fringes become darker and the dark fringes become brighter.

30 A diffraction grating has 4.00×10^5 lines per metre. A beam of light of wavelength 589×10^{-9} m is incident normally on the diffraction grating.

What is the angle between the second-order maximum and the direction of the incident beam of light?

A 13.6° **B** 27.3° **C** 28.1° **D** 56.2°

31 A length of wire RS has a circular cross-section.



At end R of the wire, the cross-sectional area is A.

At end S of the wire, the cross-sectional area is $\frac{A}{2}$.

Charge Q takes time t to pass through end R of the wire. There is a constant electric current in the wire.

How much charge will pass through end S in a time interval of $\frac{t}{4}$?

A $\frac{Q}{8}$ **B** $\frac{Q}{4}$ **C** $\frac{Q}{2}$ **D** Q

32 A power supply is connected to a component by connecting wires of total resistance $4.9\,\Omega$.

The power supply has an output power of 3.6 W and a terminal potential difference of 12 V.

How much thermal energy is dissipated in the connecting wires in a time of 1.0 hour?

A 0.44 J **B** 29 J **C** 1.6 kJ **D** 11 kJ

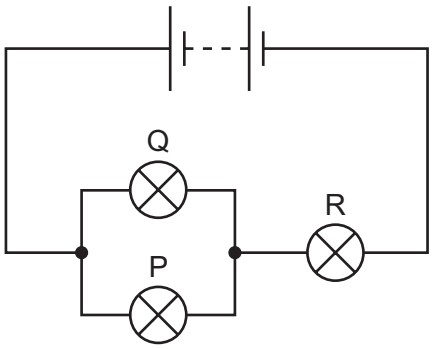
33 A copper wire is to be replaced by an aluminium alloy wire of the same length and resistance. Copper has half the resistivity of the alloy.

What is the ratio $\frac{\text{diameter of alloy wire}}{\text{diameter of copper wire}}$?

A $\sqrt{2}$ **B** 2 **C** $2\sqrt{2}$ **D** 4

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34 Three identical filament lamps, P, Q and R, are connected to a battery of negligible internal resistance, as shown.



The filament wire in lamp Q breaks so that it no longer conducts.

What are the changes in the brightness of lamps P and R?

	lamp P	lamp R
A	brighter	brighter
B	brighter	dimmer
C	dimmer	brighter
D	dimmer	dimmer

35 Which ratio has the same units as electromotive force (e.m.f.)?

A charge per unit energy transferred

B charge per unit time

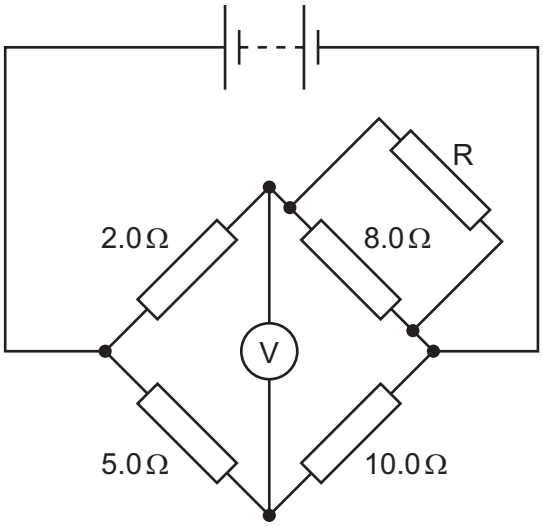
C energy transferred per unit charge

D energy transferred per unit time

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[Turn over

36 A circuit consists of a battery, a voltmeter and five fixed resistors, as shown.



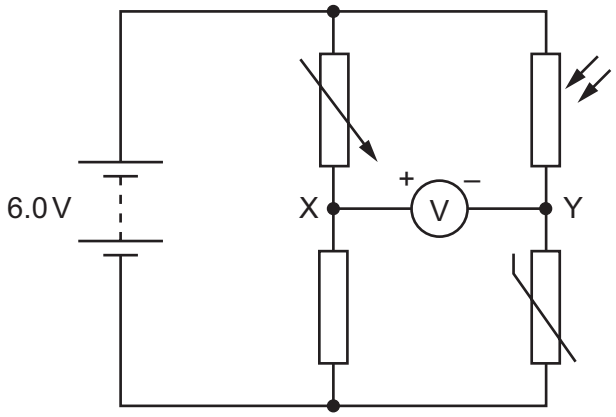
The voltmeter reading is zero.

What is the resistance of resistor R?

A 1.1 Ω **B** 2.1 Ω **C** 4.0 Ω **D** 8.0 Ω

37 A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected to a voltmeter and four other components, as shown.

The voltmeter is connected between points X and Y. The positive terminal of the voltmeter is connected to X and the negative terminal of the voltmeter is connected to Y.



Initially, the resistance of each of the four components is 1.0 kΩ.

Which change, on its own, will cause the voltmeter to show a positive reading?

A Decrease the temperature of the thermistor.

B Increase the resistance of the variable resistor.

C Reduce the intensity of light incident on the light-dependent resistor (LDR).

D Replace the fixed resistor with a 500 Ω resistor.

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38 An α-particle passes close to a gold nucleus and is deflected through an angle greater than 90°.

Which property of the α-particle changes as a result of the deflection?

A charge

B momentum

C nucleon number

D proton number

39 A nucleus of $^{238}_{92}U$ decays in stages by emitting α-particles and β^- particles, eventually forming a nucleus of $^{206}_{82}Pb$.

How many α-particles and how many β^- particles are emitted during the decay chain?

	α-particles	β^- particles
A	8	6
B	8	10
C	16	6
D	16	22

40 Which list of particles contains **only** fundamental particles?

A antineutrino, beta, meson

B baryon, neutrino, positron

C electron, neutrino, alpha

D lepton, quark, positron

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PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2022**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

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Question	Answer	Marks
1	C	**1**
2	D	**1**
3	C	**1**
4	A	**1**
5	D	**1**
6	B	**1**
7	C	**1**
8	A	**1**
9	A	**1**
10	D	**1**
11	B	**1**
12	A	**1**
13	C	**1**
14	D	**1**
15	C	**1**
16	D	**1**
17	C	**1**
18	C	**1**
19	C	**1**
20	D	**1**
21	A	**1**
22	B	**1**
23	B	**1**
24	D	**1**
25	A	**1**
26	A	**1**
27	A	**1**
28	B	**1**

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Question	Answer	Marks
29	D	**1**
30	C	**1**
31	B	**1**
32	C	**1**
33	A	**1**
34	B	**1**
35	C	**1**
36	D	**1**
37	C	**1**
38	B	**1**
39	A	**1**
40	D	**1**

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November 2022 · Paper 12

9702_w22_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2022**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

2151627545

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

IB22 11_9702_12/3RP
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[Turn over

Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which quantity is a physical quantity?

A flavour

B kelvin

C minute

D potential difference

2 What is a power of 3.7 MW when expressed in kilowatts?

A 3.7×10^{-3} kW

B 3.7×10^{-3} KW

C 3.7×10^{3} kW

D 3.7×10^{3} KW

3 A spring is suspended from a fixed point and a force is applied. The position of a pointer attached to the bottom of the spring against a vertical ruler is recorded.

Before the force is applied, the position of the pointer is (225 ± 2) mm.

After the force is applied, the position of the pointer is (250 ± 2) mm.

The extension of the spring is determined.

What is the percentage uncertainty in the extension?

A 1.6% **B** 1.8% **C** 8.0% **D** 16%

4 What is the difference between a scalar quantity and a vector quantity?

A A scalar quantity has direction but a vector quantity does not.

B A scalar quantity has magnitude but a vector quantity does not.

C A vector quantity has direction but a scalar quantity does not.

D A vector quantity has magnitude but a scalar quantity does not.

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5 A toy car travels on a circular track at a constant speed of $0.50\,m\,s^{-1}$. It passes a point on the track at time $t = 0$ and takes a time of 40 s to travel once around the track.

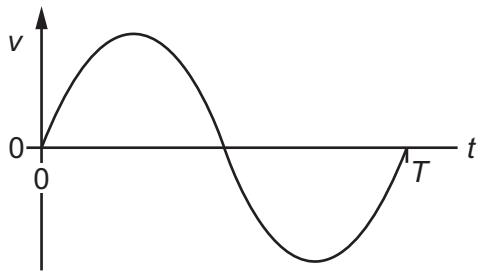
The magnitude of the average velocity of the car between $t = 0$ and $t = 20\,s$ is v_{20}.

The magnitude of the average velocity of the car between $t = 0$ and $t = 40\,s$ is v_{40}.

What are v_{20} and v_{40}?

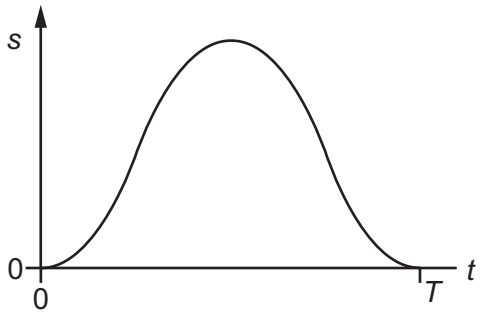
	v_{20} / m s^{-1}	v_{40} / m s^{-1}
A	0.32	0
B	0.32	0.32
C	0.50	0
D	0.50	0.50

6 The graph shows how the velocity v of an object moving in a straight line varies with time t from $t = 0$ to $t = T$.

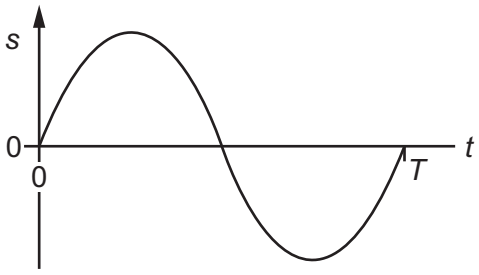


Which graph could represent the displacement s of the object from time $t = 0$ to $t = T$?

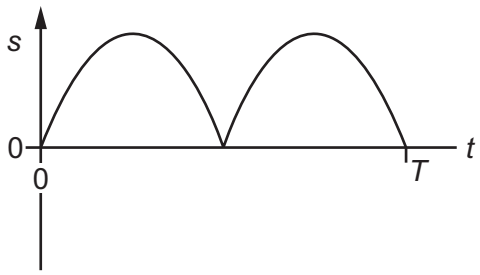
A



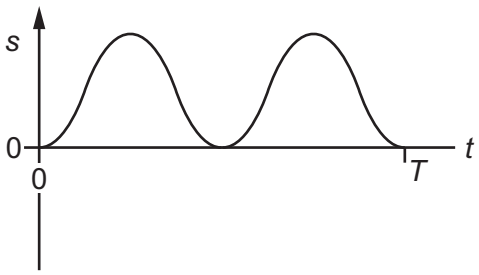
B



C



D



7 A goods train passes through a station at a steady speed of $10\,m\,s^{-1}$ at time $t = 0$. An express train is at rest at the station. The express train leaves the station with a uniform acceleration of $0.5\,m\,s^{-2}$ just as the goods train goes past. Both trains move in the same direction on straight, parallel tracks.

At which time t does the express train overtake the goods train?

A 6 s **B** 10 s **C** 20 s **D** 40 s

8 A constant resultant force F acts on an object of mass m for time t.

What is the change in momentum of the object?

A $\frac{F}{t}$ **B** $\frac{Ft}{m}$ **C** Ft **D** $\frac{F}{mt}$

9 The acceleration of free fall on the surface of planet P is one-tenth of that on the surface of planet Q.

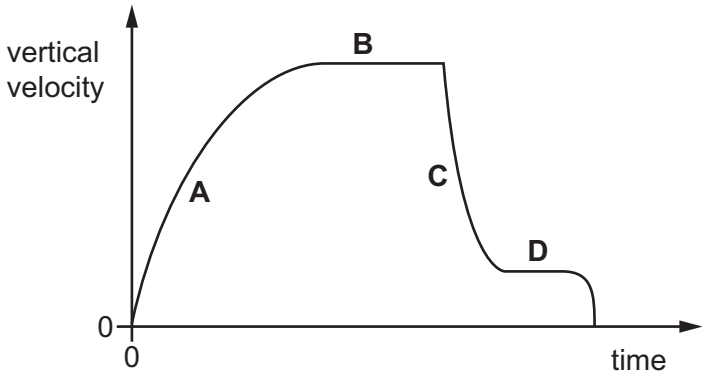
On the surface of P, an object has a mass of 1.0 kg and a weight of 1.0 N.

What are the mass and the weight of the same object on the surface of planet Q?

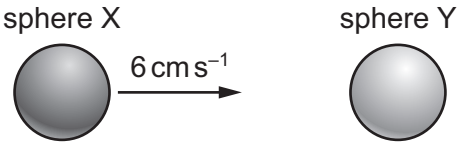
	mass on Q / kg	weight on Q / N
A	1.0	0.1
B	1.0	10
C	10	10
D	10	100

10 A parachutist falls from rest from a balloon. The variation with time of the vertical velocity of the parachutist is shown.

In which region is the force due to air resistance much greater than the weight of the parachutist?



11 Two solid spheres form an isolated system. Sphere X moves with speed $6\,cm\,s^{-1}$ in a straight line directly towards a stationary sphere Y, as shown.



The spheres have a perfectly elastic collision. After the collision, sphere X moves with speed $2\,cm\,s^{-1}$ in the same direction as before the collision.

What is the speed of sphere Y?

A $2\,cm\,s^{-1}$ **B** $4\,cm\,s^{-1}$ **C** $6\,cm\,s^{-1}$ **D** $8\,cm\,s^{-1}$

12 What is **not** a necessary requirement of the forces in a couple?

A They act in opposite directions.

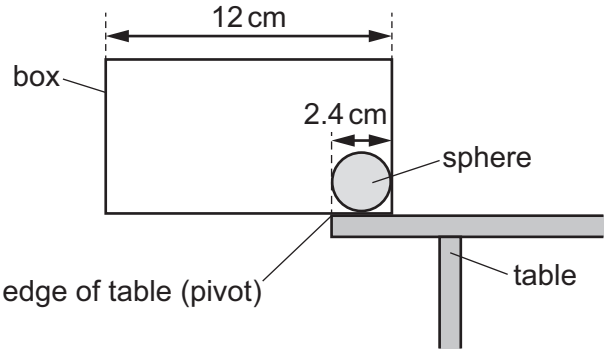
B They act along different lines.

C They have the same magnitude.

D They produce a resultant force.

13 A box of length 12 cm and weight 0.43 N is placed on a horizontal table, with the greater part of its length overhanging the edge of the table. The edge of the table acts as a pivot. The centre of gravity of the box is at its geometric centre.

To balance the box, a uniform sphere of diameter 2.4 cm is placed inside the box, touching one end, as shown.



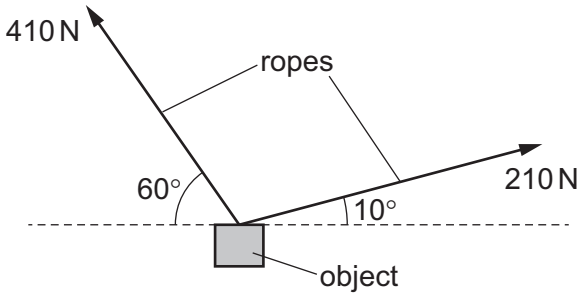
Assume that the forces acting on the box are in the plane of the diagram.

What is the minimum mass of the sphere that is needed to maintain the system in equilibrium?

A 0.066 kg **B** 0.13 kg **C** 0.22 kg **D** 1.3 kg

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14 An object is suspended by two ropes. One rope has a tension of 410 N at an angle of 60° to the horizontal. The other rope has a tension of 210 N at an angle of 10° to the horizontal.



The object is in equilibrium.

What is the mass of the object?

A 40 kg **B** 42 kg **C** 390 kg **D** 410 kg

15 A solid cube is floating in equilibrium in liquid mercury. The cube is made of iron of density 7900 kg m^{-3}.

The cube floats with 42% of its volume above the surface of the mercury.

What is the density of the mercury?

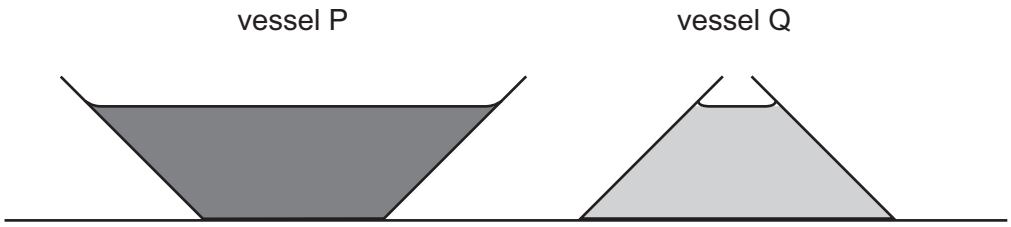
A 3300 kg m^{-3}

B 4600 kg m^{-3}

C 14 000 kg m^{-3}

D 19 000 kg m^{-3}

16 The diagram shows two vessels, P and Q, both with sides inclined at 45° to the horizontal.



Vessel P tapers outwards and vessel Q tapers inwards, as shown.

Both vessels contain a liquid. The depth of the liquid in the vessels is the same. The liquid in vessel P is twice as dense as the liquid in vessel Q.

What is the ratio $\frac{\text{pressure due to the liquid on the base of P}}{\text{pressure due to the liquid on the base of Q}}$?

A $\frac{2}{1}$ **B** $\frac{\sqrt{2}}{1}$ **C** $\frac{1}{\sqrt{2}}$ **D** $\frac{1}{2}$

17 A motor is used to lift a load vertically upwards.

The load has weight W.

The motor produces useful power output P.

The load is lifted at constant velocity v.

Which expression gives the time taken for the motor to lift the load vertically upwards through a distance d?

A $\frac{P}{Wd}$ **B** $\frac{Wv}{P}$ **C** $\frac{Wd}{P}$ **D** $\frac{Pv}{W}$

18 A lamp is switched on for 2.0 hours. The power input to the lamp is 1.0 W. The energy given out by the lamp as light is 7.0×10^3 J.

How much energy is converted to other forms by the lamp?

A 120 J **B** 200 J **C** 3400 J **D** 7200 J

19 An object of mass m is dropped onto the surface of two planets, X and Y, which have no atmosphere.

The height from which the object is dropped and the change in gravitational potential energy of the object, for each planet, are given in the table.

	height / m	change in gravitational potential energy
planet X	3	ΔE
planet Y	4	$4\Delta E$

The acceleration of free fall near the surface of planet X is g_X.

What is the acceleration of free fall near the surface of planet Y?

A $\frac{3}{4}g_X$ **B** $\frac{4}{3}g_X$ **C** $3g_X$ **D** $4g_X$

20 A known tensile force acts on a metal wire. The wire does not exceed its limit of proportionality.

Which two measurements enable the strain of the wire to be calculated?

A the unstretched length of the wire and the cross-sectional area of the wire

B the unstretched length of the wire and the extension of the wire

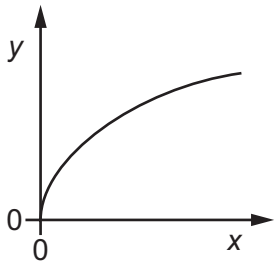
C the Young modulus of the metal and the extension of the wire

D the Young modulus of the metal and the unstretched length of the wire

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21 A wire is extended by different forces. The wire obeys Hooke's law.

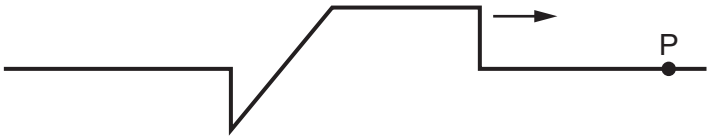
A graph is plotted to show the variation of a quantity *y* with a quantity *x*.



What could *x* and *y* represent?

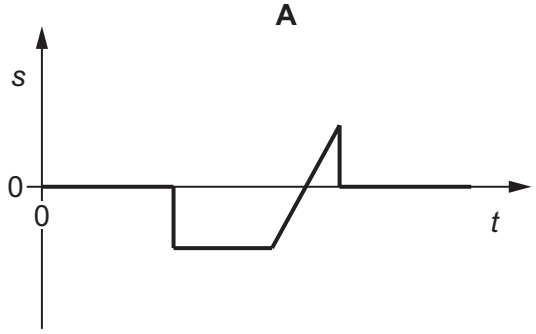
	x	*y*
A	elastic potential energy	extension
B	extension	force
C	force	extension
D	extension	elastic potential energy

22 A wave pulse moves along a stretched rope in the direction shown.

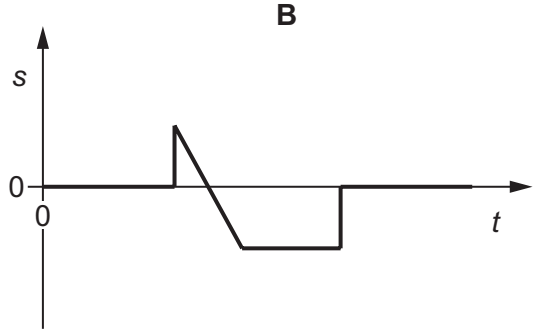


Which diagram shows the variation with time *t* of the displacement *s* of the particle P in the rope?

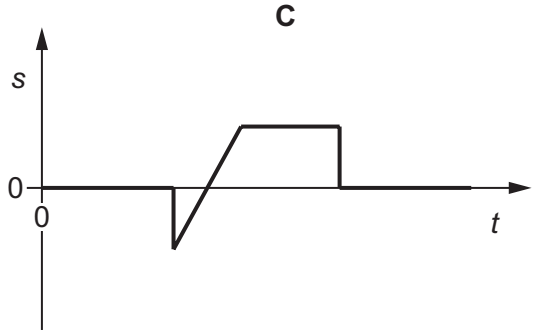
A



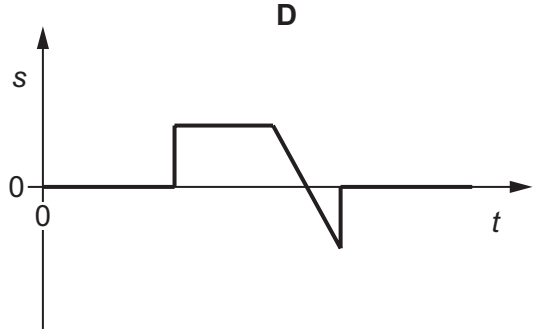
B



C



D



23 Which statement about progressive transverse and longitudinal waves is correct?

A Particles in a transverse wave have fixed equilibrium positions but those in longitudinal waves do not.

B Transverse waves can be polarised but longitudinal waves cannot.

C Transverse waves transfer energy but longitudinal waves do not.

D Two-source interference can be demonstrated with transverse waves but not with longitudinal waves.

24 A miniature loudspeaker, initially at rest, falls vertically from a window in a high building. When the speaker has fallen a distance of 10.0 m, it emits a very short pulse of sound of constant frequency 256 Hz in all directions. The pulse of sound, travelling at a speed of 330 m s^{-1}, is heard by a person leaning out of the window.

Air resistance is negligible.

What is the frequency of the pulse of sound heard by the person?

A 246 Hz **B** 249 Hz **C** 267 Hz **D** 313 Hz

25 Two electromagnetic waves have wavelengths of 5.0×10^{-7} m and 5.0×10^{-2} m in a vacuum.

Which row identifies the regions of the electromagnetic spectrum to which the waves belong?

	wavelength 5.0×10^{-7} m	wavelength 5.0×10^{-2} m
A	ultraviolet	infrared
B	visible	microwave
C	ultraviolet	microwave
D	visible	infrared

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26 The wavelength of sound in air may be determined by using stationary waves.

In one experiment, a loudspeaker produces a sound wave of constant frequency which is reflected directly back along its original path by a metal plate approximately 1 m away. A microphone connected to a cathode-ray oscilloscope (CRO) is moved between the loudspeaker and plate to identify regions of high sound intensity ('loud' spots) and low sound intensity ('quiet' spots).

The wavelength of the sound is determined using the **least** possible number of measured quantities.

Which row shows the quantities that are needed?

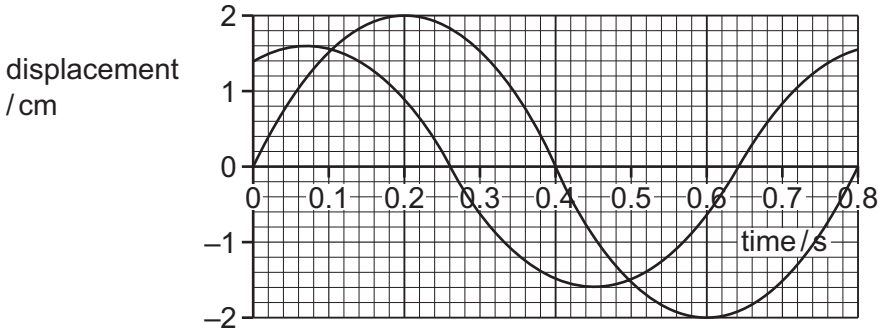
	frequency of sound	mean separation of 'quiet' spots	speed of sound in air
A	✓	✓	✗
B	✓	✗	✓
C	✓	✗	✗
D	✗	✓	✗

key

✓ = needed

✗ = not needed

27 Two progressive waves meet at a fixed point P. The variation with time of the displacement of each wave at point P is shown.



The two waves superpose at point P.

What is the resultant displacement at time 0.38 s?

A +1.0 cm **B** –1.0 cm **C** +1.8 cm **D** –1.8 cm

28 In which situation does diffraction occur?

A A wave bounces back from a surface.

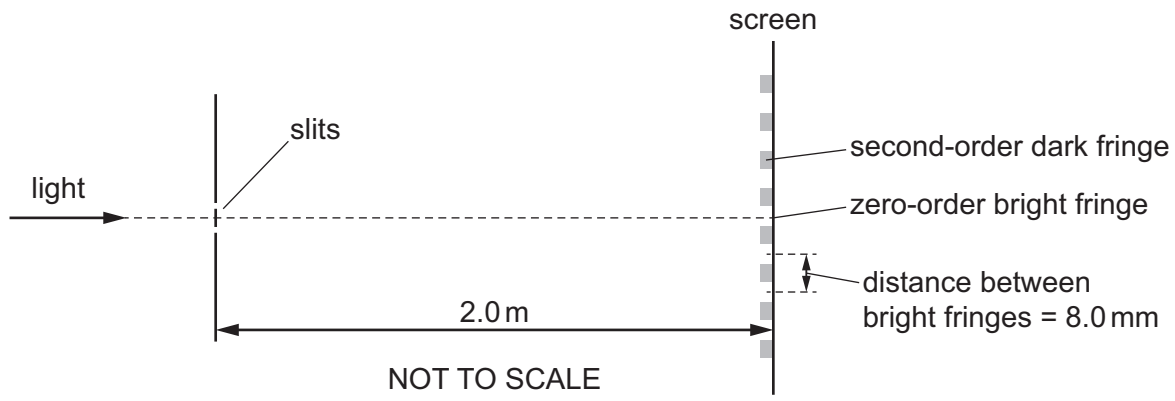
B A wave passes from one medium into another.

C A wave passes through a gap in a barrier.

D Waves from two identical sources are superposed.

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29 Light of a single frequency is incident on a pair of narrow slits that are a distance of 0.10 mm apart. A series of bright and dark fringes is observed on a screen a distance of 2.0 m away. The distance between adjacent bright fringes is 8.0 mm.



What is the path difference of the light waves from the two slits that meet at the second-order **dark** fringe?

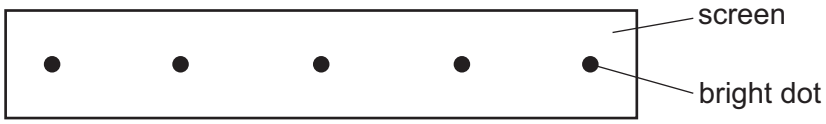
A 2.0×10^{-7} m

B 4.0×10^{-7} m

C 6.0×10^{-7} m

D 8.0×10^{-7} m

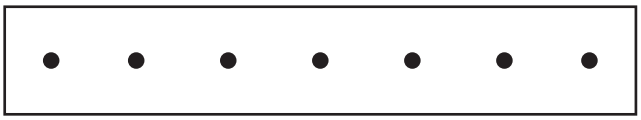
30 Red light of a single wavelength passes through a diffraction grating. Bright dots are formed on a screen, as shown.



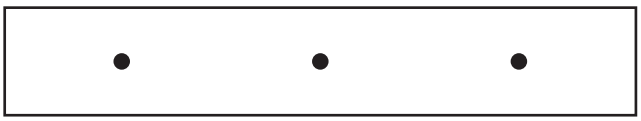
The red light is replaced with white light.

Which diagram, drawn to the same scale, shows a possible pattern of bright light on the screen?

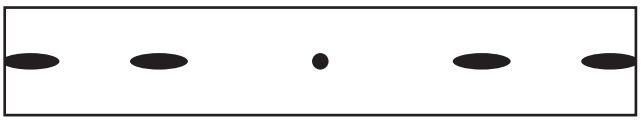
A



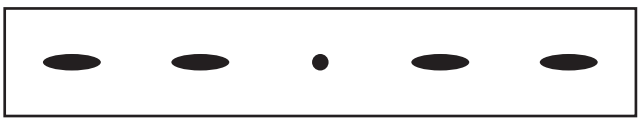
B



C



D



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31 A nichrome wire has a resistance of $15\,\Omega$ and a diameter of 3.0 mm. The number density of the free electrons in nichrome is $9.0 \times 10^{28}\,\text{m}^{-3}$.

A potential difference (p.d.) of 6.0 V is applied between the ends of the wire.

What is the average drift speed of the free electrons in the wire?

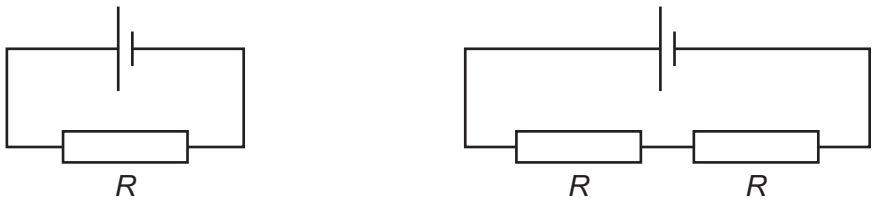
A $9.8 \times 10^{-7}\,\text{m s}^{-1}$

B $3.9 \times 10^{-6}\,\text{m s}^{-1}$

C $6.1 \times 10^{-6}\,\text{m s}^{-1}$

D $2.5 \times 10^{-5}\,\text{m s}^{-1}$

32 The diagrams show two different circuits.



The cells in each circuit have the same electromotive force (e.m.f.) and negligible internal resistance. The three resistors each have the same resistance *R*.

In the circuit on the left, the power dissipated in the resistor is *P*.

What is the total power dissipated in the circuit on the right?

A $\frac{P}{4}$ **B** $\frac{P}{2}$ **C** P **D** $2P$

33 The potential difference (p.d.) across a filament lamp is increased.

Which statement is correct?

A The resistance of the lamp decreases because the temperature decreases.

B The resistance of the lamp decreases because the temperature increases.

C The resistance of the lamp increases because the temperature decreases.

D The resistance of the lamp increases because the temperature increases.

34 A metal wire has resistance *R*.

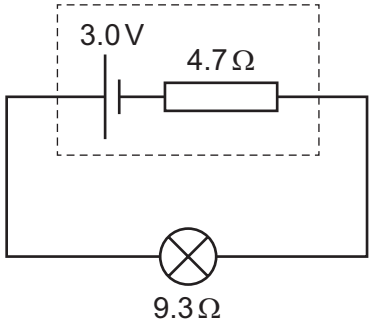
The wire is stretched so that its diameter decreases to 94.0% of the original diameter.

The volume of the wire is unchanged.

What is the resistance of the stretched wire?

A 1.06*R* **B** 1.13*R* **C** 1.20*R* **D** 1.28*R*

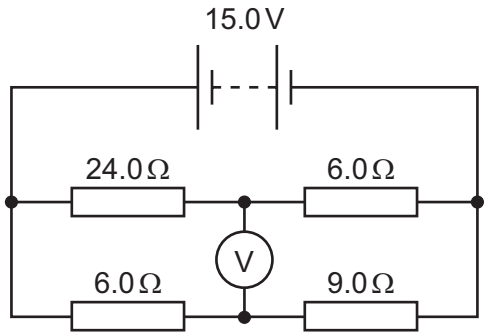
35 The diagram shows a cell of electromotive force (e.m.f.) 3.0 V and internal resistance 4.7 Ω connected across a lamp. The lamp has a resistance of 9.3 Ω.



What is the power dissipated by the internal resistance of the cell?

A 0.22 W **B** 0.43 W **C** 0.64 W **D** 1.0 W

36 A circuit consists of a battery, a high-resistance voltmeter and four fixed resistors, as shown. The battery has an electromotive force (e.m.f.) of 15.0 V and negligible internal resistance.

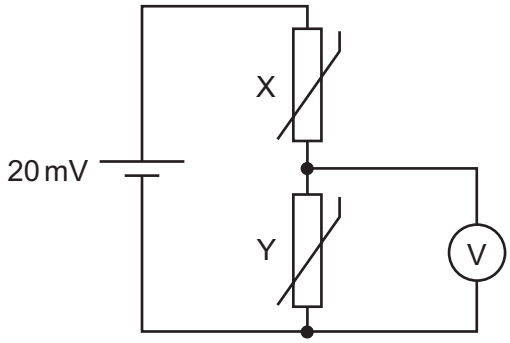


What is the reading on the voltmeter?

A 3.0 V **B** 6.0 V **C** 9.0 V **D** 12.0 V

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37 A potential divider circuit is designed to detect the difference in temperature between two different places.



The cell has electromotive force (e.m.f.) 20 mV and negligible internal resistance.

Initially, thermistors X and Y are at the same temperature and have the same resistance. The voltmeter reads 10 mV. X is then placed in a cold environment and its resistance doubles. Y is placed in a warm environment and its resistance halves.

What is the new reading on the voltmeter?

A 4 mV **B** 5 mV **C** 15 mV **D** 16 mV

38 In the α-particle scattering experiment, a beam of α-particles is aimed at a thin gold foil. Most of the α-particles go straight through or are deflected by a small angle. A very small proportion are deflected by more than 90°, effectively rebounding towards the source of the α-particles.

Which conclusion about the structure of atoms **cannot** be drawn from this experiment alone?

A Most of the atom is empty space.

B Most of the mass of an atom is concentrated in the nucleus.

C The nucleus contains both protons and neutrons.

D The nucleus is charged.

39 Which two particles have opposite charges?

A alpha-particle and helium nucleus

B antiproton and beta-plus particle

C beta-minus particle and electron

D positron and proton

40 Which particle is a lepton?

A meson

B positron

C proton

D quark

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PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2022**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

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Question	Answer	Marks
1	D	**1**
2	C	**1**
3	D	**1**
4	C	**1**
5	A	**1**
6	A	**1**
7	D	**1**
8	C	**1**
9	B	**1**
10	C	**1**
11	D	**1**
12	D	**1**
13	B	**1**
14	A	**1**
15	C	**1**
16	A	**1**
17	C	**1**
18	B	**1**
19	C	**1**
20	B	**1**
21	A	**1**
22	D	**1**
23	B	**1**
24	A	**1**
25	B	**1**
26	D	**1**
27	B	**1**
28	C	**1**

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Question	Answer	Marks
29	C	**1**
30	D	**1**
31	B	**1**
32	B	**1**
33	D	**1**
34	D	**1**
35	A	**1**
36	B	**1**
37	A	**1**
38	C	**1**
39	B	**1**
40	B	**1**

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November 2022 · Paper 13

9702_w22_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **October/November 2022**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 A train of mass 600 000 kg moves with a speed of 100 km h^{-1}.

What is the order of magnitude of the kinetic energy of the train?

A 10^6 J **B** 10^8 J **C** 10^{10} J **D** 10^{12} J

2 What are the SI base units of electromotive force (e.m.f.)?

A kg m^2 s^{-1} A^{-1}

B kg m^2 s^{-3} A^{-1}

C kg m^2 s^{-1} A

D kg m s^{-3} A^{-1}

3 A digital meter is used to measure the current in an electric circuit.

The reading on the meter fluctuates (varies) between 3.04 A and 3.08 A. The readings on the meter have an accuracy of ±1%.

What is the true value of the current, with its uncertainty?

A (3.06 ± 0.02) A

B (3.06 ± 0.04) A

C (3.06 ± 0.05) A

D (3.06 ± 0.07) A

4 Which quantity is a vector quantity?

A density

B mass

C volume

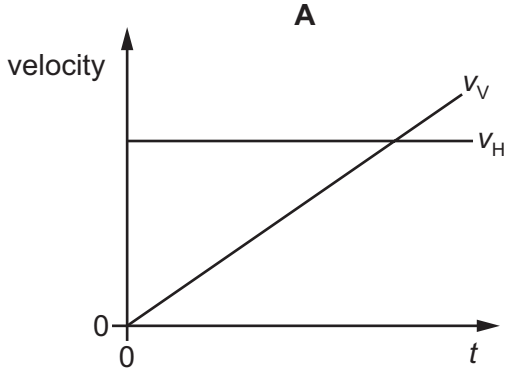
D weight

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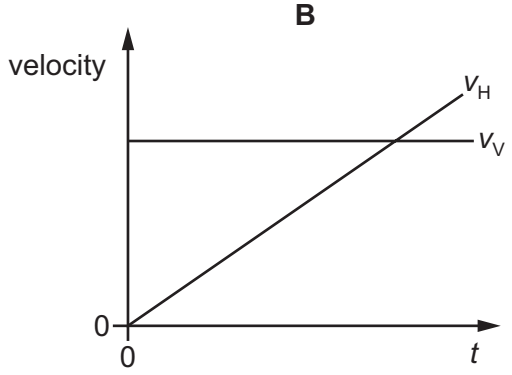
5 A stone is projected horizontally at time $t = 0$ and falls. Air resistance is negligible. The stone has a horizontal component of velocity v_H and a vertical component of velocity v_V.

Which graph shows how v_H and v_V vary with time t?

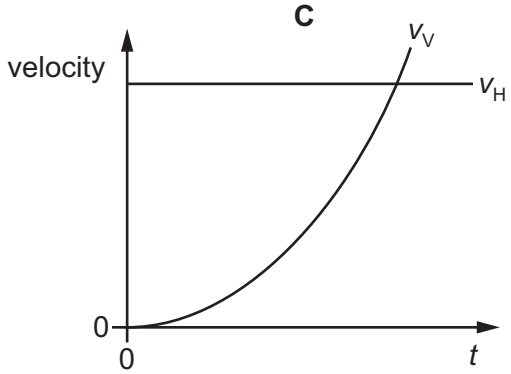
A



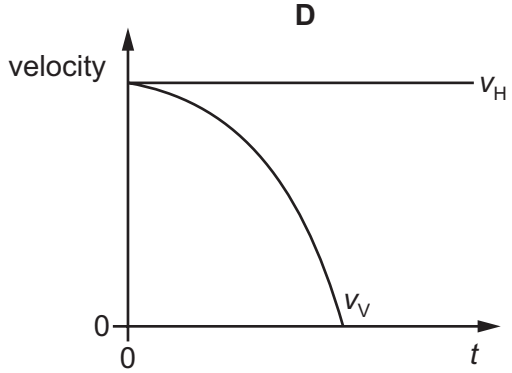
B



C



D



6 On the Earth, an object takes time T_E to fall from rest through a vertical distance h.

On the Moon, the same object takes time T_M to fall from rest through the same vertical distance h.

The ratio $\dfrac{\text{acceleration of free fall on the Earth}}{\text{acceleration of free fall on the Moon}}$ is equal to 6.

Air resistance is negligible for the object on the Earth and on the Moon.

What is the ratio $\dfrac{T_E}{T_M}$?

A $\dfrac{1}{6}$ **B** $\dfrac{1}{\sqrt{6}}$ **C** $\sqrt{6}$ **D** 6

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7 A cyclist in still air pedals as hard as she can. She reaches a maximum speed. However, after a certain time her maximum speed increases.

What could be a possible cause for this?

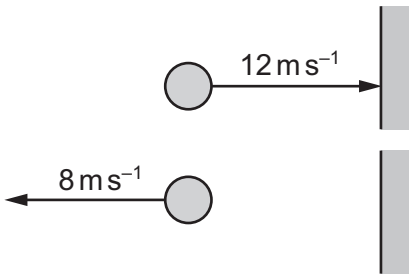
A She cycles into a wind.

B She cycles over rougher ground.

C She sits more upright on the bicycle.

D She starts to travel downhill.

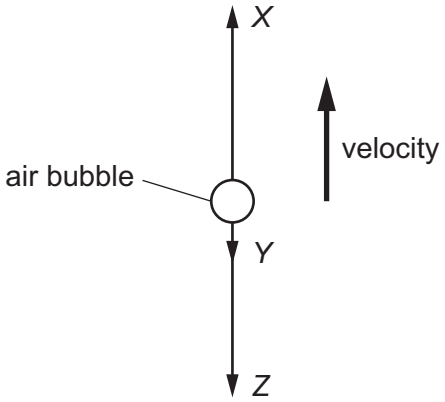
8 A ball of mass 0.5 kg hits a vertical wall at a speed of $12\,m\,s^{-1}$. It bounces back along its original path with a speed of $8\,m\,s^{-1}$. The collision lasts for 0.10 s.



What is the average force on the ball due to the collision?

A 0.2 N **B** 1 N **C** 20 N **D** 100 N

9 An air bubble in a tank of water is rising with terminal (constant) velocity. The forces acting on the bubble are *X*, *Y* and *Z*, as shown.



The upthrust on the bubble is *X*.

Which statement about the forces is correct?

A *Z* is the viscous force on the bubble, *Y* is the weight of the bubble and $X = Y + Z$.

B *Z* is the viscous force on the bubble, *Y* is the weight of the bubble and $X > Y + Z$.

C *Z* is the weight of the bubble, *Y* is the viscous force on the bubble and $X = Y + Z$.

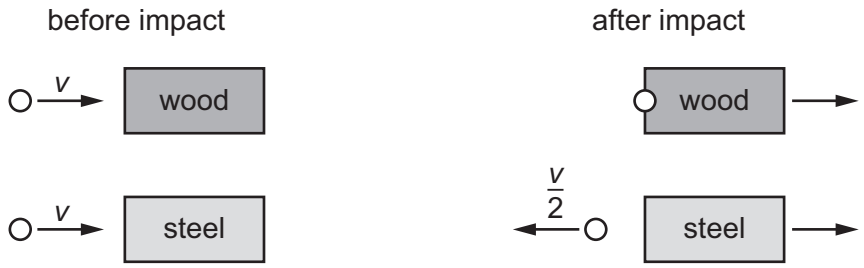
D *Z* is the weight of the bubble, *Y* is the viscous force on the bubble and $X > Y + Z$.

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10 Two blocks are at rest on a frictionless horizontal surface. One block is made of wood and the other block is made of steel.

A steel ball is fired horizontally with a speed v at the wooden block. The ball embeds itself in the block, and the ball and block move together after impact.

A second identical steel ball is fired horizontally with speed v at the steel block. The steel ball then rebounds back along its original path with speed $\frac{v}{2}$.



The wooden block and the steel block have equal mass.

Which statement about the blocks immediately after the collisions is correct?

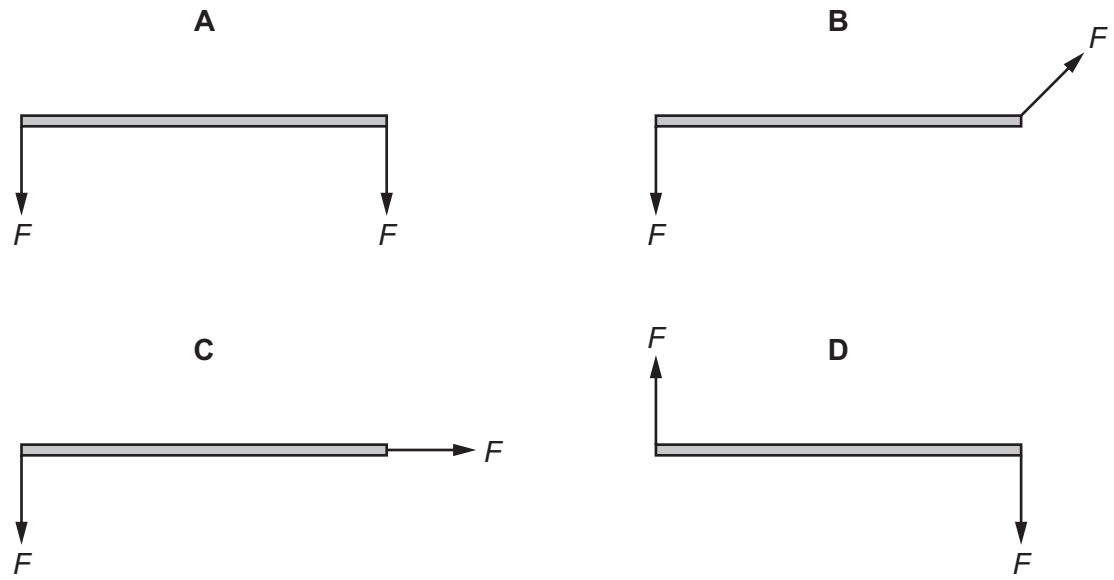
A Both blocks must travel with the same speed.

B The steel block must travel faster than the wooden block.

C The wooden block must travel faster than the steel block.

D The masses of the blocks and the steel ball are needed to determine which block travels faster.

11 Which diagram shows a couple formed by two forces, each of magnitude F, acting on a rod?



12 A student states that:

> ‘If an object is in equilibrium, the sum of the clockwise moments about a point X is equal to the sum of the anticlockwise moments about a point Y.’

Which condition would make the student’s statement correct?

A Either X or Y is the centre of gravity of the object.

B Either X or Y is the pivot of the object.

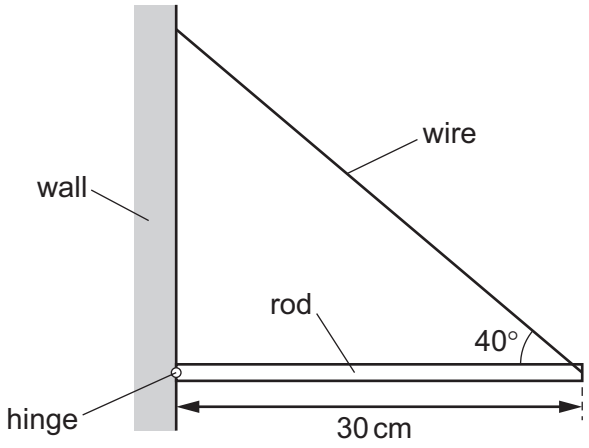
C X and Y are at opposite ends of the object.

D X and Y are the same point on the object.

13 A uniform rod of length 30 cm and weight 5.2 N is attached to a wall by a hinge at one end.

The other end of the rod is supported by a wire so that the rod is horizontal and in equilibrium.

The wire is at an angle of 40° to the horizontal.



What is the tension in the wire?

A 3.4 N **B** 4.0 N **C** 6.8 N **D** 8.1 N

14 Water is pumped through a nozzle at the end of a hose. The nozzle has a circular cross-section of diameter 50 mm. A mass of 100 kg of water takes a time of 2.0 s to move through the nozzle. The density of water is 1000 kg m^{-3}.

What is the speed of the water in the nozzle?

A 6.4 m s^{-1} **B** 13 m s^{-1} **C** 25 m s^{-1} **D** 51 m s^{-1}

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15 A solid wooden cube rests on a horizontal surface.

What gives the pressure exerted by the weight of the cube on the horizontal surface?

A the product of the acceleration of free fall and the density of the cube

B the product of the acceleration of free fall, the density and the side length of the cube

C the product of the acceleration of free fall, the density and the area of one face of the cube

D the product of the acceleration of free fall, the density and the volume of the cube

16 Which expression could be used to calculate power?

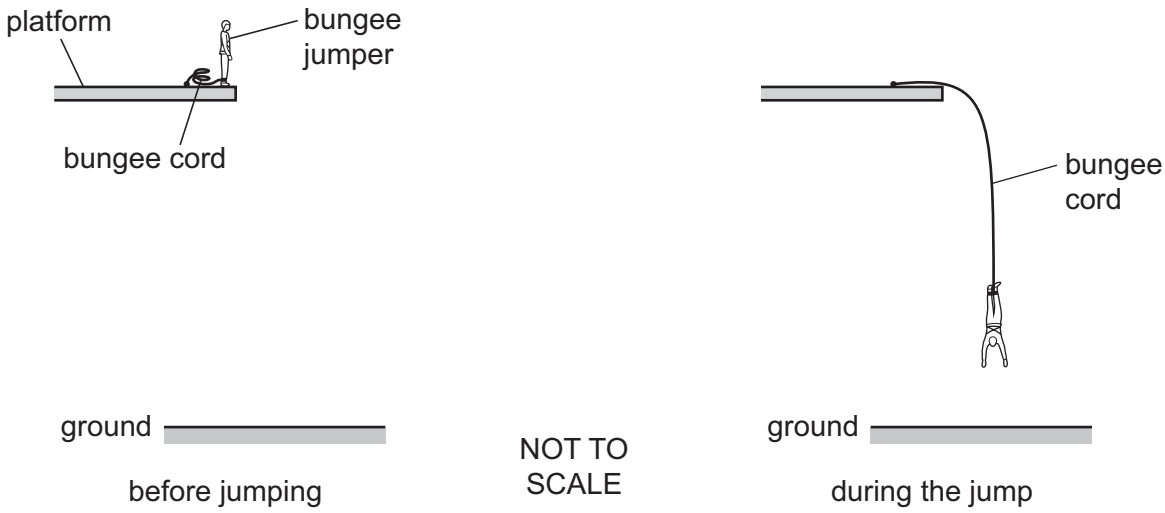
A $\dfrac{(\text{current})^2}{\text{resistance}}$

B $\dfrac{\text{force} \times \text{displacement}}{\text{time}}$

C $\text{current} \times (\text{resistance})^2$

D $\dfrac{\text{weight}}{\text{time}}$

17 A bungee jumper jumps from a platform and is decelerated by an elastic bungee cord, as shown.



When the jumper makes the jump, his initial gravitational potential energy is converted into his kinetic energy and into elastic potential energy in the cord.

At which part of the jump are all three types of energy non-zero?

A on the platform before the jump

B on the way down before the cord has started to extend

C on the way down as he decelerates

D at the bottom of the jump when he is stationary

18 An object is displaced horizontally to the right in a uniform vertical gravitational field.

Which statement describes the change in the gravitational potential energy of the object?

A It decreases in direct proportion to the displacement.

B It does not change with the displacement.

C It increases in direct proportion to the displacement.

D It increases in direct proportion to the square of the displacement.

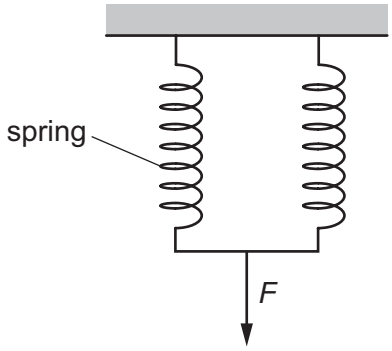
19 A copper wire of length 3.6 m and diameter 1.22 mm is stretched by a force of 37 N. The wire obeys Hooke's law. The Young modulus of copper is 1.17×10^{11} Pa.

Which extension is caused by this force?

A 0.24 mm **B** 0.76 mm **C** 0.97 mm **D** 3.1 mm

20 A spring has spring constant k. The spring obeys Hooke's law and experiences extension x when a force F is applied to it. The resulting elastic potential energy of the spring is E_P.

The diagram shows two of these springs joined together in parallel and hanging from a fixed beam.



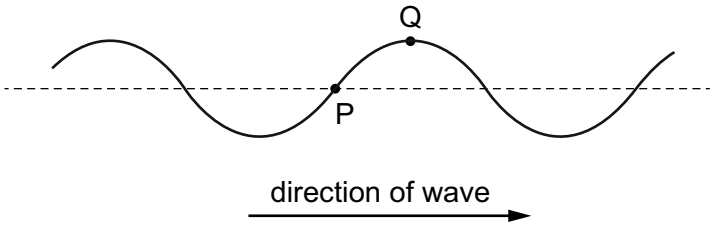
What is the extension and total elastic potential energy of this arrangement when the same force F is applied?

	extension	total elastic potential energy
A	$\frac{1}{2}x$	$\frac{1}{2}E_P$
B	$\frac{1}{2}x$	$\frac{1}{4}E_P$
C	x	$\frac{1}{2}E_P$
D	x	$\frac{1}{4}E_P$

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21 The diagram shows a transverse wave on a rope. The wave is travelling from left to right.

At the instant shown, the points P and Q on the rope have zero displacement and maximum displacement respectively.



Which row describes the direction of motion, if any, of the points P and Q at this instant?

	point P	point Q
A	downwards	stationary
B	stationary	downwards
C	stationary	upwards
D	upwards	stationary

22 The period of an electromagnetic wave in a vacuum is 1.0 ns.

What are the frequency and wavelength of the wave?

	frequency / Hz	wavelength / m
A	1.0	3.0×10^{8}
B	1.0×10^{6}	300
C	1.0×10^{9}	0.30
D	1.0×10^{12}	3.0×10^{-4}

23 An observer is standing on a railway platform. A train passes the observer at constant speed while emitting sound of constant frequency *f* from its whistle.

What does the observer hear?

A sound of a decreasing frequency as the train approaches and of an increasing frequency as it moves away

B sound of a higher frequency than *f* as the train approaches and of a lower frequency than *f* as the train moves away

C sound of a lower frequency than *f* as the train approaches and of a higher frequency than *f* as the train moves away

D sound of an increasing frequency as the train approaches and as it moves away

24 Which microorganisms have a length that is equal to the wavelength in free space of an electromagnetic wave that is visible to the human eye?

A algae of length 0.5 μm

B bacteria of length 5.0 μm

C fungi of length 50 μm

D protozoa of length 100 μm

25 Two progressive waves overlap.

What is an essential condition for the two waves to form a stationary wave?

A The waves are longitudinal.

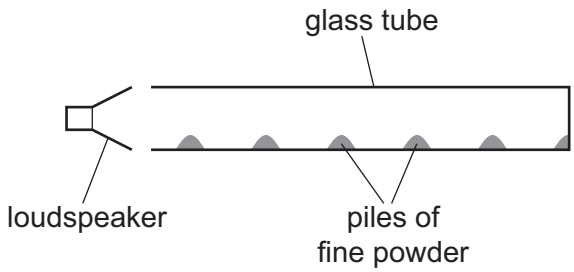
B The waves are polarised.

C The waves travel in opposite directions.

D The waves travel in the same direction.

26 In an experiment to produce a stationary sound wave in air, a fine powder is initially evenly distributed along the length of a horizontal glass tube which is closed at one end.

At the open end of the tube, a loudspeaker emits a sound wave of a constant wavelength. A stationary wave is formed and the powder accumulates in regularly spaced piles, as shown.



Which statement explains the positions of the piles of powder within the tube?

A The piles are where the air molecules vibrate with maximum amplitude.

B The piles are where the air molecules vibrate with minimum amplitude.

C The piles are where the air molecules vibrate with the highest frequency.

D The piles are where the air molecules vibrate vertically.

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27 A sound wave of wavelength 0.50 m passes through a doorway of width 1.0 m between two rooms.

Which change increases the amount of diffraction that takes place?

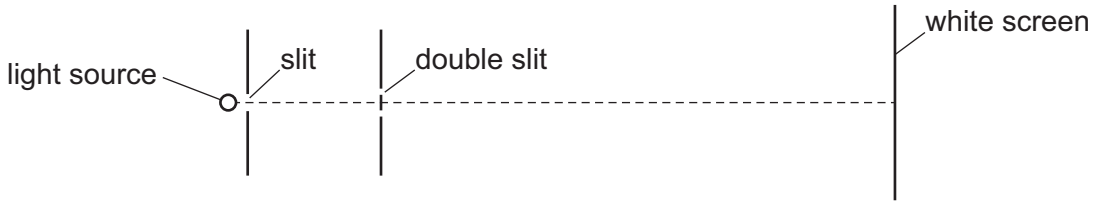
A Double the amplitude of the sound wave.

B Double the width of the doorway.

C Halve the frequency of the sound wave.

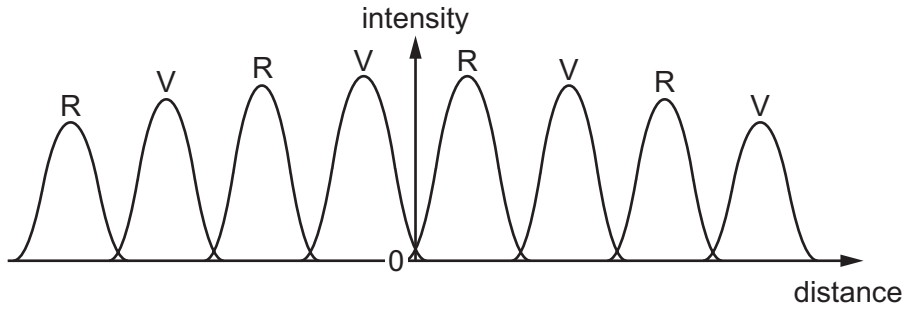
D Halve the period of the sound wave.

28 A light source consists of a vertical slit illuminated by red light (R) and violet light (V). The wavelength of R is approximately twice the wavelength of V. A parallel vertical double slit is placed nearby. A white screen is placed so that fringes are formed on it.

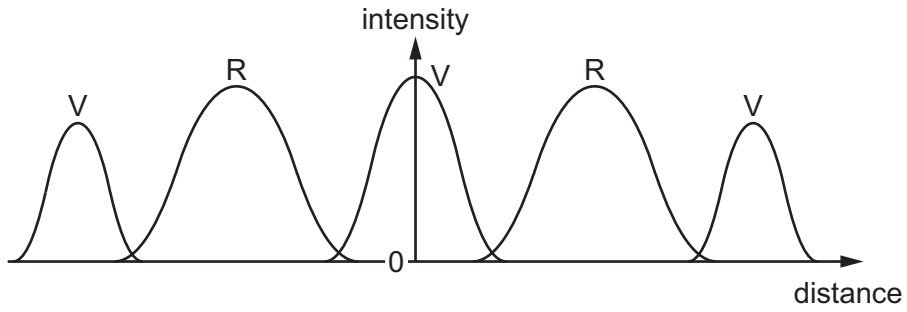


Which graph best represents the interference fringes formed on the screen?

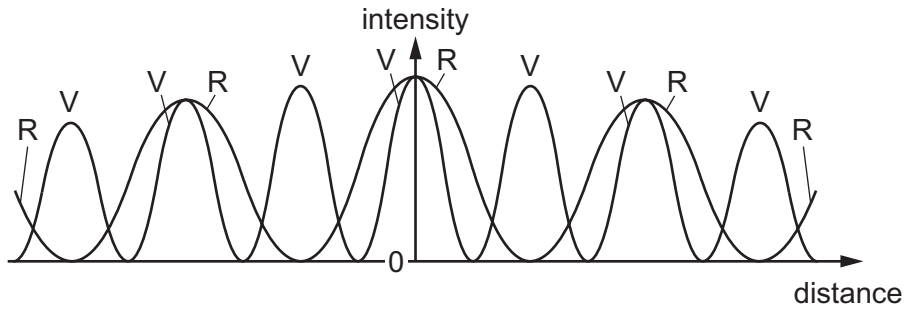
A



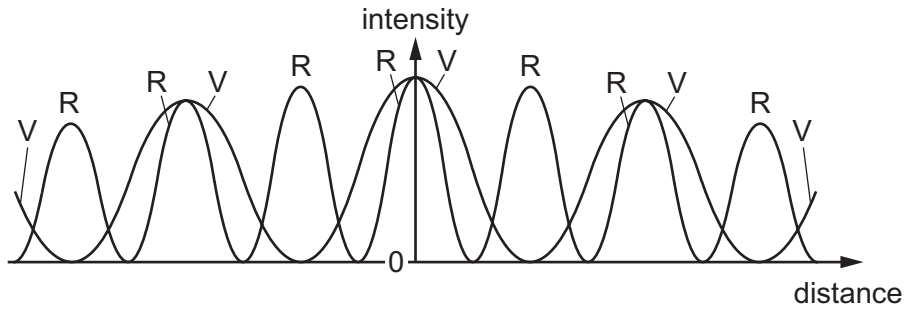
B



C



D



29 Which property of a light wave can be determined using a diffraction grating?

A amplitude

B intensity

C speed

D wavelength

30 There is an electric current in a copper wire.

Which statement describing the average drift speed of the charge carriers in the wire is correct?

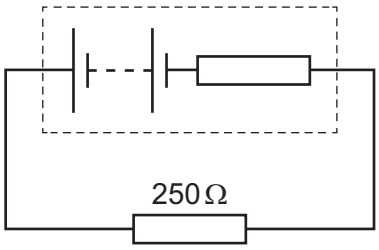
A It is nearly $3 \times 10^8\,\text{m s}^{-1}$.

B It is proportional to the cross-sectional area of the wire.

C It is proportional to the length of the wire.

D It is proportional to the magnitude of the current.

31 A battery with a constant internal resistance is connected to a resistor of resistance 250 Ω, as shown.



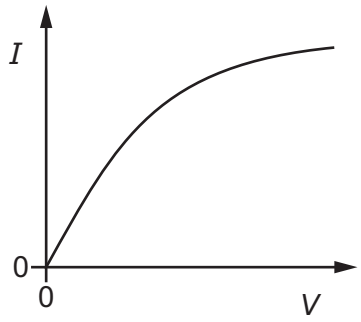
The current in the resistor is 40 mA for a time of 60 s. During this time 6.0 J of energy is dissipated by the internal resistance.

What is the energy supplied to the external resistor during the 60 s and the electromotive force (e.m.f.) of the battery?

	energy / J	e.m.f. / V
A	30	2.5
B	30	7.5
C	24	10.0
D	24	12.5

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32 Which component has the I–V graph shown?



A filament lamp

B metallic conductor at constant temperature

C resistor of fixed resistance

D semiconductor diode

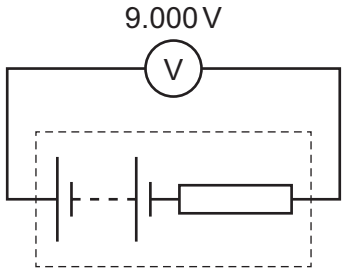
33 Two wires, P and Q, have the same resistance. Wire Q is made of material that has twice the resistivity of the material used to make wire P. The diameter of wire Q is twice the diameter of wire P.

What is the ratio $\frac{\text{length of wire P}}{\text{length of wire Q}}$?

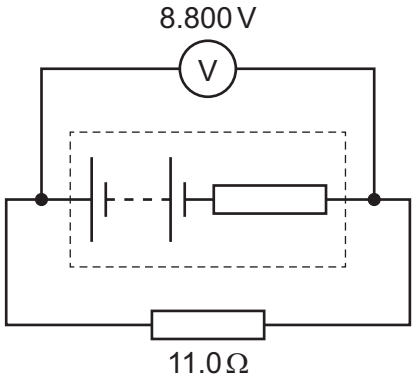
A $\frac{1}{8}$ **B** $\frac{1}{4}$ **C** $\frac{1}{2}$ **D** $\frac{2}{1}$

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34 A voltmeter reads 9.000 V when it is connected across the terminals of a battery.



When a resistor of resistance 11.0 Ω is connected in parallel with the battery, the voltmeter reading changes to 8.800 V.



What is the internal resistance of the battery?

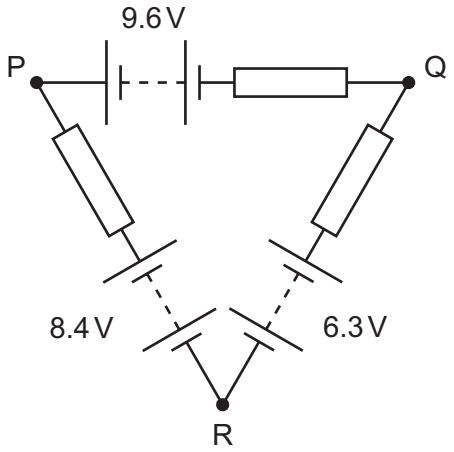
A 0.244 Ω **B** 0.250 Ω **C** 10.8 Ω **D** 11.3 Ω

35 Each of Kirchhoff's laws is linked to the conservation of a physical quantity.

Which conserved physical quantities are used in the derivation of Kirchhoff's first law and of Kirchhoff's second law?

	Kirchhoff's first law	Kirchhoff's second law
A	energy	charge
B	energy	momentum
C	charge	energy
D	momentum	energy

36 Three batteries and three identical resistors are connected in a circuit PQR, as shown.

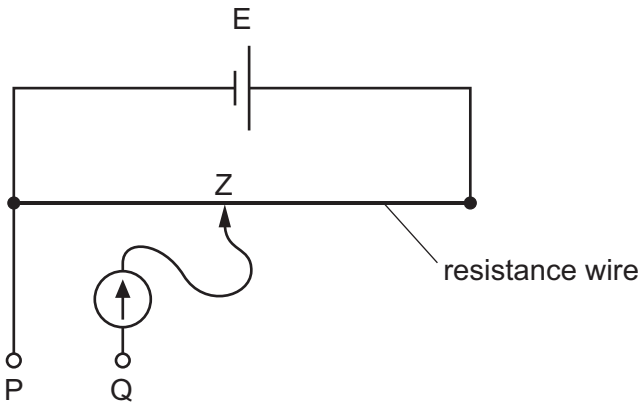


The batteries have negligible internal resistance.

What is the potential difference between points P and Q?

A 1.5 V **B** 2.1 V **C** 7.1 V **D** 12.1 V

37 A cell E, of electromotive force (e.m.f.) 2 V and negligible internal resistance, is connected to a uniform resistance wire of resistance 10 Ω and length 1.0 m.



Z is a connection that may be made at any position along the resistance wire. A galvanometer is connected between Z and a point Q.

A new source of e.m.f. of approximately 8 mV is connected between points P and Q. The e.m.f. of the new source is determined by changing the position of Z until the reading on the galvanometer is zero.

Which change to the circuit allows a much more precise value for the e.m.f. of the new source to be obtained?

A Add a resistor of resistance 0.1 Ω in series with cell E.

B Add a resistor of resistance 1000 Ω in series with cell E.

C Add a resistor of resistance 10 Ω in series with the new source.

D Add a resistor of resistance 800 Ω in series with the new source.

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38 Which statement about the alpha-particle scattering experiment provides evidence for the existence of the nucleus?

A A tiny proportion of the alpha-particles are deflected through large angles.

B Slower alpha-particles are deflected through larger angles.

C The kinetic energies of the deflected alpha-particles are unchanged.

D The number of alpha-particles deflected depends on the thickness of the foil.

39 A nucleus X undergoes β^+ decay.

The products are a nucleus Y with proton number Z, a β^+ particle and another particle P.

$$X \rightarrow {}_{Z}Y + \beta^+ + P$$

What is particle P and what is the proton number of nucleus X?

	P	proton number of nucleus X
A	antineutrino	$Z - 1$
B	antineutrino	$Z + 1$
C	neutrino	$Z - 1$
D	neutrino	$Z + 1$

40 A particle consists of two up quarks and a charm quark.

What is the charge of this particle, expressed in terms of the elementary charge e?

A $-2e$ **B** $-e$ **C** $+e$ **D** $+2e$

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Paper 1 Multiple Choice **October/November 2022**

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This document consists of **3** printed pages.

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Question	Answer	Marks
1	B	**1**
2	B	**1**
3	C	**1**
4	D	**1**
5	A	**1**
6	B	**1**
7	D	**1**
8	D	**1**
9	A	**1**
10	B	**1**
11	D	**1**
12	D	**1**
13	B	**1**
14	C	**1**
15	B	**1**
16	B	**1**
17	C	**1**
18	B	**1**
19	C	**1**
20	A	**1**
21	A	**1**
22	C	**1**
23	B	**1**
24	A	**1**
25	C	**1**
26	B	**1**
27	C	**1**
28	C	**1**

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Question	Answer	Marks
29	D	**1**
30	D	**1**
31	D	**1**
32	A	**1**
33	C	**1**
34	B	**1**
35	C	**1**
36	C	**1**
37	B	**1**
38	A	**1**
39	D	**1**
40	D	**1**

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March 2023 · Paper 12

9702_m23_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2023**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What represents a physical quantity?

A 3.0

B kilogram

C 7.0 N

D 40%

2 The relationship between the variables D and T is given by the equation

$$\frac{1}{T} = \frac{b}{\sqrt{D}} + c$$

where b and c are constants.

The unit of D is m^2 and the unit of T is s.

What are the units of b and c?

	unit of b	unit of c
A	m s	s
B	m s^{-1}	s^{-1}
C	m^{-1} s	s
D	m^{-1} s^{-1}	s^{-1}

3 A hollow cylinder, which is open at both ends, has a radius of (3.0 ± 0.1) cm and a length of (15.0 ± 0.1) cm.

What is the value, with its absolute uncertainty, of the surface area of the cylinder?

A $(280 \pm 10)\,cm^2$

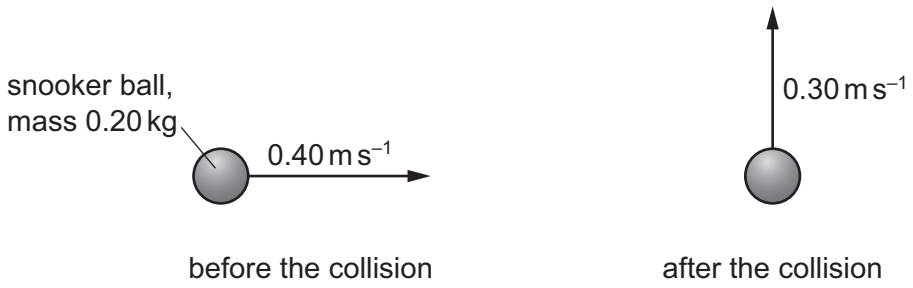
B $(282.7 \pm 0.2)\,cm^2$

C $(420 \pm 30)\,cm^2$

D $(424.1 \pm 0.3)\,cm^2$

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4 A snooker ball of mass 0.20 kg has a collision so that its direction of movement changes by an angle of 90°, as shown.



The ball has a speed of 0.40 m s^{-1} before the collision and a speed of 0.30 m s^{-1} after the collision.

What is the **magnitude** of the change in momentum of the snooker ball?

A 0.020 kg m s^{-1}

B 0.10 kg m s^{-1}

C 0.14 kg m s^{-1}

D 0.50 kg m s^{-1}

5 A ball is kicked upwards at an angle of 45° to horizontal ground. After a short flight, the ball returns to the ground.

It may be assumed that air resistance is negligible.

What is **never** zero during the flight of the ball?

A the horizontal component of the ball's acceleration

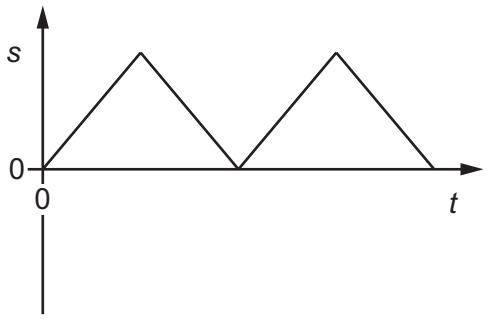
B the horizontal component of the ball's velocity

C the vertical component of the ball's momentum

D the vertical component of the ball's velocity

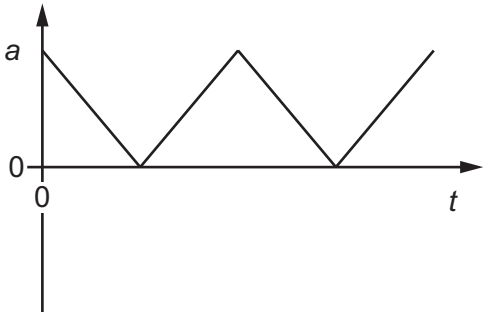
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6 The graph shows the variation with time t of the displacement s of an object.

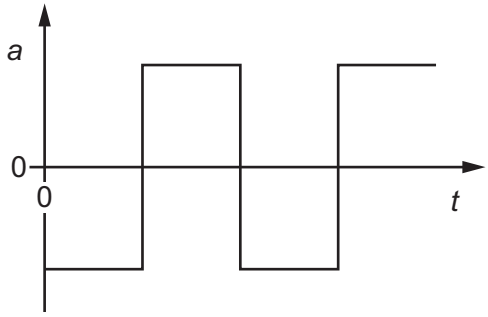


Which graph represents the variation with time t of the acceleration a of the object?

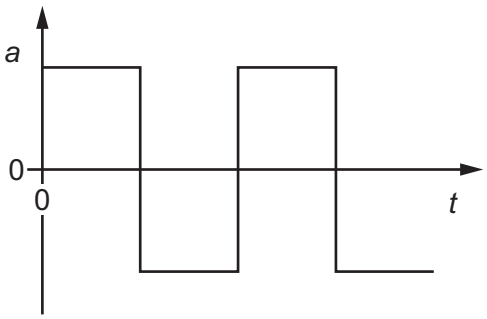
A



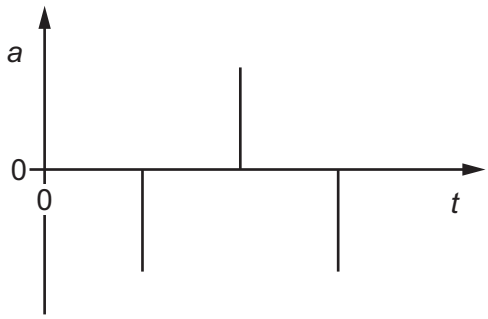
B



C



D



7 Which expression defines force?

A (mass × change in speed) × time taken

B $\dfrac{\text{mass} \times \text{change in speed}}{\text{time taken}}$

C (change of momentum) × time taken

D $\dfrac{\text{change of momentum}}{\text{time taken}}$

8 A ship of mass 8.4×10^7 kg is approaching a harbour with speed $16.4\,\text{m}\,\text{s}^{-1}$. By using reverse thrust it can maintain a constant total stopping force of 920 000 N.

How long will it take to stop?

A 15 seconds

B 150 seconds

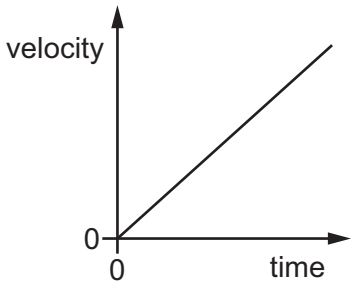
C 25 minutes

D 250 minutes

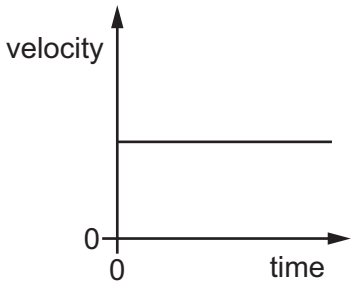
9 The velocity–time graphs of four different objects are shown.

Which graph represents an object falling from rest through a long distance in the Earth's atmosphere?

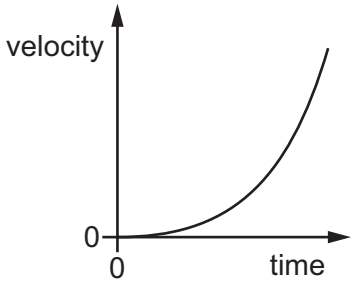
A



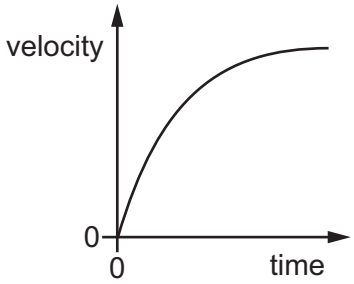
B



C



D



10 Which statement about collisions is correct?

A Kinetic energy is conserved in all collisions.

B Momentum is only conserved in perfectly elastic collisions.

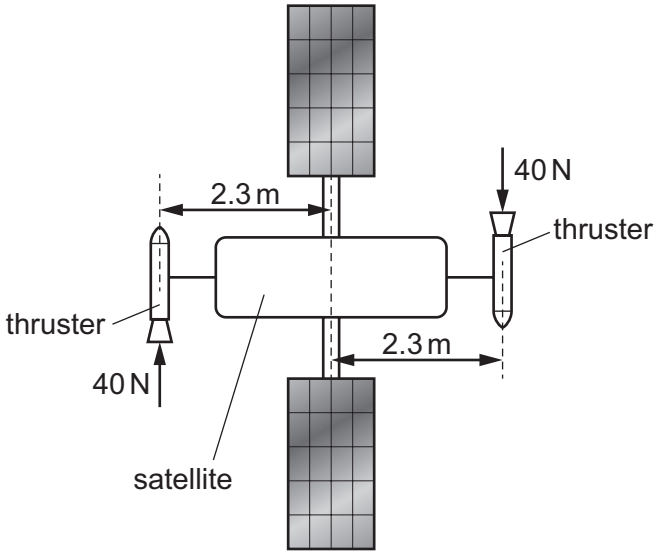
C The relative speed of approach is equal to the relative speed of separation for perfectly elastic collisions.

D When two objects of different masses collide, they exert forces of different magnitudes on each other.

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11 A satellite uses two thrusters to adjust its motion in space.

Each thruster exerts a force of 40 N on the satellite. The line of action of each force is a perpendicular distance of 2.3 m from the centre of gravity of the satellite. These two parallel forces act in opposite directions.



What are the magnitudes of the torque and the resultant force acting on the satellite due to the two thrust forces?

	torque / N m	resultant force / N
A	92	0
B	92	80
C	180	0
D	180	80

12 The diagrams show two ways of hanging the same picture.

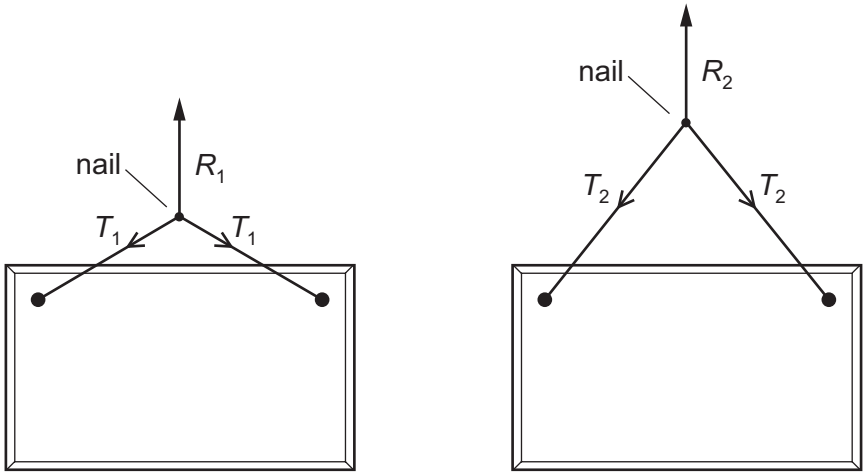


diagram 1 diagram 2

In both cases, a string is attached to the same points on the picture and looped symmetrically over a nail in a wall. The forces shown are those that act on the nail.

In diagram 1, the string loop is shorter than in diagram 2.

Which information about the magnitude of the forces is correct?

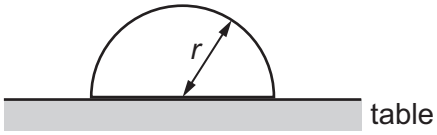
A $R_1 = R_2$ $T_1 = T_2$

B $R_1 = R_2$ $T_1 > T_2$

C $R_1 > R_2$ $T_1 < T_2$

D $R_1 < R_2$ $T_1 = T_2$

13 An object shaped as a hemisphere rests with its flat surface on a table. The object has radius r and density ρ.



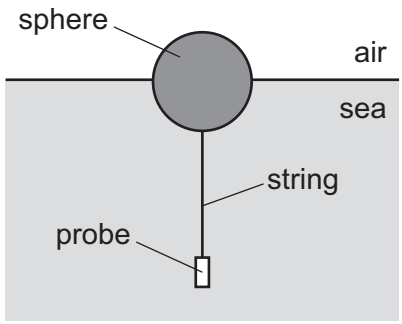
The volume of a sphere is $\frac{4}{3}\pi r^3$.

Which average pressure does the object exert on the table?

A $\frac{1}{3}\rho r^2$ **B** $\frac{1}{3}\rho r^2 g$ **C** $\frac{2}{3}\rho r$ **D** $\frac{2}{3}\rho rg$

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14 A probe is used to monitor the quality of the water in the sea. The probe is suspended by a vertical string which is attached to a sphere. The stationary sphere floats in equilibrium on the surface of the sea, as shown.



The sphere has a weight of 5.00 N. The probe and string have a combined weight of 2.00 N.

The density of the seawater is $1.03 \times 10^3\,\text{kg}\,\text{m}^{-3}$. The upthrust acting on the probe and thread is negligible.

What is the volume of the sphere below the surface of the sea?

A $1.98 \times 10^{-4}\,\text{m}^3$

B $2.97 \times 10^{-4}\,\text{m}^3$

C $4.95 \times 10^{-4}\,\text{m}^3$

D $6.93 \times 10^{-4}\,\text{m}^3$

15 What is the centre of gravity of an object?

A the geometrical centre of the object

B the point at which the weight of the object may be considered to act

C the point on the object about which there is a zero net torque

D the point where gravity acts on the object

16 A system with an efficiency of 74% wastes 230 W of power.

What is the useful output power of the system?

A 170 W **B** 310 W **C** 650 W **D** 880 W

17 A projectile of mass 0.25 kg is at a height of 30 m above horizontal ground and travelling at a speed of $15\,\text{m}\,\text{s}^{-1}$. A short time later, it is at a height of 35 m above the horizontal ground and travelling at a speed of $5.0\,\text{m}\,\text{s}^{-1}$.

How much work is done against air resistance during this time?

A 0 J **B** 13 J **C** 25 J **D** 37 J

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18 A spring has an unstretched length of 4.50 cm. The spring is fixed at one end and a force of 35.0 N is applied to the other end so that the spring extends.

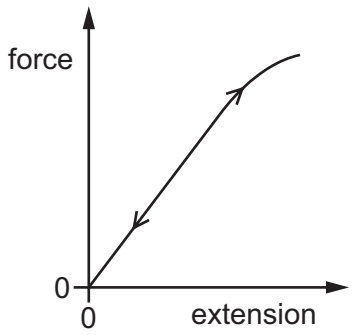
The spring obeys Hooke's law and has a spring constant of 420 N m^{-1}.

What is the strain of the extended spring?

A 0.019 **B** 0.083 **C** 1.85 **D** 2.67

19 A wire is fixed at one end and extended by a force that is applied to the other end. The force is slowly increased from zero and then slowly decreased back to zero.

The force–extension graph for the wire is shown.



The graph line for the wire being loaded is the same as the graph line for the wire being unloaded.

Which statement describes the deformation of the wire?

A It is both elastic and plastic.

B It is elastic only.

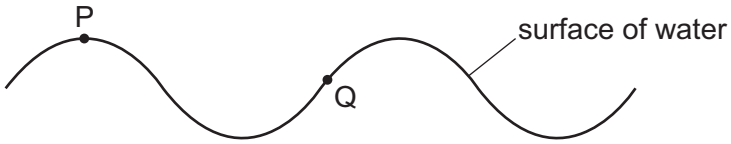
C It is neither elastic nor plastic.

D It is plastic only.

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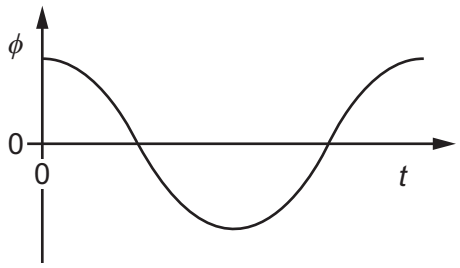
20 In a progressive water wave, two particles, P and Q, on the surface of the water, are a fixed horizontal distance apart. P and Q oscillate vertically.

At time $t = 0$, the wave is as shown.

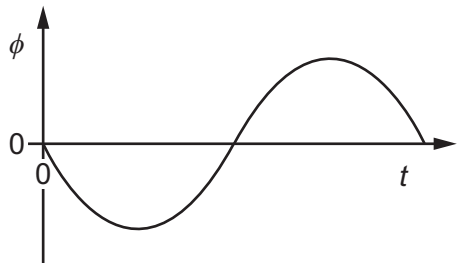


Which graph best represents the variation with time t of the phase difference ϕ between the oscillation of the water particle P and the oscillation of the water particle Q?

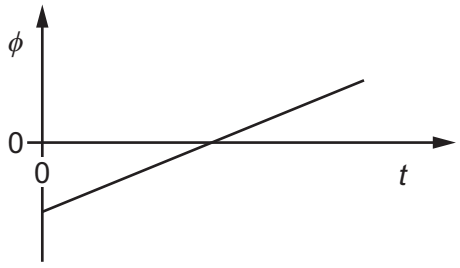
A



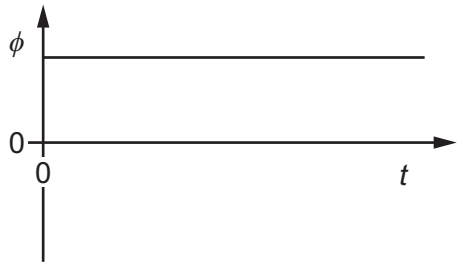
B



C



D



21 Which statement about longitudinal waves and transverse waves is **not** correct?

A Both waves can be polarised.

B Both waves can form stationary waves.

C Both waves can transfer energy as progressive waves.

D Both waves obey the equation $v = f\lambda$.

22 An observer hears a sound wave emitted from a moving source.

The observed frequency is less than the frequency of sound emitted from the source.

What could be the reason for this?

A The source is moving away from the observer.

B The source is moving towards the observer.

C The speed of the sound wave in air decreases due to the movement of the source.

D The speed of the sound wave in air increases due to the movement of the source.

23 What is the approximate range of frequencies of electromagnetic radiation visible to the human eye?

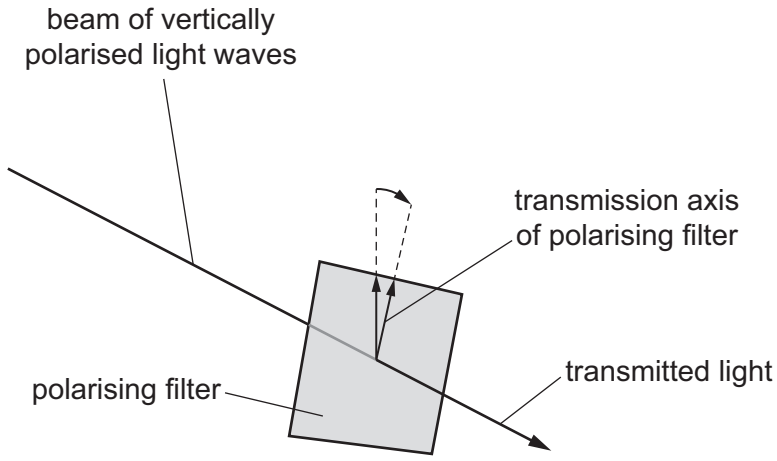
A (430–750) kHz

B (430–750) MHz

C (430–750) GHz

D (430–750) THz

24 A beam of vertically polarised light is incident normally on a polarising filter. The filter can be rotated so that it is always in a plane perpendicular to the beam. The transmission axis of the filter is initially vertical.



The filter is first rotated clockwise by an angle of 30° so that the transmitted light waves have intensity I_{30}. The filter is then rotated clockwise by a further angle of 30°.

What is the new intensity of the transmitted light waves?

A $0.25\,I_{30}$ **B** $0.33\,I_{30}$ **C** $0.75\,I_{30}$ **D** $0.87\,I_{30}$

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25 A musical instrument is made using a long tube with a mouthpiece at one end. The other end is open and flared, as shown.



A musician maintains stationary sound waves with a node at the mouthpiece and an antinode at the other end. The lowest frequency of sound that the instrument can produce is 92 Hz.

Which different frequencies of sound can be produced by the instrument?

A 92 Hz, 138 Hz, 184 Hz, 230 Hz

B 92 Hz, 184 Hz, 276 Hz, 368 Hz

C 92 Hz, 276 Hz, 460 Hz, 644 Hz

D 92 Hz, 276 Hz, 828 Hz, 1288 Hz

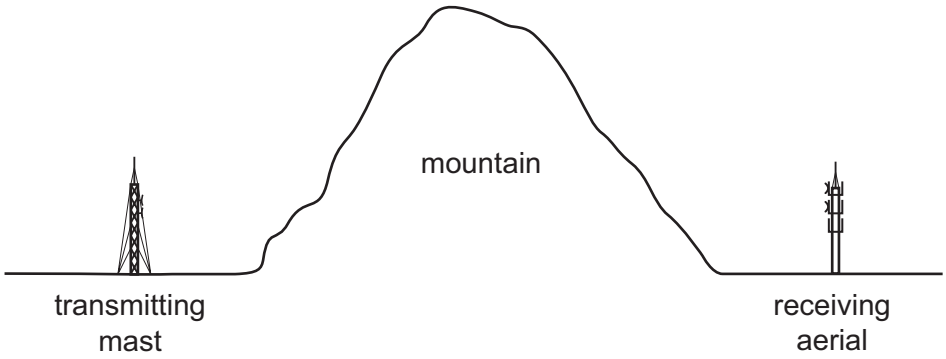
26 Two waves of equal frequency and amplitude are travelling in opposite directions along a stretched string. When they meet, they form a stationary wave with three nodes and two antinodes.

The frequency of both waves is doubled and a new stationary wave is formed.

How many antinodes are there in the new stationary wave?

A 1 **B** 2 **C** 3 **D** 4

27 A transmitting mast sends out microwaves of wavelength 1.5 cm and radio waves of wavelength 1.5 km.



NOT TO SCALE

A receiving aerial behind a mountain can detect the radio waves but not the microwaves.

What is the reason for this?

A The radio waves are coherent but the microwaves are not.

B The radio waves are diffracted around the mountain but the microwaves are not.

C The radio waves are reflected by the mountain but the microwaves are not.

D The radio waves travel at the speed of light but the microwaves do not.

28 Waves are emitted from two coherent sources.

Which statement about the waves must be correct?

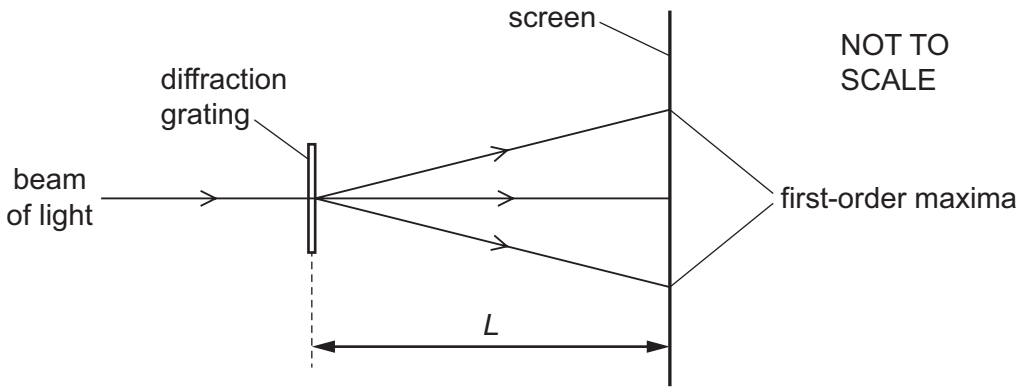
A They are in phase.

B They are transverse waves.

C They have a constant phase difference.

D They have the same amplitude.

29 The diagram shows a screen that is a distance *L* from a diffraction grating. The grating has a total number of *N* lines. Any two adjacent lines are a distance *d* apart. A beam of parallel light of wavelength λ is incident normally on the grating.



Which quantities affect the distance between the first-order diffraction maxima on the screen?

	d	λ	*L*	*N*
A	✓	✓	✓	✗
B	✓	✓	✗	✗
C	✓	✗	✓	✓
D	✗	✓	✗	✓

key

✓ = affects the distance

✗ = does not affect the distance

30 A wire carries a current of 0.10 μA. The potential difference across the wire is 10 mV.

How much energy is dissipated by the wire in a time of 10 s?

A 1.0 pJ **B** 10 pJ **C** 1.0 nJ **D** 10 nJ

31 What is the definition of the potential difference across an electrical component?

A energy transferred per unit charge

B energy transferred per unit current

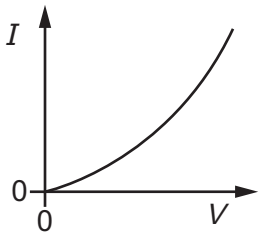
C energy transferred per unit resistance

D energy transferred per unit time

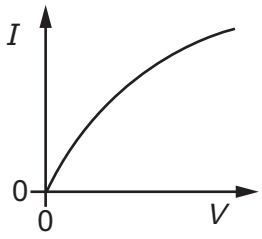
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32 Which graph represents the way the current I through a filament lamp varies with the potential difference V across it?

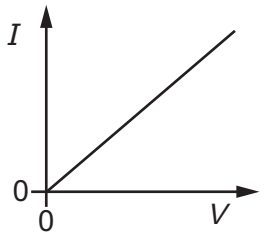
A



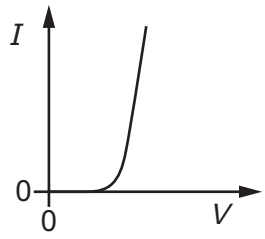
B



C



D



33 The table shows the properties of two different wires, P and Q.

	length	resistance	resistivity of material
wire P	l	R	ρ
wire Q	$2l$	$\frac{1}{4}R$	$\frac{1}{3}\rho$

Wire P has a cross-section of diameter d.

What is the diameter of the cross-section of wire Q?

A $0.41d$ **B** $1.6d$ **C** $2.7d$ **D** $7.1d$

34 A cell has a constant electromotive force.

A variable resistor is connected between the terminals of the cell.

The resistance of the variable resistor is decreased.

Which statement about the change of the cell's terminal potential difference (p.d.) is correct?

A The terminal p.d. is decreased because more work is done moving unit charge through the internal resistance of the cell.

B The terminal p.d. is decreased because the current in the variable resistor is decreased.

C The terminal p.d. is increased because more work is done moving unit charge through the variable resistor.

D The terminal p.d. is increased because the current in the variable resistor is increased.

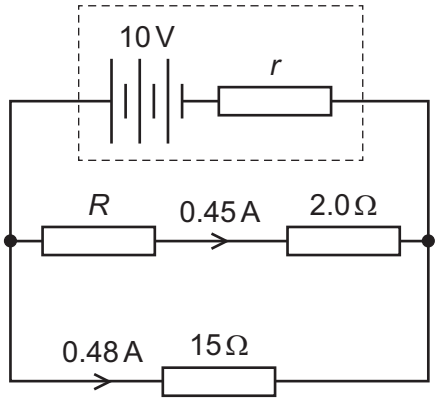
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35 Kirchhoff's two laws for electric circuits can be derived by using conservation laws.

On which conservation laws do Kirchhoff's laws depend?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	current
B	charge	energy
C	current	mass
D	energy	current

36 A battery of electromotive force (e.m.f.) 10 V and internal resistance r is connected to three resistors of resistances R, 2.0 Ω and 15 Ω, as shown. A current of 0.45 A is in the resistor of resistance 2.0 Ω and a current of 0.48 A is in the resistor of resistance 15 Ω.

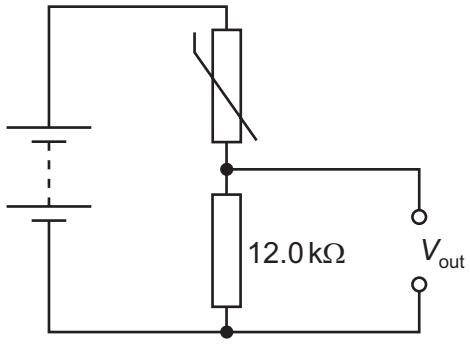


What are the values of r and R?

	r / Ω	R / Ω
A	3.0	14
B	3.0	20
C	5.8	14
D	5.8	20

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37 A battery of negligible internal resistance is connected in series with a thermistor and a fixed resistor of resistance 12.0 kΩ, as shown.



The table shows the resistance of the thermistor at two different temperatures.

temperature / °C	resistance of thermistor / kΩ
20.0	12.0
50.0	5.00

The potential difference V_{out} across the fixed resistor is 4.50 V when the thermistor is at a temperature of 20.0 °C.

What is V_{out} when the thermistor is at a temperature of 50.0 °C?

A 2.65 V **B** 3.18 V **C** 6.35 V **D** 10.8 V

38 What is a conclusion from the alpha-particle scattering experiment?

A Protons and electrons have equal but opposite charges.

B Protons have a much larger mass than electrons.

C The nucleus contains most of the mass of the atom.

D The nucleus of an atom contains protons and neutrons.

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39 Americium-241 is a radioactive nuclide used in smoke detectors. It undergoes α-decay to form nuclide X. This decay may be represented by the equation shown.

$$^{241}_{95}\text{Am} \rightarrow {}^{A}_{Z}\text{X} + \alpha$$

What are the values of A and Z?

	A	Z
A	237	93
B	239	91
C	241	94
D	241	96

40 A top quark has a charge of $+\frac{2}{3}e$, where e is the elementary charge.

What is the charge of an anti top quark?

A $-\frac{2}{3}e$ **B** $-\frac{1}{3}e$ **C** $+\frac{1}{3}e$ **D** $+\frac{2}{3}e$

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Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2023**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the February/March 2023 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

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Question	Answer	Marks
1	C	**1**
2	B	**1**
3	A	**1**
4	B	**1**
5	B	**1**
6	D	**1**
7	D	**1**
8	C	**1**
9	D	**1**
10	C	**1**
11	C	**1**
12	B	**1**
13	D	**1**
14	D	**1**
15	B	**1**
16	C	**1**
17	B	**1**
18	C	**1**
19	B	**1**
20	D	**1**
21	A	**1**
22	A	**1**
23	D	**1**
24	B	**1**
25	C	**1**
26	D	**1**
27	B	**1**
28	C	**1**

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Question	Answer	Marks
29	A	**1**
30	D	**1**
31	A	**1**
32	B	**1**
33	B	**1**
34	A	**1**
35	B	**1**
36	A	**1**
37	C	**1**
38	C	**1**
39	A	**1**
40	A	**1**

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June 2023 · Paper 11

9702_s23_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2023**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

8516742668

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which unit is **not** an SI base unit?

A A **B** kg **C** C **D** s

2 What is the best estimate of the number of atoms in a piece of metal of volume 50 cm^3?

A 5×10^{15} **B** 5×10^{25} **C** 5×10^{29} **D** 5×10^{31}

3 A copper pipe has a true diameter of 42.03 mm.

A builder measures the diameter of the pipe five times using digital calipers. The measurements are shown.

diameter / mm
48.01
47.99
48.01
48.00
47.99

What describes the builder's measurements?

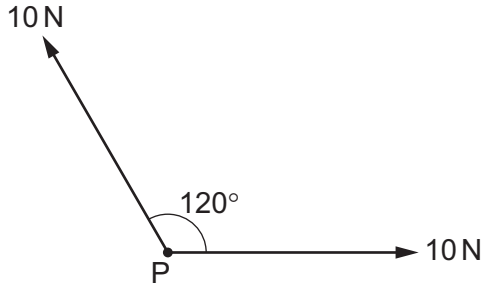
A accurate and precise

B accurate but not precise

C not precise and not accurate

D precise but not accurate

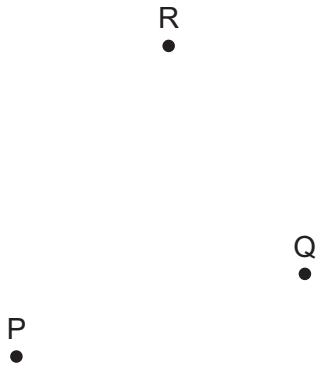
4 Two forces, each of 10 N, act at a point P, as shown. The angle between the directions of the forces is 120°.



What is the magnitude of the resultant force?

A 5 N **B** 10 N **C** 17 N **D** 20 N

5 One object moves directly from P to R.



In a shorter time, a second object moves from P to Q to R.

Which statement about the two objects is correct for the journey from P to R?

A They have the same average speed.

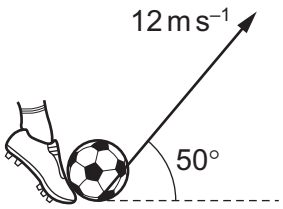
B They have the same average velocity.

C They have the same displacement.

D They travel the same distance.

6 A ball is kicked so that it has an initial velocity of $12\,m\,s^{-1}$ at an angle of 50° to horizontal ground.

Assume that air resistance is negligible.

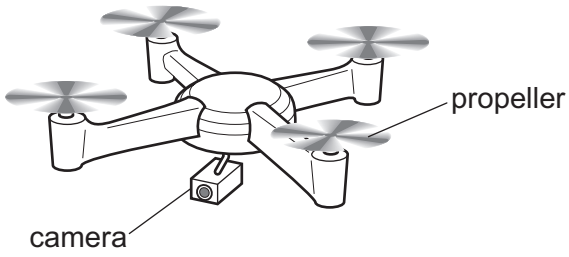


What is the maximum height above the ground that is reached by the ball?

A 0.47 m **B** 3.0 m **C** 4.3 m **D** 7.3 m

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7 A camera drone of mass 1.20 kg hovers at a fixed point above the ground. The drone has four propellers.



In a time of 1.00 s, each propeller pushes a mass of 0.400 kg of air vertically downwards.

Assume that the air above the propellers is stationary.

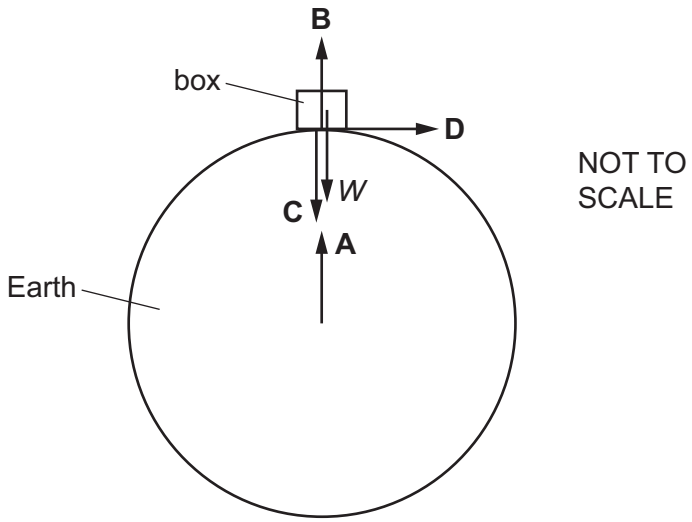
What is the speed of the air leaving each propeller?

A $0.750\,m\,s^{-1}$ **B** $3.00\,m\,s^{-1}$ **C** $7.36\,m\,s^{-1}$ **D** $29.4\,m\,s^{-1}$

8 A box rests on the Earth, as shown.

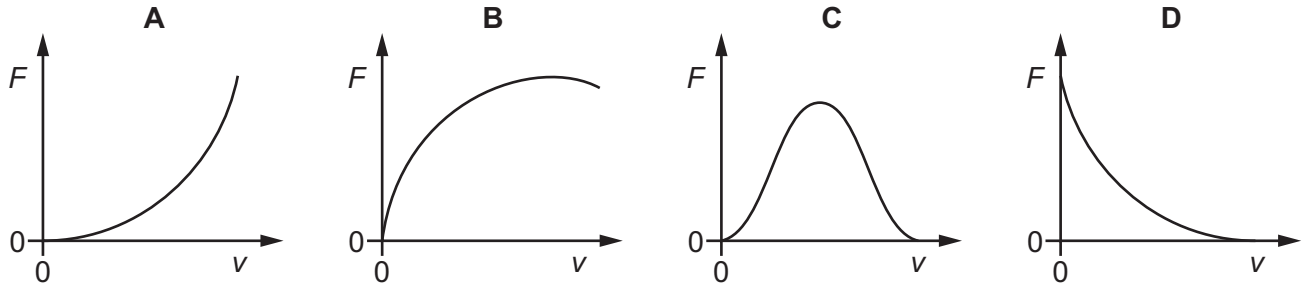
Newton's third law describes how forces of the same type act in pairs. One of the forces of a pair is the weight *W* of the box.

Which arrow represents the other force of this pair?



9 A small ball is held at the surface of liquid oil in a container. The ball is released from rest and falls through the oil. The ball has velocity v. A viscous (drag) force F acts on the ball.

Which graph could show the variation with v of F?



10 An object of mass m, moving at speed u along a frictionless horizontal surface, collides head-on with a stationary object of mass $4m$.



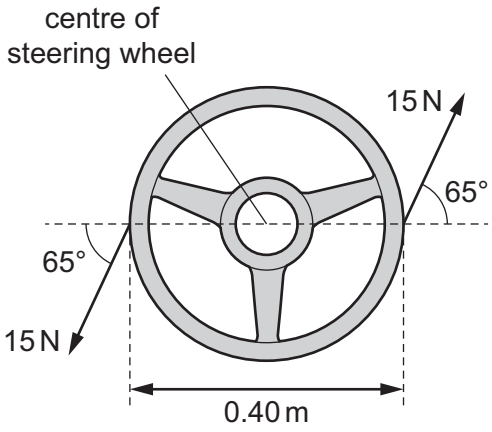
before the collision

After the collision, the object of mass m rebounds along its initial path with $\frac{1}{4}$ of its kinetic energy before the collision.

What is the speed of the object of mass $4m$ after the collision?

A $\frac{u}{8}$ **B** $\frac{3u}{16}$ **C** $\frac{5u}{16}$ **D** $\frac{3u}{8}$

11 The driver of a car applies two parallel forces to a steering wheel, as shown.



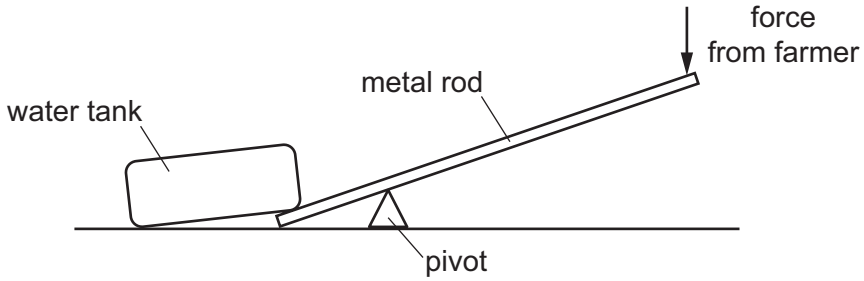
Each force has a magnitude of 15 N and acts in the direction shown. The steering wheel has a diameter of 0.40 m.

What is the torque exerted on the steering wheel?

A 1.3 N m **B** 2.5 N m **C** 2.7 N m **D** 5.4 N m

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12 A farmer is trying to lift the corner of a large water tank. She uses a metal rod as a lever.



The vertical force from the farmer is constant and is always applied to the end of the rod.

Which change **must** increase the upward force on the water tank?

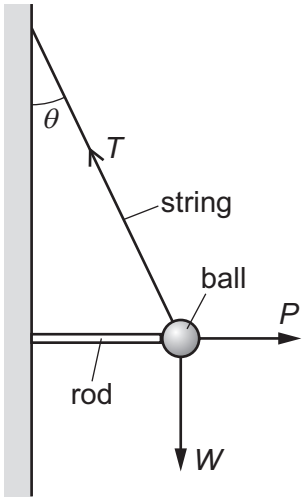
A using a longer rod and moving the pivot closer to the tank

B using a longer rod and moving the pivot further away from the tank

C using a shorter rod and moving the pivot closer to the tank

D using a shorter rod and moving the pivot further away from the tank

13 The diagram shows a ball of weight W hanging in equilibrium from a string.



The string is at an angle θ to the vertical. The tension in the string is T. The ball is held away from the wall by a horizontal force P from a metal rod.

Which relationship between the magnitudes of T, P and W is correct?

A $P = T\cos\theta$ and $W = T\sin\theta$

B $T = P + W$

C $T^2 = P^2 + W^2$

D $W = P\tan\theta$ and $W = T\cos\theta$

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14 Which expression for pressure is correct?

A force per unit area

B force per unit volume

C mass per unit area

D mass per unit volume

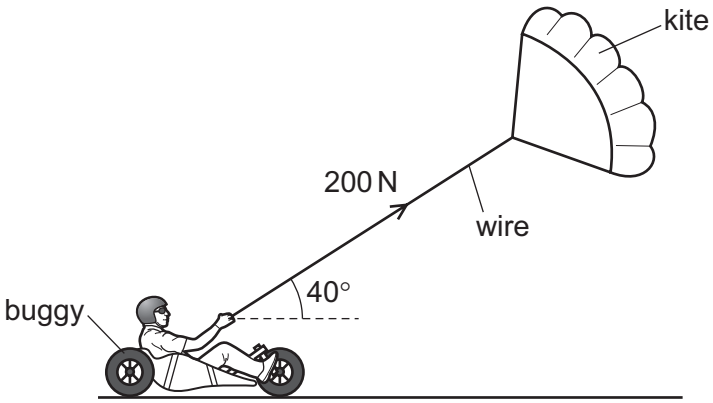
15 A ball has a mass of 0.50 kg and a volume of $1.3 \times 10^{-3}\,m^3$. The ball is floating in equilibrium on still water. The two forces that act on the ball are its weight and the upthrust due to the water.

The density of the water is $1.0 \times 10^{3}\,kg\,m^{-3}$.

What is the percentage of the volume of the ball above the surface of the water?

A 3.9% **B** 38% **C** 62% **D** 96%

16 A man sits on a buggy that is pulled along by a wire attached to a kite. The wire is at an angle of 40° to the horizontal and has a constant tension of 200 N. The man and buggy travel a distance of 20 m along a straight horizontal path. The wire and the path of the buggy are in the same vertical plane.



What is the work done by the tension force on the man and buggy?

A 2.6 kJ **B** 3.1 kJ **C** 3.4 kJ **D** 4.0 kJ

17 A ball is thrown vertically upwards from the surface of the Earth.

Which statement describes the energy of the ball as it rises through the air?

A The kinetic energy of the ball decreases as the gravitational potential energy decreases.

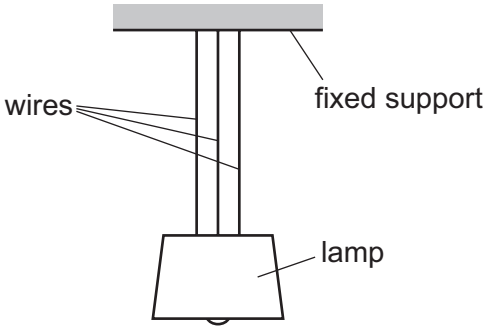
B The kinetic energy of the ball decreases as the gravitational potential energy increases.

C The kinetic energy of the ball increases as the gravitational potential energy decreases.

D The total energy of the ball increases.

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18 A lamp is suspended in equilibrium from a fixed support by three long identical wires.



The weight of the lamp causes each wire to have an extension of 0.40 cm. The height *h* of the lamp above the floor is measured.

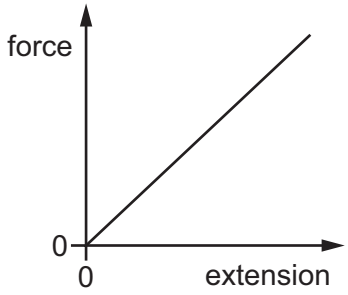
The middle wire suddenly breaks and the lamp falls a small distance as the extensions of the remaining two wires increase. The wires obey Hooke's law.

When the lamp is in equilibrium, the height *h* of the lamp above the floor is measured again.

What is the difference between the two values of *h*?

A 0.20 cm **B** 0.27 cm **C** 0.40 cm **D** 0.60 cm

19 The force–extension graph for a spring is shown.



What represents the work done to extend the spring?

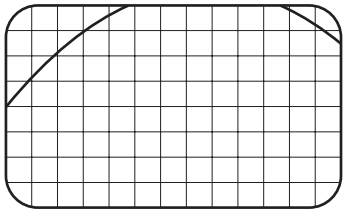
A the area under the graph

B the gradient of the graph

C the reciprocal of the gradient of the graph

D twice the area under the graph

20 In an experiment, a student uses a microphone and a cathode-ray oscilloscope (CRO) to analyse a sound wave. The diagram shows the trace on the screen of the CRO.

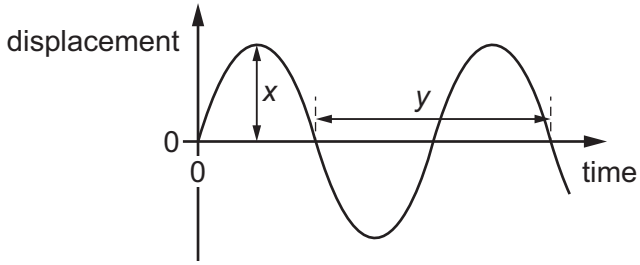


The student is expecting a sinusoidal waveform to be shown on the screen.

Which changes should the student make to the time-base and the *y*-gain of the CRO so that the screen shows a continuous trace for one complete cycle of the waveform?

	time-base	*y*-gain
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

21 The graph shows the variation with time of the displacement of a particle in a progressive wave.



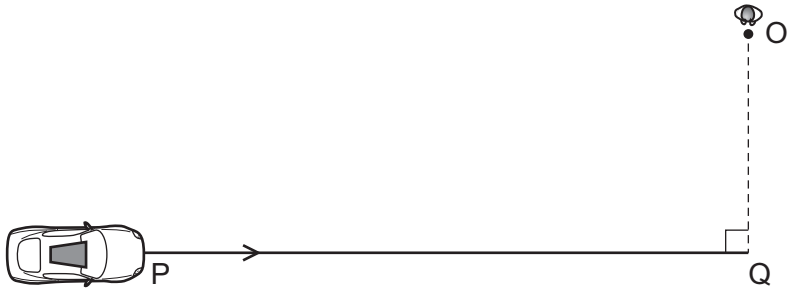
Two measurements, *x* and *y*, are labelled on the graph.

What do *x* and *y* represent?

	x	*y*
A	amplitude	period
B	frequency	period
C	amplitude	wavelength
D	frequency	wavelength

22 A car travels at a constant speed along a straight line PQ.

A loudspeaker attached to the car emits sound of constant frequency *f*. A stationary observer is at point O.



What does the observer hear as the car moves from P towards Q?

A a frequency less than *f* that decreases as the car moves from P towards Q

B a frequency less than *f* that increases as the car moves from P towards Q

C a frequency more than *f* that decreases as the car moves from P towards Q

D a frequency more than *f* that increases as the car moves from P towards Q

23 Some sources of electromagnetic waves in free space are listed.

1 a radio wave transmitter

2 a source of X-rays

3 a 30 mm wavelength radar transmitter

4 a light-emitting diode that emits red light

Which list gives the sources in order of increasing wavelength, from left to right, of the waves emitted by the sources?

A $1 \rightarrow 3 \rightarrow 4 \rightarrow 2$

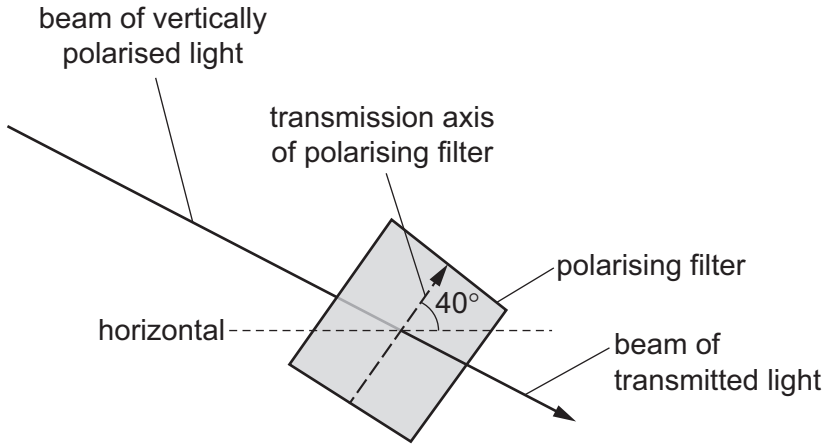
B $2 \rightarrow 4 \rightarrow 1 \rightarrow 3$

C $2 \rightarrow 4 \rightarrow 3 \rightarrow 1$

D $3 \rightarrow 1 \rightarrow 4 \rightarrow 2$

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24 A vertically polarised beam of light is incident normally on a polarising filter. The transmission axis of the filter is at an angle of 40° to the horizontal.



What is the ratio $\frac{\text{amplitude of transmitted beam}}{\text{amplitude of incident beam}}$?

A 0.41 **B** 0.59 **C** 0.64 **D** 0.77

25 Two progressive waves meet at a point.

Which condition must be met for superposition of the waves to occur?

A The waves must be coherent.

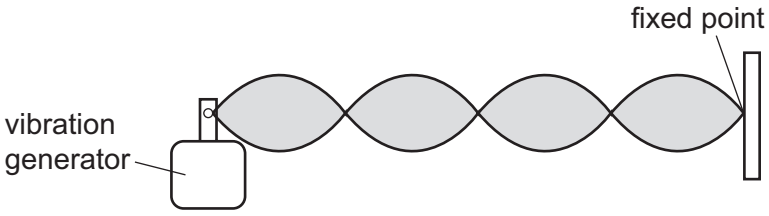
B The waves must be of the same type.

C The waves must be travelling in opposite directions.

D The waves must meet in phase.

26 A string is stretched between a vibration generator and a fixed point.

When the vibration generator is vibrating at a frequency *f*, a stationary wave with five nodes is created on the stretched string, as shown. There is a node at the end of the string that is attached to the vibration generator.



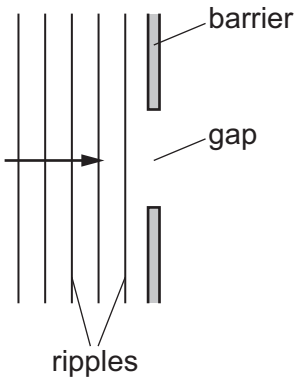
The frequency of vibration of the vibration generator is slowly increased.

What is the next frequency that produces a stationary wave on the string?

A 1.25*f* **B** 1.50*f* **C** 1.75*f* **D** 2.00*f*

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27 A ripple tank and a barrier with a single gap are used to demonstrate the diffraction of surface ripples on water. Initially, the wavelength of the ripples is five times smaller than the gap in the barrier.



Which change increases the amount of diffraction observed?

A double the amplitude of the ripples

B double the width of the gap

C halve the frequency of the ripples

D halve the wavelength of the ripples

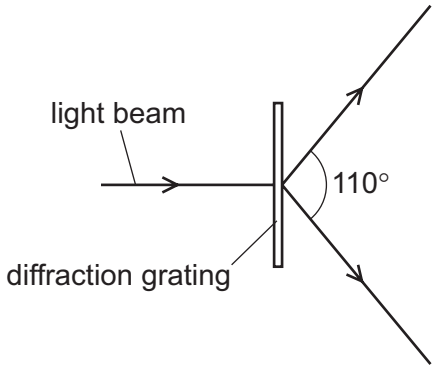
28 A laser produces a beam of light of wavelength 650 nm. The beam is incident normally on two slits that are a distance of 0.12 mm apart.

A screen is placed parallel to the slits. The bright interference fringes on the screen have a separation of 7.5 cm.

What is the distance between the screen and the two slits?

A 1.4 m **B** 2.8 m **C** 7.0 m **D** 14 m

29 A beam of light from a laser is incident normally on a diffraction grating.



The diagram shows only the **second**-order maxima that are produced.

The grating has a line spacing of 1.0×10^{-6} m. The angle between the two second-order maxima is 110°.

What is the wavelength of the light?

A 4.1×10^{-7} m

B 4.7×10^{-7} m

C 8.2×10^{-7} m

D 9.4×10^{-7} m

30 The electric current in a metal wire is 4.0 mA.

How many electrons pass a fixed point in the wire in a time of 10 hours?

A 2.5×10^{17} **B** 2.5×10^{20} **C** 9.0×10^{20} **D** 9.0×10^{23}

31 What is the definition of the potential difference across an electrical component?

A the charge per unit time passing through the component

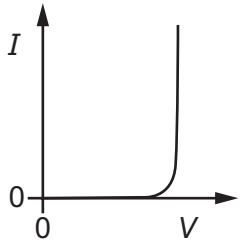
B the energy transferred per unit charge

C the force per unit charge

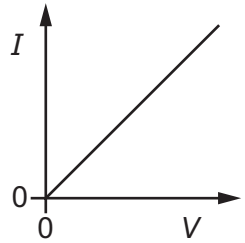
D the resistance per unit current

32 Which graph shows the *I*–*V* characteristic of a filament lamp?

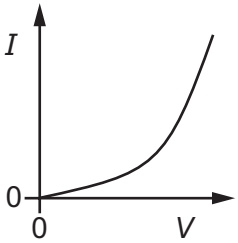
A



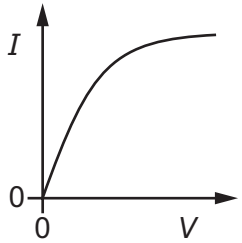
B



C



D



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33 A metal wire has a length of 2.50 m and a cross-sectional area of $4.50 \times 10^{-6}\,m^2$. The resistivity of the metal is $3.50 \times 10^{-7}\,\Omega m$.

The wire is stretched so that its length increases to 2.65 m. The wire remains cylindrical and the **volume** of the wire remains constant.

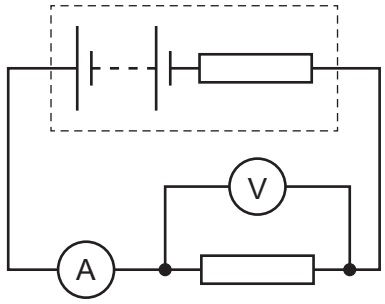
What is the change in the resistance of the wire?

A $0.012\,\Omega$ **B** $0.024\,\Omega$ **C** $0.19\,\Omega$ **D** $0.22\,\Omega$

34 Which circuit symbol represents a microphone?



35 A battery with internal resistance is connected to a fixed resistor, an ammeter and a voltmeter, as shown.



The battery is replaced by a different battery that has the same electromotive force (e.m.f.) but a greater internal resistance.

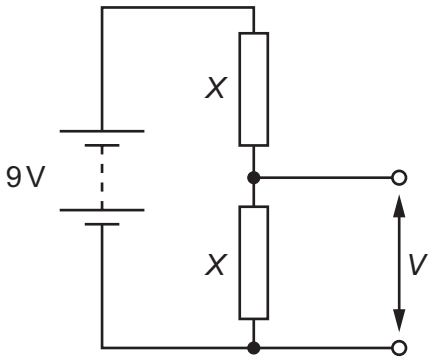
What happens to the readings on the ammeter and voltmeter?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

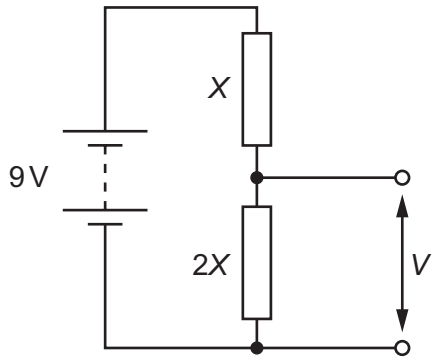
36 Four potential divider circuits each consist of a battery of electromotive force (e.m.f.) 9 V and negligible internal resistance connected to a combination of resistors. Each of the resistors in the circuits has a resistance of X or $2X$.

Which circuit has the largest output voltage V?

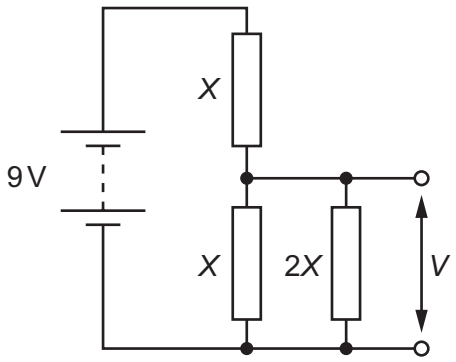
A



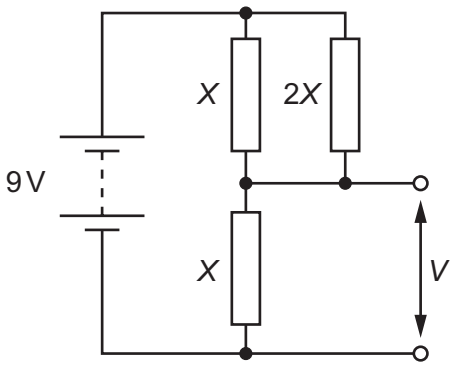
B



C

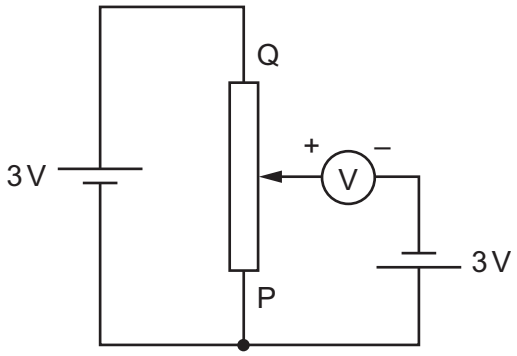


D



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37 A voltmeter is connected into a circuit with the polarity shown.



The sliding contact is moved to end P of the potentiometer and then to end Q.

What are the two readings of the voltmeter?

	sliding contact at end P	sliding contact at end Q
A	0 V	3 V
B	0 V	6 V
C	3 V	3 V
D	3 V	6 V

38 A proton has mass m_p and charge $+e$.

What are the mass and charge of an antiproton?

	mass	charge
A	$-m_p$	$+e$
B	$-m_p$	$-e$
C	m_p	$+e$
D	m_p	$-e$

39 A uranium nucleus has 92 protons and 143 neutrons.

The nucleus emits a total of 3 α-particles and 4 β^- particles to form nucleus X.

How can nucleus X be represented?

A $^{131}_{90}X$ **B** $^{219}_{87}X$ **C** $^{223}_{82}X$ **D** $^{223}_{90}X$

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40 Which statement about a proton is correct?

A A proton is a baryon but not a meson.

B A proton is a meson but not a hadron.

C A proton is both a hadron and a meson.

D A proton is both a meson and a baryon.

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Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2023**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Question	Answer	Marks
1	C	**1**
2	B	**1**
3	D	**1**
4	B	**1**
5	C	**1**
6	C	**1**
7	C	**1**
8	A	**1**
9	A	**1**
10	D	**1**
11	D	**1**
12	A	**1**
13	C	**1**
14	A	**1**
15	C	**1**
16	B	**1**
17	B	**1**
18	A	**1**
19	A	**1**
20	D	**1**
21	A	**1**
22	C	**1**
23	C	**1**
24	C	**1**
25	B	**1**
26	A	**1**
27	C	**1**
28	D	**1**

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Question	Answer	Marks
29	A	**1**
30	C	**1**
31	B	**1**
32	D	**1**
33	B	**1**
34	C	**1**
35	A	**1**
36	B	**1**
37	D	**1**
38	D	**1**
39	D	**1**
40	A	**1**

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June 2023 · Paper 12

9702_s23_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2023**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 A stone sinks in water.

What is a possible value for the density of the stone?

A $8 \times 10^2\,\text{kg m}^{-3}$

B $2 \times 10^3\,\text{kg m}^{-3}$

C $8 \times 10^3\,\text{N m}^{-3}$

D $2 \times 10^4\,\text{N m}^{-3}$

2 Gm, Tm, μm and pm are all units of length.

Which unit is the largest and which unit is the smallest?

	largest unit	smallest unit
A	Gm	μm
B	Gm	pm
C	Tm	μm
D	Tm	pm

3 Two measurements for a solid sphere are shown.

$$\text{mass} = (32.5 \pm 0.1)\,\text{g}$$

$$\text{diameter} = (1.87 \pm 0.04)\,\text{cm}$$

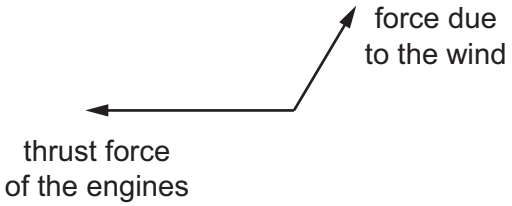
These values are used to determine the density of the sphere.

What is the percentage uncertainty in the density?

A 2.4% **B** 4.6% **C** 6.1% **D** 6.7%

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4 An aircraft travels along a horizontal path. Two of the forces that act horizontally on the aircraft are the thrust force of the engines and the force due to the wind. The vector diagram for these forces is shown.



Which vector represents the resultant horizontal force acting on the aircraft due to these two forces?

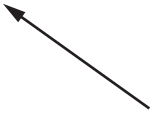
A



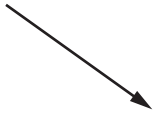
B



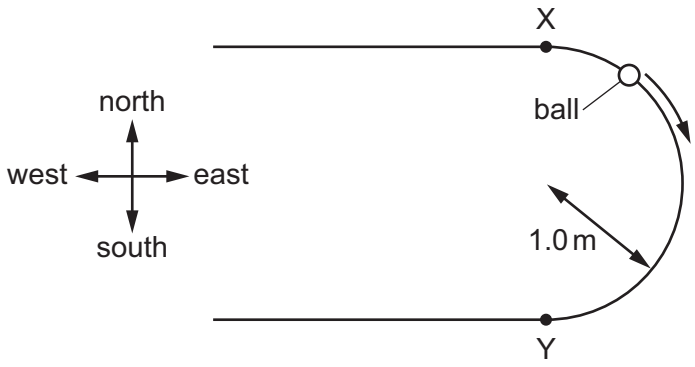
C



D



5 A ball travels from point X to point Y around a semicircular track of radius 1.0 m, as shown.



What is the displacement of the ball from X to Y?

A 2.0 m

B 2.0 m due south

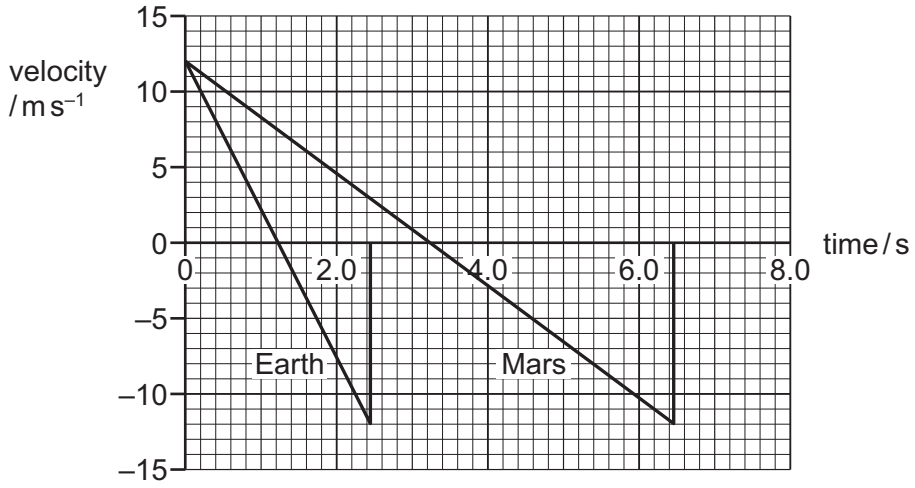
C 3.1 m

D 3.1 m due south

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6 A rock is launched vertically upwards from the surface of the Earth and an identical rock is launched vertically upwards from the surface of Mars. Each rock is launched with an initial velocity of $12\,m\,s^{-1}$.

Each rock then reaches its maximum height above the surface before returning back down to the surface. The velocity–time graph for each rock is shown. In both cases, air resistance is negligible.



What is the difference in the maximum heights of the two rocks?

A 12 m **B** 15 m **C** 19 m **D** 24 m

7 Which statement describes the mass of an object?

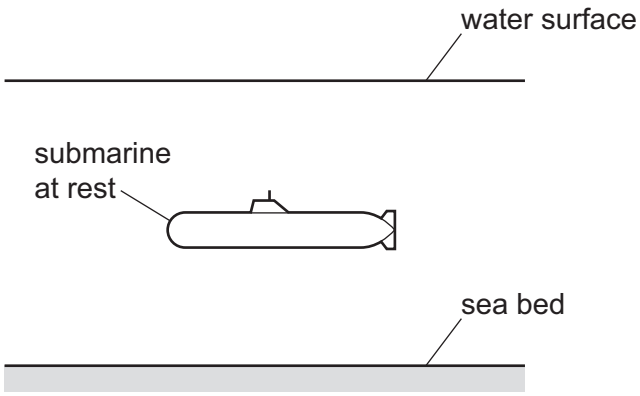
A the force the object experiences due to gravity

B the momentum of the object before a collision

C the resistance of the object to changes in motion

D the weight of the object as measured by a balance

8 A submarine of total mass 3200 kg is at rest underwater.



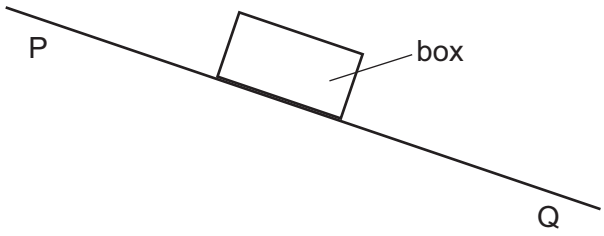
The total mass of the submarine is suddenly decreased by 200 kg by pumping water out of the submarine horizontally in a negligible time. The upthrust acting on the submarine is unchanged.

The change in the total weight of the submarine causes it to accelerate vertically upwards.

What is the initial upwards acceleration of the submarine?

A $0.613\,m\,s^{-2}$ **B** $0.654\,m\,s^{-2}$ **C** $9.81\,m\,s^{-2}$ **D** $10.5\,m\,s^{-2}$

9 A box in air slides with increasing speed down a rough slope from point P to point Q.



The slope surface exerts a constant frictional force on the box.

As the box moves from P to Q, there are changes to the magnitudes of its acceleration and the total resistive force acting on it.

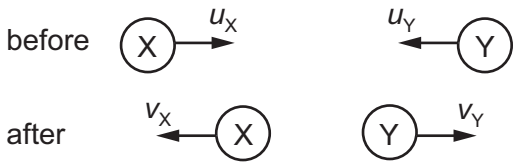
Which row describes the changes?

	magnitude of acceleration	magnitude of total resistive force
A	increases	decreases
B	decreases	decreases
C	increases	increases
D	decreases	increases

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10 Two balls, X and Y, approach each other along the same straight line and collide. The collision is perfectly elastic.

Their initial speeds are u_X and u_Y respectively. After the collision they move apart with speeds v_X and v_Y respectively. Their directions are shown.



Which equation is correct?

A $u_X + u_Y = v_X + v_Y$

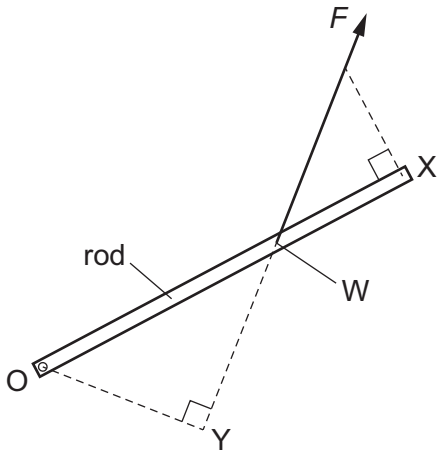
B $u_X + u_Y = v_X - v_Y$

C $u_X - u_Y = v_X + v_Y$

D $u_X - u_Y = v_X - v_Y$

11 A rod is pivoted at point O.

A force F is applied to the rod at point W, as shown.



What is the moment of the force F about O?

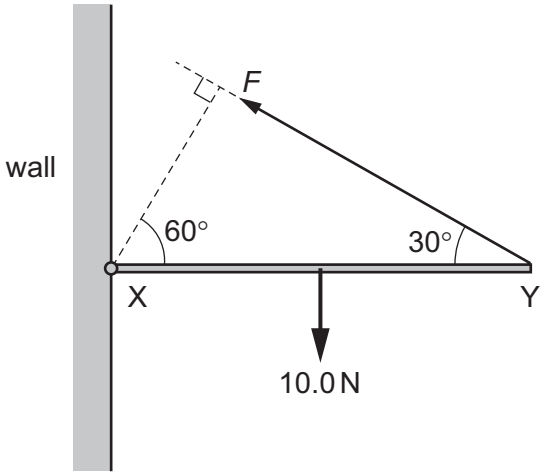
A $F \times$ distance OX

B $F \times$ distance OY

C $F \times$ distance WO

D $F \times$ distance WX

12 A uniform rod XY of weight 10.0 N is freely hinged to a wall at X. It is held horizontal by a force F acting from Y at an angle of 30° to the horizontal, as shown.



What is the value of F?

A 5.0 N **B** 8.7 N **C** 10 N **D** 20 N

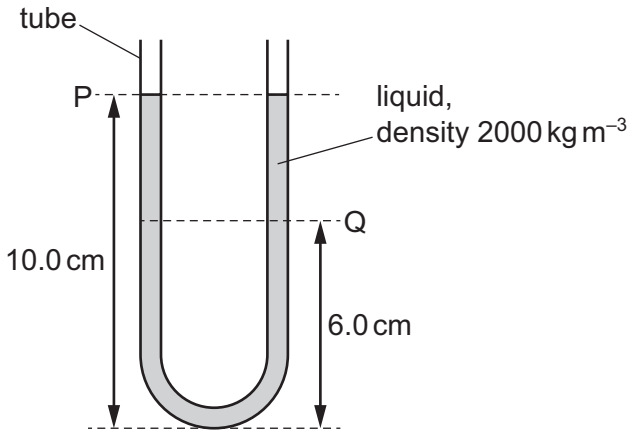
13 Two solid cubes X and Y are made of material of the same density. Cube X has twice the mass of cube Y.

Cube X has sides of length x. Cube Y has sides of length y.

What is the ratio $\frac{x}{y}$?

A 1.26 **B** 1.41 **C** 2.00 **D** 8.00

14 A U-shaped glass tube contains liquid of density 2000 kg m^{-3}, as shown.

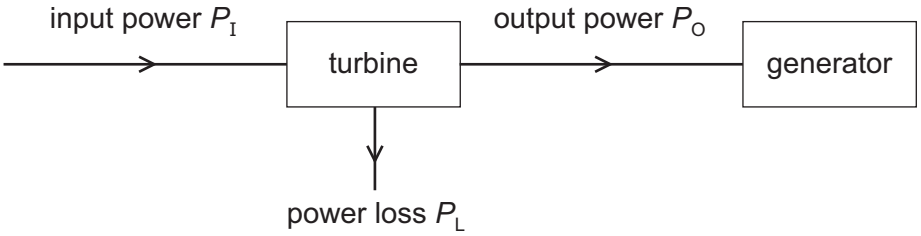


What is the difference in pressure due to the liquid between levels P and Q?

A 780 Pa **B** 1200 Pa **C** 1600 Pa **D** 2000 Pa

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15 A steam turbine is used to drive a generator. The input power to the turbine is P_I and the output power is P_O. The power loss in the turbine is P_L, as shown.

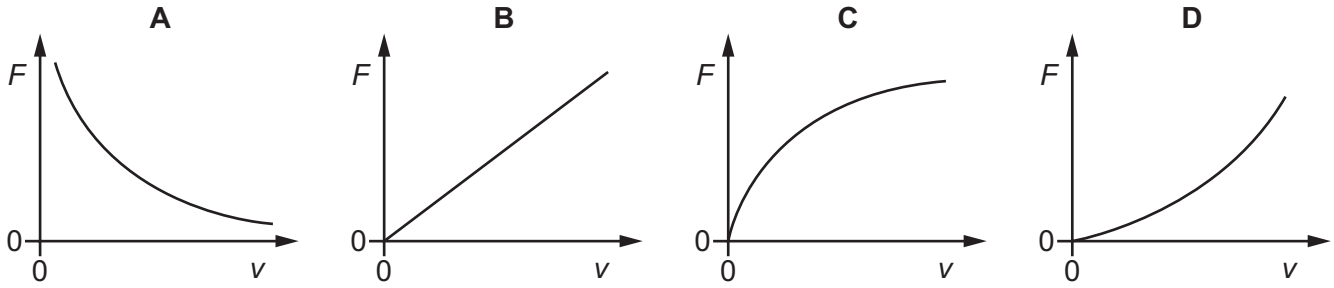


What is the efficiency of the turbine?

A $\frac{P_L}{P_O}$ **B** $\frac{P_I}{P_O}$ **C** $\frac{P_L}{P_I}$ **D** $\frac{P_O}{P_I}$

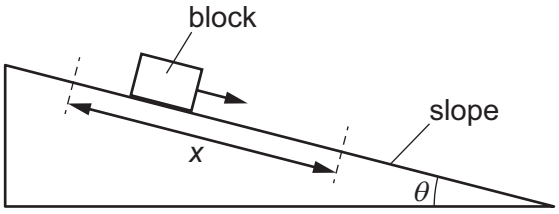
16 A variable force is applied to ensure that a constant power is supplied to a train.

Which graph best shows the variation of the force F applied with the velocity v of the train?



17 A block is released from rest and slides a distance x down a straight slope.

The slope is at an angle θ to the horizontal.



The slope is frictionless and air resistance is negligible.

The acceleration of free fall is g.

Which expression gives the final speed of the block?

A $\sqrt{\frac{gx\sin\theta}{2}}$ **B** $\sqrt{2gx\sin\theta}$ **C** $\sqrt{2gx}$ **D** $2gx\sin\theta$

18 What is meant by the spring constant of a spring?

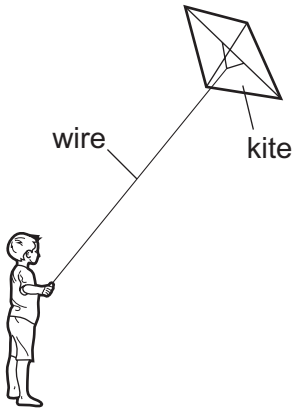
A extension per unit force

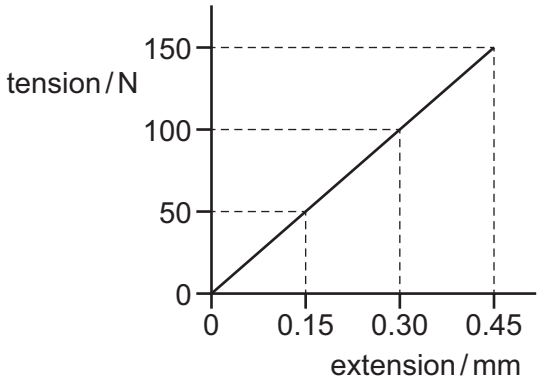
B $\frac{1}{2} \times$ force $\times$ extension

C force $\times$ extension

D force per unit extension

19 A child holds a thin metal wire that is attached to a kite. The graph shows how the extension of the wire varies with its tension.





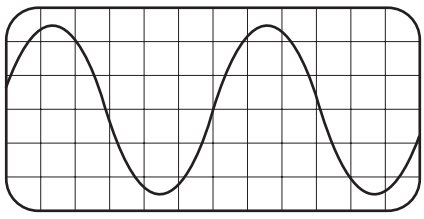
A gust of wind increases the tension from 100 N to 150 N.

What is the change in the elastic potential energy of the wire caused by the gust of wind?

A 3.8 mJ **B** 19 mJ **C** 34 mJ **D** 38 mJ

20 A microphone detects a sound wave. The microphone is connected to a cathode-ray oscilloscope (CRO).

The shape of the trace on the screen of the CRO is shown.



Which property of the sound wave can be determined by using only the measurement of a horizontal distance on the screen and the value of a control setting of the CRO?

A amplitude

B frequency

C speed

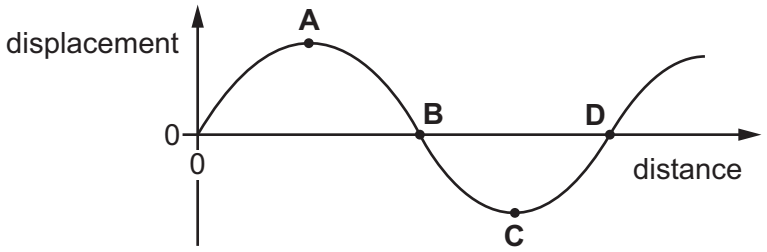
D wavelength

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21 A longitudinal wave is travelling from left to right. The graph shows the variation of the displacement of the particles with distance along the wave at one instant in time.

Displacements to the right are positive; displacements to the left are negative.

Which labelled point represents a compression?



22 A source X emits a sound wave of constant frequency *f*.

The wave is subsequently received at a stationary detector Y.

The frequency of the wave that is detected by Y is less than *f*.

What could be the reason for this?

A Between X and Y, the wave undergoes diffraction.

B Between X and Y, the wave undergoes reflection.

C X is moving away from Y.

D X is moving towards Y.

23 A beam of visible light is in a vacuum.

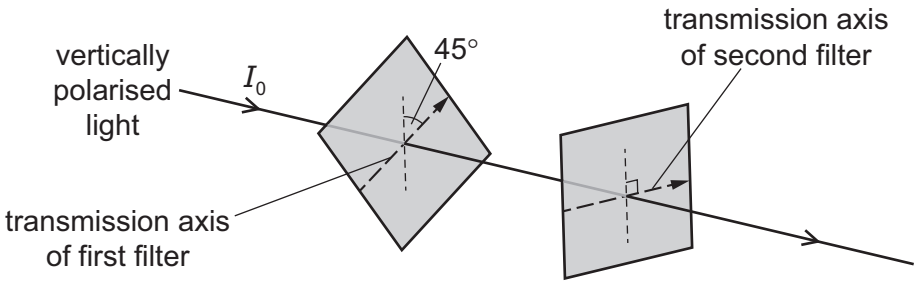
What could be the frequency of the light?

A 5.0×10^{5} Hz **B** 5.0×10^{8} Hz **C** 5.0×10^{11} Hz **D** 5.0×10^{14} Hz

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24 A vertically polarised beam of light of intensity I_0 is incident normally on a polarising filter.

The transmission axis of the filter is at 45° to the vertical. The beam of light transmitted by this filter is then incident normally on a second filter. The transmission axis of the second filter is horizontal.



What is the intensity of the beam of light after transmission through the second filter?

A 0 **B** $\frac{1}{4}I_0$ **C** $\frac{1}{2}I_0$ **D** I_0

25 A stationary wave is formed by two progressive waves travelling in opposite directions along the same line of travel.

Which statement about the two progressive waves is **not** correct?

A They must have a constant phase difference.

B They must have the same frequency.

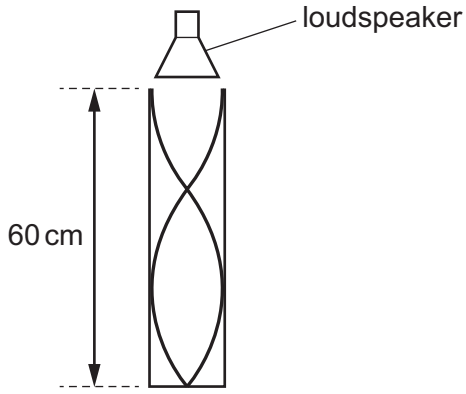
C They must have the same wavelength.

D They must travel at the same speed.

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26 The sound from a loudspeaker placed above a tube causes resonance of the air in the tube.

A stationary wave is formed with two nodes and two antinodes, as shown.



The tube has height 60 cm.

The speed of sound in the air is 340 m s^{-1}.

What is the frequency of the sound?

A 430 Hz **B** 570 Hz **C** 850 Hz **D** 1700 Hz

27 A continuous progressive water wave in a ripple tank passes through a gap in a barrier and diffracts.

The width of the gap is greater than the wavelength of the wave.

Which change causes the wave to spread over a larger angle as it passes through the gap?

A a small decrease in the wavelength of the wave

B a small decrease in the width of the gap

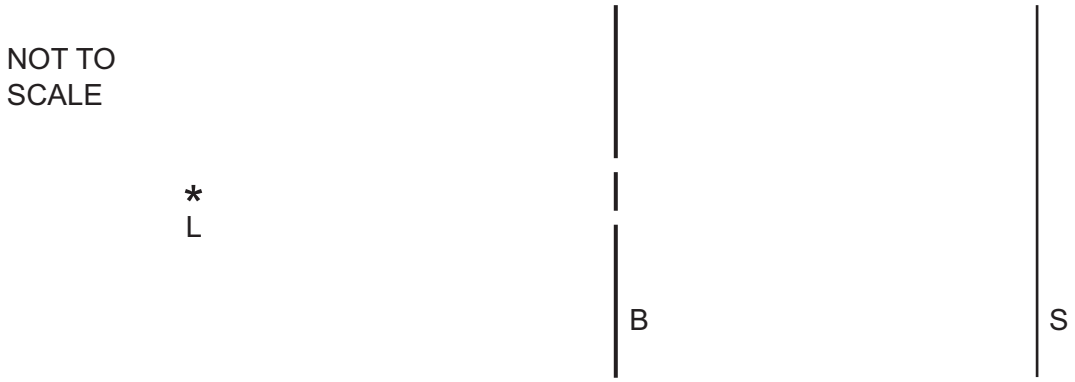
C a small increase in the frequency of the wave

D a small increase in the height of the barrier

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28 The diagram shows a view from above of a double-slit interference demonstration.

L is a monochromatic light source with a vertical filament. B is a barrier with two narrow vertical slits and S is a screen upon which interference fringes form.



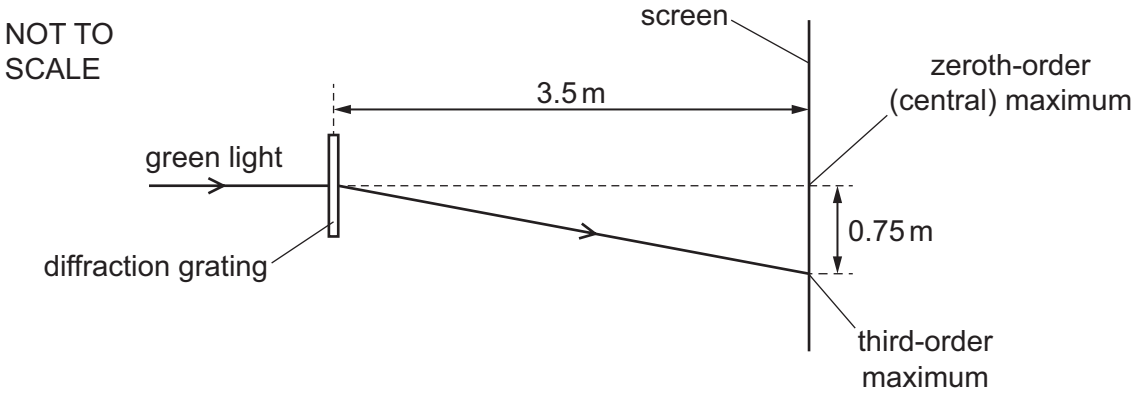
The intensity is I at the point on the screen where the centre of the fringe pattern forms.

When one of the slits is covered, what is the intensity at the same point on the screen?

A $\frac{I}{\sqrt{2}}$ **B** $\frac{I}{2}$ **C** $\frac{I}{2\sqrt{2}}$ **D** $\frac{I}{4}$

29 Green light of wavelength 550 nm is incident normally on a diffraction grating and produces a diffraction pattern on a screen placed 3.5 m from the diffraction grating.

The **third**-order maximum on the screen is a distance of 0.75 m from the zeroth-order (central) maximum.



What is the distance between two adjacent slits in the diffraction grating?

A 2.6×10^{-6} m

B 7.7×10^{-6} m

C 7.9×10^{-6} m

D 1.0×10^{-5} m

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30 Which charge can be carried by a charge carrier?

A 1.1×10^{-19} C

B 4.0×10^{-19} C

C 4.8×10^{-19} C

D 9.1×10^{-19} C

31 A resistor of resistance R is connected across a cell of electromotive force (e.m.f.) E and negligible internal resistance.

Which single change to the circuit would lead to the largest increase in the power dissipated in the resistor?

A doubling the value of E

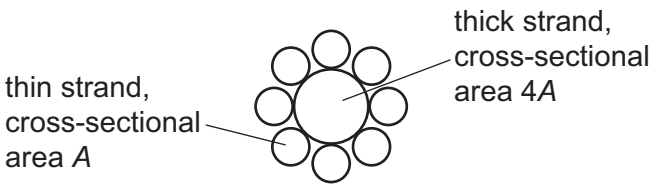
B doubling the value of R

C halving the value of E

D halving the value of R

32 An electrical cable is made up of one thick strand of copper wire that is surrounded by eight thin strands of copper wire. All nine strands of wire are connected in parallel with each other.

A cross-section of the cable is shown.



Each thin strand of wire has cross-sectional area A and length L.

The thick strand of wire has cross-sectional area $4A$ and length L.

The cable has total resistance R.

Which expression gives the resistivity of copper?

A $\frac{4A}{33RL}$ **B** $\frac{12A}{RL}$ **C** $\frac{4AR}{L}$ **D** $\frac{12AR}{L}$

33 A car has sensors for detecting the light intensity and temperature of its environment.

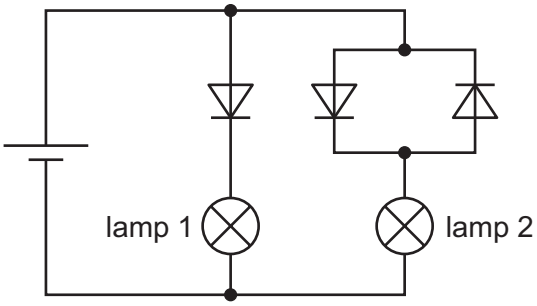
The sensors make use of a light-dependent resistor (LDR) and a thermistor.

The car moves from a warm and dark environment into a cold and bright environment.

What are the changes to the resistances of the LDR and thermistor?

	resistance of LDR	resistance of thermistor
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

34 A student assembles the circuit shown.



Which row describes the state of the two lamps?

	lamp 1	lamp 2
A	off	off
B	on	off
C	off	on
D	on	on

35 Kirchhoff's first law is a consequence of the conservation of which quantity?

A charge

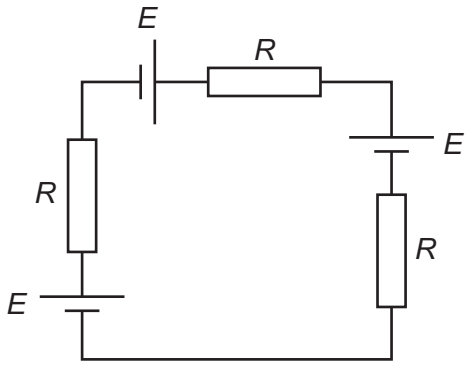
B energy

C momentum

D potential difference

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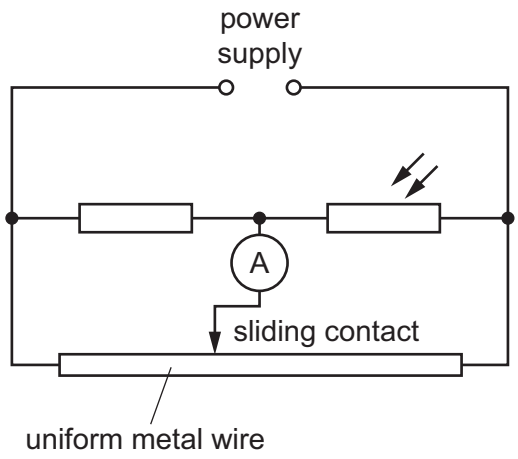
36 Three identical cells each have electromotive force (e.m.f.) E and negligible internal resistance. The cells are connected to three identical resistors, each of resistance R, as shown.



What is the potential difference across one of the resistors?

A 0 **B** $\frac{E}{3}$ **C** $\frac{2E}{3}$ **D** E

37 In the potentiometer circuit shown, the reading on the ammeter is zero.



The light-dependent resistor (LDR) is then covered and the ammeter gives a non-zero reading.

Which change could return the ammeter reading to zero?

A decreasing the supply voltage

B increasing the supply voltage

C moving the sliding contact to the left

D moving the sliding contact to the right

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38 What are isotopes?

A nuclei of different elements with the same number of neutrons

B nuclei of different elements with the same number of nucleons

C nuclei of the same element with different numbers of neutrons

D nuclei of the same element with different numbers of protons

39 A neutron ${}^{1}_{0}\mathrm{n}$ is fired at a ${}^{235}_{92}\mathrm{U}$ nucleus. The neutron is absorbed by the nucleus which then splits to form nuclei of ${}^{141}_{56}\mathrm{Ba}$ and ${}^{92}_{36}\mathrm{Kr}$.

What is the number of free neutrons emitted when the ${}^{235}_{92}\mathrm{U}$ nucleus splits?

A 0 **B** 1 **C** 2 **D** 3

40 A π^+ meson has a charge of $+e$, where e is the elementary charge. It consists of an up quark and one other quark.

What could be the other quark in the π^+ meson?

A anti-down

B anti-up

C bottom

D charm

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Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2023**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	B	**1**
2	D	**1**
3	D	**1**
4	C	**1**
5	B	**1**
6	A	**1**
7	C	**1**
8	B	**1**
9	D	**1**
10	A	**1**
11	B	**1**
12	C	**1**
13	A	**1**
14	A	**1**
15	D	**1**
16	A	**1**
17	B	**1**
18	D	**1**
19	B	**1**
20	B	**1**
21	B	**1**
22	C	**1**
23	D	**1**
24	B	**1**
25	A	**1**
26	A	**1**
27	B	**1**
28	D	**1**

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Question	Answer	Marks
29	C	**1**
30	C	**1**
31	A	**1**
32	D	**1**
33	C	**1**
34	D	**1**
35	A	**1**
36	B	**1**
37	C	**1**
38	C	**1**
39	D	**1**
40	A	**1**

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June 2023 · Paper 13

9702_s23_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2023**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What **must** be included in a record of a physical quantity?

A an integer value for the quantity

B an SI unit

C a numerical value for the quantity

D a unit expressed in base units

2 What is the ohm expressed in SI base units?

A $kg\,m^2\,s^{-3}\,A^{-2}$ **B** $kg^{-1}\,m^{-2}\,s^3\,A^2$ **C** $J\,C^{-1}\,A^{-1}$ **D** $W\,A^{-2}$

3 A desk has a true width of 50.0 cm.

Two students, X and Y, measure the width of the desk.

Student X uses a tape measure and records a width of (49.5 ± 0.5) cm.

Student Y uses a metre rule and records a width of (51.4 ± 0.1) cm.

Which statement about the measurement of student X is correct?

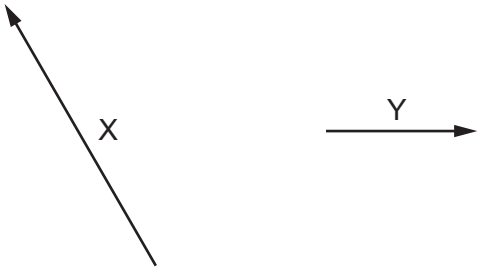
A It is less accurate and less precise than the measurement of student Y.

B It is less accurate but more precise than the measurement of student Y.

C It is more accurate and more precise than the measurement of student Y.

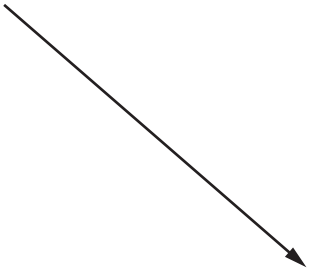
D It is more accurate but less precise than the measurement of student Y.

4 The diagram shows two vectors, X and Y, drawn to scale.

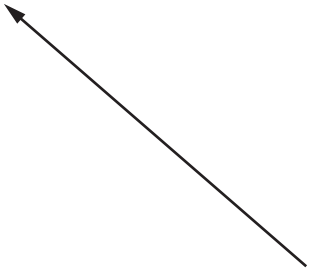


If X = Y – Z, which diagram represents the vector Z?

A



B



C

D

5 Which equation, representing uniformly accelerated motion in a straight line, can be determined using **only** the definition of acceleration?

A $s = ut + \frac{1}{2}at^2$

B $s = \frac{1}{2}(u + v)t$

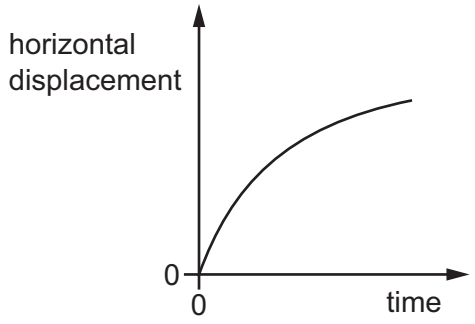
C $v = u + at$

D $v^2 = u^2 + 2as$

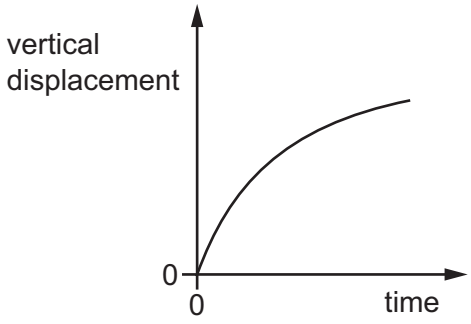
6 An object moves from rest with uniform velocity horizontally and uniform acceleration vertically.

Which graph showing the variation with time of the displacement of the object from its initial position is correct?

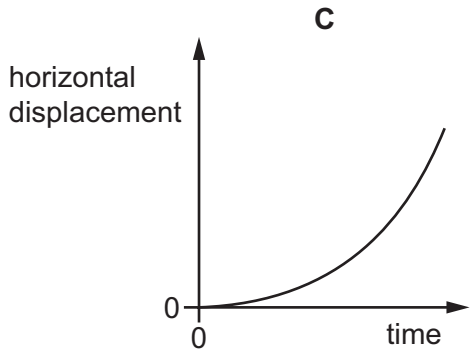
A



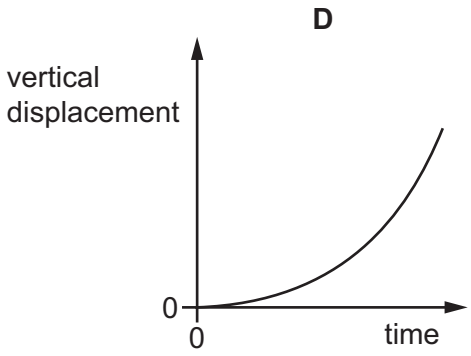
B



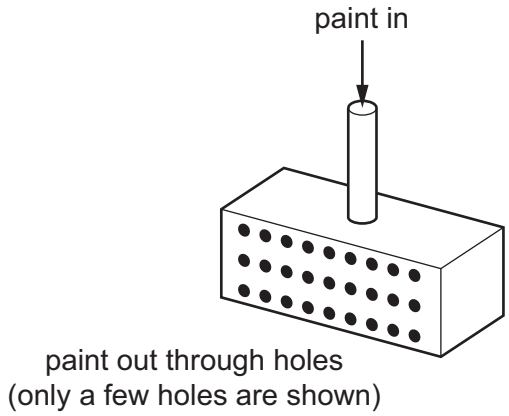
C



D



7 A device for spraying paint consists of a box with its faces horizontal and vertical. One of its vertical faces contains small holes. Paint is fed into the box under pressure via a vertical tube and exits through the holes as fine streams moving horizontally.



The paint is ejected at a speed of $2.5\,m\,s^{-1}$ through 400 holes, each of area $0.4\,mm^2$. The density of the paint is $900\,kg\,m^{-3}$.

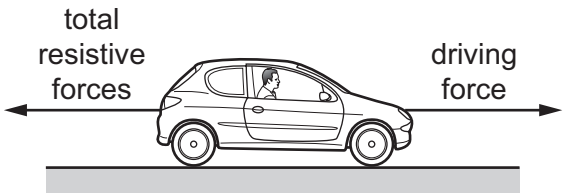
What is the horizontal force required to hold the device stationary as it ejects the paint?

A 0.36 N **B** 0.90 N **C** 2.3 N **D** 900 N

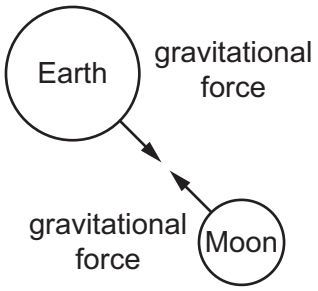
8 Each diagram illustrates a pair of forces of equal magnitude.

Which diagram gives an example of a pair of forces that is described by Newton's third law of motion?

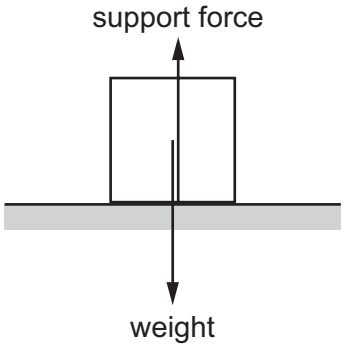
A



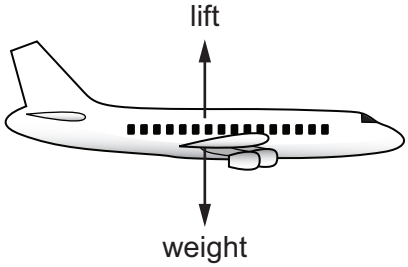
B



C



D



9 Two balls of identical shape and size but different masses are falling through the same liquid.

The sum of the drag force and upthrust acting on each ball is equal to its weight.

Which statement about the two balls is correct?

A The heavier ball has a larger acceleration than the lighter ball.

B The heavier ball has a smaller deceleration than the lighter ball.

C The heavier ball is falling at the same speed as the lighter ball.

D The heavier ball is falling at a larger speed than the lighter ball.

10 A perfectly elastic collision occurs between two objects X and Y. The mass of X is m and the mass of Y is $4m$. Object X travels at speed v before the collision and speed $\frac{3v}{5}$ in the opposite direction after the collision. Object Y is stationary before the collision.



What is the kinetic energy of Y after the collision?

A $\frac{8}{10}mv^2$ **B** $\frac{34}{50}mv^2$ **C** $\frac{16}{50}mv^2$ **D** $\frac{1}{5}mv^2$

11 What is **not** a requirement for two forces to act as a couple?

A The two forces act in opposite directions.

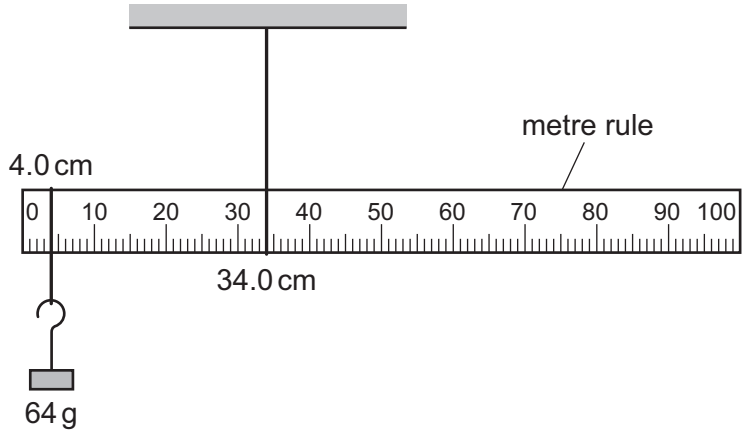
B The two forces act through the same point.

C The two forces combine to produce zero resultant force.

D The two forces have equal magnitude.

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12 A uniform metre rule is pivoted at the 34.0 cm mark, as shown.



The rule balances when a 64 g mass is hung from the 4.0 cm mark.

What is the mass of the metre rule?

A 38 g **B** 44 g **C** 120 g **D** 136 g

13 A volume of 1.5 m^3 of water is mixed with 0.50 m^3 of alcohol. The density of water is 1000 $kg\,m^{-3}$ and the density of alcohol is 800 $kg\,m^{-3}$.

The volume of the mixture is 2.0 m^3.

What is the density of the mixture?

A 850 $kg\,m^{-3}$ **B** 900 $kg\,m^{-3}$ **C** 940 $kg\,m^{-3}$ **D** 950 $kg\,m^{-3}$

14 An object is falling at a constant speed through a viscous liquid. F_U is the upthrust on the object due to the liquid. W_L is the weight of the liquid displaced by the object. W_O is the weight of the object.

Which equation **must** be correct?

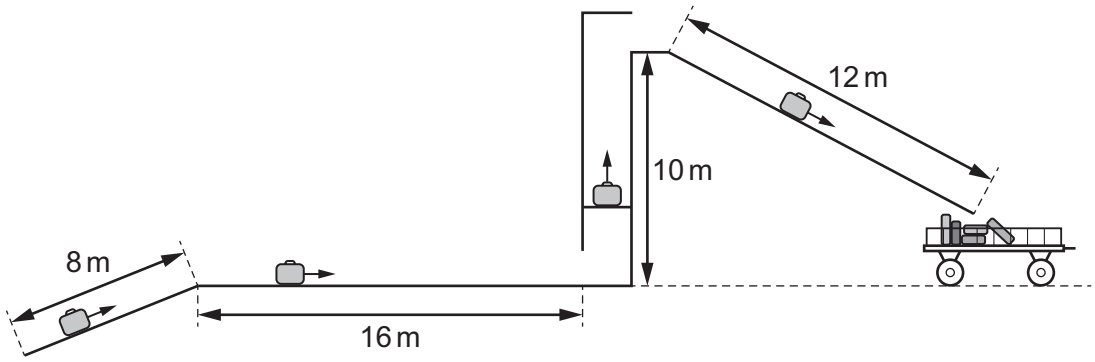
A $F_U = W_L$

B $F_U = W_O - W_L$

C $F_U = W_O$

D $F_U = W_O + W_L$

15 An airport has a mechanical system for moving luggage. The system uses a horizontal conveyor belt, a sloping conveyor belt, a lift and a frictionless slide. A suitcase is moved around the airport using this system.



Resistive forces opposing the motion of the suitcase are negligible.

For which movement of the suitcase is the net work done on the suitcase greatest?

A moving the suitcase a distance of 8 m at a constant speed up the sloping conveyor belt

B moving the suitcase a distance of 16 m at a constant speed along the horizontal conveyor belt

C moving the suitcase a distance of 10 m at a constant speed vertically upwards on the lift

D moving the suitcase a distance of 12 m at increasing speed downwards on the slide

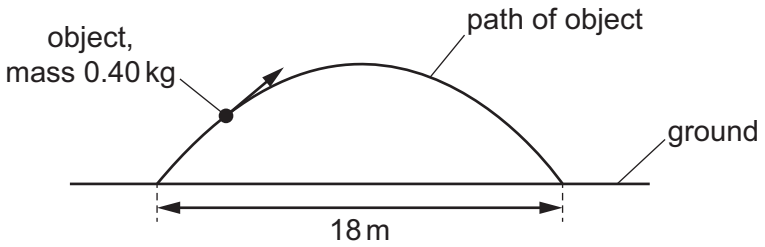
16 A car moves along a horizontal road with a constant velocity v against a resistive force F.

The engine of the car has an efficiency of 25%.

What is the input power to the engine?

A $\frac{Fv}{4.0}$ **B** $\frac{4.0}{Fv}$ **C** $4.0Fv$ **D** $\frac{4.0F}{v}$

17 An object of mass 0.40 kg is projected into the air and follows a curved path above horizontal ground.



The object takes a time of 1.5 s to move along its path. The object lands a horizontal distance of 18 m from its initial position. Air resistance is negligible.

What is the kinetic energy of the object at its maximum height?

A 0 J **B** 2.4 J **C** 11 J **D** 29 J

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18 A metal wire has length 5.2 m and diameter 1.0 mm. The metal has Young modulus 360 GPa.

The wire is fixed at one end and a force is applied to the other end. The force extends the wire by 7.2 mm. The wire obeys Hooke's law.

What is the force applied to the wire?

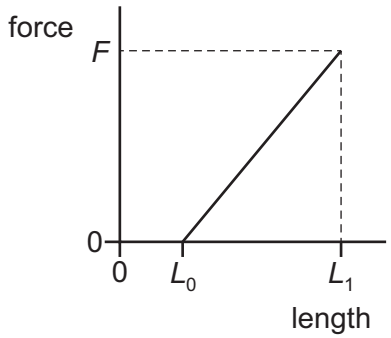
A 1.2×10^2 N

B 3.9×10^2 N

C 5.0×10^2 N

D 1.6×10^3 N

19 The graph shows how the length of a spring varies with the force applied to it.



The spring has unstretched length L_0. When a force F is applied, the spring has length L_1.

What is the work done in stretching the spring to length L_1?

A $\frac{1}{2}FL_1$ **B** $\frac{1}{2}F(L_1 - L_0)$ **C** FL_1 **D** $F(L_1 - L_0)$

20 A progressive wave of frequency 1.5 kHz travels in a medium at a speed of 340 m s^{-1}.

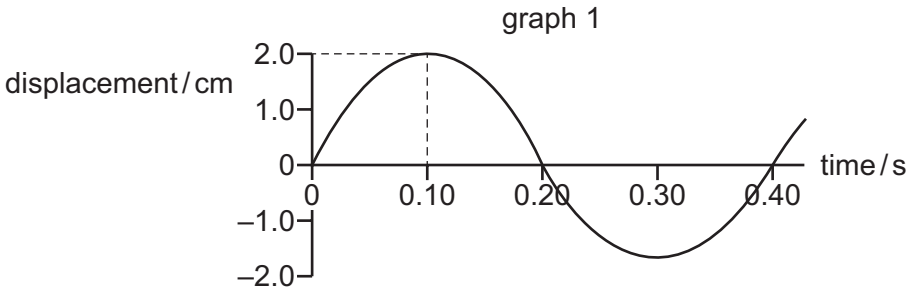
What is the minimum distance between two points on the wave that have a phase difference of 70°?

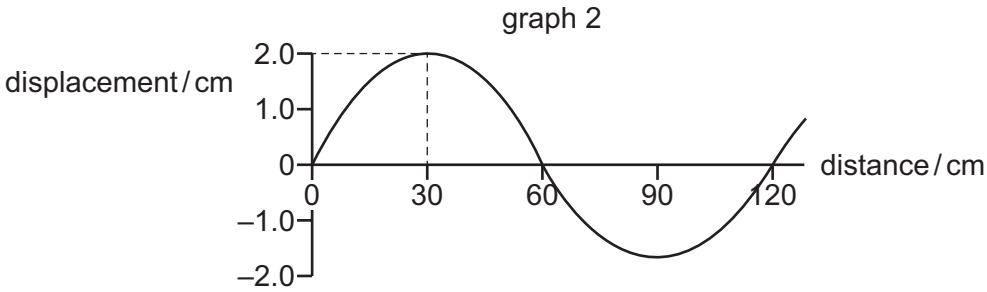
A 4.4 cm **B** 8.8 cm **C** 18 cm **D** 23 cm

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21 Graph 1 shows the variation with time of displacement at a fixed distance along a progressive wave.

Graph 2 represents the same wave and shows the variation with distance of displacement at an instant in time.





What is the speed of the wave?

A $5.0\,cm\,s^{-1}$ **B** $48\,cm\,s^{-1}$ **C** $150\,cm\,s^{-1}$ **D** $300\,cm\,s^{-1}$

22 A vehicle moves with constant velocity along a road directly towards an observer. The observed frequency of the sound from the vehicle changes as the vehicle moves past the observer.

Which phenomenon explains the change in frequency?

A diffraction

B interference

C polarisation

D the Doppler effect

23 An electromagnetic wave has a wavelength of 2.1 cm in a vacuum.

Which region of the electromagnetic spectrum contains this wave?

A infrared

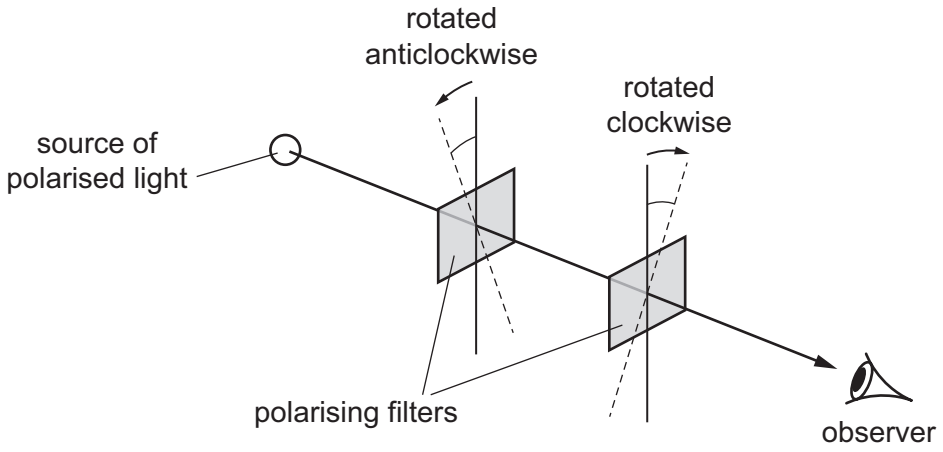
B microwave

C radio wave

D visible light

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24 A source of plane polarised light is observed through two polarising filters.



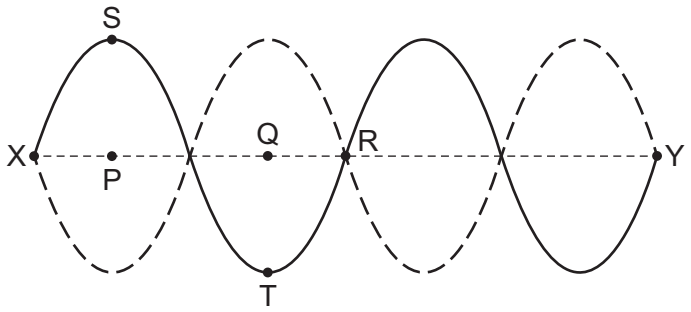
The filters are positioned so that the source appears at its brightest. One of the filters is then rotated clockwise and the other filter is rotated anticlockwise through the same angle.

How does the source appear when both filters have been rotated 90° and 180° from their initial positions?

	90°	180°
A	brightest	brightest
B	brightest	darkest
C	darkest	brightest
D	darkest	darkest

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25 The diagram shows a string stretched between fixed points X and Y. There is a stationary wave on the string.



The solid curve shows the string at a position of maximum displacement at time t_0. The dashed curve shows the other position of maximum displacement. The straight central dashed line shows the mean position of the string.

Point S on the string is directly above point P. Point T on the string is directly below point Q.

Which statement is correct?

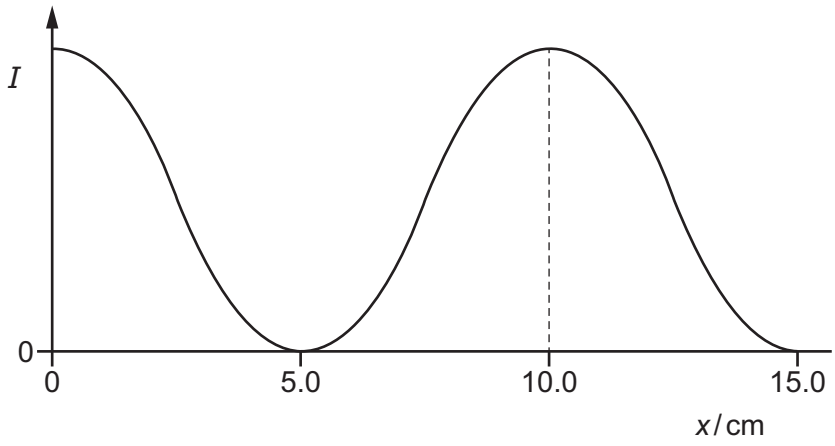
A A short time after t_0, point R on the string will be displaced.

B A short time after t_0, points S and T on the string move in opposite directions.

C The distance between P and Q is one wavelength.

D Two moving points on the string that are equal distances from point R vibrate in phase.

26 The variation with distance x of the intensity I along a stationary sound wave in air is shown.



The speed of sound in air is $340\,\text{m}\,\text{s}^{-1}$.

What is the frequency of the sound wave?

A 1700 Hz **B** 2300 Hz **C** 3400 Hz **D** 6800 Hz

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27 An experiment is set up to demonstrate the diffraction of water waves in a ripple tank.

The waves pass through a gap of width w and some diffraction of the waves is observed.

The wavelength of the waves is now doubled.

What is the new gap width needed to cause the same amount of diffraction as before?

A $0.5w$ **B** w **C** $2w$ **D** $4w$

28 Two coherent electromagnetic waves are travelling in a vacuum. The two waves meet at a point. At this point, the two waves have different intensities.

Which statement about the waves is **not** correct?

A They have a constant phase difference at the point.

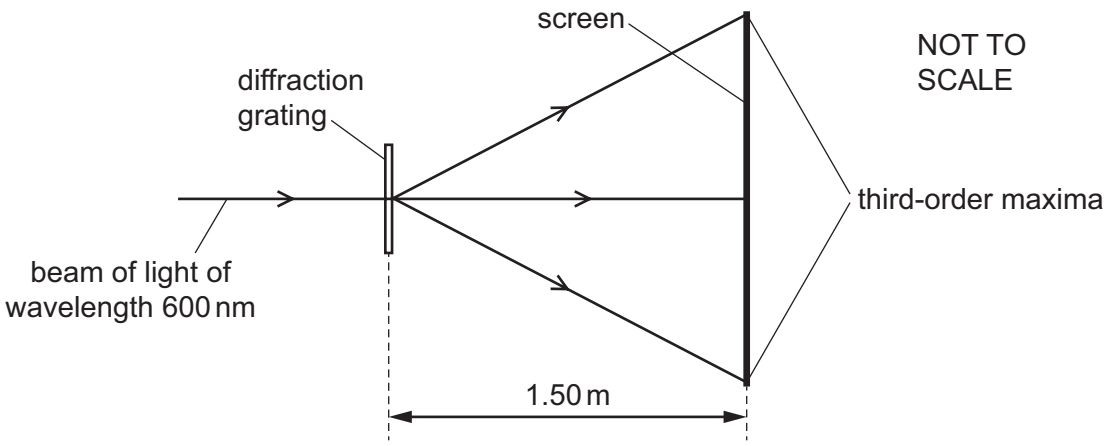
B They have the same amplitude at the point.

C They have the same frequency.

D They travel at the same speed.

29 A parallel beam of light of wavelength 600 nm is incident normally on a diffraction grating.

The distance between adjacent slits in the grating is 2.0×10^{-6} m. A screen is placed parallel to the grating, at a distance of 1.50 m from the grating. **Third**-order diffraction maxima are observed at the two ends of the screen, as shown.



What is the distance between the two ends of the screen?

A 1.4 m **B** 2.7 m **C** 3.1 m **D** 6.2 m

30 A metal wire is connected between the terminals of a cell so that there is a current in the wire.

Which statement is correct?

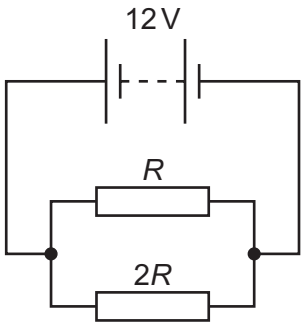
A Negatively charged electrons in the wire move from the negative terminal to the positive terminal.

B Negatively charged nuclei in the wire move from the negative terminal to the positive terminal.

C Positively charged electrons in the wire move from the positive terminal to the negative terminal.

D Positively charged nuclei in the wire move from the positive terminal to the negative terminal.

31 Two resistors of resistances R and $2R$ are connected in parallel with a battery of electromotive force (e.m.f.) 12 V and negligible internal resistance.



The total power dissipated by the two resistors is 36 W.

What is the value of R?

A 0.50 Ω **B** 2.7 Ω **C** 4.0 Ω **D** 6.0 Ω

32 A wire has a length of 3.0 m and is made of metal of resistivity $4.9 \times 10^{-7}\,\Omega\,\text{m}$.

A potential difference (p.d.) of 12 V is applied across the wire so that it has a current of 1.4 A.

What is the cross-sectional area of the wire?

A $1.2 \times 10^{-7}\,\text{m}^2$

B $1.7 \times 10^{-7}\,\text{m}^2$

C $1.1 \times 10^{-6}\,\text{m}^2$

D $1.3 \times 10^{-5}\,\text{m}^2$

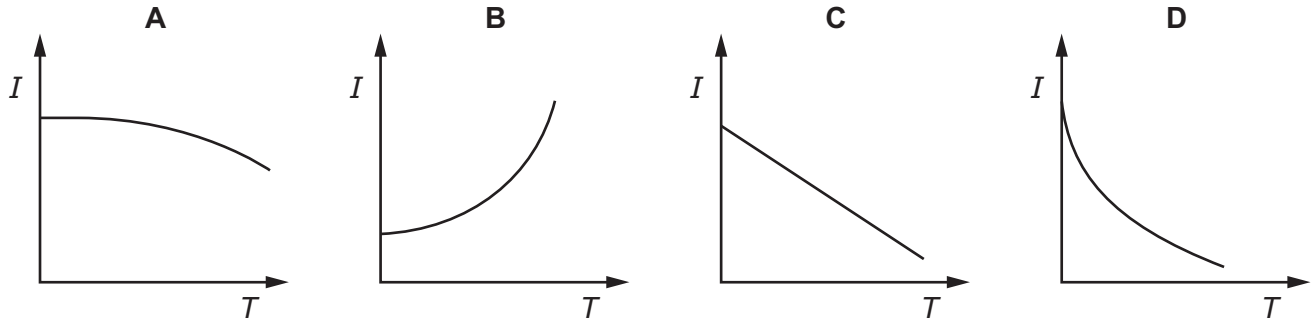
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33 A cell of negligible internal resistance is connected in series with a thermistor, a fixed resistor and an ammeter.

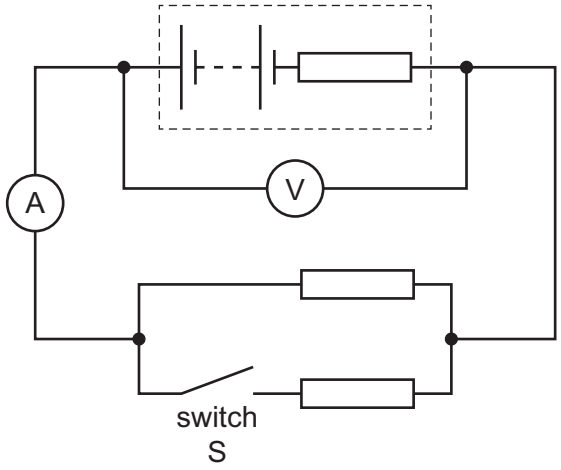
The thermistor is placed in a beaker of water and the temperature of the water is slowly increased.

A graph of current I against the temperature T of the thermistor is plotted.

Which graph could show the variation of I with T?



34 A battery with internal resistance is connected to a parallel arrangement of two resistors and a switch S, as shown.



Initially, switch S is open.

What happens to the voltmeter and ammeter readings when switch S is closed?

	voltmeter reading	ammeter reading
A	decreases	increases
B	decreases	decreases
C	increases	increases
D	increases	decreases

35 Kirchhoff's first law states that the sum of the currents entering a junction in a circuit is equal to the sum of the currents leaving it.

The law is based on the conservation of a physical quantity.

What is this physical quantity?

A charge

B energy

C mass

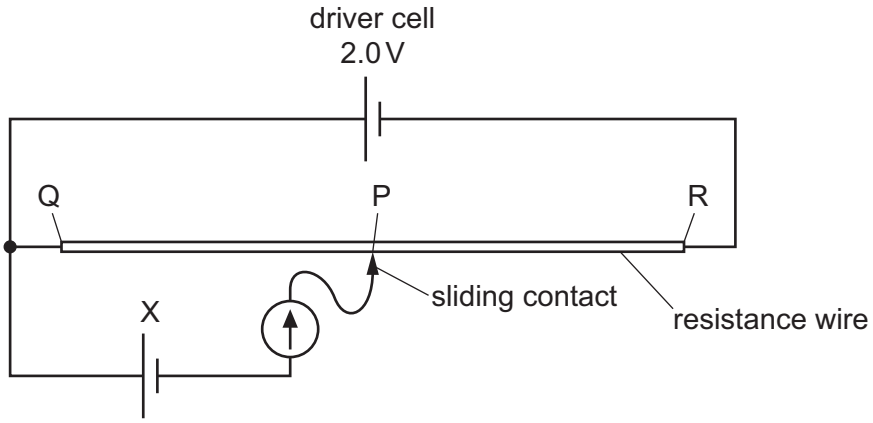
D momentum

36 Two resistors have a combined resistance of 34 Ω when connected in series. The same resistors have a combined resistance of 7.4 Ω when connected in parallel.

What is the resistance of one of the resistors?

A 15 Ω **B** 17 Ω **C** 23 Ω **D** 27 Ω

37 A potentiometer circuit is used to investigate the electromotive force (e.m.f.) of a cell X.



The e.m.f. of cell X is known to be approximately 0.50 V.

The driver cell has negligible internal resistance and an e.m.f. of 2.0 V. The sliding contact is moved along the uniform resistance wire between ends Q and R to a point P where the reading on the galvanometer is zero.

What is an expression for the approximate length QP?

A $\frac{QR}{4}$ **B** $\frac{QR}{3}$ **C** $\frac{2QR}{3}$ **D** $\frac{3QR}{4}$

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38 The table contains data for four different nuclei P, Q, R and S.

nucleus	number of neutrons	nucleon number
P	5	10
Q	6	10
R	6	14
S	8	16

Which two nuclei are isotopes of the same element?

A P and Q **B** P and S **C** Q and R **D** R and S

39 Which statement about β^- decay is correct?

A A neutron changes to a proton in the nucleus and an electron and an antineutrino are emitted.

B A neutron changes to a proton in the nucleus and an electron and a neutrino are emitted.

C A proton changes to a neutron in the nucleus and an electron and an antineutrino are emitted.

D A proton changes to a neutron in the nucleus and an electron and a neutrino are emitted.

40 What is the quark composition of a hydrogen-3 nucleus, ${}^{3}_{1}\mathrm{H}$?

	number of quarks	
	up	down
A	4	5
B	5	4
C	5	7
D	7	5

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Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2023**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Question	Answer	Marks
1	C	**1**
2	A	**1**
3	D	**1**
4	A	**1**
5	C	**1**
6	D	**1**
7	B	**1**
8	B	**1**
9	D	**1**
10	C	**1**
11	B	**1**
12	C	**1**
13	D	**1**
14	A	**1**
15	C	**1**
16	C	**1**
17	D	**1**
18	B	**1**
19	B	**1**
20	A	**1**
21	D	**1**
22	D	**1**
23	B	**1**
24	C	**1**
25	B	**1**
26	A	**1**
27	C	**1**
28	B	**1**

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Question	Answer	Marks
29	D	**1**
30	A	**1**
31	D	**1**
32	B	**1**
33	B	**1**
34	A	**1**
35	A	**1**
36	C	**1**
37	A	**1**
38	D	**1**
39	A	**1**
40	A	**1**

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November 2023 · Paper 11

9702_w23_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2023**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What is a reasonable estimate of the cross-sectional area of the wire in a paper clip?

A $1 \times 10^{-3}\,m^2$ **B** $8 \times 10^{-5}\,m^2$ **C** $8 \times 10^{-7}\,m^2$ **D** $1 \times 10^{-9}\,m^2$

2 Which quantity is **not** an SI base quantity?

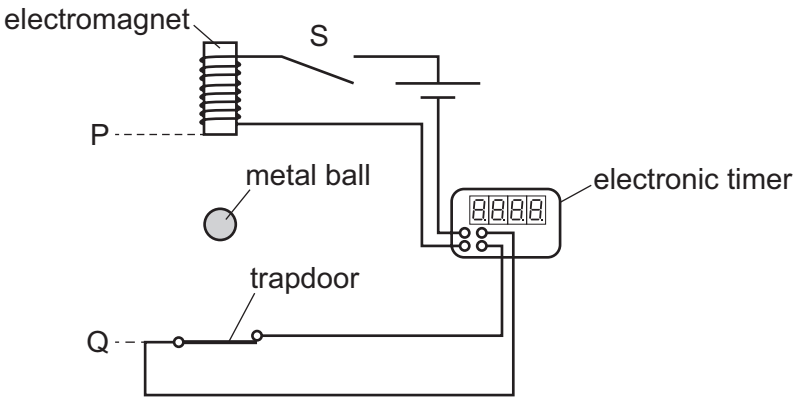
A charge

B mass

C temperature

D time

3 A student determines the acceleration of free fall by using a small metal ball, as shown.



When switch S is opened, the ball is released from an electromagnet and an electronic timer is started. The ball then falls vertically downwards. The timer stops when the ball hits a trapdoor. The student measures the distance PQ between the electromagnet and the trapdoor. This distance and the reading on the timer are then used to calculate the acceleration of free fall.

Which statement about errors in the experiment is correct?

A The random error can be reduced by adding the diameter of the ball to the distance PQ.

B The random error can be reduced by subtracting the diameter of the ball from the distance PQ.

C The systematic error can be reduced by adding the diameter of the ball to the distance PQ.

D The systematic error can be reduced by subtracting the diameter of the ball from the distance PQ.

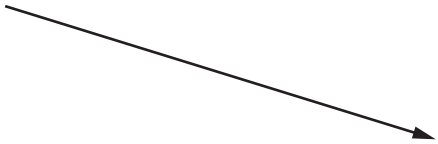
4 The diagram shows two coplanar forces, P and Q, drawn to scale.



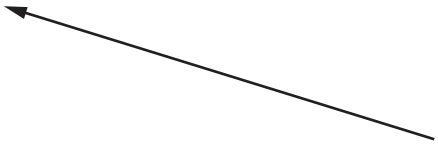
Force R is given by $R = Q - P$.

Which diagram represents R?

A



B



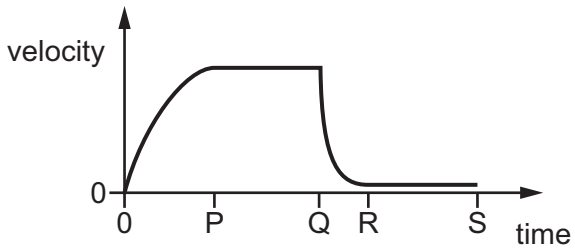
C



D

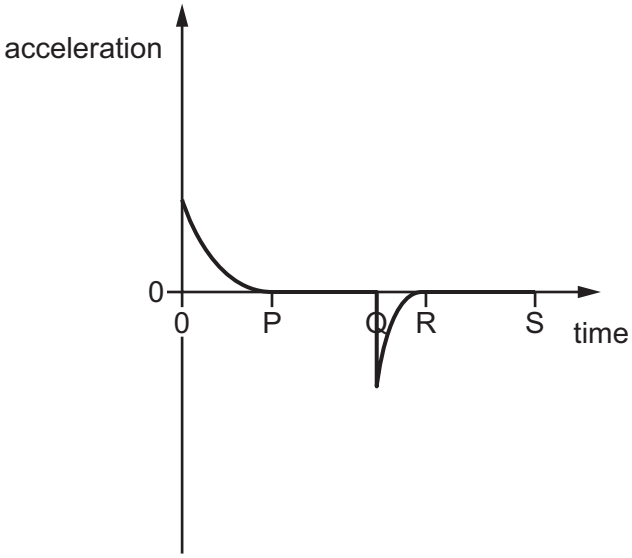


5 A parachutist falls from a stationary balloon at time $t = 0$. The velocity–time graph for the parachutist from time $t = 0$ until the time when he is just above the ground is shown.

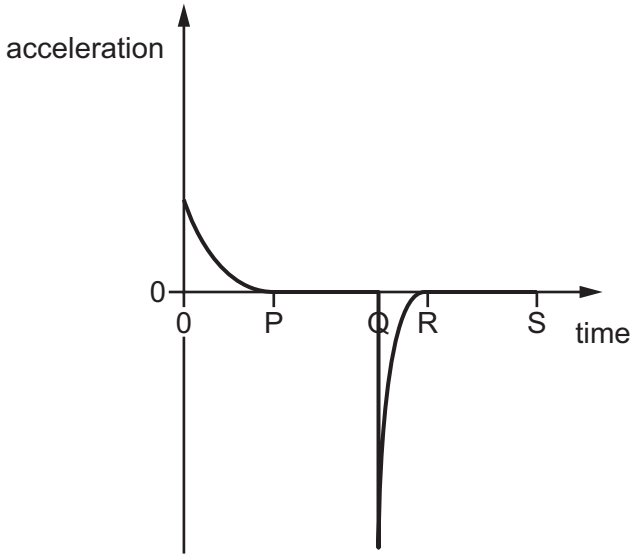


Which graph best shows the variation with time of the acceleration of the parachutist?

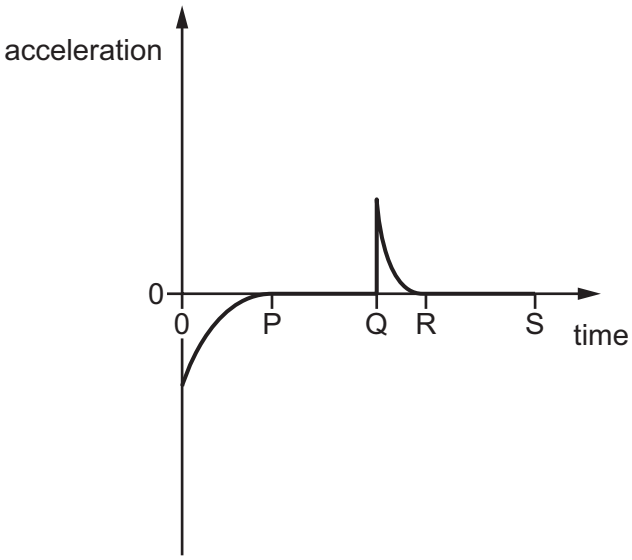
A



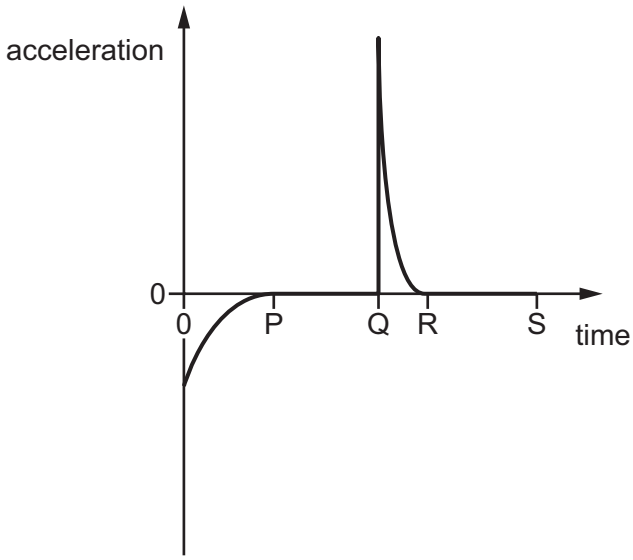
B



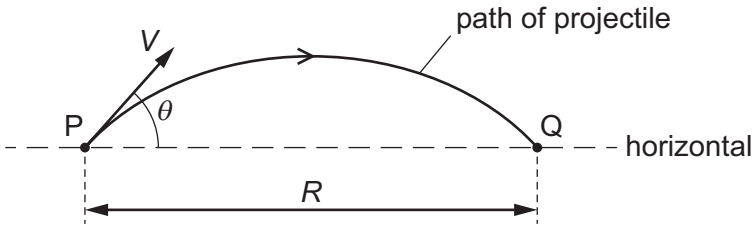
C



D



6 A projectile is fired from point P with velocity V at an angle θ to the horizontal. It lands at point Q, a horizontal distance R from P, after time T.



The acceleration of free fall is g. Air resistance is negligible.

Which equation is correct?

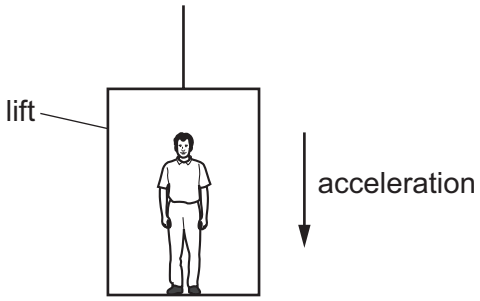
A $R = VT\cos\theta$

B $R = VT\sin\theta$

C $R = VT\cos\theta - \frac{1}{2}gT^2$

D $R = VT\sin\theta - \frac{1}{2}gT^2$

7 A man stands in a lift that is accelerating vertically downwards, as shown.



Which statement describes the force exerted by the man on the floor?

A It is equal to the weight of the man.

B It is greater than the force exerted by the floor on the man.

C It is less than the force exerted by the floor on the man.

D It is less than the weight of the man.

8 A ball of mass 200 g is thrown horizontally with a speed of $20\,m\,s^{-1}$ against a vertical wall.

The ball is in contact with the wall for a time of 0.10 s before rebounding back along its original path with a speed of $10\,m\,s^{-1}$.

What is the average force exerted by the wall on the ball during the collision?

A 20 N **B** 60 N **C** 20 kN **D** 60 kN

9 In an experiment, a metal ball is dropped into a viscous liquid. The terminal velocity of the ball in the liquid is measured.

The experiment is repeated four times. For each repeat, a change is made to one of the following.

1 the density of the metal of the ball

2 the height from which the ball is dropped

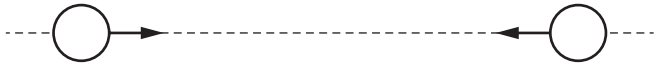
3 the density of the liquid

4 the depth of the liquid

Which two changes separately affect the terminal velocity of the ball in the liquid?

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

10 Two objects move towards each other along the same straight line.



After colliding, the two objects stick together and are stationary.

Which statement **must** be correct?

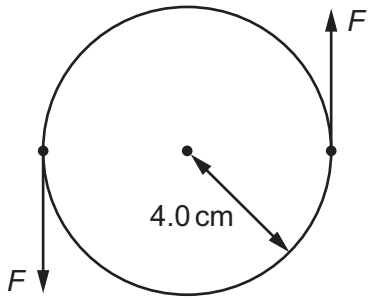
A The total kinetic energy of the two objects does not change during the collision.

B The total momentum of the two objects before the collision is zero.

C The two objects have equal mass.

D The two objects have the same speed before the collision.

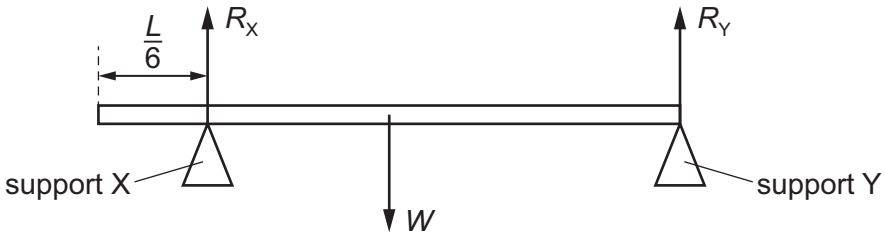
11 A minimum torque of 20 N m must be applied to the lid of a jar for it to open. The radius of the lid is 4.0 cm.



What is the minimum force *F* that must act on each side of the lid in order to open it?

A 2.5 N **B** 5.0 N **C** 250 N **D** 500 N

12 A uniform bar of length L and weight W rests horizontally on two supports X and Y.



Support X exerts a vertical force R_X at a distance of $\frac{L}{6}$ from one end of the bar.

Support Y exerts a vertical force R_Y at the other end of the bar.

The bar is in equilibrium.

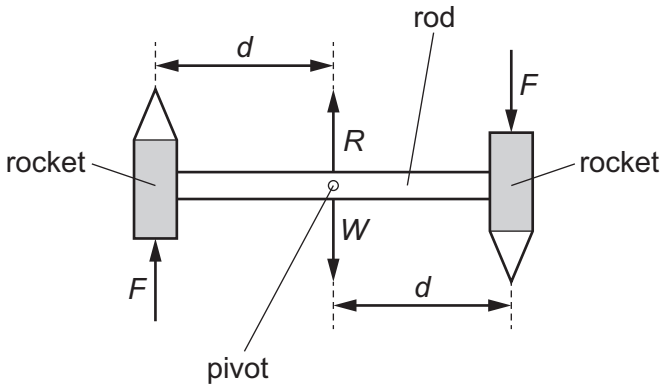
What is the ratio $\frac{R_X}{R_Y}$?

A $\frac{3}{2}$ **B** $\frac{2}{3}$ **C** $\frac{3}{5}$ **D** $\frac{2}{5}$

13 A type of firework is made by connecting two rockets, facing in opposite directions, to a rod, as shown.

The rod is attached to a frictionless pivot so that the firework can rotate in a vertical plane.

The firework has weight W. The pivot exerts a force R on the rod that is equal and opposite to W.



Each rocket exerts a force of magnitude F on the rod at a perpendicular distance d from the pivot. The forces exerted by the rockets are always in opposite directions.

Air resistance is negligible.

Which statement is correct?

A The firework is in equilibrium because the resultant force acting on it is zero.

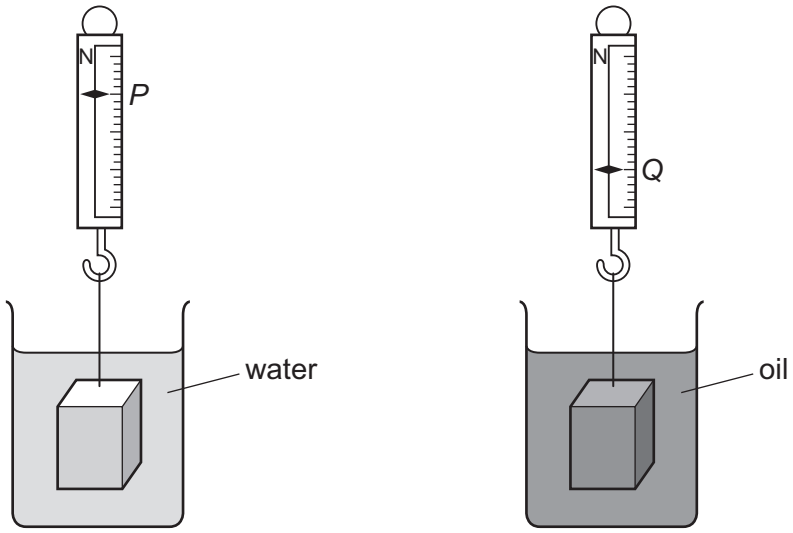
B The firework is in equilibrium because the resultant torque acting on it is zero.

C The firework is not in equilibrium because the resultant force acting on it is not zero.

D The firework is not in equilibrium because the resultant torque acting on it is not zero.

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14 An object of weight W is suspended from a newton meter. When the object is completely immersed in water, the newton meter reads P. When the object is completely immersed in oil, the newton meter reads Q.



What is the ratio $\frac{\text{density of oil}}{\text{density of water}}$?

A $\frac{W-P}{Q-P}$ **B** $\frac{Q-P}{W-P}$ **C** $\frac{W-P}{W-Q}$ **D** $\frac{W-Q}{W-P}$

15 A crate of mass 50 kg is pushed a distance of 6.0 m along a horizontal surface against a constant resistive force of 70 N. The crate moves at a constant speed. It is then lifted, at a constant speed, through a vertical distance of 1.2 m onto the back of a lorry.

What is the total work done in this process?

A 420 J **B** 480 J **C** 590 J **D** 1000 J

16 The input power to a television is P_{in}. The useful sound and light power emitted by the television is P_{out}.

What is the efficiency of the television?

A $\frac{P_{out}}{P_{in}}$ **B** $\frac{P_{in}-P_{out}}{P_{in}}$ **C** $\frac{P_{in}}{P_{out}}$ **D** $\frac{P_{out}}{P_{in}-P_{out}}$

17 A builder holding a brick of mass 3000 g drops the brick on his foot.

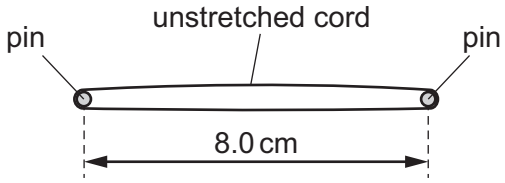
What is a reasonable estimate of the change in gravitational potential energy of the brick?

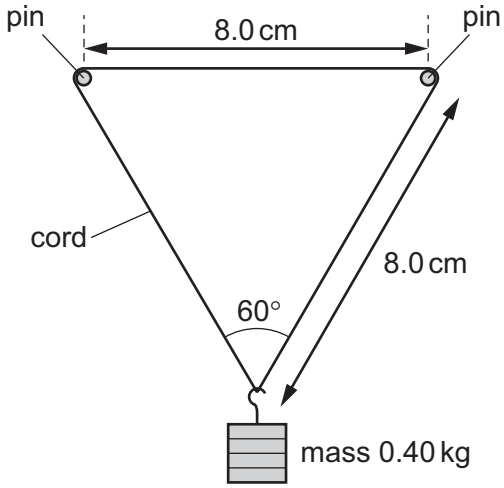
A 30 J **B** 300 J **C** 3000 J **D** 30 000 J

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18 An elastic cord of unstretched total length 16.0 cm and cross-sectional area $2.0 \times 10^{-6}\,m^2$ is held horizontally by two smooth pins a distance 8.0 cm apart.

The cord obeys Hooke's law. A load of mass 0.40 kg is suspended centrally on the cord. The angle between the two sides of the cord supporting the load is 60°.



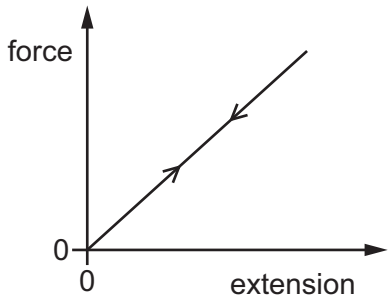


What is the Young modulus of the cord material?

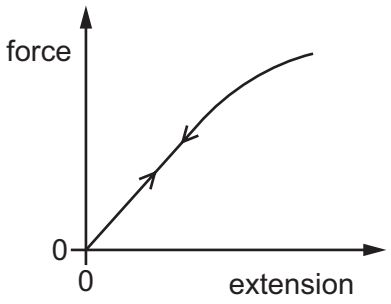
A $5.7 \times 10^5\,Pa$ **B** $1.1 \times 10^6\,Pa$ **C** $2.3 \times 10^6\,Pa$ **D** $3.9 \times 10^6\,Pa$

19 Which force–extension graph shows plastic deformation of a sample of material?

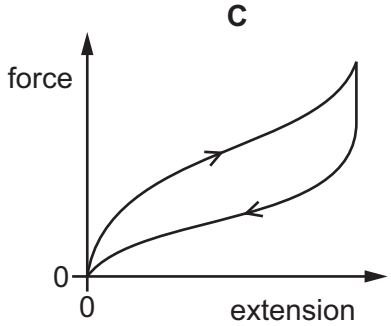
A



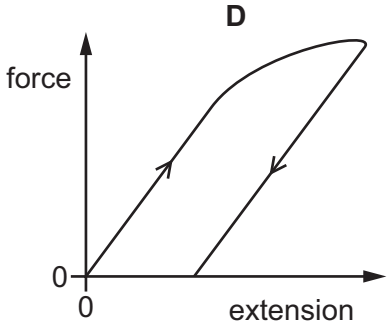
B



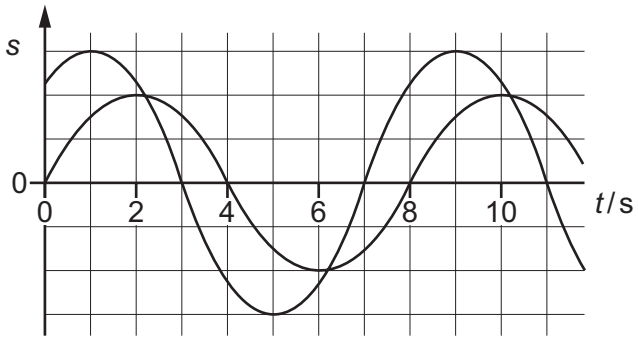
C



D



20 Two waves pass through a point P. The graph shows the variation with time t of the displacement s of the two waves at point P.



What is the phase difference between the two waves at point P?

A 0° **B** 45° **C** 90° **D** 180°

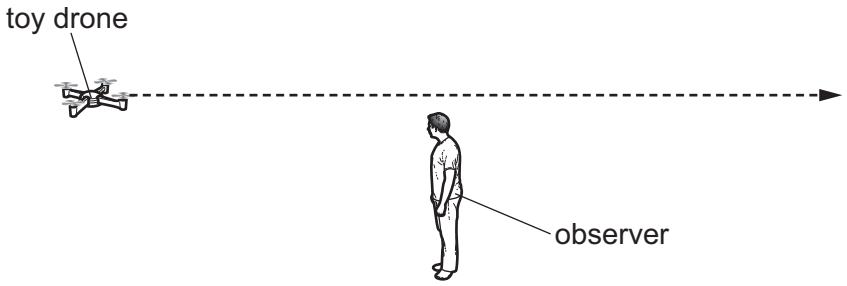
21 Which row is correct for both progressive transverse waves and progressive longitudinal waves?

	transverse waves	longitudinal waves
A	contain compressions and rarefactions	some can travel in a vacuum
B	can be polarised	contain compressions and rarefactions
C	vibrations are perpendicular to the direction of travel of the wave energy	can be polarised
D	some can travel in a vacuum	vibrations are perpendicular to the direction of travel of the wave energy

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22 A toy drone emits a sound of constant frequency 800 Hz. The speed of the sound in the air is $330\,m\,s^{-1}$.

The drone moves along a straight path directly towards an observer and then continues in a straight line directly away from the observer. The speed of the drone is constant.



What is the velocity of the drone when the frequency of the sound heard by the observer is 850 Hz?

	magnitude of velocity / $m\,s^{-1}$	direction of velocity
A	19	away from the observer
B	21	away from the observer
C	19	towards the observer
D	21	towards the observer

23 Which statement about electromagnetic waves in a vacuum is correct?

A Infrared waves have shorter wavelengths than visible light waves.

B Microwaves have longer wavelengths than radio waves.

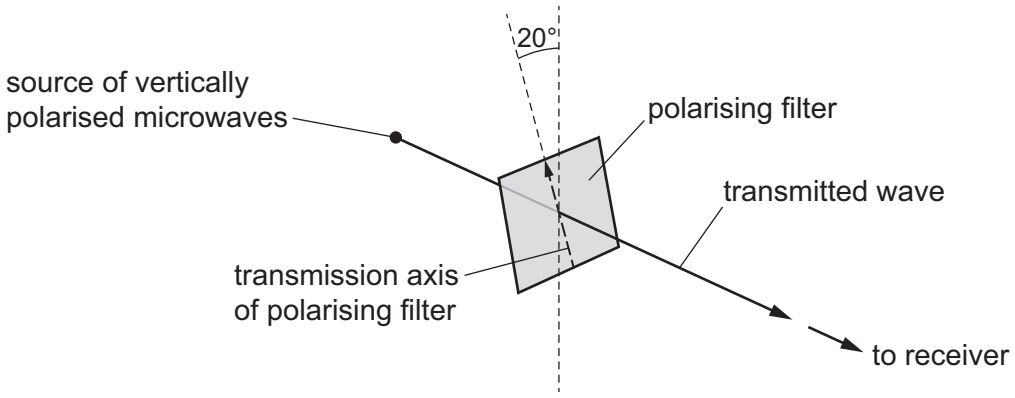
C Ultraviolet waves have higher frequencies than visible light waves.

D γ-rays have lower frequencies than X-rays.

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24 Vertically polarised microwaves are emitted from a source. The microwaves are detected by a receiver that is connected to a cathode-ray oscilloscope (CRO). The waveform displayed on the screen of the CRO has an amplitude of 2.6 cm.

A metal wire grid that acts as a polarising filter is now placed between the source and the receiver. The filter is orientated so that the plane of polarisation of the transmitted wave is at an angle of 20° to the vertical.



The distance between the source and receiver is unchanged. The settings on the CRO are also unchanged.

What is now the amplitude of the waveform displayed on the screen of the CRO?

A 0.30 cm **B** 0.89 cm **C** 2.3 cm **D** 2.4 cm

25 In an experiment, a stationary wave is formed on a string stretched horizontally between two fixed points.

Which statement about the experiment is correct?

A At certain times, the string between two nodes is horizontal with all points having zero displacement.

B Each point on the string between two antinodes has an oscillation of the same amplitude.

C The number of nodes is equal to the number of antinodes.

D Two adjacent antinodes oscillate in phase.

26 A musical organ produces notes by blowing air into a set of pipes that are open at one end and closed at the other.

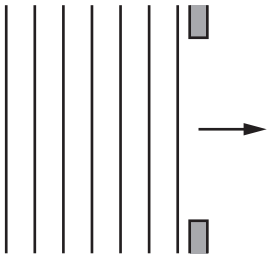
The speed of sound in the air in the pipes is 320 $m\,s^{-1}$.

What is the lowest frequency of sound produced by a pipe of length 10 m?

A 4 Hz **B** 8 Hz **C** 16 Hz **D** 32 Hz

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27 In an experiment, water waves in a ripple tank are incident on a gap, as shown.



Some diffraction of the water waves is observed.

Which change to the experiment would provide a better demonstration of diffraction?

A Increase the amplitude of the waves.

B Increase the frequency of the waves.

C Increase the wavelength of the waves.

D Increase the width of the gap.

28 Light of wavelength λ is emitted from two point sources R and S and falls onto a distant screen.



At point P on the screen, the light intensity is zero.

What could explain the zero intensity at P?

A Light from the two sources is emitted 180° out of phase and the path difference to P is $\frac{1}{2}\lambda$.

B Light from the two sources is emitted in phase and the path difference to P is λ.

C Light from the two sources is emitted 90° out of phase and the path difference to P is λ.

D Light from the two sources is emitted in phase and the path difference to P is $\frac{1}{2}\lambda$.

29 A beam of red light of wavelength 720 nm is incident normally on a diffraction grating and produces a diffraction pattern on a screen placed parallel to the grating.

The beam of red light is replaced with a beam of electromagnetic radiation of wavelength *X*, which is incident normally on the same diffraction grating.

The third-order maximum for the electromagnetic radiation of wavelength *X* is at the same position on the screen as the second-order maximum for the red light.

What is wavelength *X*?

A 480 nm **B** 540 nm **C** 960 nm **D** 1100 nm

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30 The current I in a conductor is given by the equation shown.

$$I = Anvq$$

What does the letter n represent in this equation?

A charge carried per charge carrier

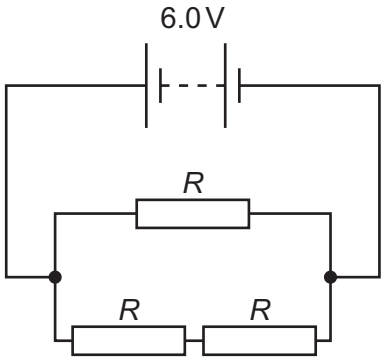
B number of charge carriers per unit area

C number of charge carriers per unit volume

D total mass of charge carriers per unit volume

31 In the circuit shown, the battery has an electromotive force (e.m.f.) of 6.0 V and negligible internal resistance.

The three resistors each have resistance R.

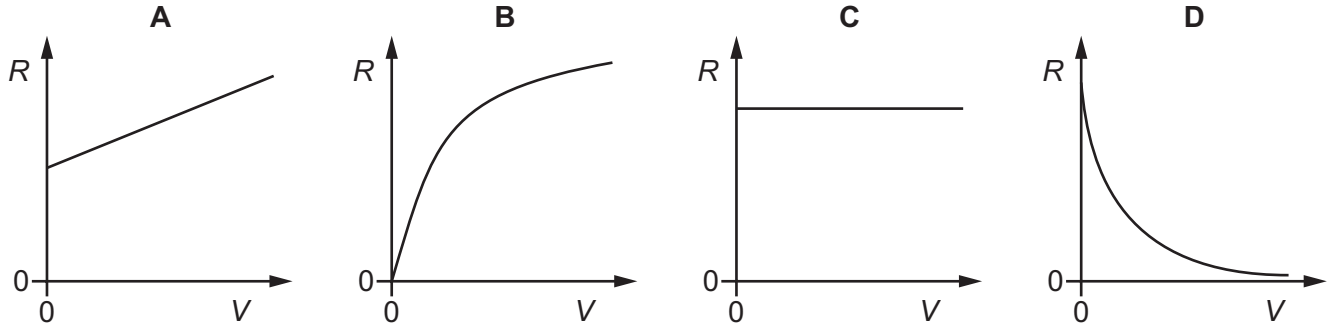


The total power dissipated in the resistor network is 24 W.

What is the value of R?

A 0.50 Ω **B** 1.0 Ω **C** 1.5 Ω **D** 2.3 Ω

32 Which graph could show how the resistance R of a filament lamp varies with the applied potential difference (p.d.) V, as V is increased to the normal operating p.d.?



33 A piece of conducting putty is in the shape of a cylinder of length 60 mm and diameter 20 mm. The resistance between the ends of the cylinder is 20 Ω.

What is the resistivity of the putty?

A 0.033 Ω m **B** 0.10 Ω m **C** 0.42 Ω m **D** 5.2 Ω m

34 Which statement about the electromotive force (e.m.f.) of a cell is **always** correct?

A The e.m.f. is the energy converted from electrical to other forms in the cell.

B The e.m.f. is the energy provided by the cell per unit charge passing through it.

C The e.m.f. is the potential difference across the internal resistance of the cell.

D The e.m.f. is the potential difference across the terminals of the cell.

35 Kirchhoff's first and second laws are a consequence of the conservation of which quantities?

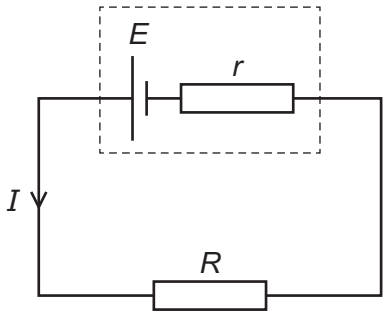
A charge and energy

B charge and resistance

C mass and energy

D mass and resistance

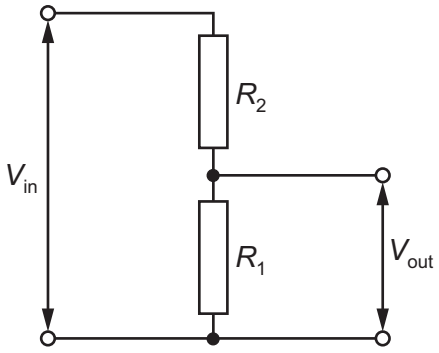
36 A circuit contains a cell of electromotive force E and internal resistance r connected to a resistor of resistance R. The current in the circuit is I.



Which equation is correct?

A $E - Ir = IR$ **B** $E = Ir - IR$ **C** $E + Ir = IR$ **D** $E = IR$

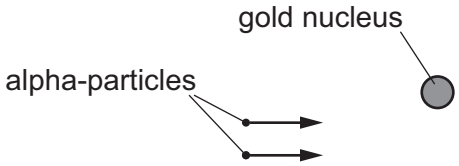
37 A potential divider consists of two resistors of resistances R_1 and R_2 connected in series across a source of potential difference (p.d.) V_{in}. The p.d. across R_1 is V_{out}.



Which changes to R_1 and R_2 will increase the value of V_{out}?

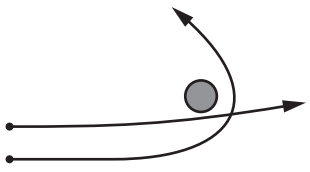
	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

38 Two alpha-particles with the same kinetic energy are moving towards, and are then deflected by, a gold nucleus.

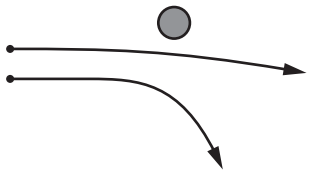


Which diagram could show the paths of the two alpha-particles?

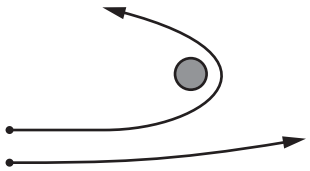
A



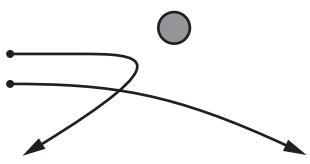
B



C



D



39 Which nuclide is formed when $^{10}_{6}C$ undergoes β^+ decay?

A $^{11}_{6}C$ **B** $^{9}_{6}C$ **C** $^{10}_{5}B$ **D** $^{10}_{7}N$

40 A particular hadron is composed of three quarks and has zero charge.

What is a possible quark composition of the hadron?

A down, down, strange

B up, down, strange

C up, up, down

D up, up, strange

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PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2023**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	C	**1**
2	A	**1**
3	D	**1**
4	C	**1**
5	B	**1**
6	A	**1**
7	D	**1**
8	B	**1**
9	B	**1**
10	B	**1**
11	C	**1**
12	A	**1**
13	D	**1**
14	D	**1**
15	D	**1**
16	A	**1**
17	A	**1**
18	C	**1**
19	D	**1**
20	B	**1**
21	B	**1**
22	C	**1**
23	C	**1**
24	D	**1**
25	A	**1**
26	B	**1**
27	C	**1**
28	D	**1**

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Question	Answer	Marks
29	A	**1**
30	C	**1**
31	D	**1**
32	A	**1**
33	B	**1**
34	B	**1**
35	A	**1**
36	A	**1**
37	B	**1**
38	D	**1**
39	C	**1**
40	B	**1**

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November 2023 · Paper 12

9702_w23_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2023**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 A student estimates the maximum speed of some different moving objects.

Which maximum speed is **not** a reasonable estimate?

A container ship: $10\,\text{m}\,\text{s}^{-1}$

B Olympic sprinter: $0.1\,\text{km}\,\text{s}^{-1}$

C racing car: $9000\,\text{cm}\,\text{s}^{-1}$

D snail: $0.01\,\text{km}\,\text{h}^{-1}$

2 Which quantity is an SI base quantity?

A force

B newton

C second

D time

3 A student takes measurements to determine the constant acceleration of a model car moving from rest in a straight line. The measured values with their absolute uncertainties are shown.

quantity	measured value	uncertainty
displacement	16.5 m	± 0.1 m
time	15.0 s	± 1.0 s

The student uses the equation $s = \frac{1}{2}at^2$ to calculate the acceleration of the car.

What is the acceleration and its absolute uncertainty?

A $(0.11 \pm 0.01)\,\text{m}\,\text{s}^{-2}$

B $(0.11 \pm 0.02)\,\text{m}\,\text{s}^{-2}$

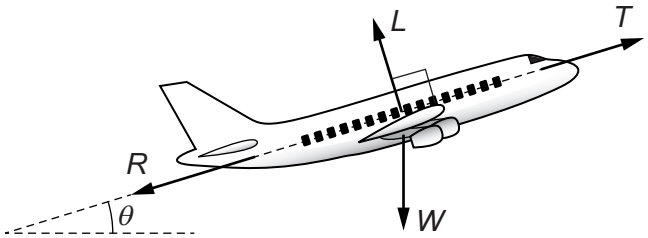
C $(0.15 \pm 0.01)\,\text{m}\,\text{s}^{-2}$

D $(0.15 \pm 0.02)\,\text{m}\,\text{s}^{-2}$

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4 An aeroplane is moving at a constant speed in a straight line at an angle θ to the horizontal.

Four forces act on the aeroplane: thrust force T, weight W, lift force L and resistive force R.



Which two equations must be correct?

A $L = W\cos\theta$ and $T = R + W\sin\theta$

B $L = W\sin\theta$ and $T = R + W\cos\theta$

C $L = W\cos\theta$ and $T = R - W\sin\theta$

D $L = W\sin\theta$ and $T = R - W\cos\theta$

5 What is the definition of acceleration?

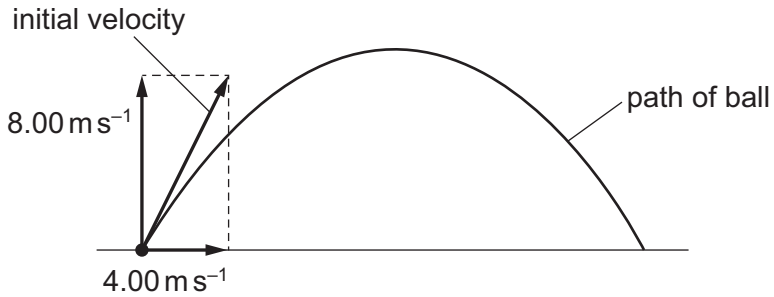
A the rate of change of displacement

B the rate of change of kinetic energy

C the rate of change of momentum

D the rate of change of velocity

6 An astronaut on the Moon, where there is no air resistance, throws a ball. The ball's initial velocity has a vertical component of $8.00\,m\,s^{-1}$ and a horizontal component of $4.00\,m\,s^{-1}$, as shown.



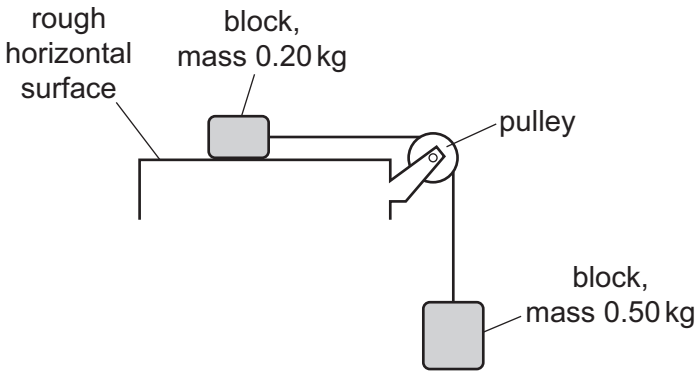
The acceleration of free fall on the Moon is $1.62\,m\,s^{-2}$.

What is the speed of the ball 9.00 s after being thrown?

A $6.58\,m\,s^{-1}$ **B** $7.70\,m\,s^{-1}$ **C** $10.6\,m\,s^{-1}$ **D** $14.6\,m\,s^{-1}$

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7 Two blocks, of mass 0.20 kg and 0.50 kg, are connected by a light inextensible string that passes over a frictionless pulley.



The blocks are initially held stationary. The block of mass 0.20 kg rests on a rough horizontal surface.

The block of mass 0.50 kg is suspended in air. Air resistance is negligible.

When the blocks are released, they have an acceleration of magnitude 2.0 m s^{-2}.

What is the magnitude of the frictional force between the block of mass 0.20 kg and the rough surface?

A 3.5 N **B** 3.9 N **C** 4.5 N **D** 6.3 N

8 A resultant force causes an object to accelerate.

What is equal to the resultant force?

A the acceleration of the object per unit mass

B the change in kinetic energy of the object per unit time

C the change in momentum of the object per unit time

D the change in velocity of the object per unit time

9 An object falls from a stationary helicopter and reaches terminal velocity.

What happens to the acceleration of the object between leaving the helicopter and reaching terminal velocity?

A It decreases to 9.81 m s^{-2}.

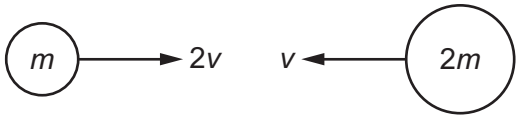
B It decreases to zero.

C It increases to 9.81 m s^{-2}.

D It remains constant at 9.81 m s^{-2}.

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10 Two balls, of masses m and $2m$, travelling in a vacuum with initial velocities $2v$ and v respectively, collide with each other head-on, as shown.

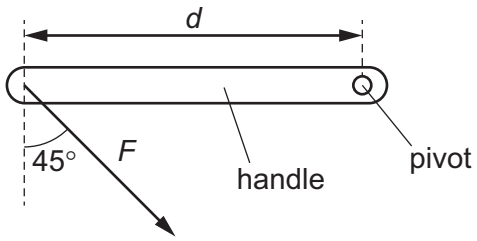


After the collision, the ball of mass m rebounds to the left with velocity v.

What is the loss of kinetic energy in the collision?

A $\frac{3}{4}mv^2$ **B** $\frac{3}{2}mv^2$ **C** $\frac{9}{4}mv^2$ **D** $\frac{9}{2}mv^2$

11 A force F is applied at an angle of 45° to a door handle at a distance d from the pivot of the handle, as shown.

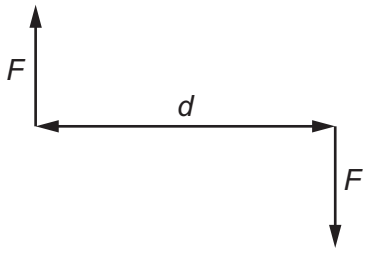


What is the moment of the force about the pivot?

A $\frac{Fd}{\sqrt{2}}$ **B** Fd **C** $Fd\sqrt{2}$ **D** $2Fd$

12 A couple consists of two forces, each of magnitude F, that act in opposite directions in the same plane.

The perpendicular distance between the two forces is d.

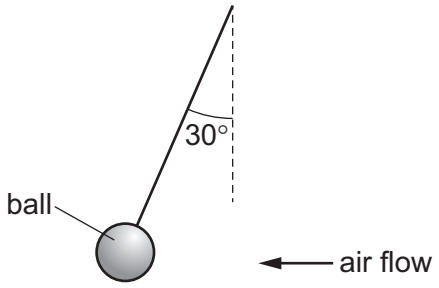


What is the torque of the couple?

A $\frac{Fd}{2}$ **B** $\frac{F}{d}$ **C** Fd **D** $2Fd$

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13 The diagram shows an experiment to determine the force exerted on a ball by a horizontal air flow.



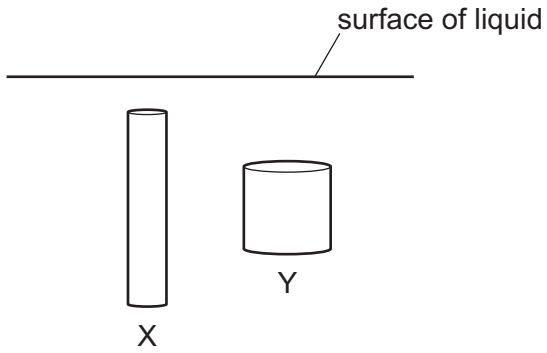
The ball is suspended by a light string and weighs 0.15 N.

The deflection of the string from vertical is 30°. The ball is in equilibrium.

What is the force on the ball from the air flow?

A 0.075 N **B** 0.087 N **C** 0.26 N **D** 0.30 N

14 Two solid cylindrical objects X and Y are held fully submerged in a liquid, as shown.



The objects have the same volume. The density of the material of Y is twice the density of the material of X. Both objects are stationary.

Which statement is correct?

A The force due to the liquid acting on the top surface of X is greater than that acting on the top surface of Y.

B The pressure difference due to the liquid between the top and bottom surfaces of X is the same as that for Y.

C The upthrust acting on X is the same as the upthrust acting on Y.

D The weight of X is the same as the weight of Y.

15 An electric car travels at a constant speed of 70 km h^{-1} for 80 km on a straight horizontal road and uses energy E from its battery.

The total resistive force acting on the car is proportional to (speed)^2. Assume that the electric motor is 100% efficient.

How much energy is used from the battery when the car travels at a constant speed of 60 km h^{-1} for 80 km on the straight horizontal road?

A 0.73E **B** 0.86E **C** 1.2E **D** 1.4E

16 What is meant by the efficiency of a system?

A the total energy input to the system divided by the useful energy output by the system

B the useful energy output from the system divided by the energy wasted by the system

C the useful energy output from the system divided by the total energy input to the system

D the energy wasted by the system divided by the total energy input to the system

17 When an object of mass m is raised through a vertical height Δh, the gain of its gravitational potential energy is ΔE_P.

ΔE_P and Δh are related by the equation

$$\Delta E_P = mg\Delta h,$$

where g is the acceleration of free fall.

The definition of which physical quantity is needed to derive this equation?

A acceleration

B momentum

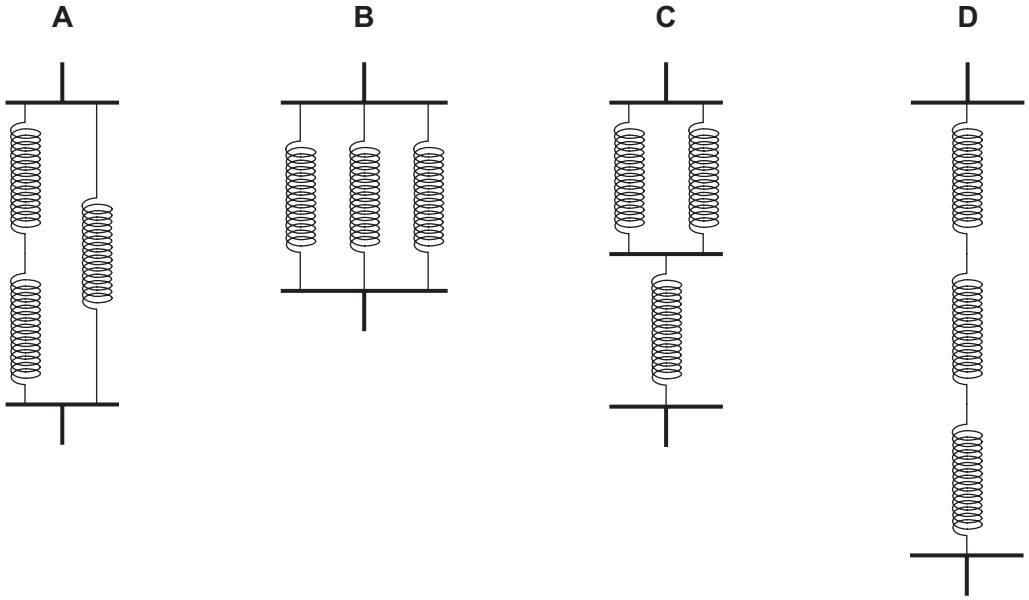
C power

D work done

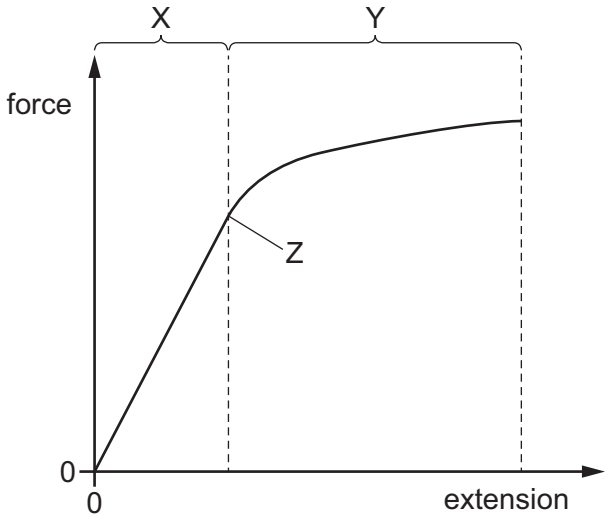
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18 Three identical springs, each with the same spring constant, are connected together in four different arrangements, as shown.

Which arrangement has the largest combined spring constant?



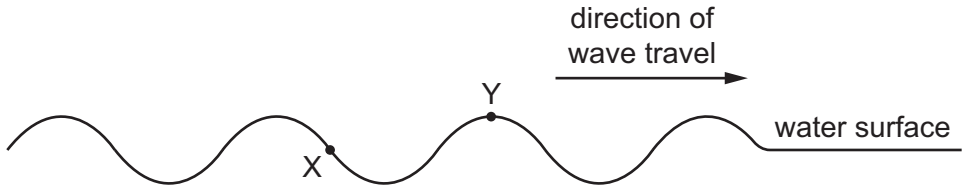
19 The force–extension graph for a wire is shown.



Which row could identify the labels X, Y and Z?

	limit of proportionality	region of elastic deformation	region of plastic deformation
A	X	Y	Z
B	Z	Y	X
C	Y	Z	X
D	Z	X	Y

20 X and Y are two points on the surface of water in a ripple tank. A source of constant frequency generates a wave which travels past X and Y, causing them to oscillate vertically.

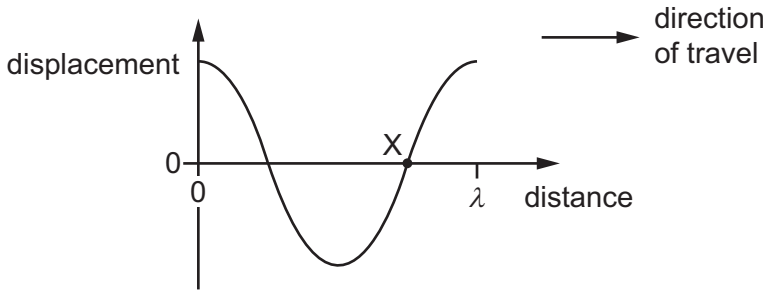


What is the phase difference between X and Y?

A 45° **B** 135° **C** 180° **D** 270°

21 A transverse wave on a rope has wavelength λ and period T.

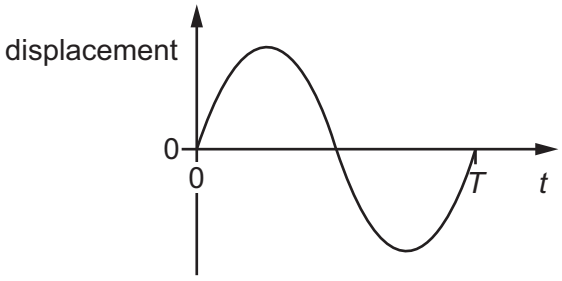
The graph shows the variation of the displacement of the particles of the rope with distance in the direction of travel of the wave at time $t = 0$.



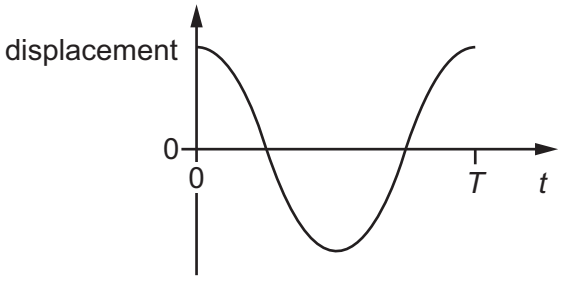
A particle X is labelled.

Which graph shows the variation of the displacement of particle X with time t?

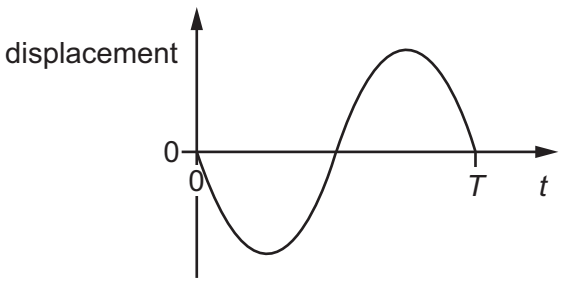
A



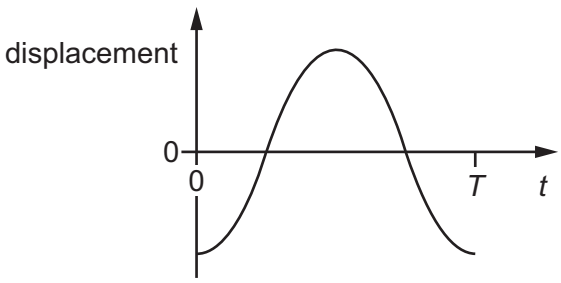
B



C



D



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22 A source of sound waves is moving at a constant speed directly towards a stationary observer.

The sound waves have a speed of $340\,m\,s^{-1}$ and a frequency of 480 Hz. The observer hears sound waves of frequency 650 Hz.

What is the speed of the source?

A $89\,m\,s^{-1}$ **B** $120\,m\,s^{-1}$ **C** $250\,m\,s^{-1}$ **D** $340\,m\,s^{-1}$

23 A student is investigating two electromagnetic waves, X and Y, in a vacuum.

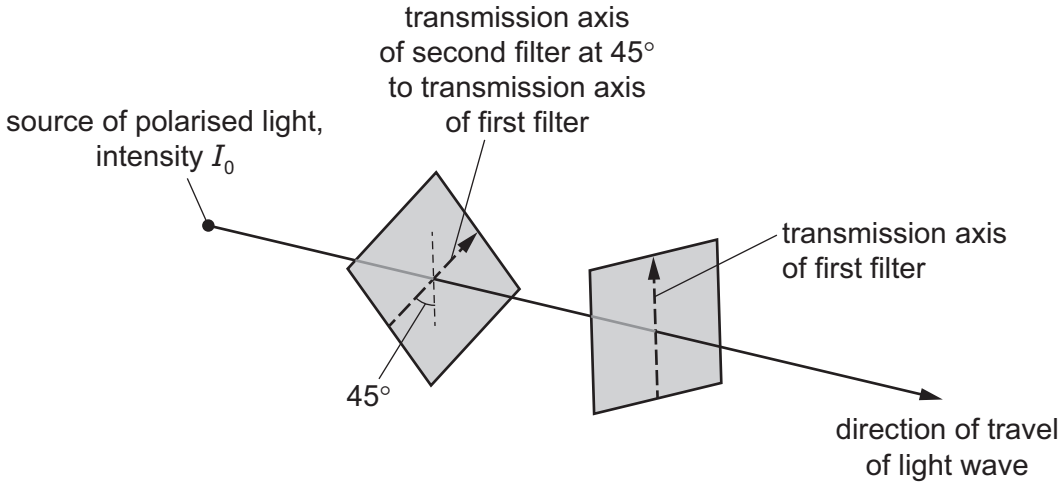
Wave X has a wavelength of 5.2×10^{-7} m. Wave Y has a frequency of 9.4 GHz.

Which principal regions of the electromagnetic spectrum contain waves X and Y?

	X	Y
A	radio wave	ultraviolet
B	ultraviolet	visible
C	visible	microwave
D	microwave	radio wave

24 A plane polarised light wave of intensity I_0 is incident normally on a polarising filter. The initial intensity of the transmitted wave is 0.

A second polarising filter is then inserted between the source and the first filter. Its transmission axis is at 45° to the transmission axis of the first filter, as shown.



What is the intensity of the transmitted wave from the filter combination?

A 0 **B** $\frac{I_0}{8}$ **C** $\frac{I_0}{4}$ **D** $\frac{I_0}{2}$

25 What can explain how stationary waves are formed from progressive waves?

A diffraction

B polarisation

C superposition

D the Doppler effect

26 A pipe has a length of 2.0 m. It is open at one end and closed at the other end.

A stationary sound wave is set up within the pipe. There are four nodes (N) and four antinodes (A) within the length of the pipe.



What is the wavelength of the sound wave?

A 0.57 m **B** 1.1 m **C** 1.3 m **D** 1.6 m

27 A teacher is explaining diffraction to a group of students.

Which piece of apparatus is most appropriate for the teacher to use to demonstrate diffraction?

A a long spring

B a ripple tank

C a rope

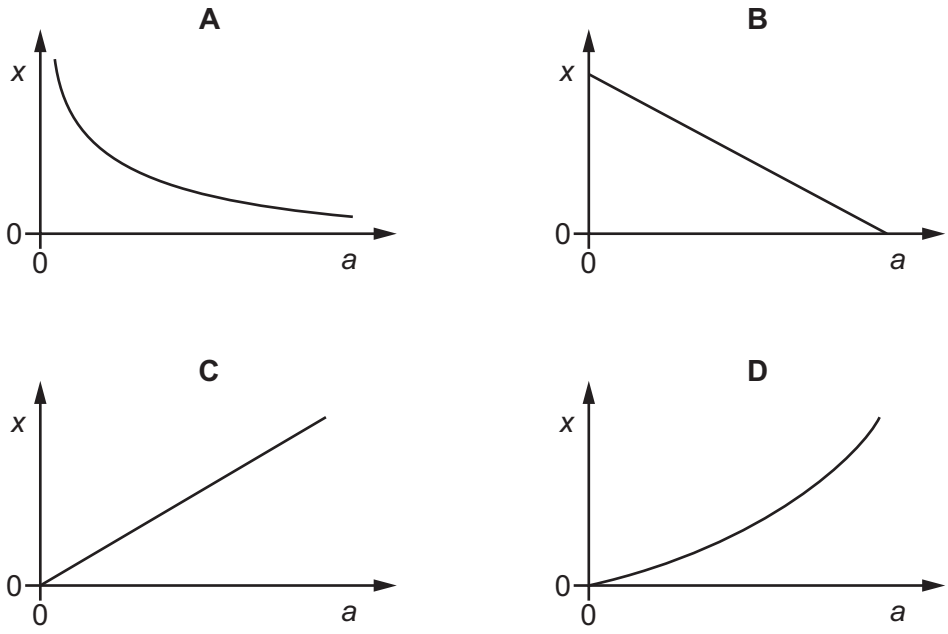
D a stretched string

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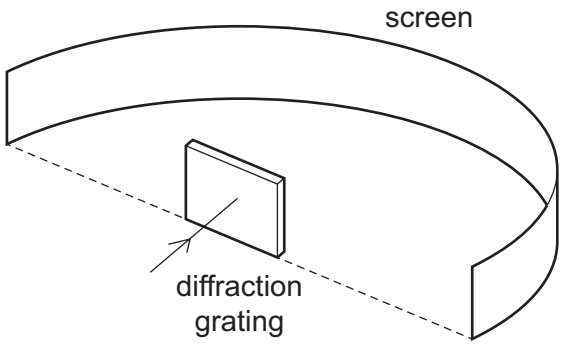
28 Coherent light of constant wavelength is incident normally on a double slit. Interference fringes are formed on a screen that is a fixed distance from the double slit. The screen is parallel to the double slit.

The separation of the slits is varied.

Which graph best shows the variation with slit separation *a* of the spacing *x* of the interference fringes?



29 Light of wavelength 690 nm passes through a diffraction grating with 300 lines per mm, producing a series of bright spots (maxima) on a screen.



What is the total number of bright spots that are produced?

A 4 **B** 5 **C** 8 **D** 9

30 A fine mist of oil droplets is sprayed into air. As the oil droplets leave the nozzle of the spraying device they can become electrically charged.

What is **not** a possible value for the charge on an oil droplet?

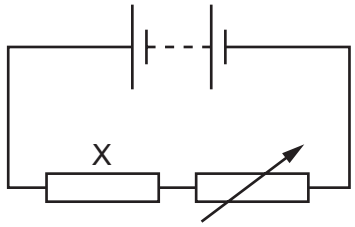
A zero

B 1.0×10^{-19} C

C 4.8×10^{-19} C

D 8.0×10^{-19} C

31 In the circuit shown, a fixed resistor X is connected in series with a battery and a variable resistor.



The power dissipated in resistor X is 7.2 W when a current of 3.0 A passes through it.

The variable resistor is adjusted so that the power dissipated in X increases by 50%.

What is the new current in the circuit?

A 2.4 A **B** 3.7 A **C** 4.5 A **D** 14 A

32 The potential difference across a metal wire is kept constant. The length l and the diameter d of the wire are both varied. The type of metal is kept the same.

How is the current in the wire related to l and d?

A It is directly proportional to l and inversely proportional to d.

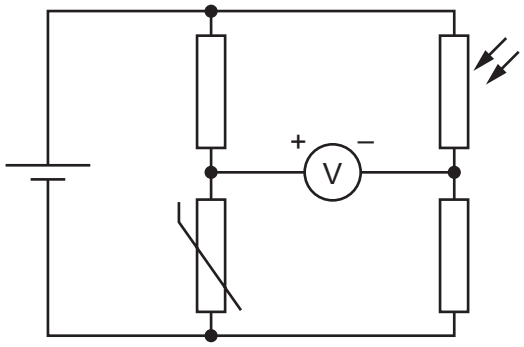
B It is directly proportional to l and inversely proportional to d^2.

C It is inversely proportional to l and directly proportional to d.

D It is inversely proportional to l and directly proportional to d^2.

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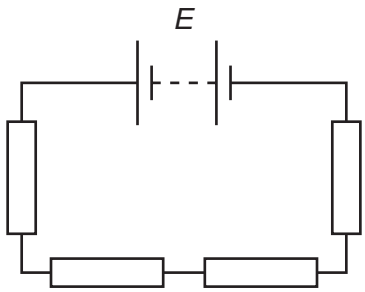
33 A student sets up a circuit. The circuit diagram shows how the positive and negative terminals of a voltmeter are connected to the circuit. The voltmeter has an initial reading that is positive.



Which changes, if any, in temperature and light intensity would cause the voltmeter reading to **decrease**?

	temperature	light intensity
A	increase	increase
B	no change	decrease
C	decrease	no change
D	decrease	decrease

34 Some resistors and a battery of electromotive force (e.m.f.) *E* and negligible internal resistance are connected in series, as shown.



Which statement is correct?

A The e.m.f. across each resistor equals the potential difference across the battery.

B The potential difference across each resistor equals the e.m.f. *E* of the battery.

C The sum of the e.m.f.s across the resistors equals the potential difference across the battery.

D The sum of the potential differences across the resistors equals the e.m.f. *E* of the battery.

35 Kirchhoff's first law is a consequence of the conservation of which physical quantity?

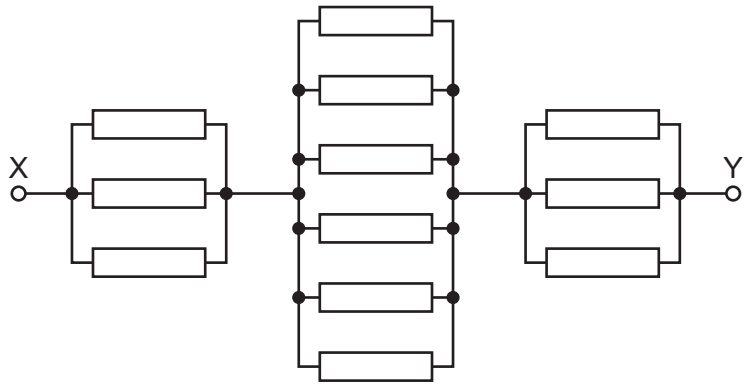
A charge

B energy

C linear momentum

D potential difference

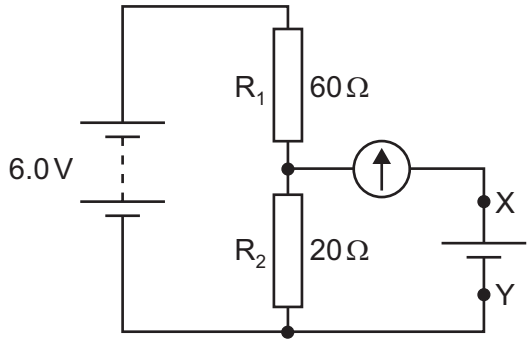
36 The diagram shows a network of resistors. Each resistor has a resistance of 6.0 Ω.



What is the total resistance of the network between points X and Y?

A 3.0 Ω **B** 5.0 Ω **C** 7.2 Ω **D** 18 Ω

37 In the circuit shown, a battery of negligible internal resistance is connected in series with a pair of fixed resistors R_1 and R_2.



The circuit is to be used to test whether the electromotive force (e.m.f.) of a particular cell is 1.5 V. The cell is connected between terminals X and Y in parallel with R_2 and in series with a galvanometer.

Which statement about the test is correct?

A Any non-zero reading on the galvanometer means the cell has an e.m.f. of 1.5 V.

B The battery does not need to have an e.m.f. of 6.0 V.

C The cell may be connected either way round between X and Y.

D The galvanometer does not need a scale calibrated in amperes.

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38 Two nuclides are different isotopes of the same element.

Which statement about the nuclides is correct?

A Neutral atoms of the nuclides have the same number of electrons.

B Nuclei of the nuclides have different numbers of protons.

C Nuclei of the nuclides have the same number of nucleons.

D Nuclei of the nuclides have the same number of neutrons.

39 The charge-to-mass ratio r of a particle is given by the equation shown.

$$r = \frac{\text{charge on particle}}{\text{mass of particle}}$$

The value of r is determined for an α-particle, a β^+ particle and a proton p.

Which list shows the particles in order of increasing magnitude of r from left to right?

A $\alpha \rightarrow \beta^+ \rightarrow \text{p}$

B $\alpha \rightarrow \text{p} \rightarrow \beta^+$

C $\text{p} \rightarrow \alpha \rightarrow \beta^+$

D $\text{p} \rightarrow \beta^+ \rightarrow \alpha$

40 Which combination of up (u) and down (d) quarks forms a neutron?

A u u u **B** u u d **C** u d d **D** d d d

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Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2023**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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Question	Answer	Marks
1	B	**1**
2	D	**1**
3	D	**1**
4	A	**1**
5	D	**1**
6	B	**1**
7	A	**1**
8	C	**1**
9	B	**1**
10	C	**1**
11	A	**1**
12	C	**1**
13	B	**1**
14	C	**1**
15	A	**1**
16	C	**1**
17	D	**1**
18	B	**1**
19	D	**1**
20	D	**1**
21	C	**1**
22	A	**1**
23	C	**1**
24	C	**1**
25	C	**1**
26	B	**1**
27	B	**1**
28	A	**1**

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Question	Answer	Marks
29	D	**1**
30	B	**1**
31	B	**1**
32	D	**1**
33	A	**1**
34	D	**1**
35	A	**1**
36	B	**1**
37	D	**1**
38	A	**1**
39	B	**1**
40	C	**1**

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November 2023 · Paper 13

9702_w23_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **October/November 2023**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

IB23 11_9702_13/RP
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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What is the best estimate of the wavelength of green light?

A 260 nm **B** 540 nm **C** 780 nm **D** 920 nm

2 In an electric circuit, an ammeter reads 2 μA.

In a second circuit, the ammeter reads 1 mA.

How many times larger is the current in the second circuit compared with the current in the first circuit?

A 500 **B** 5000 **C** 500 000 **D** 5 000 000

3 A set of repeated measurements is made of a fixed quantity. An average of these measurements is calculated.

What is the effect of averaging on the random error and the systematic error in the measurements?

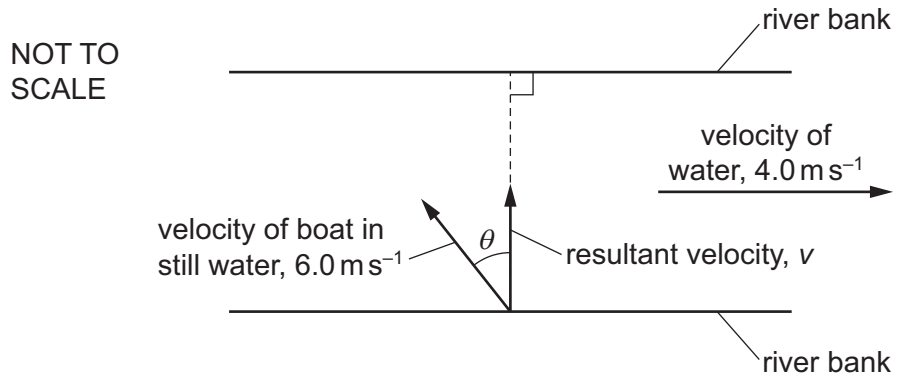
A Random error and systematic error are both reduced.

B Random error and systematic error are both unaffected.

C Random error is reduced but systematic error is unaffected.

D Random error is unaffected but systematic error is reduced.

4 A boat is crossing a river in which the water is moving at a speed of $4.0\,m\,s^{-1}$ from left to right.



In still water, the speed of the boat is $6.0\,m\,s^{-1}$. The boat is directed at an angle θ to a line perpendicular to the river banks. The resultant velocity v of the boat is in a direction perpendicular to the river banks.

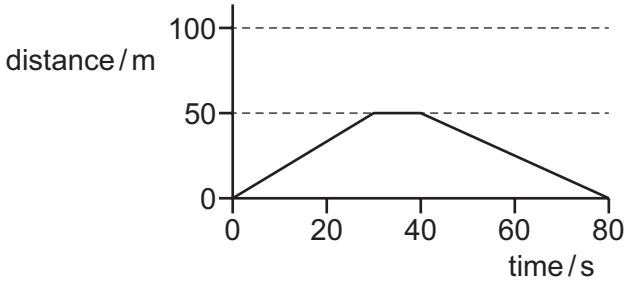
What are the values of θ and v?

	θ / °	v / m s^{-1}
A	42	4.5
B	42	7.2
C	48	4.5
D	48	7.2

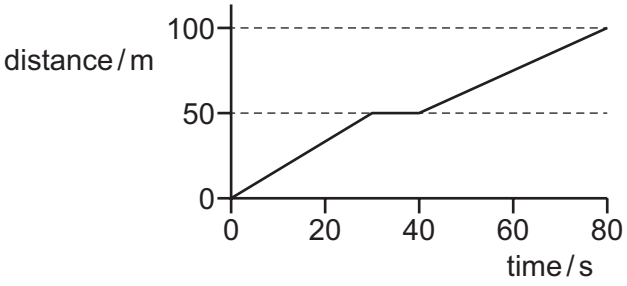
5 A student walks at a constant speed for a distance of 50 m in a time of 40 s. The student rests for a time of 10 s and then walks back to the starting point at a constant speed in a time of 30 s.

What is the distance–time graph for the motion of the student?

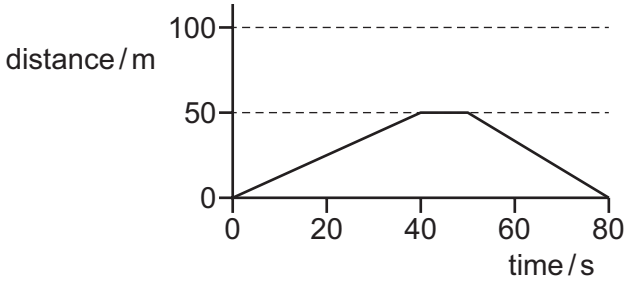
A



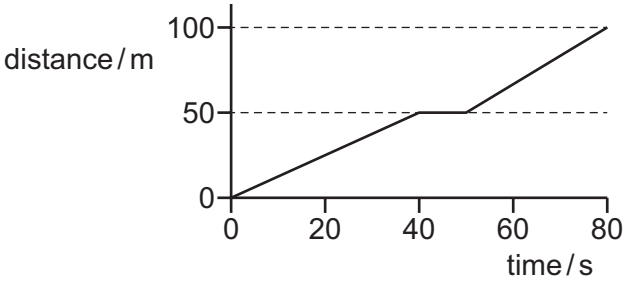
B



C



D



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6 The time taken for an object to fall from rest through a certain distance on Mars is T_M. The time taken for the same object to fall from rest through the same distance on Earth is T_E. The acceleration of free fall on Mars is $3.71\,m\,s^{-2}$.

Assume that air resistance is negligible on both Earth and Mars.

What is the ratio $\frac{T_M}{T_E}$?

A 0.378 **B** 0.615 **C** 1.63 **D** 2.64

7 Which statement about mass is correct?

A Mass has a magnitude and a direction.

B Mass resists changes in motion.

C The greater the mass of an object, the greater its acceleration when falling in a vacuum.

D The mass of an object depends on its location.

8 A snooker ball has a mass of 200 g. It hits the cushion of a snooker table and rebounds along its original path.

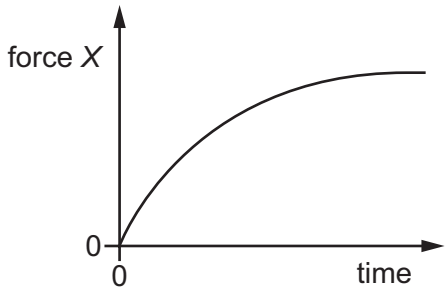
The ball arrives at the cushion with a speed of $14.0\,m\,s^{-1}$ and then leaves it with a speed of $7.0\,m\,s^{-1}$. The ball and the cushion are in contact for a time of 0.60 s.

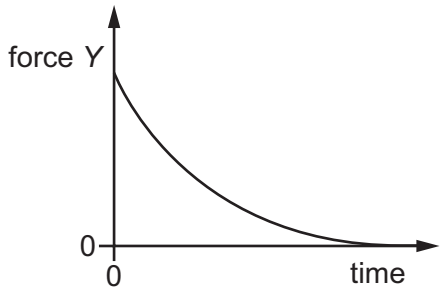
What is the average force exerted on the ball by the cushion?

A 1.4 N **B** 2.3 N **C** 4.2 N **D** 7.0 N

9 A ball falls from rest through air and eventually reaches a constant velocity.

For this fall, forces *X* and *Y* vary with time as shown.



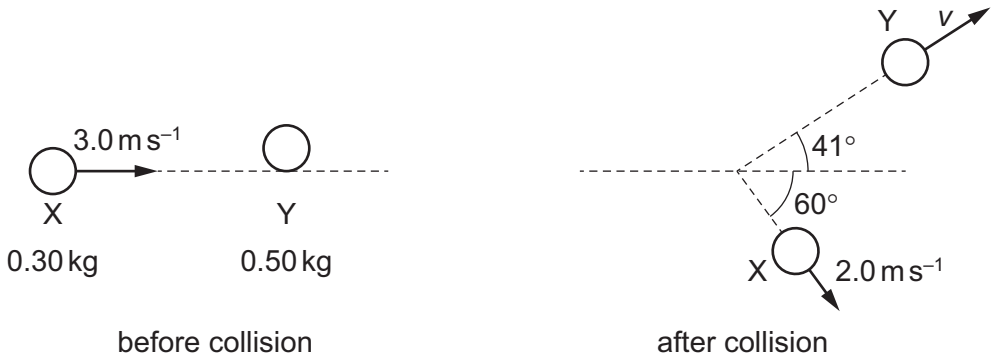


What could be forces *X* and *Y*?

	force *X*	force *Y*
A	air resistance	resultant force
B	air resistance	weight
C	upthrust	resultant force
D	upthrust	weight

10 An object X of mass 0.30 kg is travelling in a straight line at a constant velocity of $3.0\,m\,s^{-1}$ on a horizontal frictionless surface. Object X collides with a stationary object Y of mass 0.50 kg.

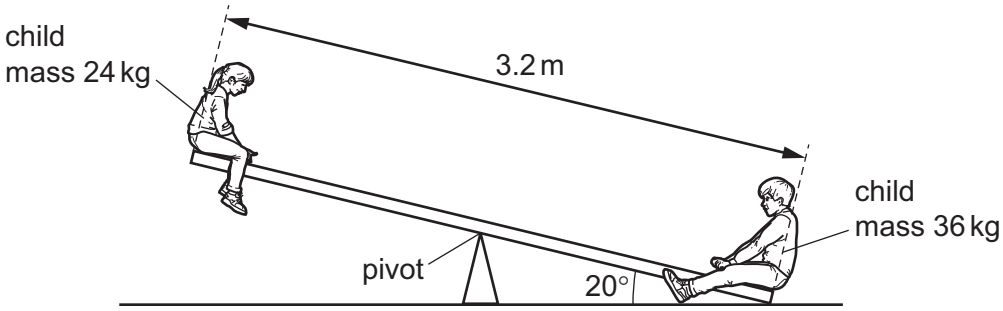
After the collision, X moves with a velocity of $2.0\,m\,s^{-1}$ at an angle of 60° to its direction before the collision. Object Y moves with a velocity v at an angle of 41° to the direction of X before the collision, as shown.



What is the value of v?

A $0.80\,m\,s^{-1}$ **B** $1.2\,m\,s^{-1}$ **C** $1.6\,m\,s^{-1}$ **D** $1.8\,m\,s^{-1}$

11 A uniform rigid beam of length 3.2 m is pivoted at its centre. Two children sit at the opposite ends of the beam, as shown.



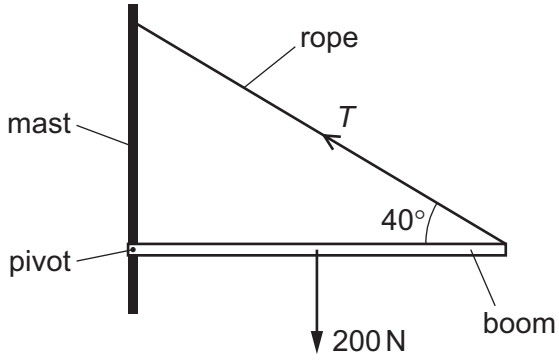
One child has a mass of 24 kg. The other child has a mass of 36 kg. The heavier child causes one end of the beam to permanently rest on the ground, so that the beam makes an angle of 20° to the horizontal ground.

What is the moment of the weight of the 24 kg child about the pivot?

A 72 N m **B** 130 N m **C** 350 N m **D** 380 N m

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12 Two parts of a sailing boat are the mast and the boom. The mast is a vertical rigid beam and the boom is a horizontal rigid beam. One end of the boom is attached to the mast by a pivot. The other end of the boom is connected to the mast by a rope, as shown.



The rope is at an angle of 40° to the horizontal and exerts a tension force T on the boom. The weight of the boom is 200 N. The mass of the boom is uniformly distributed along its length. The boom is in equilibrium.

What is the magnitude of T?

A 130 N **B** 160 N **C** 260 N **D** 310 N

13 Full-fat milk is made up of fat-free milk mixed with fat.

A volume of $1.000 \times 10^{-3}\,m^3$ of full-fat milk has a mass of 1.035 kg. It contains 4.00% fat by volume.

The density of fat-free milk is $1.040 \times 10^{3}\,kg\,m^{-3}$.

What is the density of fat?

A $1.25 \times 10^{2}\,kg\,m^{-3}$

B $9.15 \times 10^{2}\,kg\,m^{-3}$

C $9.28 \times 10^{2}\,kg\,m^{-3}$

D $1.16 \times 10^{3}\,kg\,m^{-3}$

14 Which expression for pressure is correct?

A force per unit area

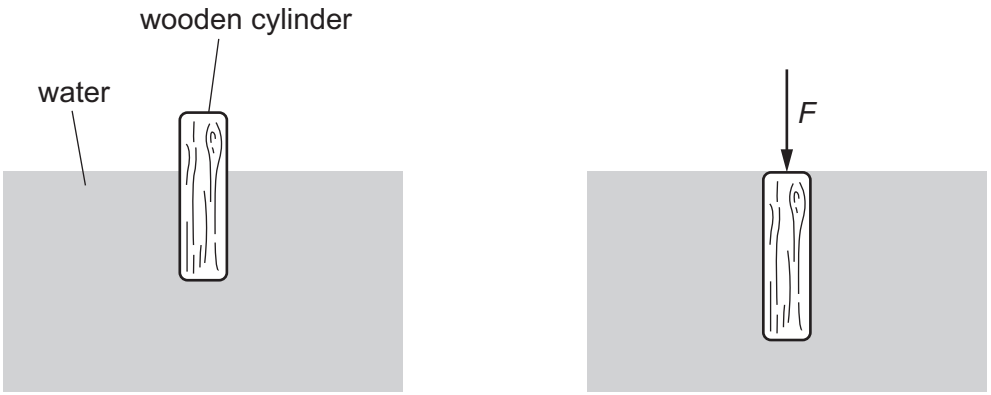
B mass per unit area

C mass per unit volume

D weight per unit volume

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15 A wooden cylinder floats partially submerged in a bath of water. A force F is applied to the cylinder until it is just fully submerged.



Which statement is **not** correct?

A Some of the water gains gravitational potential energy.

B The cylinder loses gravitational potential energy.

C Work is done by force F on the cylinder.

D Work is done by the upthrust on the cylinder.

16 A system has a useful power output of 4.0 W and a wasted power of 16 W.

What is the efficiency of the system?

A 5.0% **B** 20% **C** 25% **D** 80%

17 A parachutist is falling towards the ground at a constant speed v. The rate at which she is losing gravitational potential energy is R.

The acceleration of free fall is g.

What is the mass of the parachutist?

A $\frac{gv}{R}$ **B** $\frac{R}{gv}$ **C** $\frac{2R}{v^2}$ **D** $\frac{v^2}{2R}$

18 A copper wire of diameter 1.6 mm is stretched within its limit of proportionality by a tensile force of 430 N.

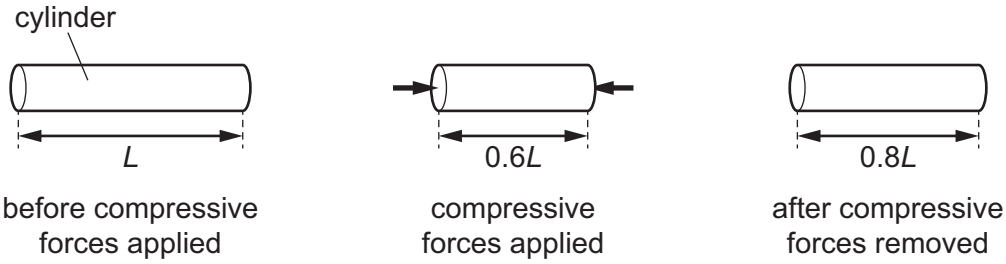
The Young modulus of copper is 130 GPa.

What is the strain in the wire?

A 4.1×10^{-4} **B** 1.3×10^{-3} **C** 1.6×10^{-3} **D** 5.2×10^{-3}

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19 Compressive forces are applied normally to the end faces of a cylinder of initial length L. The cylinder is compressed by the forces so that its length decreases to $0.6L$. After the compressive forces are removed, the cylinder's length increases to $0.8L$.



What describes the deformation of the cylinder when its length was $0.6L$?

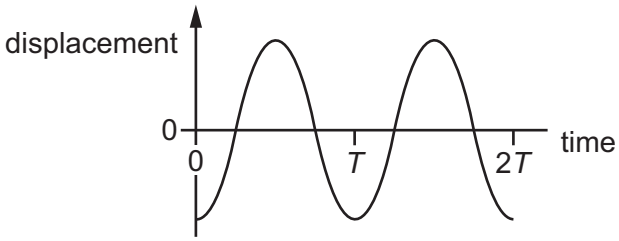
A both elastic and plastic

B elastic only

C plastic only

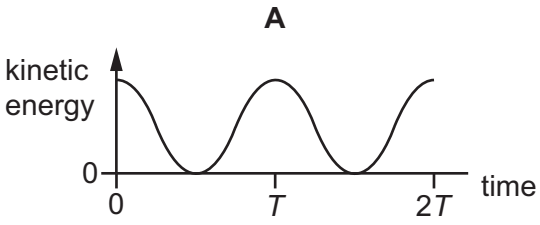
D neither elastic nor plastic

20 When sound travels through air, the air particles vibrate. A graph of displacement against time for a single air particle is shown.

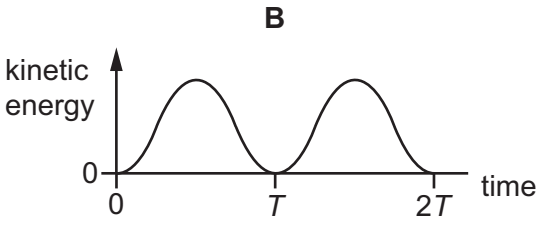


Which graph shows how the kinetic energy of the air particle varies with time?

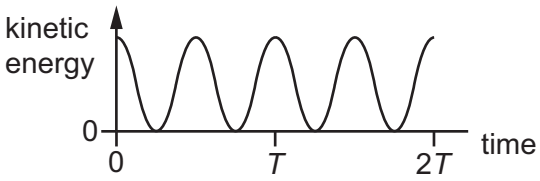
A



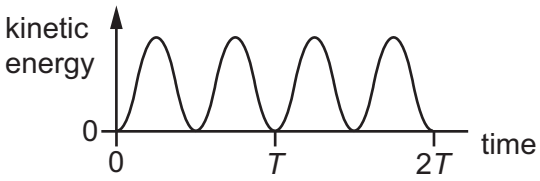
B



C

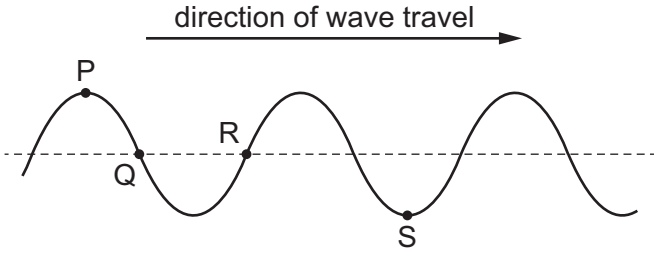


D



21 A wave travels on the surface of water. P, Q, R and S are four particles of water on the surface.

The diagram shows the positions of the particles at one instant. The direction of travel of the wave is from left to right.



Which two particles are about to move upwards?

A P and R **B** P and S **C** R and S **D** Q and S

22 A loudspeaker emits sound of frequency f_s. The loudspeaker is attached to a car that moves with increasing speed directly towards a stationary observer.

Which statement describes the frequency of the sound heard by the observer?

A a frequency greater than f_s and increasing

B a frequency greater than f_s but decreasing

C a frequency less than f_s and decreasing

D a frequency less than f_s but increasing

23 Which statement about electromagnetic waves in a vacuum is correct?

A Amplitude is inversely proportional to velocity.

B Frequency is inversely proportional to wavelength.

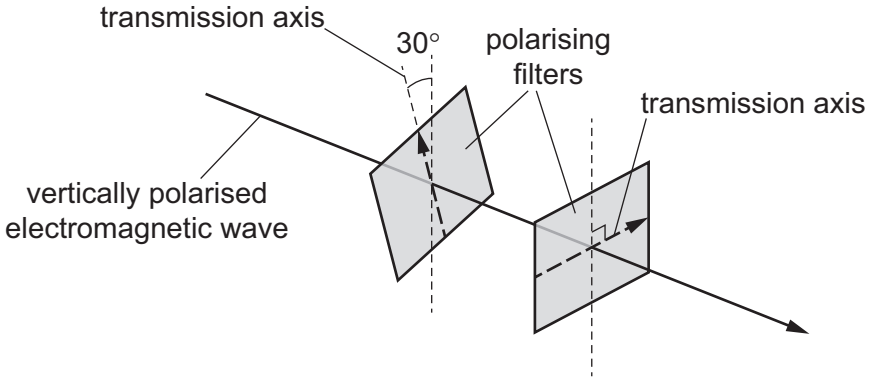
C Intensity is proportional to amplitude.

D Velocity is proportional to wavelength.

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24 A vertically polarised electromagnetic wave of intensity I_0 is incident normally on a polarising filter. The transmission axis of the filter is at an angle of 30° to the vertical.

The transmitted wave from the first filter is then incident normally on a second polarising filter. The transmission axis of this filter is at an angle of 90° to the vertical.



What is the intensity of the wave after passing through the second filter?

A 0 **B** $0.063\,I_0$ **C** $0.19\,I_0$ **D** $0.56\,I_0$

25 Which statement concerning a stationary wave is correct?

A All the particles between two adjacent nodes oscillate in phase.

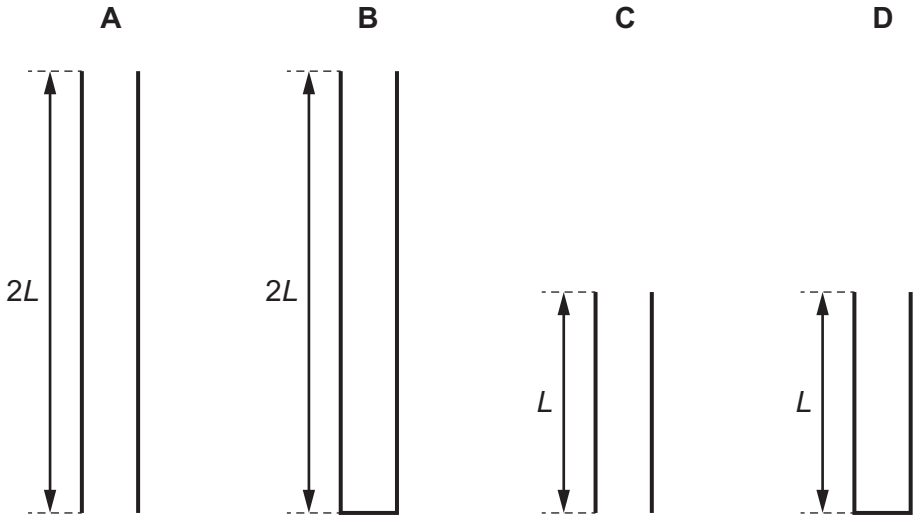
B The amplitude of the stationary wave is equal to the amplitude of one of the waves creating it.

C The wavelength of the stationary wave is equal to the separation of two adjacent nodes.

D There is no displacement of a particle at an antinode at any time.

26 Stationary sound waves can be formed in the air columns of pipes. One type of pipe is closed at one end and open at the other end. Another type of pipe is open at both ends.

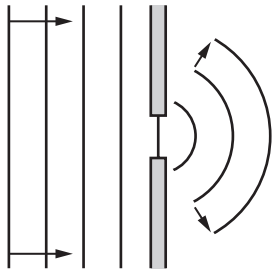
Which pipe can form a stationary sound wave with the lowest frequency?



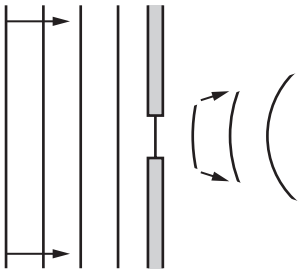
27 Water waves in a ripple tank are made to pass through a small gap, as shown.

Which diagram shows the waves after they have passed through the gap?

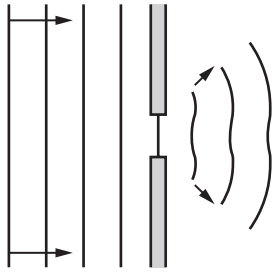
A



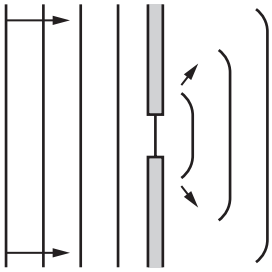
B



C



D



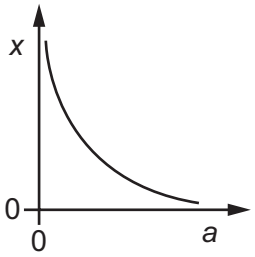
28 Light of a single wavelength is incident normally on a double slit. The slit separation can be varied.

A screen is placed a fixed distance away from the double slit. The screen and double slit are parallel. A pattern of bright interference fringes is observed on a screen.

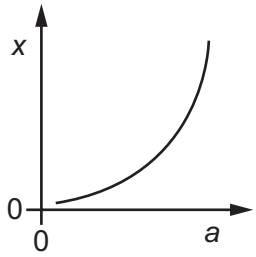


Which graph best shows the variation of the separation x of the bright interference fringes with the slit separation a?

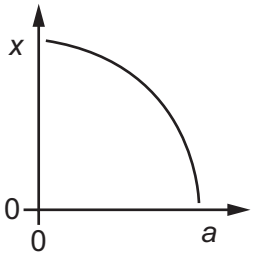
A



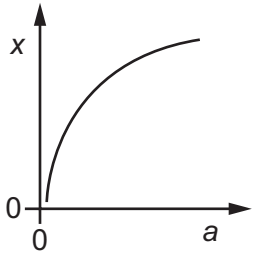
B



C

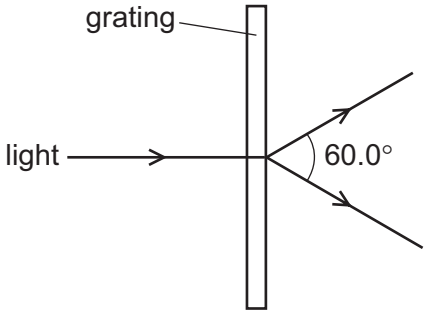


D



29 A diffraction grating is used to measure the wavelength of light.

The spacing of the slits in the grating is 1.15×10^{-6} m. The angle between the first-order diffraction maxima is 60.0°, as shown.



What is the wavelength of the light?

A 288 nm **B** 498 nm **C** 575 nm **D** 996 nm

30 What could **not** be used to create an electric current?

A alpha-particles

B beta-particles

C neutrons

D protons

31 What is the definition of the potential difference (p.d.) across a component?

A the energy transferred per unit charge

B the energy transferred per unit current

C the power transferred per unit charge

D the power transferred per unit current

32 The resistance of a filament lamp increases as the current in it increases.

What is the reason for this?

A The charge of each charge carrier increases.

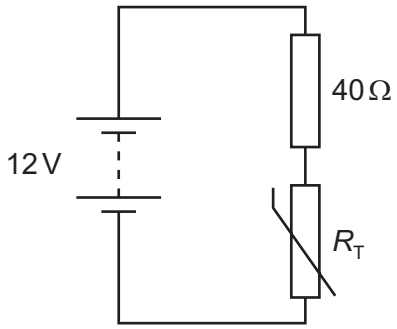
B The potential difference across the filament decreases.

C The power dissipated by the filament decreases.

D The temperature of the filament increases.

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33 A battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected to a fixed resistor of resistance 40 Ω and a thermistor of resistance R_T, as shown.



Initially, the temperature of the thermistor is 15 °C and the current in the circuit is 0.10 A.

The temperature of the thermistor then changes, which causes the current to increase to 0.12 A.

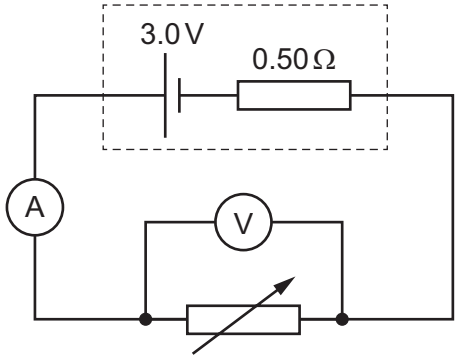
How does the temperature of the thermistor change and what is R_T at the new temperature?

	temperature of thermistor	R_T at new temperature / Ω
A	increases	60
B	decreases	60
C	increases	100
D	decreases	100

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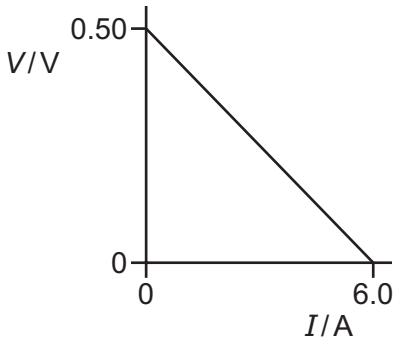
34 A cell of electromotive force (e.m.f.) 3.0 V and internal resistance 0.50 Ω is connected to a variable resistor, a voltmeter and an ammeter, as shown. The resistance of the variable resistor is varied.

The reading on the ammeter I and the reading on the voltmeter V are recorded.

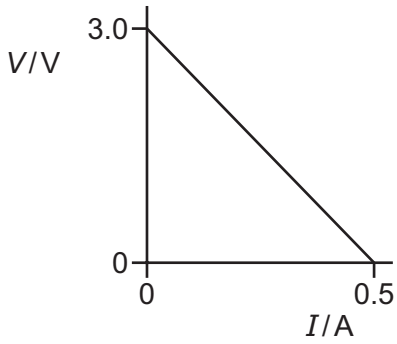


Which graph shows how V varies with I?

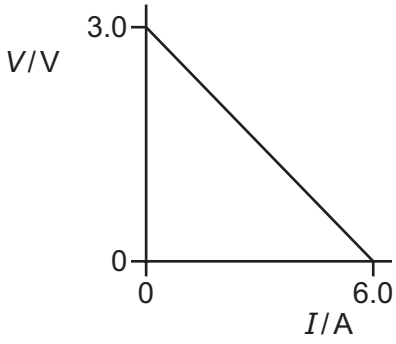
A



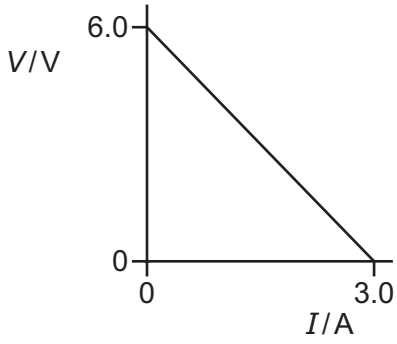
B



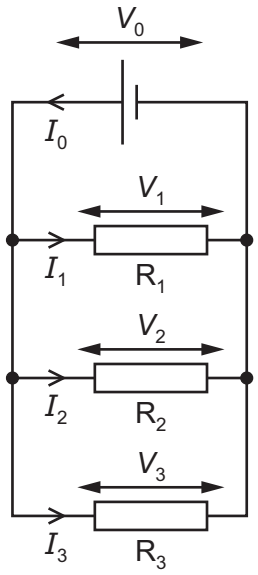
C



D



35 Three resistors, R_1, R_2 and R_3, are connected in parallel to a cell. The currents in the resistors are I_1, I_2 and I_3. The potential differences across the resistors are V_1, V_2 and V_3. The current in the cell is I_0. The potential difference across the cell is V_0, as shown.



Which equation can be obtained by applying Kirchhoff's second law to the circuit?

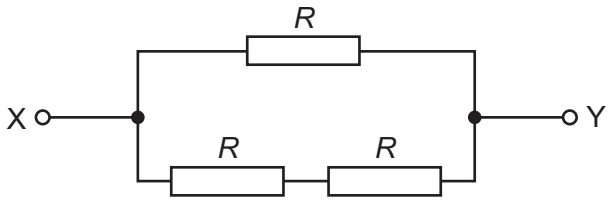
A $I_0 = I_1 = I_2 = I_3$

B $I_0 = I_1 + I_2 + I_3$

C $V_0 = V_1 = V_2 = V_3$

D $V_0 = V_1 + V_2 + V_3$

36 Three resistors, each of resistance R, are connected in a network, as shown.



The total resistance between points X and Y is 8.0 Ω.

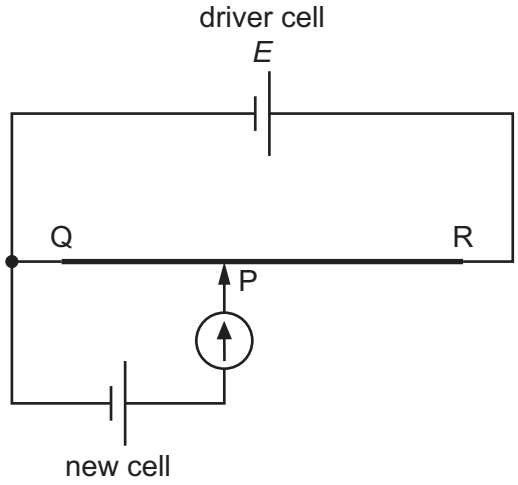
What is the value of R?

A 2.7 Ω **B** 4.0 Ω **C** 5.3 Ω **D** 12 Ω

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37 A potentiometer and a driver cell of electromotive force (e.m.f.) *E* are used to measure the e.m.f. of a new cell.

A sliding contact at P is moved along a resistance wire QR until the reading on the galvanometer is zero.



What is an essential requirement for the e.m.f. of the new cell to be measured accurately?

A The e.m.f. of the driver cell must be less than the e.m.f. of the new cell.

B The galvanometer must have a large resistance.

C The internal resistance of the new cell must be zero.

D The resistance per unit length of the wire QR must be constant.

38 The table shows the number of nucleons and the total number of particles (protons, neutrons and electrons) in neutral atoms of four nuclides W, X, Y and Z.

	number of nucleons	total number of particles
W	19	30
X	19	31
Y	21	32
Z	22	31

Which two nuclides are isotopes of each other?

A W and X **B** W and Y **C** X and Z **D** Y and Z

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39 When a sample of a radioactive isotope decays by α-particle emission, the α-particles emitted have a single discrete energy.

When a sample of a radioactive isotope decays by β^- particle emission, the β^- particles emitted have a continuous range of energies.

What is the explanation for this?

A An antineutrino is emitted with a β^- particle but not with an α-particle.

B An antineutrino is emitted with an α-particle but not with a β^- particle.

C The α-particles have much more energy than the β^- particles.

D The β^- particles have much more energy than the α-particles.

40 Some particles are a combination of three quarks.

Which combination of quarks does **not** result in a particle with a charge of either $+1.6 \times 10^{-19}$ C or zero?

A up, down, down

B up, strange, strange

C up, up, down

D up, up, up

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PHYSICS **9702/13**

Paper 1 Multiple Choice **October/November 2023**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	B	**1**
2	A	**1**
3	C	**1**
4	A	**1**
5	D	**1**
6	C	**1**
7	B	**1**
8	D	**1**
9	A	**1**
10	C	**1**
11	C	**1**
12	B	**1**
13	B	**1**
14	A	**1**
15	D	**1**
16	B	**1**
17	B	**1**
18	C	**1**
19	A	**1**
20	D	**1**
21	D	**1**
22	A	**1**
23	B	**1**
24	C	**1**
25	A	**1**
26	B	**1**
27	A	**1**
28	A	**1**

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Question	Answer	Marks
29	C	**1**
30	C	**1**
31	A	**1**
32	D	**1**
33	A	**1**
34	C	**1**
35	C	**1**
36	D	**1**
37	D	**1**
38	B	**1**
39	A	**1**
40	D	**1**

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